Univariate Real Analysis

Muchang Bahng

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Contents

Let's first talk about why we need analysis in general in the first place. Algebra allows us to define certain algebraic structures, which are essentially sets with operations. These operations are defined to have a finite number of arguments. For example, let's take a look at the negation $x \mapsto -x$ and the addition $x, y \mapsto x + y$ operations in a group G. We can compose these operations up to any finite length n, removing the parentheses due to associativity, but note that the "sum" below is not a single operation. It is a composition of $n - 1$ operations.

$$x_1 + x_2 + \ldots + x_n \in G \tag{1}$$
$$-(-(\ldots(-x))) \in G \tag{2}$$

This is still well defined due to closure, but what if we wanted to do this an infinite number of times?

$$x_1 + x_2 + \ldots =? \tag{3}$$
$$\ldots(-(-x)) =? \tag{4}$$

For someone who has learned about sequences and series in high school, this may not be a big jump in logic, but it is. The objects above are not even well-defined and trying to define them with algebraic tools is manifested in the famous Zeno's paradox. So we simply need to add more tools in order to define these new mathematical objects, which we call *series*. To define series, we need to first define sequences. Can we do this with algebra? Yes,[1] since we can simply model it as a function.

Definition 0.1 (Sequence)

A sequence is a function $f : \mathbb{N} \to X$. We usually denote a sequence by writing out the first few terms of the sequence, followed by an ellipsis.

$$a_1 = f(1), a_2 = f(2), \ldots \tag{5}$$

or as an indexed set over the naturals $\{a_i\}_{i \in \mathbb{N}}$.

Therefore, we can consider series as a sequence of finite sums, each element which is well-defined.

$$x_1, x_1 + x_2, x_1 + x_2 + x_3, \ldots \tag{6}$$

For any $n \in \mathbb{N}$, we can get the value of $a_n = \sum_{i=1}^n a_i$, but can we say something about the limiting behavior of a_n? That is, maybe we can just slap a value x onto this series such that it doesn't "break" any of the rules we have in the finite sense. Unfortunately, it is not possible to define such values for all series, but it is possible for some of them, which we call *convergent series*. To rigorously determine which ones are convergent and which ones are not, we need the tools of topology. Defining the concept of sequences that model infinitely composed operations is what allows us to define differentiation and integration.

Great, we've motivated the need for analysis, but before jumping straight into real analysis, let's talk about what analysis in general works with. It studies functions of the form $f : X \to Y$, and minimally both X, Y must be *Banach spaces*, i.e. complete normed vector spaces over some field $\mathbb{F}$. Almost all flavors of analysis, including real ($\mathbb{R}$), complex ($\mathbb{C}$), multivariate ($\mathbb{R}^n$), p-adic, and functional (infinite-dimensional Banach spaces) analysis require *at least* a Banach space structure. Why are Banach spaces so great? Well if we were to define convergence in X or Y, then it only makes sense to talk about convergence with respect to a topology. So X, Y must at least be topological spaces. It would also be bad if we were to take a sequence in X and find out that it converges to some element outside of X. Therefore, we want a notion of *completeness* in the sense that all sequences that "get closer," i.e. Cauchy sequences, actually converge in X. Unfortunately, while convergence of sequences is preserved under homeomorphisms (and is thus a topological property), convergence of Cauchy sequences is not.[2] Furthermore, the notion of uniform convergence is a metric space

[1] In fact, we don't even need algebra, just set theory.

[2] Consider the sequence $a_n = 1/(n+1)$ in $(0, 1)$ and the map $f(x) = 1/x$ to the set $(1, +\infty)$. a_n is Cauchy but $f(a_n)$ is not.

property, not a topological one. Therefore, the concept of distances is crucial to the construction of analysis. As for the norm, I'm still not sure why we need this.[3]

But in college courses such as real and complex analysis, why do we say we work over the *fields* $\mathbb{R}$ and $\mathbb{C}$ rather than the Banach spaces $\mathbb{R}$ and $\mathbb{C}$? This is because of the following theorem.

Theorem 0.1

Every field $\mathbb{F}$ is a 1-dimensional vector space over itself.

Therefore, when we talk about the *field* $\mathbb{R}$, we are really treating it as a vector space $\mathbb{R}$ over the field $\mathbb{R}$.[4] Every other structure beyond this is a "bonus" property that gives us extra tools to prove stronger properties. The most notable is the total ordering on $\mathbb{R}$, which allows us to define upper/lower bounds and other real-analysis specific theorems like the intermediate value theorem or the mean value theorem. Other structures include the inner product or the measure.

Now that we've taken in the big picture, for each type of analysis, we should construct the underlying relevant Banach space. At the very least, we can with the tools of set theory and algebra define the rationals $\mathbb{Q}$ as an ordered field over the quotient space $\mathbb{Z} \times \mathbb{Z} / \sim$. Furthermore, $\mathbb{Q}$ itself is a normed vector space (over $\mathbb{Q}$)[5] and the only thing we need now is completeness.

1. If the norm on $\mathbb{Q}$ is defined as the normal absolute value (Euclidean norm), completing it gives $\mathbb{R}$ as an ordered field which also has a compatible order as that of $\mathbb{Q}$. We study functions mapping to and from $\mathbb{R}$ with *single-variable real analysis*.
2. If we take the *p-adic* norm, then completing it with respect to this gives the *p-adic numbers*, which also forms a field but loses the ordering. We deal with functions over the p-adics with *p-adic analysis*.
3. We can construct $\mathbb{C}$ by taking $\mathbb{R}^2$ and endowing it with a bit more structure. We get *complex analysis*.
4. We can construct $\mathbb{R}^n$ and $\mathbb{C}^n$ by easily defining its vector space structure and then endowing it with a norm, and showing that it is complete with respect to the norm-induced metric. This is known as *multivariate analysis*.
5. With all these defined, we can define Banach function spaces like L^p and perform analysis on operators $f : L^p \to L^q$. This is *functional analysis*.

[3] Aspinwall and Ng told me this, but I'm not sure why. The Frechet derivative seems like it can be purely defined with a metric.

[4] Thanks to Prof. Lenny Ng for clarifying this.

[5] Note that while we define the norm and metric to usually map to $\mathbb{R}^+$, $\mathbb{R}$ isn't even defined yet and so to avoid circular definitions, we define the norm on the rationals to have codomain $\mathbb{Q}$.

1 The Real Numbers

By constructing $\mathbb{Q}$ and its topology in my algebra and topology notes, we can talk about convergence. The first question to ask (if you were the first person inventing the reals) is "how do I know that there exists some other numbers at all?" The first clue is trying to find the side length of a square with area 2. As we see, this number is not rational.

Theorem 1.1 ($\sqrt{2}$ is Not Rational)

There exists no $x \in \mathbb{Q}$ s.t. $x^2 = 2$.

Proof. Assume such a number $x = p/q$ exists, where $\gcd(p, q) = 1$. Then, by the field axioms of $\mathbb{Q}$, we can deduce that

$$2 = \frac{p^2}{q^2} \implies 2q^2 = p^2 \tag{7}$$

This implies that $2 \mid p$, so we have $p = 2p_0$, and we can write $2q^2 = 4p_0^2$. Dividing both sides by 2, we get $q^2 = 2p_0^2$, which implies that $2 \mid q$. This contradicts the fact that p and q are coprime.

We can "imagine" that a square with area 2 certainly exists, but the fact that its side length is undefined is certainly unsettling. I don't know about you, but I would try to "invent" $\sqrt{2}$. We can do this in 4 distinct ways, though some may be more similar than others.

1. *Dedekind Completeness.* I would try and generate new elements by taking a specific *cut*—a partition into two sets such that elements of one set is always greater than that of the other—and seeing which elements lie right in between these cuts. We will often see that we can always find a cut for every rational, but there are additional cuts for each irrational number. This is the idea behind *Dedekind completeness.*

2. *Cauchy Completeness.* I write out the decimal expansion one by one, which gives our first exposure to sequences.

$$1, 1.4, 1.41, 1.414, \ldots \tag{8}$$

   It is clear that on $\mathbb{Q}$, this sequence does not converge. Our intuition tells that that if the terms get closer and closer to each other, they must be getting closer and closer to *something*, though that something is not in $\mathbb{Q}$. This motivates the definition for *Cauchy completeness.*

3. *Nested Interval Completeness.* I would write out maybe some nested intervals so that $\sqrt{2}$ *must* lie within each interval.

$$[1, 2] \supset [1.4, 1.5] \supset [1.41, 1.42] \supset \ldots \tag{9}$$

   This motivates the definition of *nested-interval completeness.*

4. *Least Upper Bound Completeness.* I would define the set of all rationals such that $x^2 < 2$, and try to define $\sqrt{2}$ as the max or supremum of this set. We will quickly find that neither the max nor the supremum exists in $\mathbb{Q}$, and this motivates the definition for *least upper bound completeness.*

All four of these methods points at the same intuition that there should not be any "gaps" or "missing points" in the set that we will construct to be $\mathbb{R}$, which is the general notion of *completeness.* This contrasts with the rational numbers, whose corresponding number line has a "gap" at each irrational value. Even though constructing the reals with one method is sufficient, knowing the different flavors in which completeness manifests is very useful. It allows us to view properties of $\mathbb{R}$ through different lens.

The main division between these four properties is that the first two are methods to directly *construct* the reals from $\mathbb{Q}$, while the latter two are more *axiomatic properties* that we use to verify completeness for a given set. In the next two sections, we will take the rationals and add on extra elements using Dedekind cuts and Cauchy sequences. However, it isn't as conventional (though possible) to construct them as the single point contained in a sequence of nested intervals nor as a supremum of all upper-bounded sets. In fact, an alternative way to construct the reals is to define it axiomatically—as a totally ordered field with either the LUB property or the nested interval property.[6]

Therefore, in the following sections, we will

1. first define the relevant notion of completeness,
2. show that the rationals are not complete
3. and then construct the completed version of the rationals as a version of the reals.

Once we have done this for all three versions, we will unify them by proving they are all equivalent.

There is a second essential property of the reals that is not talked about as often is the Archimidean principle.

Definition 1.1 (Archimidean Principle)

An ordered ring $(X, +, \cdot, \leq)$ that embeds the naturals $\mathbb{N}^a$ is said to obey the **Archimedean principle** if given any $x, y \in X$ s.t. $x, y > 0$, there exists an $n \in \mathbb{N}$ s.t. $\iota(n) \cdot x > y$. Usually, we don't care about the canonical injection and write $nx > y$.

[a] as in, there exists an ordered ring homomorphism $\iota : \mathbb{N} \to X$

Lemma 1.2 (Rationals are Archimidean)

$\mathbb{Q}$ satisfies the Archimidean principle.

Proof. Take any $x = p_1/q_1, y = p_2/q_2 \in \mathbb{Q}$. Then, take $n = q_1 p_2$, and we get

$$nx = q_1 p_2 \frac{p_1}{q_1} = p_1 p_2 > \frac{p_2}{q_2} = y \iff p_1 > \frac{1}{q_2} \tag{10}$$

which must be true since $p_1 \geq 1 \geq \frac{1}{q_2}$.

Usually, we just construct $\mathbb{R}$ right out of $\mathbb{Q}$, and it turns out that the Archimidean principle just trivially follows. However, if we construct $\mathbb{R}$ axiomatically (without the rationals), it needs to be stated. In this axiomatic formulation, we will find that certain types of completeness—like Dedekind completeness—actually *implies* the Archimidean principle, while others—like Cauchy and nested-intervals completeness—does not. Therefore, in a sense, Dedekind completeness is the "strongest" form of completeness.

1.1 Dedekind Completeness

This is an explicit construction from the rationals.

Definition 1.2 (Dedekind Cut)

A **Dedekind cut** is a partition of the rationals $\mathbb{Q} = A \sqcup A'$ satisfying the three properties.[a]

1. $A \neq \emptyset$ and $A \neq \mathbb{Q}$.[b]
2. $x < y$ for all $x \in A, y \in A'$.
3. The maximum element of A does not exist in $\mathbb{Q}$.

The minimum of A' may exist in $\mathbb{Q}$, and if it does, the cut is said to be **generated** by $\min A'$.

[a] This can really be defined for any totally ordered set.
[b] By relaxing this property, we can actually complete $\mathbb{Q}$ to the extended real number line.

[6] In fact, you also need the Archimidean principle, but we'll talk about this later.

Note that in $\mathbb{Q}$, there will be two types of cuts:

1. ones that are generated by rational numbers, such as
$$A = \{x \in \mathbb{Q} \mid x < 2/3\}, A' = \{x \in \mathbb{Q} \mid x \geq 2/3\} \tag{11}$$
2. and the ones that are not
$$A = \{x \in \mathbb{Q} \mid x^2 < 2\}, A' = \{x \in \mathbb{Q} \mid x^2 \geq 2\} \tag{12}$$

We can intuitively see that the set of all Dedekind cuts (A, A') will "extend" the rationals into a bigger set. We can then define some operations and an order to construct this into an ordered field, and finally it will have the property that we call "completeness."

Definition 1.3 (Dedekind Completeness)

A totally ordered algebraic field $\mathbb{F}$ is **Dedekind-complete** if every Dedekind cut of $\mathbb{F}$ is generated by an element of $\mathbb{F}$.

Lemma 1.3 (Rationals are Not Dedekind-Complete)

$\mathbb{Q}$ is not Dedekind-complete.

Proof. Take a look at the cut
$$A = \{x \in \mathbb{Q} \mid x^2 < 2\}, A' = \{x \in \mathbb{Q} \mid x^2 \geq 2\} \tag{13}$$
We wish to show that A' has no minimum. Assume that it does, and call it p. Then, define
$$q \coloneqq p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} \tag{14}$$
We can see that $p > 2 \implies p^2 - 2 > 0 \implies q < p$. But we also see that
$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} \implies q^2 > 2 \tag{15}$$
Therefore, we have found a $q \in A'$ that is smaller than p, a contradiction.

These Dedekind cuts are simply subsets of the power set of $\mathbb{Q}$, which always exists due to the power set axiom. Therefore, we can simply use the axiom of restricted comprehension (?) to create a well-defined set of Dedekind cuts.

Definition 1.4 (Reals as the Dedekind-Completion of Rationals)

Let $\mathbb{R}_D$ be the set of all Dedekind cuts A^a of $\mathbb{Q}$.
$$\mathbb{R}_D \coloneqq \{A \in 2^{\mathbb{Q}} \mid (A, A^c) \text{ is a Dedekind cut}\} \tag{16}$$
By doing this we can intuitively think of a real number as being represented by the set of all smaller rational numbers. Let $A, B \in \mathbb{R}_D$ two Dedekind cuts. Then, we define the following order and operations.

1. *Order.* $A \leq_{\mathbb{R}} B \iff A \subset B$.
2. *Addition.* $A +_{\mathbb{R}} B \coloneqq \{a +_{\mathbb{Q}} b \mid a \in A, b \in B\}$.
3. *Additive Identity.* $0_{\mathbb{R}} \coloneqq \{x \in \mathbb{Q} \mid x < 0\}$.

4. *Additive Inverse.* $-B := \{a - b \mid a < 0, b \in (\mathbb{Q} \setminus B)\}$.
5. *Multiplication.* If $A, B \geq 0$, then we define $A \times_{\mathbb{R}} B := \{a \times_{\mathbb{Q}} b \mid a \in A, b \in B, a, b \geq 0\} \cup 0_{\mathbb{R}}$. If A or B is negative, then we use the identity $A \times B = -(A \times_{\mathbb{R}} -B) = -(-A \times_{\mathbb{R}} B) = (-A \times_{\mathbb{R}} -B)$ to convert A, B to both positives and apply the previous definition.
6. *Multiplicative Identity.* $1_{\mathbb{R}} = \{x \in \mathbb{Q} \mid x < 1\}$.
7. *Multiplicative Inverse.* If $B > 0$, $B^{-1} := \{a \times_{\mathbb{Q}} b^{-1} \mid a \in 1_{\mathbb{R}}, b \in (\mathbb{Q} \setminus B)\}$. If B is negative, then we compute $B^{-1} = -((-B)^{-1})$ by first converting to a positive number and then applying the definition above.

We claim that $(\mathbb{R}, +_{\mathbb{R}}, \times_{\mathbb{R}}, \leq_{\mathbb{R}})$ is a Dedekind-complete totally ordered field, and the canonical injection $\iota : \mathbb{Q} \to \mathbb{R}$ defined

$$\iota(q) = \{x \in \mathbb{Q} \mid x < q\} \tag{17}$$

is an ordered field isomorphism.

[a]For convenience we can uniquely represent (A, A') with just A since $A' = \mathbb{Q} \setminus A$.

Proof.

By the canonical injections $\mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R}_D$, we can talk about whether this set has the Archimedean property. By construction, Archimidean is trivial since $\mathbb{R}_D$ contains $\mathbb{Q}$ which is Archimidean.

Theorem 1.4 (Dedekind Reals is Archimedean)

$\mathbb{R}_D$ satisfies the Archimedean principle.

Proof. $\mathbb{R}_D$ contains $\mathbb{Q}$.

Definition 1.5 (Axiomatic Construction of Dedekind-Reals)

$\mathbb{R}'_D$ is a totally ordered field that is Dedekind complete.[a]

[a]Just for this section, I will denote $\mathbb{R}'$ as reals constructed axiomatically.

Theorem 1.5 (Axiomatic Dedekind Reals is Archimedean)

$\mathbb{R}'_D$ satisfies the Archimedean principle.

Proof. Assume that this property doesn't hold. Then for any fixed x, $nx < y$ for all $n \in \mathbb{N}$. Consider the set

$$A = \bigcup_{n \in \mathbb{N}} (-\infty, nx), \qquad B = \mathbb{R} \setminus A \tag{18}$$

A by definition is nonempty, and B is nonempty since it contains y. Then, we can show that $a \in A, b \in B \implies a < b$ using proof by contradiction. Assume that there exists $a' \in A, b' \in B$ s.t. $a' > b'$. Since $a' \in A$, there exists a $n' \in \mathbb{N}$ s.t. $a' \in (-\infty, n'x) \iff a' < n'x$. But by transitivity of order, this means $b' < n'x \iff b' \in (-\infty, n'x) \implies b' \in A$.

Going back to the main proof, we see that A is upper bounded by y, and so by the least upper bound property it has a supremum $z = \sup A$.

1. If $z \in A$, then by the induction principle[a] $z + x \in A$, contradicting that z is an upper bound.
2. If $z \notin A$, then by the induction principle[b] $z - x \notin A \implies z - x \in B$. Since every element of B upper bounds A and since $x > 0$, this means that $z - x < z$ is a smaller upper bound of A, contradicting that z is a least upper bound.

Therefore, it must be the case that $nx > y$ for some $n \in \mathbb{N}$.

[a]Note that $\mathbb{N}$ is defined recursively as $1 \in \mathbb{N}$ and if $n \in \mathbb{N}$, then $n+1 \in \mathbb{N}$.
[b]The contrapositive of the recursive definition of $\mathbb{N}$ is: if $n \notin \mathbb{N}$, then $n-1 \notin \mathbb{N}$.

1.2 Cauchy Completeness

In many cases we are not working with ordered fields, and so different types of completeness may be more useful. In actual practice, you tend to use Cauchy completeness which only assumes a metric structure.

Definition 1.6 (Cauchy Sequence)

A sequence $(x_n)_n$ in a metric space (X, d) is a **Cauchy sequence** if $\forall \epsilon > 0$, $\exists N \in \mathbb{N}$ s.t.

$$n, m \geq N \implies d(x_n, x_m) < \epsilon \tag{19}$$

To motivate this definition, note that in a general topological space X, we can define convergence of a sequence $x_n \to x$ perfectly fine. However, take some subset $U \subset X$, and let x be a limit point of U. In this case, x_n does not converge in U, but it does converge to something outside of U—namely, $x \in X$. This is similar to $\mathbb{Q} \subset \mathbb{R}$, where x is an irrational point. However, we are trying to *construct* $\mathbb{R}$, so Cauchy convergence allows us to speak of convergence without actually referring to *what* a sequence is converging to.

Note that it is not sufficient to say that a sequence is Cauchy by claiming that each term becomes arbitrarily close to the preceding term. That is,

$$\lim_{n \to \infty} d(x_{n+1}, x_n) = 0 \tag{20}$$

Example 1.1 (Adjacent Terms Converging Doesn't Imply Sequence is Cauchy)

For example, look at the sequence

$$a_n = \sqrt{n} \implies a_{n+1} - a_n = \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{2\sqrt{n}} \tag{21}$$

However, it is clear that a_n gets arbitrarily large, meaning that a finite interval can contain at most a finite number of terms in $\{a_n\}$.

It is often more convenient to think of the limit of the *diameter* of rest of the sequence. That is, a sequence is Cauchy if

$$\lim_{n \to \infty} \operatorname{diam}\{x_m\}_{m \geq n} = 0 \tag{22}$$

It is trivial that convergence implies Cauchy convergence, but the other direction is not true. Therefore, we would like to work in a space where these two are equivalent, and this is called completeness. Therefore, we can construct the reals as equivalence classes over Cauchy sequences. Rather than using the order, we take advantage of the metric.

Definition 1.7 (Cauchy Completeness)

A metric space (X, d) is **Cauchy complete** if every Cauchy sequence in that space converges to an element in X.

$\mathbb{Q}$ is not Cauchy-complete. Let a_n be the largest number x up to the nth decimal expansion such that x^2

does not exceed 2. The first few terms are

$$1.4, 1.41, 1.414, \ldots \tag{23}$$

In this case, we can see that this is Cauchy since at the nth element and on, the first n decimal places are kept fixed and so the most that the rest of the sequence can deviate by is 10^{-n}.

Definition 1.8 (Reals as the Cauchy-Completion of the Rationals)

Let $\mathbb{R}_C$ be the quotient space of all Cauchy (under the Euclidean metric) sequences (x_n) of rational numbers with the equivalence relation $(x_n) = (y_n)$ iff their difference tends to 0.[a] That is, for every rational $\epsilon > 0$, there exists an integer N s.t. for all naturals $n > N$, $|x_n - y_n| < \epsilon$.

1. *Order.* $(x_n) \leq_{\mathbb{R}} (y_n)$ iff $x = y$ or there exists $N \in \mathbb{N}$ s.t. $x_n \leq_{\mathbb{Q}} y_n$ for all $n > N$.
2. *Addition.* $(x_n) + (y_n) \coloneqq (x_n + y_n)$.
3. *Additive Identity.* $0_{\mathbb{R}} \coloneqq (0_{\mathbb{Q}})$.
4. *Additive Inverse.* $-(x_n) \coloneqq (-x_n)$.
5. *Multiplication.* $(x_n) \times_{\mathbb{R}} (y_n) = (x_n \times_{\mathbb{Q}} y_n)$.
6. *Multiplicative Identity.* $1_{\mathbb{R}} \coloneqq (1)$.
7. *Multiplicative Inverse.* $(x_n)^{-1} \coloneqq (x_n^{-1})$.

We claim that $(\mathbb{R}, +_{\mathbb{R}}, \times_{\mathbb{R}}, \leq_{\mathbb{R}})$ is a totally ordered field, and the canonical injection $\iota : \mathbb{Q} \to \mathbb{R}$ defined

$$\iota(q) = (q) \tag{24}$$

is an ordered field isomorphism. Finally, by construction $\mathbb{R}$ is Cauchy-complete.

[a] This equivalence class reflects that the same real number can be approximated in many different sequences. In fact, this shows *by definition* that $1, 1, \ldots$ and $0.9, 0.99, 0.999, \ldots$ are the same number!

Proof.

Theorem 1.6 (Cauchy Reals is Archimidean)

$\mathbb{R}_C$ satisfies the Archimedean principle.

Proof. $\mathbb{R}_C$ contains $\mathbb{Q}$, which is Archimidean.

The best thing about Cauchy completeness is that we can just take $\mathbb{Q}^n$ to create $\mathbb{R}^n$. It becomes quite general. However, note that first, Cauchy completion depends on *which* metric you use to complete it.

Example 1.2 (P-adic Numbers)

Let p be a prime number. For a non-zero rational $x = p^k \cdot \frac{a}{b}$ where $p \nmid a, b$, define the **p-adic norm**[a] as

$$|x|_p \coloneqq p^{-k}, \qquad |0|_p \coloneqq 0 \tag{25}$$

The **p-adic numbers** $\mathbb{Q}_p$ are the Cauchy completion of $\mathbb{Q}$ with respect to the p-adic metric $d_p(x, y) = |x - y|_p$. This set not only does not satisfy the Archimedean principle; it doesn't even have a natural ordering!

[a] This measures divisibility by p: the more p divides x, the smaller $|x|_p$. For example, $|8|_2 = 2^{-3} = 1/8$ and $|3|_2 = 1$.

Definition 1.9 (Axiomatic Construction of Cauchy-Reals)

$\mathbb{R}'_D$ is a totally ordered field that is Cauchy complete and that satisfies the Archimidean principle.

Note that we require the extra Archimidean assumption in the axiomatic construction.

Example 1.3 (Ordered Cauchy-Complete Fields that are Not Archimidean)

Provide examples of ordered, Cauchy-complete fields that are not Archimedean. Hyperreals?

1.3 Least Upper Bound Completeness

Definition 1.10 (Least Upper Bound Property)

A totally ordered algebraic field $\mathbb{F}$ (must it be a field?) is **least-upper-bound complete**, or is said to satisfy the **least upper bound (LUB) property**, if every nonempty set of $\mathbb{F}$ having an upper bound must have a least upper bound (supremum) in $\mathbb{F}$.

Theorem 1.7 (LUB is Equivalent to GLB)

A set $(\mathbb{F}, \leq)$ has the least upper bound property iff it has the *greatest lower bound property*, which states that every set bounded below has a greatest lower bound.

Proof. We will prove one direction since the other is the same logic. Let $S \subset X$ be a nonempty set that is bounded below by some $l \in X$. Let $L \subset X$ be the set of all lower bounds of S. Since l exists, it is nonempty. Furthermore, L is bounded above by any element of S. Due to LUB property L has a least upper bound, call it $z = \sup L$. We claim that $z = \inf S$.

1. z is a lower bound of S. Assume that it is not. Then there exists $s \in S$ s.t. $s < z$. But by construction s is an upper bound for L and so z s not the *least* upper bound, a contradiction.
2. z is a *greatest* lower bound. Assume that z is not. Then there exists a $z' \in X$ s.t. $z < z' \leq s$ for all $s \in S$. But since z', z are lower bounds, this means $z, z' \in L$ by definition and $z < z'$ contradicts the fact that z is an upper bound of L.

We are done.

$\mathbb{Q}$ does not satisfy the least upper bound property, but proving this can be tricky for the first time. We state this as a lemma.

Theorem 1.8 (Rationals Doesn't Satisfy LUB Property)

$\mathbb{Q}$ does not satisfy the LUB property.

Proof. Assume it does, and let us denote

$$p \coloneqq \sup\{x \mid \mathbb{Q} \mid x^2 < 2\} \in \mathbb{Q} \tag{26}$$

The key here is to find another rational that we can always "squeeze" in between p and 2. This can be done with the Archimidean principle, which is already satsified in $\mathbb{Q}$. Since we have proved that there exists no rational that squares to 2, we only need to consider the two cases.

1. $p^2 < 2$. Take $\epsilon \in \mathbb{Q}$ so small that

$$p^2 + 2p\epsilon + \epsilon^2 = (p + \epsilon)^2 < 2 \tag{27}$$

To show complete steps, we can see that by density of reals, there exists some rational r s.t. $0 < r < 2 - p^2$. Therefore, we can invoke Archimidean principle to find a $n \in \mathbb{N}$ s.t. $\epsilon = p/n < 2p/n < r$. Therefore, p is not an upper bound, so this cannot be true.

2. $p^2 > 2$. We can again take an $\epsilon \in \mathbb{Q}$ so small that

$$p^2 - 2p\epsilon + \epsilon^2 = (p - \epsilon)^2 > 2 \tag{28}$$

Therefore, p is not least, so this cannot be true.

Definition 1.11 (Axiomatic Construction of Reals with LUB Property)

$\mathbb{R}'_I$ is a totally ordered field that satisfies the least upper bound property.

Note that we don't need to explicitly assume Archimidean principle here. The LUB property is strong enough that it implies Archimidean!

Theorem 1.9 (LUB Property Implies Archimidean)

$\mathbb{R}'_I$ is Archimidean.

Now let's see how our previous constructions of the reals relate to the LUB property.

Theorem 1.10 (Dedekind Completed Reals Satisfies LUB Property)

$\mathbb{R}_D$ satisfies the least upper bound property.

Proof. Let $\mathcal{A}$ be a nonempty subset of $\mathbb{R}_D$ bounded from above by T. Then, $\forall A \in \mathcal{A}$, $A \coloneqq (A, A^c)$ is a Dedekind cut, and we can define

$$(B, B') \coloneqq \bigg(\bigcup_{A \in \mathcal{A}} A, \bigcap_{A \in \mathcal{A}} A^c \bigg) \tag{29}$$

We claim that this is a Dedekind cut.

1. First, it is nonempty set $\mathcal{A} \neq \emptyset$ and so for each $A \in \mathcal{A}$, $A \subset \mathbb{Q}$ is nonempty. It is also not all of $\mathbb{Q}$ since T is an upper bound of $\mathcal{A}$, and so $T \geq a \; \forall a \in A \; \forall A \in \mathcal{A}$, which implies that

$$T \notin \bigcup_{A \in \mathcal{A}} A \tag{30}$$

2. Now let $x \in B, y \in B'$. Then, $x \in A_0$ for some $A_0 \in \mathcal{A}$, and since y is in the intersection of all the corresponding A^c, it must be in the corresponding A_0^c. Therefore we invoke the Dedekind cut property of (A_0, A_0^c) and see $x < y$.
3. Finally, for the sake of contradiction, let $m \in \mathbb{Q}$ be the maximum of B. Then, $m \in A^*$ for some $A^* \in \mathcal{A}$. But m is an upper bound for the whole B, so this means that $m = \max\{A^*\}$, which contradicts the fact that lower cut cannot have a rational maximum.

Now we claim that B is the supremum. It is an upper bound since

$$A \subset B = \bigcup_{A \in \mathcal{A}} A \quad \forall A \in \mathcal{A} \tag{31}$$

To prove least, we should see that if (S, S^c) is another upper bound of $\mathcal{A}$, then $A \subset S$ for all $A \in \mathcal{A} \implies B = \bigcup_{A \in \mathcal{A}} A \subset S$, which establishes that B is least.

1.4 Nested Intervals Completeness

The next flavor we present is nested-intervals completeness. This is the least popular way to construct the reals, and it is used more as a post-hoc tool to analyze the reals after you construct it using either of the two previous methods.

Definition 1.12 (Nested Interval Completeness)

Let $\mathbb{F}$ be a totally ordered algebraic field. Let $I_n = [a_n, b_n]$ $(a_n < b_n)$ be a sequence of decreasing nested intervals that are

1. closed,
2. bounded,
3. nested, $I_1 \supset I_2 \supset I_3 \supset \ldots$
4. and decreasing to 0 in the sense that $\lim_{n\to\infty} b_n - a_n = 0$.

$\mathbb{F}$ is **nested-interval complete** if the intersection of all of these intervals I_n contains exactly one point.

$$\bigcup_{n=1}^{\infty} I_n \in \mathbb{F} \tag{32}$$

Note that defining nested intervals requires only an ordered field. One may look at this and try to ask if this is a specific instance of the following conjecture: The intersection of a nested sequence of nonempty closed sets in a topological space has exactly 1 point. This claim may not even make sense, actually. If we define nested in terms of proper subsets, then for a finite topological space a sequence cannot exist since we will run out of open sets and so this claim is vacuously true and false. If we allow $S_n = S_{n+1}$ then we can just select $X \supset X \supset \ldots$, which is obviously not true. However, a slightly weaker claim is that every proper nested non-empty closed sets has a non-empty intersection is a consequence of compactness.

Theorem 1.11 (Rationals are Not Nested-Interval Complete)

$\mathbb{Q}$ is not nested-interval complete.

Proof. This is a nice proof that uses a class method of algorithmically selecting nested intervals that satisfy a following property. This trick will be used many times in analysis.

For the sake of contradiction, let us assume that $\mathbb{Q}$ satisfies nested intervals completeness. Since $\mathbb{Q}$ is countable, enumerate it as $q_1, q_2, \ldots$.

1. Choose any closed interval I_1 of length 1 with rational endpoints that doesn't contain q_1.
2. Now partition I_1 into two segments of equal length, and choose I_2 to be the segment that doesn't contain q_2.

At this point, I_n is a closed interval with rational endpoints of length 2^{-n} that cannot contain q_n. Therefore, $\cap_n I_n$ cannot contain any $q_n \in \mathbb{Q}$. But this contradicts our assumption of nested interval completeness.

One may ask: what is the relationship between LUB property and nested-intervals completeness? It turns out that they are equivalent.

Theorem 1.12 (LUB is Equivalent to Nested Interval Completeness in Reals)

Listed.

1. $\mathbb{R}'_L$ satisfies nested intervals completeness.
2. $\mathbb{R}'_I$ satisfies least upper bound completeness.

Proof. Listed.

1. Note that $\{a_n\}$ is bounded above by b_1. Therefore by LUB property it must have a supremum, call it $x = \sup_n\{a_n\}$. Then, we see that $a_n \leq x \leq b_n$ for all n, and so x is in the intersection.

Theorem 1.13 (Cantor's Intersection Theorem)

$\mathbb{R}$ is nested-interval complete.

Proof. We prove this by first proving the claim that given nested, closed, and bounded sets I_n (not even necessarily intervals), then

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset \tag{33}$$

Suppose this is not true. For every $x \in \mathbb{R}$, there exists a n_x s.t. $x \notin I_n$ (and all later I_m for $m > n$). Let $O_n = I_n^c$ open sets. Then, $\mathbb{R} \subset \cup_n O_n$ In particular, $I_1 \subset \cup_n O_n$. But I_1 is closed and bounded. So we can extract a finite subcover. $O_{n_1}, O_{n_2}, \ldots, O_{n_m}$ (ordered $n_1 < n_2 < \ldots < n_m$). Then since O_n are increasing, $I_1 \subset O_{n_m} = I_{n_m}^c$. But $I_{n_m} \subset I_1$, a contradiction.
Now with this, we know that because the limits of the endpoints of the intervals go to 0, then there cannot be more than 2 points in the intersection. Thus there must be 1 unique point.

Definition 1.13 (Axiomatic Construction of Reals with Nested Intervals)

$\mathbb{R}_I$ is a totally ordered field that

1. satisfies the nested intervals completeness, and
2. satisfies the Archimidean principle.

1.5 Properties of the Real Line

Perfect, now all that remains is to unite the two constructions of the reals.

Theorem 1.14 (Dedekind and Cauchy Complete Reals are Isomorphic)

Given that $\mathbb{R}_D$ is the Dedekind-completed version of the rationals and $\mathbb{R}_C$ is the Cauchy-completed version of the rationals, we claim that the two are isomorphic as ordered fields.

$$\mathbb{R}_D \simeq \mathbb{R}_C \tag{34}$$

Proof.

Therefore, it doesn't really matter which one we talk about, and we can refer to *the* real numbers as a single set. Great! Now we can finally feel satisfied about defining metrics, norms, and inner products as mappings to the codomain $\mathbb{R}$.

Definition 1.14 (Reals (as Construction from Rationals))

The **reals** $\mathbb{R}$ is the totally ordered complete Archimedean field constructed as the completion[a] of $\mathbb{Q}$.

[a]either one

So far, we have taken the completion of the rationals as our main mode of construction. However, we can take an axiomatic approach, and it turns out that there is only one set, up to isomorphism, that satisfies all these properties.

Theorem 1.15 (Axiomatic Definition of Reals)

The **real numbers**, denoted $\mathbb{R}$, is any totally ordered complete Archimedean field. $\mathbb{R}$ is unique up to field isomorphism. That is, if two individuals construct two ordered complete Archimedean fields $\mathbb{R}_A$ and $\mathbb{R}_B$, then

$$\mathbb{R}_A \simeq \mathbb{R}_B \tag{35}$$

Proof. The proof is actually much longer than I expected, so I draw a general outline.[a] We want to show how to construct an isomorphism $f : \mathbb{R}_A \to \mathbb{R}_B$.

1. Realize that there are unique embeddings of $\mathbb{N}$ in $\mathbb{R}_A$ and $\mathbb{R}_B$ that preserve the inductive principle, the order, closure of addition, and closure of multiplication, the additive identity, and the multiplicative identity. Call these ordered doubly-monoid (since it's a monoid w.r.t. $+$ and $\times$) homomorphisms ι_A, ι_B.
2. Construct an isomorphism $f_1 : \iota_A(\mathbb{N}) \to \iota_B(\mathbb{N})$ that preserves the inductive principle, order, addition, and multiplication. This is easy to do by just constructing $f_1 = \iota_B \circ \iota_A^{-1}$.
3. Extend f_1 to the ordered ring isomorphism f_2 by explicitly defining what it means to map additive inverses, i.e. negative numbers.
4. Extend f_2 to the ordered field isomorphism f_3 by explicitly defining what it means to map multiplicative inverses, i.e. reciprocals.
5. Extend f_3 to the ordered field isomorphism on the entire domain $\mathbb{R}_A$ and codomain $\mathbb{R}_B$. There is no additional operations that we need to support, but we should explicitly show that this is both injective and surjective, which completes our proof.

[a]Followed from here.

It seems that the real numbers is *any* set that satisfies the definition above. Therefore, a line $\mathbb{L}$ with $+$ associated with the translation of $\mathbb{L}$ along itself and $\cdot$ associated with the "stretching/compressing" of the line around the additive origin 0 is a valid representation of the reals. $\mathbb{R}$ can also be represented as an uncountable list of numbers with possibly infinite decimal points, known as the decimal number system.

$$\ldots, -2.583\ldots, \ldots, 0, \ldots, 1.2343\ldots, \ldots, \sqrt{2}, \ldots \tag{36}$$

The first property we should know is that the reals are uncountable.

Theorem 1.16 (Cantor's Diagonalization)

$\mathbb{R}$ is uncountable.

Proof. We proceed by contradiction. Suppose the real numbers are countable. Then there exists a bijection $f : \mathbb{N} \to \mathbb{R}$. This means we can list all real numbers in $[0, 1]$ as an infinite sequence.[a]

$$f(1) = 0.a_{11}a_{12}a_{13}\ldots \tag{37}$$
$$f(2) = 0.a_{21}a_{22}a_{23}\ldots \tag{38}$$
$$f(3) = 0.a_{31}a_{32}a_{33}\ldots \tag{39}$$
$$\vdots \tag{40}$$

where each a_{ij} is a digit between 0 and 9.

Now construct a new real number $r = 0.r_1 r_2 r_3 \ldots$ where:

$$r_n = \begin{cases} 1 & \text{if } a_{nn} \neq 1 \\ 2 & \text{if } a_{nn} = 1 \end{cases} \tag{41}$$

This number r is different from $f(n)$ for every $n \in \mathbb{N}$, since r differs from $f(n)$ in the nth decimal place. Therefore $r \in [0, 1]$ but $r \notin \text{range}(f)$, contradicting that f is surjective. Thus our assumption that the real numbers are countable must be false.

[a] This must be explicitly proven, but we can take the set of all Cauchy sequences of rationals in their decimal expansion and construct the reals this way.

With this, we can add the inner product, metric, and topology.

1.6 Exponentials, Roots, and Logarithms

Now we will focus on some other operations that become well-defined in the reals. We know that x^n for $n \in \mathbb{N}$ denotes repeated multiplication and x^{-1} denotes the multiplicative inverse. We need to build up on this notation. As a general outline, we will show that x^{-n} is well defined, then $x^q, q \in \mathbb{Q}$ is well-defined, and finally $x^r, r \in \mathbb{R}$ is well-defined. For the naturals, we have defined x^n as the repeated multiplication of n. It is trivial that the canonical injection $\iota_0 : \mathbb{N} \to \mathbb{R}$ commutes with the exponential map of naturals. We prove that $\iota_1 : \mathbb{Z} \to \mathbb{R}$ also commutes.

Lemma 1.17 (Integer Exponents)

We have

1. For $x_1, \ldots, x_n \in \mathbb{R}$, $(x_1 \ldots x_n)^{-1} = x_n^{-1} \ldots x_1^{-1}$.
2. For $x \in \mathbb{R}$, $x > 0$, $(x^n)^{-1} = (x^{-1})^n$. This value is denoted x^{-n}.
3. For $x \in \mathbb{R}$ and $w, z \in \mathbb{Z}$, $x^{w+z} = x^w x^z$.
4. For $w, z \in \mathbb{Z}$, $x^{wz} = (x^z)^w = (x^w)^z$.

Proof. Listed.

1. The proof is trivial, but for $n = 2$ and $x_1 = x, x_2 = y$, we see that by associativity, $(x^{-1}x^{-1})(xy) = y^{-1}(x^{-1}x)y = y^{-1}y = 1$ and we know inverses are unique.
2. Set $x_i = x$ using (1).
3. If $w, z > 0$ this is trivial by the associative property. If either or both are negative, say $w < 0 < z$, then we set $w' = -w > 0$ and using (2) we know that

$$x^w x^z = (x^{-1})^{w'} x^z = x^{-w'+z} = x^{w+z} \tag{42}$$

by associativity in the second last equality.

Therefore, we have successfully defined x^z for all $z \in \mathbb{Z}$, and if z is negative, we're allowed to "swap" the -1 and $|z|$ in the exponents. Now we want to extend this into rational exponents, first by proving the existence and uniqueness of nth roots for any real. The proof is a little involved, but the general idea is that we want to use the LUB property to define the nth root as the supremum of a set.

Theorem 1.18 (Existence of Nth Roots)

For any real $x > 0$ and every $n \in \mathbb{N}$ there is one and only one positive real $y \in \mathbb{R}$ s.t. $y^n = x$. This is denoted $x^{1/n}$.

Proof. Let E be the set consisting of all reals $t \in \mathbb{R}$ s.t. $t^n < x$. We show that

1. it is nonempty. Consider $t = x/(1+x)$. Then $0 \leq t < 1 \implies t^n \leq t < x$. Thus $t \in E$ and E is nonempty.
2. it is bounded. Consider any number $s = 1 + x$. Then $s^n \geq s > x$, so $s \notin E$, and $s = 1 + x$ is an upper bound of E.

Therefore, E is a nonempty set that is upper bounded, so it has a least upper bound, called $y = \sup E$. We claim that $y^n = x$, proving by contradiction. For both cases, we use the fact that the identity $b^n - a^n = (b-a)(b^{n-1} + b^{n-2}a + \ldots + a^{n-1})$ gives the inequality

$$b^n - a^n < (b-a)nb^{n-1} \text{ for } 0 < a < b \tag{43}$$

1. Assume $y^n < x$. Then we choose a fixed $0 < h < 1$ s.t.
$$h < \frac{x - y^n}{n(y+1)^{n-1}} \tag{44}$$
Then by putting $a = y, b = y + h$, we have
$$(y+h)^n - y^n < hn(y+h)^{n-1} < hn(y+1)^{n-1} < x - y^n \tag{45}$$
and thus $y^n < (y+h)^n < x$. This means that $y + h \in E$, and so y is not an upper bound.
2. Assume $y^n > x$. Then we set a fixed number
$$k = \frac{y^n - x}{ny^{n-1}} \tag{46}$$
Then $0 < k < y$. If we take any $t \in \mathbb{R}$ s.t. $t \geq y - k$, this implies that $t^n \geq (y-k)^n \implies -t^n \geq -(y-k)^n$, and so
$$y^n - t^n \leq y^n - (y-k)^n < kny^{n-1} = y^n - x \tag{47}$$
Thus $t^n > x$ and $t \notin E$. So it must be the case that $t < y - k$, and so $y - k$ is an upper bound of E, contradicting that y is least.

At this point, rooting has been introduced as sort of an independent map from exponentiation. We show that they have the nice property of commuting.

Lemma 1.19 (Rooting and Exponentiation Commute)

For $p \in \mathbb{Z}, q \in \mathbb{N}$ and $x \in \mathbb{R}$ with $x > 0$, we have

$$(x^p)^{1/q} = (x^{1/q})^p \tag{48}$$

Proof. If $p > 0$, then let $r = (x^p)^{1/q}$. By definition $r^q = x^p$. Let $s = x^{1/q}$ By definition $s^q = x$. Therefore $r^q = (s^q)^p = s^{qp}$ from the lemma on integer exponents. But since roots are well-defined and unique

$$r = (r^q)^{1/q} = (s^{qp})^{1/q} = s^p \implies (x^p)^{1/q} = (x^{1/q})^p \tag{49}$$

If $p = 0$, this is trivially 0, and if $p < 0$ the by the same logic with $p = -p'$ for $p' > 0$ and $y = x^{-1} > 0$. we know

$$(x^p)^{1/q} = \big((y^{-1})^{-p'}\big)^{1/q} = (y^{-(-p')})^{1/q} = (y^{p'})^{1/q} \tag{50}$$
$$= (y^{1/q})^{p'} = ((x^{-1})^{1/q})^{p'} = (x^{1/q})^{-p'} = (x^{1/q})^p \tag{51}$$

Theorem 1.20 (Rational Exponential Function)

Given $m, p \in \mathbb{Z}$ and $n, q \in \mathbb{N}$, prove that

$$(b^m)^{1/n} = (b^p)1/q \tag{52}$$

Hence it makes sens to define $b^r = (b^m)^{1/n}$, since every element of the equivalence class r of each rational number maps to the same value.

Proof. Since $m/n = p/q \implies mq = np$,

$$\begin{aligned} b^{mq} = b^{np} &\implies (b^m)^q = (b^p)^n && (53) \\ &\implies b^m = ((b^m)^q)^{1/q} = ((b^p)^n)^{1/q} && (54) \\ &\implies b^m = ((b^p)^{1/q})^n && (55) \\ &\implies (b^m)^{1/n} = (b^p)^{1/q} && (56) \end{aligned}$$

Therefore we can define for any $r \in \mathbb{Q}$

$$x^r = x^{m/n} = (x^m)^{1/n} = (x^{1/n})^m \tag{57}$$

where the final equality holds from the commutativity of rooting and exponentiation.

It turns out that this is a homomorphism.

Corollary 1.21 (Rational Exponential Function is a Homomorphism)

The rational exponential function is a homomorphism. That is, given $r, s \in \mathbb{Q}$ and $x \in \mathbb{R}$,

$$x^{r+s} = x^r \cdot x^s \tag{58}$$

Proof. Let $r = m/n, s = p/q$. Then

$$\begin{aligned} x^{r+s} = x^{m/n+p/q} &= x^{\frac{mq+np}{nq}} && \text{(addition on } \mathbb{Q}) \\ &= (x^{mq+np})^{1/nq} && \text{(exp and roots commute)} \\ &= (x^{mq} + x^{np})^{1/nq} && \text{(int exp lemma)} \\ &= (x^{mq})^{1/nq}(x^{np})^{1/nq} && \text{(int exp lemma)} \\ &= x^{mq/nq}x^{np/nq} && \text{(exp and roots commute)} \\ &= x^{m/n}x^{p/q} && \text{(relation from } \mathbb{Q}) \end{aligned}$$

With rational exponents defined, we can use the least upper bound property to define a consistent extension of a real exponent.

Lemma 1.22

If $r \in \mathbb{Q}$ with $r \geq 0$, then for $x \in \mathbb{R}$, $x > 1$, $1 \leq b^r$.

Proof. Let $r = m/n$. Then $x^r = x^{m/n} = (x^m)^{1/n}$. Since $1 < x$, and $m \geq 0$, we have

$$1 \leq x \leq x^2 \leq \ldots \leq x^m \implies 1 \leq b^m \tag{59}$$

Now set $y = x^{m/n}$ and assume that $y < 1$. Then

$$x^m = y^n < y^{n-1} < \ldots < y < 1 \tag{60}$$

and so $x^m < 1$, which is a contradiction. So it must be the case that $y > 1$.

Lemma 1.23 (Monotonicity of Rational Exponents)

If $x, y \in \mathbb{R}$, then for any rational $r \in \mathbb{Q}$ with $r < x + y$, there exists a $p, q \in \mathbb{Q}$ s.t. $p < x, q < y$ and $p + q = r$. The converse is true as well.

Proof. $r < x + y \implies r - y < x$. By density of $\mathbb{Q}$ in $\mathbb{R}$, we can choose $r - y < p < x$. Then $-r + y > -p > x \implies r - r + y > r - p > r - x \implies y > r - p > r - x$, and we set $q = r - p$. We are done. The converse is trivial since given $p, q \in \mathbb{Q}$ with $p < x, q < y$, then by the ordered field properties $p + q < x + y$.

Corollary 1.24 (Real Exponential Function)

Given $x \in \mathbb{R}$, we define

$$B(x) \coloneqq \{x^q \in \mathbb{R} \mid q \in \mathbb{Q},\ q \leq x\} \tag{61}$$

We claim that given $r \in \mathbb{R}$,

$$x^r \coloneqq \sup B(r) \tag{62}$$

is well-defined and is a homomorphism extension of the rational exponential function. That is,

$$\sup B(x + y) = \sup B(x) \cdot \sup B(y) \tag{63}$$

Proof. To show that $x^r \coloneqq \sup B(r)$ where $B(r) = \{x^t \in \mathbb{R} \mid t \in \mathbb{Q}, t \leq r\}$,

1. We show it's an upper bound. Assume it wasn't. Then $x^r < x^t$ for some $t \in \mathbb{Q}$ satisfying $t \leq r$. But $t \leq r \implies 0 \leq r - t$, and by the previous lemma, $1 \leq x^{r-t}$. So $1 \leq x^{r-t} = x^r x^{-t} = x^r (x^t)^{-1} \implies x^t \leq x^r$, which is a contradiction.
2. We show that it is least. Assume that it is not. Then $\exists r' \in \mathbb{Q}$ s.t. $x^t \leq x^{r'}$ and $r' < r$. Now let $s \in \mathbb{Q}$ be an element between r' and r, which is guaranteed to exist due to density of rationals in reals. But $s < r$, so by definition $x^s \in B(r)$, but

$$\begin{aligned} 0 < s - r' &\implies 1 < b^{s-r'} && (64)\\ &\implies b^{r'}(b^{r'})^{-1} < b^s(b^{-r'}) && (65)\\ &\implies 1 < b^s(b^{r'})^{-1} && (66)\\ &\implies b^{r'} < b^s && (67) \end{aligned}$$

   and so $b^{r'}$ is not an upper bound for $B(r)$. By contradiction, b^r is least.

Since this is defined, the analogous definition for real numbers is consistent with that of hte rationals, and it is upper bounded by the Archimedean principle, so such a supremum must exist. Note that t is rational. For the second part, from the previous lemma and the homomorphism properties of the rational exponent,

$$\begin{aligned} B(x + y) = B'(x + y) &\coloneqq \{b^{p+q} \in \mathbb{R} \mid p, q \in \mathbb{Q}, p \leq x, q \leq y\} && (68)\\ &= \{b^p b^q \in \mathbb{R} \mid p, q \in \mathbb{Q}, p \leq x, q \leq y\} && (69)\\ & && (70) \end{aligned}$$

Therefore we can treat B and B' as the same set.

1. Prove upper bound $\sup B(x+y) \leq \sup B(x) \sup B(y)$. Given $\alpha \in B'(x+y)$, there exists $p_\alpha, q_\alpha \in \mathbb{Q}$ (with $p_\alpha < x$, $q_\alpha < y$) s.t. $b^{p_\alpha} b^{q_\alpha} = \alpha$. But
$$b^{p_\alpha} b^{q_\alpha} \leq \sup_{p_\alpha}\{b^{p_\alpha}\} \cdot \sup_{q_\alpha}\{b^{q_\alpha}\} = \sup B(x) \sup B(y) \tag{71}$$
2. To prove least, assume there exists $K \in \mathbb{R}$ s.t. $\sup B'(x+y) \leq K < \sup B(x) \sup B(y)$. Then, since the image of b^x is always positive, we assume $0 < K$. We bound its factors as so: $K < \sup B(x) \sup B(y) \implies K/\sup B(x) < \sup B(y)$. By density of the rationals, there exists a $\beta \in \mathbb{Q}$, s.t.
$$\frac{K}{\sup B(x)} < \beta < \sup B(y) \tag{72}$$
This means $K/\beta < \sup B(x)$ and $\beta < \sup B(y)$. But this means that there exists $\phi, \gamma \in B(x), B(y)$ s.t. $K/\beta < \phi, \beta < \gamma \implies K = (K/\beta) \cdot \beta < \phi\gamma \implies \phi\gamma \in B'(x+y)$ by definition. So K is not an upper bound.

Furthermore, this is an isomorphism, and the inverse is defined. Let's define this analytically.

Theorem 1.25 (Logarithm)

For $b > 1$ and $y > 0$, there is a unique real number x s.t. $b^x = y$. We claim
$$x = \sup\{w \in \mathbb{R} \mid b^w < y\} \tag{73}$$
x is called the **logarithm of y to the base b**.

Proof. We use the inequality $b^n - 1 \leq n(b-1)$ for all $n \in \mathbb{N}$.[a] By substituting $b = b^{1/n}$ (valid since $b > 1 \iff b^{1/n} > 1$) so $b - 1 \geq n(b^{1/n} - 1)$. Now set some $t > 1$, and by Archimidean principle, we can choose some $n \in \mathbb{N}$ s.t. $n > \frac{b-1}{t-1}$. Then $n(t-1) > b-1$, and with the inequality derived we get
$$n(t-1) > b - 1 \geq n(b^{1/n} - 1) \implies t > b^{1/n} \tag{74}$$
This allows us to prove 2 things.

1. If w satisfies $b^w < y$, then $b^{w+(1/n)} < y$ for sufficiently large n. Setting $t = yb^{-w}$ (which is greater than 1 since $b^w < y$) gives $y \cdot b^{-w} > b^{1/n} \implies b^w b^{1/n} < y \implies b^{w+(1/n)} < y$.
2. If w satisfies $b^w > y$, then $b^{w-(1/n)} > y$ for sufficiently large n. Setting $t = b^w y^{-1}$ (which is greater than 1 since $b^w > y$) gives $b^w y^{-1} > b^{1/n} \implies b^{w-(1/n)} > y$.

Now we can prove existence. Let A the set of all w s.t. $b^w < y$. We claim that $x = \sup A$.

1. Assume that $b^x < y$. We know that there exists $n \in \mathbb{N}$ s.t. $b^{x+(1/n)} < y \implies x + (1/n) \in A$, contradicting that x is an upper bound.
2. Assume that $b^x >$. We know that there exists $n \in \mathbb{N}$ s.t. $b^{x-(1/n)} > y \implies x - (1/n)$ is also an upper bound for A, contradicting that x is least. Therefore $b^x = y$.

We now prove uniqueness. Assume that there are two such x's , call them x, x'. By total ordering and $x \neq x'$, WLOG let $x > x' \implies x - x' > 0 \implies b^{x-x'} > 1$. By density of rationals, since we can choose $r \in \mathbb{R}$ s.t. $0 < r < x - x'$, we have $1 < b^r < b^{x-x'}$ and so $B(r) \subset B(x - x')$. Since $1 < b^{x-x'} \implies 1 \cdot b^{x'} < b^{x-x'} \cdot b^{x'} = b^x$, we have $b^{x'} < b^x$ and they cannot both by y. So $x = x'$.

[a] We prove by induction. For $n = 1$ $b^1 - 1 \leq 1(b-1)$. Assume that this holds for some n. Then $b^{n+1} - 1 = b^{n+1} - b + b - 1 = b(b^n - 1) + (b-1) \geq bn(b-1) + (b-1) = (bn+1)(b-1) \geq (n+1)(b-1)$, where the last step follows since $b \geq 1 \implies bn \geq n \implies bn + 1 \geq n + 1$.

1.7 Extended Reals

Often, we deal with numbers that are not finite, and we would like to have a system to incorporate $\pm\infty$ into the real line. Most first courses glaze over this, but it's important to see the construction as well. The problem is that we can't really add in these numbers without breaking a lot of the algebraic properties, but let's see for ourselves. It should be pretty obvious that we want (note the strict inequalities)

$$-\infty < x < +\infty \quad \forall x \in \mathbb{R} \tag{75}$$

To define addition, we can't make $x + \infty$ a finite number since then

$$\infty \leq x + \infty = y \tag{76}$$

which is a contradiction. So $x + \infty = +\infty$. We keep doing this but the main problem comes in with trying to define $\infty - \infty$ or ∞/∞, which are known as **indeterminate forms**. These are particularly bad since we cannot deduce $x = y$ from $x + \infty = y + \infty$ or from $x \cdot \infty = y \cdot \infty$. The solution to this is to *simply avoid them*[7] by making these indeterminate terms undefined.

Definition 1.15 (Extended Real Number Line)

The **extended real number line** is the set $\mathbb{R} \cup \{\pm\infty\}$ with the following operations.

1. *Order.* $-\infty \leq x$ and $x \leq +\infty$ for all $x \in \overline{\mathbb{R}}$.
2. *Addition.*

$$\forall x \in \mathbb{R}, x + \infty = \infty + x = +\infty \tag{77}$$
$$\forall x \in \mathbb{R}, x - \infty = \infty - x = -\infty \tag{78}$$
$$+\infty + \infty = +\infty \tag{79}$$
$$-\infty - \infty = -\infty \tag{80}$$
$$+\infty - \infty, -\infty + \infty \text{ are undefined} \tag{81}$$

3. *Multiplication.*[a]

$$\forall x \in \mathbb{R} \setminus \{0\}, x \cdot +\infty = +\infty \cdot x = \begin{cases} +\infty \text{ if } x > 0 \\ -\infty \text{ if } x < 0 \end{cases} \tag{82}$$
$$\forall x \in \mathbb{R} \setminus \{0\}, x \cdot -\infty = -\infty \cdot x = \begin{cases} -\infty \text{ if } x < 0 \\ -\infty \text{ if } x > 0 \end{cases} \tag{83}$$
$$0 \cdot +\infty = +\infty \cdot 0 = 0 \tag{84}$$
$$0 \cdot -\infty = -\infty \cdot 0 = 0 \tag{85}$$
$$+\infty \cdot +\infty = -\infty \cdot -\infty = +\infty \tag{86}$$
$$+\infty \cdot -\infty = -\infty \cdot +\infty = +\infty \tag{87}$$
$$\tag{88}$$

[a]The fact that $0 \cdot \infty = 0$ might sound odd. Look at the extension into hyperreals later.

It turns out that this is still Dedekind-complete, which is nice. Unfortunately this is not even a field since the multiplicative inverse of $\pm\infty$ is not defined. Furthermore, we lose the Archimedean property.

The general rule of thumb is that if one wishes to use cancellation, this is only safe if one can guarantee that the numbers we work with are all finite. If we must work with infinity, another way is to work with the nonnegative reals.

[7]as far as I know

Definition 1.16 (Extended Real Number Line)

The **extended nonnegative reals** is the set $\mathbb{R}_{\geq 0} \cup \{+\infty\}$ with the following operations.

1. *Order.* $x \leq +\infty$ for all $x \in \overline{\mathbb{R}}$.
2. *Addition.*
$$\forall x \in [0, +\infty], +\infty + x = x + \infty = +\infty \tag{89}$$
3. *Multiplication.*
$$\forall x \in (0, +\infty], +\infty \cdot x = x \cdot +\infty = +\infty \tag{90}$$
$$0 \cdot +\infty = +\infty \cdot 0 = 0 \tag{91}$$

There is a tradeoff here: we can work with infinity, or negative numbers, but not both. Also, note that if we define $\infty \cdot 0 = 0$, the multiplication becomes *upward continuous*, but not *downwards continuous*. This leads to an asymmetry when defining integrals, but in univariate analysis we will only work with bounded functions, and this will not hinder us until measure theory.

1.8 Hyperreals

The loss of the field property of the extended reals is quite bad, and we might want to recover this. Therefore, we can add more elements that serve to be the multiplicative inverse of infinity. We call these inverses *infinitesimals* and the new set the *hyperreal numbers*.

Theorem 1.26 (Hyperreals)

The **hyperreals**

In fact, when Newton first invented calculus, the hyperreals were what he worked with, and you can surprisingly build a good chunk of calculus with this. Even though this is a dead topic at this point, a lot of modern notation is based off of this number system, so it's good to see how it works. For example, when we write the integral
$$\int f(x)\,dx \tag{92}$$
we are saying that we are taking the uncountable sum of the terms $f(x)\,dx$, the multiplication of the real number $f(x)$ and the infinitesimal number dx living in the hyperreals. Unfortunately, we cannot fully construct a rigorous theory of calculus with only infinitesimals. However, in practice (especially physics) people tend to manipulate and do algebra with infinitesimals, so having a good foundation on what you can and cannot do with them is practical. While the focus won't be on *smooth infinitesimal analysis (SIA)*, I will include some alternate constructions later purely with infinitesimals.

1.9 Some Algebraic Inequalities

We also introduce various inequalities that may be useful for producing future results. The following lemmas can be proved with elementary algebra on the field of reals.

Lemma 1.27 (Bernoulli's Inequality)

For any real $x \geq -1$ and $n \in \mathbb{N}$, we have
$$(1 + x)^n \geq 1 + nx \tag{93}$$

Proof. We prove by induction. For $n = 1$, it is trivial. Now given that the inequality is satisfied for some $n \in \mathbb{N}$, we have

$$\begin{aligned}(1+x)^{n+1} &= (1+x)^n(1+x) && (94)\\ &\geq (1+nx)(1+x) && (95)\\ &= 1 + nx + x + nx^2 && (96)\\ &= 1 + (n+1)x && (97)\end{aligned}$$

Lemma 1.28 (Young's Inequalities)

If $a > 0$ and $b > 0$, and the numbers p and p are such that $p \neq 0, 1$ and $q \neq 0, 1$, and $\frac{1}{p} + \frac{1}{q} = 1$, then

$$a^{\frac{1}{p}} b^{\frac{1}{q}} \leq \frac{1}{p} a + \frac{1}{q} b \text{ if } p > 1 \tag{98}$$

$$a^{\frac{1}{p}} b^{\frac{1}{q}} \geq \frac{1}{p} a + \frac{1}{q} b \text{ if } p < 1 \tag{99}$$

and equality holds in both cases if and only if $a = b$.

Proof.

Lemma 1.29 (Holder's Inequalities)

Let $x_i \geq 0, y_i \geq 0$ for $i = 1, 2, ..., n$, and let $\frac{1}{p} + \frac{1}{q} = 1$. Then,

$$\sum_{i=1}^n x_i y_i \leq \bigg(\sum_{i=1} x_i^p \bigg)^{\frac{1}{p}} \bigg(\sum_{i=1} y_i^q \bigg)^{\frac{1}{q}} \text{ for } p > 1 \tag{100}$$

$$\sum_{i=1}^n x_i y_i \geq \bigg(\sum_{i=1} x_i^p \bigg)^{\frac{1}{p}} \bigg(\sum_{i=1} y_i^q \bigg)^{\frac{1}{q}} \text{ for } p < 1, p \neq 0 \tag{101}$$

Proof.

Lemma 1.30 (Minkowski's Inequalities)

Let $x_i \geq 0, y_i \geq 0$ for $i = 1, 2, ..., n$. Then,

$$\bigg(\sum_{i=1}^n (x_i + y_i)^p \bigg)^{\frac{1}{p}} \leq \bigg(\sum_{i=1}^n x_i^p \bigg)^{\frac{1}{p}} + \bigg(\sum_{i=1}^n y_i^p \bigg)^{\frac{1}{p}} \text{ when } p > 1 \tag{102}$$

$$\bigg(\sum_{i=1}^n (x_i + y_i)^p \bigg)^{\frac{1}{p}} \geq \bigg(\sum_{i=1}^n x_i^p \bigg)^{\frac{1}{p}} + \bigg(\sum_{i=1}^n y_i^p \bigg)^{\frac{1}{p}} \text{ when } p < 1, p \neq 0 \tag{103}$$

Proof.

1.10 Exercises

Exercise 1.1 (Math 531 Spring 2025, PS2.1)

Prove that the set of all matrices of the form:

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}, \tag{104}$$

with $a, b \in \mathbb{R}$ forms a field with the usual sum and product operations of matrices. What does this field resemble? Give extensions to 3×3 and 4×4 matrices.

Solution.

Exercise 1.2 (Math 531 Spring 2025, PS2.2)

Why can't the field of complex numbers (with its usual operations) be made into an ordered field?

Solution. Solution is shown as theorem.

Exercise 1.3 (Math 531 Spring 2025, PS2.3)

Prove there are no finite ordered fields.

Solution. Solution is shown as theorem.

Exercise 1.4 (Math 531 Spring 2025, PS2.4)

Prove that if x and n are natural numbers, then

$$x^n - 1 = (x-1)(1 + x + x^2 + \ldots + x^{n-1}). \tag{105}$$

Solution. We use the commutative addition and multiplication, plus distributive property in $\mathbb{Z}$.

$$\begin{aligned}
(x-1)\left(\sum_{i=0}^{n-1} x^i\right) &= x\sum_{i=0}^{n-1} x^i - \sum_{i=0}^{n-1} x_i && (106)\\
&= \sum_{i=1}^{n} x^i - \sum_{i=0}^{n-1} x^i && (107)\\
&= x^n + \sum_{i=1}^{n-1} x^i - \sum_{i=1}^{n-1} x^i - 1 && (108)\\
&= x^n - 1 && (109)
\end{aligned}$$

Exercise 1.5 (Math 531 Spring 2025, PS2.7)

Prove that there is no $q \in \mathbb{Q}$ for which

$$q^2 + q = 4. \tag{110}$$

Solution. Assume that such a $q \in \mathbb{Q}$ in canonical form exists. Then by the field properties, since $\frac{1}{4} \in \mathbb{Q}$,

$$q^2 + q + \frac{1}{4} = 4 + \frac{1}{4} = \frac{17}{4} \in \mathbb{Q} \tag{111}$$

But by distributive properties, $(q + \frac{1}{2})^2 \in \mathbb{Q}$. We claim that there exists no rational $x = a/b$ (a, b coprime) s.t. $x^2 = 17/4$. If there were, then clearly $a \neq 0$ and

$$\begin{aligned}
\frac{a^2}{b^2} = \frac{17}{4} &\implies 4a^2 = 17b^2 && (112)\\
&\implies 2|b, \text{ and so } b = 2b' \text{ for some } b' \in \mathbb{N} && (113)\\
&\implies a^2 = 17(b')^2 && (114)\\
&\implies 17|a \text{ and so } a = 17a' \text{ for some } a' \in \mathbb{Z} && (115)\\
&\implies 17(a')^2 = (b')^2 && (116)
\end{aligned}$$

which implies that $17|b'$, but this contradicts the assumption that a, b are coprime. Therefore $q + \frac{1}{2} \notin \mathbb{Q} \implies q \notin \mathbb{Q}$.

Exercise 1.6 (Math 531 Spring 2025, PS2.8)

Let X be an ordered set with the least upper bound property. Prove that X has the greatest lower bound property.

Solution. Shown in theorem above.

Exercise 1.7 (Math 531 Spring 2025, PS2.9)

Prove that if $x, y \in \mathbb{Q}$ we have that

$$||x| - |y|| \leq |x - y|. \tag{117}$$

Solution. By subadditivity of the norm we have

$$\begin{aligned}
|x| \leq |x - y| + |y| &\implies |x| - |y| \leq |x - y| && (118)\\
|y| \leq |y - x| + |x| &\implies |y| - |x| \leq |y - x| && (119)
\end{aligned}$$

But $|y - x| = |-1(x - y)| = |-1| \cdot |x - y| = |x - y|$, and so

$$\max\{|x| - |y|, |y| - |x|\} \leq |x - y| \tag{120}$$

and the LHS is the definition of the norm $||x| - |y||$ in $\mathbb{Q}$.

Exercise 1.8 (Rudin 1.1)

If r is rational ($r \neq 0$) and x is irrational, prove that $r + x$ and rx are irrational.

Solution. If we assume that $rx = t$ and $r + x = s$ are rational, then this violates the field axioms of $\mathbb{Q}$ since then $x = tr^{-1}$ and $x = s + (-r)$ are rational.

Exercise 1.9 (Rudin 1.2)

Prove that there is no rational number whose square is 12.

Solution. Assume that there exists a number p/q such that p and q are both not even. Then,

$$\left(\frac{p}{q}\right)^2 = 12 \implies p^2 = 12q^2 = 3(2q)^2 \tag{121}$$

So p much be even $p = 2p'$. Therefore, $p'^2 = 3q^2$, and q must be odd. This means that p' must be odd. We can rewrite the equation

$$p'^2 - q^2 = 2q^2 \implies (p' + q)(p' - q) = 2q^2 \tag{122}$$

where the left hand side is divisible by 4 but the right hand side is divisible by at most 2, leading to a contradiction.

Exercise 1.10 (Rudin 1.3)

Prove that the axioms of multiplication imply the following.

1. If $x \neq 0$ and $xy = xz$, then $y = z$.
2. If $x \neq 0$ and $xy = x$, then $y = 1$.
3. If $x \neq 0$ and $xy = 1$, then $y = x^{-1}$.
4. If $x \neq 0$, then $(x^{-1})^{-1} = x$.

Solution. Listed.

1. $xy = xz \implies \frac{1}{x} \cdot xy = \frac{1}{x}xz \implies y = z$
2. $xy = x \implies \frac{1}{x}xy = \frac{1}{x}x \implies y = 1$
3. $xy = 1 \implies \frac{1}{x}xy = \frac{1}{x}1 \implies y = \frac{1}{x}$
4. $(x^{-1})^{-1} \cdot x^{-1} = 1 \implies (x^{-1})^{-1} \cdot x^{-1} \cdot x = 1 \cdot x \implies (x^{-1})^{-1} = x$

Exercise 1.11 (Rudin 1.4)

Let E be a nonempty subset of an ordered set; suppose α is a lower bound of E and β is an upper bound of E. Prove that $\alpha \leq \beta$.

Solution. Since E is nonempty, we choose any $x \in E$. By definition, $\alpha \leq x$ and $x \leq \beta$, and by transitive property of orderings, we have $\alpha \leq \beta$.

Exercise 1.12 (Rudin 1.5)

Let A be a nonempty set of real numbers which is bounded below. Let $-A$ be the set of all numbers $-x$, where $x \in A$. Prove that

$$\inf A = -\sup(-A) \tag{123}$$

Solution. We would like to prove that $\inf A \leq -\sup(-A)$ and $\inf A \geq -\sup(-A)$. For the first part, we start off with the definition of the infimum.

$$\begin{aligned}\inf A \leq x \;\forall x \in A &\implies -\inf A \geq -x \;\forall x \in A\\ &\implies -\inf A \geq x \forall x \in -A\\ &\implies -\inf A \geq \sup(-A)\\ &\implies \inf A \leq -\sup(-A)\end{aligned}$$

For the second part, we start with the definition of the supremum.

$$\begin{aligned}\sup(-A) \geq x \forall x \in -A &\implies \sup(-A) \geq -x \forall x \in A\\ &\implies -\sup(-A) \geq x \forall x \in A\\ &\implies -\sup(-A) \leq \inf A\end{aligned}$$

Exercise 1.13 (Rudin 1.6)

Fix $b > 1$.

1. If m, n, p, q are integers, $n > 0$, $q > 0$, and $r = m/n = p/q$, prove that
$$(b^m)^{1/n} = (b^p)^{1/q}. \tag{124}$$
Hence it makes sense to define $b^r = (b^m)^{1/n}$.
2. Prove that $b^{r+s} = b^r b^s$ if r and s are rational.
3. If x is real, define $B(x)$ to be the set of all numbers b^t, where t is rational and $t \leq x$. Prove that
$$b^r = \sup B(r) \tag{125}$$
when r is rational. Hence it makes sense to define
$$b^x = \sup B(x) \tag{126}$$
for every real x.
4. Prove that $b^{x+y} = b^x b^y$ for all real x and y.

Solution. Proved in theorem above.

Exercise 1.14 (Rudin 1.7)

Fix $b > 1$, $y > 0$, and prove that there is a unique real x such that $b^x = y$, by completing the following outline. (This x is called the logarithm of y to the base b.)

1. For any positive integer n, $b^n - 1 \geq n(b - 1)$.
2. Hence $b - 1 \geq n(b^{1/n} - 1)$.
3. If $t > 1$ and $n > (b - 1)/(t - 1)$, then $b^{1/n} < t$.
4. If w is such that $b^w > y$, then $b^{-(1/n)} < y$ for sufficiently large n; to see this, apply part (c) with $t = y \cdot b^{-w}$.
5. If $b^w > y$, then $b^{w-(1/n)} > y$ for sufficiently large n.
6. Let A be the set of all w such that $b^w < y$, and show that $x = \sup A$ satisfies $b^x = y$.
7. Prove that this x is unique.

Solution. Proved in theorem above.

Exercise 1.15 (Rudin 1.8)

Prove that no order can be defined in the complex field that turns it into an ordered field.

Solution. Note that if $x \geq 0$, then $-x \leq 0$ for all x of any ordered field. Since if $x \geq 0$ and $-x > 0$, then $x - x > 0$, which is absurd. Therefore, one of either i or $-i$ should be greater than 0. But $i^2 = (-i)^2 = -1$, so this means that $-1 > 0$, which implies that $0 < 1$. But either 1 or -1 must ≥ 0.

Exercise 1.16 (Rudin 1.9)

Equip $\mathbb{C}$ with the dictionary order. That is, given $z = a + bi$ and $w = c + di$, $z < w$ if $a < c$, or if $a = c$ and $b < d$. Does this ordered set have a least upper bound property?

Solution. No it does not. Consider the set $S = \{a + bi \in \mathbb{C} \mid a \leq 3\}$. S is bounded by 4, but it doesn't have a least upper bound. Given any $3 + bi$, this is not an upper bound since we can construct $3 + (b + \epsilon)i \in S$. Given any $a + bi$ where $a > 3$, we can always find a lower bound of form $a + (b - \epsilon)i$ that also bounds S.

Exercise 1.17 (Rudin 1.10)

Suppose $z = a + bi$, $w = u + iv$, and

$$a = \left(\frac{|w| + u}{2} \right)^{1/2} \text{ and } b = \left(\frac{|w| - u}{2} \right)^{1/2} \tag{127}$$

Prove that $z^2 = w$ if $v \geq 0$ and that $(\bar{z})^2 = w$ if $v \leq 0$. Conclude that every complex number (with one exception!) has two complex square roots.

Solution. We can calculate

$$z^2 = (a^2 - b^2) + 2abi = u + \sqrt{v^2} i = \begin{cases} u + vi & \text{if } v \geq 0 \\ u - vi & \text{if } v \leq 0 \end{cases} \tag{128}$$

Since if we assume $v \geq 0$, then we have $z^2 = w$. We also get

$$\bar{z}^2 = (a^2 - b^2) - 2abi = u - \sqrt{v^2} i = \begin{cases} u - vi & \text{if } v \geq 0 \\ u + vi & \text{if } v \leq 0 \end{cases} \tag{129}$$

and assuming $v \leq 0$, we have $\bar{z}^2 = w$. Therefore, every complex number w has both $\pm z$ as its square root if $v \geq 0$, $\pm \bar{z}$ if $v \leq 0$, and just one root if $z = 0$.

Exercise 1.18 (Rudin 1.11)

If z is a complex number, prove that there exists an $r \geq 0$ and a complex number w with $|w| = 1$ s.t. $z = rw$. Are w and r always uniquely determined by z?

Solution. If $z = 0$, then $r = 0$ and there is no unique w. If $z = a + bi \neq 0$, then define

$$r = |z| = (a^2 + b^2)^{1/2}, \;\; w = \frac{1}{r} z \tag{130}$$

which proves existence. As for uniqueness, assume that there are two forms

$$z = rw = r'w' \tag{131}$$

Then, $w = \frac{r'}{r} w' \implies |w| = \left|\frac{r'}{r}\right| |w'| = 1$, which implies that $r'/r = 1$ and so $r = r'$. This means that $w = w'$.

Exercise 1.19 (Rudin 1.12)

If $z_1, \ldots, z_n$ are complex, prove that

$$|z_1 + z_2 + \ldots + z_n| \leq |z_1| + \ldots + |z_n| \tag{132}$$

Solution. By induction, it suffices to prove $|z_1 + z_2| \leq |z_1| + |z_2|$. We have

$$\begin{aligned}
|z_1 + z_2|^2 &= (z_1 + z_2)(\overline{z_1 + z_2}) \\
&= (z_1 + z_2)(\bar{z_1} + \bar{z_2}) \\
&= z_1 \bar{z_1} + z_1 \bar{z_2} + z_2 \bar{z_1} + z_2 \bar{z_2} \\
&= |z_1|^2 + |z_2|^2 + z_1 \bar{z_2} + z_2 \bar{z_1} \\
&= |z_1|^2 + |z_2|^2 + 2(ac + bd) \\
&\leq |z_1|^2 + |z_2|^2 + 2\sqrt{a^2 + b^2}\sqrt{c^2 + d^2} \quad \text{(Schwartz)} \\
&= |z_1|^2 + |z_2|^2 + 2|z_1||z_2| \\
&= (|z_1| + |z_2|)^2
\end{aligned}$$

since both sides are positive, we can take their square root to get the desired result.

Exercise 1.20 (Rudin 1.13)

If x, y are complex, prove that

$$\big||x| - |y|\big| \leq |x - y| \tag{133}$$

Solution. Since both sides are nonnegative, we can square both sides. Note that due to Cauchy Schwartz inequality, $2|x||y| \geq x\bar{y} + y\bar{x}$ since expanding them gives

$$2\sqrt{a^2 + b^2}\sqrt{c^2 + d^2} \geq 2(ac + bd) \tag{134}$$

Therefore, the following inequality is true:

$$|x|^2 + |y|^2 - 2|x||y| \leq x\bar{x} + y\bar{y} - x\bar{y} - y\bar{x} \tag{135}$$

which reduces to form $(|x| - |y|)^2 \leq |x - y|^2$.

Exercise 1.21 (Rudin 1.14)

If z is a complex number s.t. $|z| = 1$, that is such that $z\bar{z} = 1$, compute

$$|1 + z|^2 + |1 - z|^2 \tag{136}$$

Solution. Compute.

$$(1+z)(1+\bar{z}) + (1-z)(1-\bar{z}) = 1 + z + \bar{z} + z\bar{z} + 1 - z - \bar{z} + z\bar{z} = 4 \tag{137}$$

Exercise 1.22 (Rudin 1.15)

Under what conditions does equality hold in the Schwarz inequality?

Solution. If they are antiparallel, since

$$\langle x, y \rangle = ||x||||y|| \cos\theta \tag{138}$$

Exercise 1.23 (Rudin 1.16)

Suppose $k \geq 3$, $\mathbf{x}, \mathbf{y} \in \mathbb{R}^k$, $|\mathbf{x} - \mathbf{y}| = d > 0$, and $r > 0$. Prove:

a) If $2r > d$, there are infinitely many $\mathbf{z} \in \mathbb{R}^k$ s.t.

$$|\mathbf{z} - \mathbf{x}| = |\mathbf{z} - \mathbf{y}| = r \tag{139}$$

b) If $2r = d$, there is exactly one such $\mathbf{z}$.

c) If $2r < d$, there is no such $\mathbf{z}$.

Solution.

Exercise 1.24 (Rudin 1.17)

Prove that

$$|\mathbf{x} + \mathbf{y}|^2 + |\mathbf{x} - \mathbf{y}|^2 = 2|\mathbf{x}|^2 + 2|\mathbf{y}|^2 \tag{140}$$

Solution. This is trivial if we simply expand

$$\begin{aligned}
|\mathbf{x} + \mathbf{y}|^2 + |\mathbf{x} - \mathbf{y}|^2 &= \langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle + \langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y} \rangle && (141) \\
&= \langle \mathbf{x}, \mathbf{x} \rangle + 2\langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{x} \rangle - 2\langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle && (142) \\
&= 2\langle \mathbf{x}, \mathbf{x} \rangle + 2\langle \mathbf{y}, \mathbf{y} \rangle && (143) \\
&= 2|\mathbf{x}|^2 + 2|\mathbf{y}|^2 && (144)
\end{aligned}$$

Exercise 1.25 (Rudin 1.18)

If $k \geq 2$ and $\mathbf{x} \in \mathbb{R}^k$, prove that there exists $\mathbf{y} \in \mathbb{R}^k$ s.t. $\mathbf{y} \neq \mathbf{0}$, but $\mathbf{x} \cdot \mathbf{y} = 0$. Is this also true if $k = 1$?

Solution. Let $x \in \mathbb{R}^k$ and $\ell \in \mathbb{R}^{k*}$, the dual space. By Riesz representation theorem, we can define the canonical isomorphism $\ell \mapsto y$ between these two spaces as

$$\ell(x) = (x, y) \tag{145}$$

Since $y \neq 0$ by assumption, $\ell \neq 0$, and so its rank is at least 1. Since ℓ maps to $\mathbb{R}$, the rank has to be 1. By rank nullity theorem, we have

$$\dim N(\ell) = k - \operatorname{rank}(\ell) = k - 1 \tag{146}$$

and so there exists nontrivial annihilators ℓ of x, which can be mapped to a nontrivial $y \in \mathbb{R}^k$.

Exercise 1.26 (Rudin 1.19)

Suppose $\mathbf{a}, \mathbf{b} \in \mathbb{R}^k$. Find $\mathbf{c} \in \mathbb{R}^k$ and $r > 0$ s.t.

$$|\mathbf{x} - \mathbf{a}| = 2|\mathbf{x} - \mathbf{b}| \tag{147}$$

if and only if $|\mathbf{x} - \mathbf{c}| = r$.

Solution. If we draw out the circle, it must contain two points on the line drawn by connecting A and B. Since it must be symmetric, its center and radius can then be easily calculated to be

$$r = \frac{2}{3}|b - a|, \quad c = \frac{1}{3}(4b - a) \tag{148}$$

Exercise 1.27 (Zorich 2.2.1)

Using the principle of induction, show that
1. the sum $x_1 + \ldots + x_n$ of real numbers is defined independently of the insertion of parentheses to specify the order of addition.
2. the same is true of the product $x_1 \ldots x_n$
3. $|x_1 + \ldots + x_n| \leq |x_1| + \ldots + |x_n|$
4. $|x_1 \ldots x_n| \leq |x_1| \ldots |x_n|$
5. For any $n, m \in \mathbb{N}$ such that $m < n$, $(n - m) \in \mathbb{N}$.
6. $(1 + x)^n \geq 1 + nx$ for $x > -1$ and $n \in \mathbb{N}$, equality holding for when $n = 1$ or $x = 0$.
7. $(a + b)^n = a^n +_n C_1 a^{n-1} b^1 + \ldots + b^n$ (aka binomial theorem).

Solution. Listed.
1. Let n denote the number of elements in the sum. We prove by strong law of induction. The base case for when $n = 1, 2, 3$ is trivially true.

$$\begin{aligned} x_1 &= x_1 && \text{(identity)} \\ x_1 + x_2 &= x_1 + x_2 && \text{(identity)} \\ (x_1 + x_2) + x_3 &= x_1 + (x_2 + x_3) && \text{(associativity)} \end{aligned}$$

   Then, the sum of $n = k$ parameters is defined by $k - 2$ pairs of parentheses defining the order of the sum. These parentheses define a sequence of $k - 1$ 2-fold additions. Now, assume that the claim is true for

$$S_n \equiv x_1 + \ldots x_n \text{ for } n = 1, 2, \ldots, k \tag{149}$$

   Then, for a specific sum S_{k+1} of $k + 1$ elements with $k - 1$ parentheses, we can reduce the sum to its final 2-fold addition

$$S_{k+1} \equiv \underbrace{(x_1 + \ldots + x_i)}_{\varphi_1} + \underbrace{(x_{i+1} + \ldots + x_{k+1})}_{\varphi_2} \tag{150}$$

   Since $i, k - i + 1 < k$, by the strong law φ_1 and φ_2 are independent of the order of sum.
2. Exactly identical to (a).
3. By the triangle inequality $|x_1 + x + 2| \leq |x_1| + |x_2|$. Now, assume for $n = k$ is true. Then, let

$S_k = x_1 + \ldots + x_k$, so

$$|x_1 + \ldots + x_k + x_{k+1}| = |S_k + x_{k+1}| \leq |S_k| + |x_{k+1}| \leq \sum_{i=1}^{k+1} |x_i| \tag{151}$$

4. Same as (c).
5. Let us fix m to be any element of $\mathbb{N}$. Then, the base case is for $n = m+1$ (which is in $\mathbb{N}$ since it is inductive), so
$$n - m = (m+1) - m = 1 \in \mathbb{N} \tag{152}$$
Now, given that for some integer $n \geq m+1$, $n - m \in \mathbb{N}$ is true, we have
$$\begin{aligned} (n+1) - m &= n + (1 - m) && \text{(associativity)} \\ &= n + (-m + 1) && \text{(commutativity)} \\ &= (n - m) + 1 && \text{(associativity)} \end{aligned}$$
where $(n-m)+1 \in \mathbb{N}$ by inductive property of $\mathbb{N}$.
6. We prove by induction. For $n = 1$, it is trivial that $(1+x)^1 \geq 1 + 1 \cdot x$. Now assume that the claim is true for some $k \in \mathbb{N}$. Then,
$$\begin{aligned} (1+x)^{k+1} = (1+x)^k(1+x) &\geq (1+kx)(1+x) \\ &= 1 + (k+1)x + kx^2 \\ &\geq 1 + (k+1)x \end{aligned}$$
where equality holds if $x = 0 \implies 1^{k+1} = 1^k \cdot 1 = 1$ or $n = 1 \implies$ trivial case.
7. The base case for $n = 1$ is trivial since $(a+b)^1 = \binom{1}{0}a + \binom{1}{1}b$. We introduce Newton's identity.
$$\begin{aligned} \binom{k}{j-1} + \binom{k}{j} &= \frac{k!}{(j-1)!(k-j+1)!} + \frac{k!}{j!(k-j)!} \\ &= k!\bigg(\frac{j}{j!(k-j+1)!} + \frac{k-j+1}{j!(k-j+1)!}\bigg) \\ &= k! \cdot \frac{k+1}{j!(k-j+1)!} \\ &= \frac{(k+1)!}{j!(k-j+1)!} = \binom{k+1}{j} \end{aligned}$$
Now assuming that the binomial formula holds for some $n = k$, we have
$$\begin{aligned} (a+b)^{k+1} &= (a+b)^k(a+b) && (153) \\ &= \bigg(\sum_{j=0}^{k} \binom{k}{j} a^j b^{k-j}\bigg)(a+b) && (154) \\ &= \sum_{j=0}^{k} \binom{k}{j} a^{j+1} b^{k-j} + \sum_{j=0}^{k} \binom{k}{j} a^j b^{k-j+1} && (155) \\ &= \binom{k}{0} a^0 b^{k+1} + \binom{k}{k} a^{j+1} b^0 + \sum_{j=0}^{k-1} \binom{k}{j} a^{j+1} b^{k-j} + \sum_{j=1}^{k} \binom{k}{j} a^j b^{k-j+1} && (156) \\ &= \binom{k+1}{0} a^0 b^{k+1} + \binom{k+1}{k+1} a^{j+1} b^0 + \sum_{j=1}^{k} \bigg[\binom{k}{j-1} + \binom{k}{j}\bigg] a^j b^{k-j+1} && (157) \\ &= \sum_{j=0}^{k+1} \binom{k+1}{j} a^j b^{k-j+1} && (158) \end{aligned}$$

Exercise 1.28 (Zorich 2.2.3)

Show that an inductive set is not bounded above.

Solution. Assume that a X is a nonempty inductive set that is bounded above. By definition, there exists a number $B \in \mathbb{R}$ such that $\max X < B$. Then, this means that there exists no numbers in $[B, B+1)$. Since X is inductive, this means that there cannot exist any elements of X in the interval $[B-1, B)$, and similarly for the interval $[B-2, B)$, and so on, meaning that if $x \in X$, then $x \notin [B-k, B-k+1)$ for all $k \in \mathbb{Z}$. By the Archimidean principle, this implies that $X = \emptyset$, contradicting our assumption.

Exercise 1.29 (Zorich 2.2.4)

Prove the following.

1. An inductive set is infinite (that is, equipollent with one of its subsets different from itself).
2. The set $E_n = \{x \in \mathbb{N} \mid x \leq n\}$ is finite.

Solution. Listed.

1. Assume that an inductive set X is finite $\implies$ X is bounded above (we can choose upper bound $B = \max X + 1$). But from 2.2.3, an inductive set cannot be bounded above, contradicting our assumption.
2. It is trivial that $E_1 = \{1\}$ is finite since $\mathrm{card} E_1 = 1$. Now, if for some k, E_k is finite with cardinality e_k, then $\mathrm{card} E_{k+1} = e_k + 1$, which implies finiteness.

Exercise 1.30 (Zorich 2.2.5)

Listed.

1. Let $m, n \in \mathbb{N}$ and $m > n$. Their greatest common divisor $\gcd(m, n) = d \in \mathbb{N}$ can be found in a finite number of steps using the following algorithm of Euclid involving successive divisions with remainder.

$$\begin{aligned} m &= q_1 n + r_1 \\ n &= q_2 r_1 + r_2 \\ r_1 &= q_3 r_2 + r_3 \\ \ldots &= \ldots \\ r_{k-2} &= q_k r_{k-1} + r_k \\ r_{k-1} &= q_{k+1} r_k + 0 \end{aligned}$$

   Then $d = r_k$
2. If $d = \gcd(m, n)$, one can choose numbers $p, q \in \mathbb{Z}$ such that $pm + qn = d$.

Solution. Listed.

1.
2. Letting $n = r_0$, notice that the equations above satisfy for $i = 0, 1, \ldots$

$$r_i = q_{i+2} r_{i+1} + r_{i+2} \implies r_i - q_{i+2} r_{i+1} = r_{i+2} \tag{1}$$

   Note that the second-to-last equation allows us to write r_k as a linear combination of r_{k-2} and r_{k-1}: $r_k = r_{k-2} - q_k r_{k-1}$. Now by applying (1), we can reduce the above to a linear combination of r_{k-3} and r_{k-2}.

$$\begin{aligned} r_k &= r_{k-2} - q_k r_{k-1} \\ &= r_{k-2} - q_k(r_{k-3} - q_{k-1} r_{k-3}) \\ &= (1 + q_{k-1} q_k) r_{k-2} - q_k r_{k-3} \end{aligned}$$

and repeatedly doing this allows us to reduce r_k to a linear combination $q_0 r_0 + q_1 r_1$. By the ring properties of $\mathbb{Z}$, the new linear coefficients are also in $\mathbb{Z}$. Reducing one last time using the first equation in the Euclidean algorithm gives

$$\begin{aligned} r_k &= q_0 r_0 + q_1 r_1 \\ &= q_0 n + q_1(m - q_1 n) \\ &= q_1 m + (q_0 - q_1) n \\ &= pm + qn \end{aligned}$$

Exercise 1.31 (Zorich 2.2.9)

Show that if the natural number n is not of the form k^m, where $k, m \in \mathbb{N}$, then the equation $x^m = n$ has no rational roots.

Solution. Assume that there is a rational solution $x = p/q$, with $p, q \in \mathbb{N}$ of the equation. Then,

$$\left(\frac{p}{q}\right)^m = \frac{p^m}{q^m} = n \implies p^m = q^m n \tag{159}$$

By the fundamental theorem of arithmetic, the exponents of the prime factors of p^m must all be multiples of m, and so it must be so for the right hand side $\implies$ x must be of form $x = k^m$ for some k. This is a contradiction.

Exercise 1.32 (Zorich 2.2.12)

Knowing that $\frac{m}{n} \equiv m \cdot n^{-1}$ by definition, where $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, derive the "rules" for addition, multiplication, and division of fractions, and also the condition for two fractions to be equal.

Solution. We can construct a $\mathbb{Q}$ as a quotient space $\mathbb{Z} \times \mathbb{N} / \sim$, where $\sim$ is an equivalent relation where

$$(q_1, p_1) \sim (q_2, p_2) \text{ iff } q_1 p_2 = p_1 q_2 \tag{160}$$

which is the familiar equivalence relation from "simplifying" a fraction. We define addition and multiplication as the following

$$\begin{aligned} (a, b) + (c, d) &= (ad + bc, bd) \\ (a, b) \cdot (c, d) &= (ac, bd) \end{aligned}$$

which turns out to be algebraically closed in $\mathbb{Q}$. The additive identity is the equivalence class $0 = \{(0, c) \mid c \in \mathbb{N}\}$, and the multiplicative identity is the equivalence class $1 = \{(c, c) \mid c \in \mathbb{N}\}$. It is easy to check that $+$ is commutative, the additive inverse is $-(a, b) = (-a, b)$, and the multiplicative inverse is $(a, b)^{-1} = (b, a)$. We can subtract and divide these elements of $\mathbb{Q}$, called "fractions," as such:

$$\begin{aligned} (a, b) - (c, d) &= (a, b) + (-(c, d)) = (a, b) + (-c, d) = (ad - bc, bd) \\ (a, b) \div (c, d) &= (a, b) \cdot (c, d)^{-1} = (a, b) \cdot (d, c) = (ad, bc) \end{aligned}$$

Exercise 1.33 (Zorich 2.2.13)

Verify that the rational numbers $\mathbb{Q}$ satisfy all the axioms for real numbers except for the axiom of completeness.

Solution. From continuing the steps of 2.2.14, we can prove $\mathbb{Q}$ is an algebraic field (associativity, commutativity of addition and multiplication, along with distributive property).We can actually define the order relation $\leq_{\mathbb{Q}}$ in two ways:

1. $(a, b) \leq (c, d)$ iff $ad \leq_{\mathbb{Z}} bc$, where $\leq_{\mathbb{Z}}$ is the order relation on $\mathbb{Z}$ (which can be defined much more simply).
2. Recognizing that $\mathbb{Q} \subset \mathbb{R}$, we define the canonical injection map $i : \mathbb{Q} \longrightarrow \mathbb{R}$ and by abuse of language, endow the relation $\leq_{\mathbb{Q}}$ as the restriction of $\leq_{\mathbb{R}}$ onto $\mathbb{Q}$. That is, for $(a, b), (c, d) \in \mathbb{Q}$,

$$(a, b) \leq_{\mathbb{Q}} (c, d) \text{ iff } i(a, b) \leq_{\mathbb{R}} i(c, d) \tag{161}$$

The ordering for the 1st step can be checked for consistency.

1. $(a, b) \leq (a, b)$ since $ab \leq ab$ (true in $\mathbb{Z}$)
2. $(a, b) \leq (c, d), (c, d) \leq (a, b)$ means that $ad \leq bc$ and $bc \leq ad \implies ad = bc$ (true in $\mathbb{Z}$)
3. $(a, b) \leq (c, d) \leq (e, f)$ implies $ad \leq bc, cf \leq de$. Multiplying positive (important that $f > 0$!) to the first inequality gives $adf \leq bcf$, and multiplying positive b to the second gives $bcf \leq bde$, and by interpreting $\leq$ as the ordering defined on $\mathbb{Z}$, we use transitive property of $\leq_{\mathbb{Z}}$ to get $adf \leq bde \implies af \leq be \iff (a, b) \leq (e, f)$.
4. For any $(a, b), (c, d) \in \mathbb{Q}$, $(a, b) \leq (c, d)$ or $(a, b) \geq (c, d)$, which is equivalent to $ad \leq bc$ or $ad \geq bc$, which is true in $\mathbb{Z}$.

It is easy to prove $(a, b) \leq (c, d) \implies (a, b) + (p, q) \leq (c, d) + (p, q)$, and $0_{\mathbb{Q}} \leq (a, b), (c, d) \implies 0_{\mathbb{Q}} \leq (a, b) \cdot (c, d)$. However, $\mathbb{Q}$ is **not** complete. We prove this by showing that the subset $X = \{x \in \mathbb{Q} \mid x^2 \leq 2\} \subset \mathbb{Q}$ does not satisfy the least upper bound property. Assume that there is a least upper bound $c \in \mathbb{Q}$. $c \neq \sqrt{2}$ (you should know how to prove irrationality of $\sqrt{2}$!), we have either $c > \sqrt{2}$ or $c < \sqrt{2}$.

1. Let $c < \sqrt{2} \iff c - \sqrt{2} > 0$. By the Archimidean principle, there exists a $k \in \mathbb{N}$ such that $0 < \frac{1}{k} < c - \sqrt{2}$. Then, $\frac{1}{k} \in \mathbb{Q}$ and $\mathbb{Q}$ is a field, so $c - \frac{1}{k} \in \mathbb{Q}$.

$$c - \frac{1}{k} < c - c + \sqrt{2} = \sqrt{2} \tag{162}$$

   So c is not least and so it must be the case that $c < \sqrt{2}$.
2. Let $c < \sqrt{2} \iff \sqrt{2} - c > 0$. By the Archimidean principle, there exists a $k \in \mathbb{N}$ such that $0 < \frac{1}{k} < \sqrt{2} - c$. Then, $c + \frac{1}{k} \in \mathbb{Q}$ and

$$c + \frac{1}{k} < c + (\sqrt{2} - c) = c \tag{163}$$

   So c is not an upper bound.

Note that given a well-defined $c = \sup X$ and in the case where $c < \sqrt{2}$, we have $2 - c^2 > 0$, so we can choose a well-defined δ satisfying (by Archimidean principle)

$$0 < \delta < \min\left\{1, \frac{2 - c^2}{2c + 1}\right\} \tag{164}$$

which gives us

$$\begin{aligned}(c + \delta)^2 &= c^2 + \delta(2c + \delta) \\ &< c^2 + \delta(2c + 1) && (\delta < 1) \\ &< c^2 + (2 - c^2) = 2\end{aligned}$$

meaning that c is not an upper bound. Similarly for when $c > \sqrt{2}$.

Exercise 1.34 (Zorich 2.2.15)

Prove the equivalence of these two statements.

1. If X and Y are nonempty sets of $\mathbb{R}$ having the property that $x \leq y$ for every $x \in X, y \in Y$, then there exists $c \in \mathbb{R}$ such that $x \leq c \leq y$ for all $x \in X$ and $y \in Y$.
2. Every set $X \subset \mathbb{R}$ that is bounded above has a least upper bound.

Solution. Let S_1 be the first statement and S_2 the second.

1. ($S_2 \implies S_1$). Let $X \subset \mathbb{R}$ be a set that is bounded above, and Y is a set such that $x \leq y$ for all $x \in X, y \in Y$. Then, by LUB principle, there exists $c = \sup X \in \mathbb{R}$. Now, we claim that $c \leq y$ for all $y \in Y$. Assume it doesn't: then there exists $y' \in Y$ such that $y' < c$. But since we assumed $x \leq y$ for all $x \in X, y \in Y$, we have $x \leq y'$ for all $x \in X$, which means that y' is an upper bound of X. But $y' < c$, contradicting the given fact that c was the least upper bound.
2. ($S_1 \implies S_2$). Given a nonempty set $X \subset \mathbb{R}$, we wish to show the existence of $\sup X$. We are guaranteed the existence of nonempty set $Y \subset \mathbb{R}$ such that $x \leq y$ for all $x \in X, y \in Y$, which implies that X must be bounded above. Then, by S_1, there must exist a $c \in \mathbb{R}$ such that
$$x \leq c \leq y \text{ for all } x \in X, y \in Y \tag{165}$$
We claim that $c = \sup X$. It is an upper bound of X since $x \leq c$ for all $x \in X$. It is least since the set of all upper bounds of X is Y, and $c \leq y$ for all $y \in Y$.

Exercise 1.35 (Olmsted 1.15)

Prove **Dedekind's Theorem**: Let the real numbers be divided into two nonempty sets A and B such that (i) if $x \in A$ and if $y \in B$, then $x < y$ and (ii) if $x \in \mathbb{R}$ then either $x \in A$ or $x \in B$, then there exists a number c (which may belong to either A or B) such that any number less than c belongs to A and any number greater than c belongs to B.

Solution. This is really the same statement as Zorich 2.2.15.a, the original statement of completeness, but with the extra condition that the sets $A = X, B = Y$ must be disjoint.

Exercise 1.36 (Olmsted 1.7)

If x is an irrational number, under what conditions on the rational numbers a, b, c, d is $(ax+b)/(cx+d)$ rational?

Solution. Note that a trivial solution is $a = b = c = d = 1$ which gives 1. Since
$$\frac{ax+b}{cx+d} = \frac{acx + ad - ad + bc}{cx+d} = a + \frac{bc - ad}{cx+d} \tag{166}$$
for the above to be rational it is necessary that $1/(cx+d)$ is rational. But this cannot be the case, which leaves us with the condition that $bc = ad$.

Exercise 1.37 (Olmsted 1.8)

Prove that the system of integers satisfies the axiom of completeness.

Solution. Let $S \subset \mathbb{Z}$ be bounded from above. It must have a maximum element (justify?), call it c. Then we claim that $c \in \mathbb{Z}$ is the least upper bound. Being the maximum, it is an upper bound, and c is least since the next smallest element is $c-1$, which is less than $c \in S$, and therefore cannot be an upper bound.

Exercise 1.38 (Zorich 2.2.16/Olmsted 1.16)

Prove the following.

1. If $A \subset B \subset \mathbb{R}$, then $\sup A \leq \sup B$ and $\inf A \geq \inf B$.
2. Let $\mathbb{R} \supset X \neq \emptyset$ and $R \supset Y \neq \emptyset$. If $x \leq y$ for all $x \in X, y \in Y$, then X is bounded above , Y is bounded below, and $\sup X \leq \inf Y$.
3. If the sets X, Y in (b), are such that $X \cup Y = \mathbb{R}$, then $\sup X = \inf Y$.
4. If X and Y are the sets defined in (c), then either X has a maximal element or Y as a minimal element.
5. Show that Dedekind's theorem is equivalent to the axiom of completeness.

Solution. Listed.

1. Let
$$\begin{aligned} A' &= \{x \in \mathbb{R} \mid x \geq a \; \forall a \in A\} \\ B' &= \{x \in \mathbb{R} \mid x \geq b \; \forall b \in B\} \end{aligned}$$
where we can easily verify that $B' \subset A'$. By definition, we get $\sup B = \min B'$ and $\sup A = \min A'$. But since $B' \subset A'$, for any $b' \in B'$, there exists an $a' \in A'$ such that $a' \leq b'$, which implies that $\sup B = \min B' \leq \min A' = \sup A$.
2. X is bounded above by any element of Y. Y is bounded below by any element of X. By the completion axiom, there exists a $c \in \mathbb{R}$ such that
$$x \leq c \leq y \text{ for all } x \in X, y \in Y \tag{167}$$
Since c is an upper bound of X, $\sup X \leq c$ by definition, and since c is a lower bound of Y, $\inf Y \geq c$ by definition. Therefore, $\sup X \leq c \leq \inf Y$.
3. From completeness there exists a $c \in \mathbb{R}$ such that $x \leq c \leq y$ for all $x \in X, y \in Y$. Y is, by definition, the set of *all* upper bounds of X (i.e. *every* upper bound of X is in Y, unlike Y defined in 2.2.16.b). Since $c \leq y$ for all $y \in Y$, c is minimal and so $c = \sup X$. X is the set of all lower bounds of Y by definition, so $c \geq x$ for all $x \in X \implies c = \inf Y$. So, $\inf Y = c = \sup X$.
4. We know that there exists $c = \inf Y = \sup X$. Since $X \cup Y = \mathbb{R}$, c must be in at least X or Y. If $c \in X$, then $c = \sup X = \max X$, and if $c \in Y$, then $c = \inf Y = \min Y$.
5. This is the same statement as Zorich 2.2.15.a (an iff equivalence, not just one way implying).

Exercise 1.39 (Olmsted 1.13)

Let S be a nonempty set of numbers bounded above, and let x be the least upper bound of S. Prove that x has the two properties corresponding to an arbitrary positive number ϵ:

1. every element $s \in S$ satisfies the inequality $s < x + \epsilon$
2. at least one element $s \in S$ satisfies the inequality $s > x - \epsilon$

Solution. Listed.

1. x is an upper bound $\implies$ $s \leq x$ for all $s \in S$, which implies that $s \leq x < x + \epsilon$.
2. By definition, $x - \epsilon$ cannot be an upper bound, so $x - \epsilon \geq s$ for all $s \in S$ is not true. Therefore, there must exist one $s \in S$ such that $s > x - \epsilon$.

Exercise 1.40 (Zorich 2.2.18)

Let $-A$ be the set of numbers of the form $-a$, where $a \in A \subset \mathbb{R}$. Show that $\sup(-A) = -\inf(A)$.

Solution. If A is unbounded below, then $-\inf A = \infty$ and $-A$ is unbounded above, implying that $\sup A = \infty$. Now assume that A is bounded below, then by completeness, it must have a greatest lower bound. Let us define the set $B = \{b \in \mathbb{R} \mid b \leq a \; \forall a \in A\}$. From 2.2.16.b, we have $b \leq \inf A \leq a$ for all $a \in A, b \in B$. Multiplying by -1 gives $-b \geq -\inf A \geq -a$ for all $a \in A, b \in B$, which is equivalent to saying

$$a \leq -\inf A \leq b \text{ for all } a \in -A, b \in -B \tag{168}$$

by definition of $-A, -B$. $-\inf A$ is clearly an upper bound of $-A$, and since

$$\begin{aligned} B &= \{b \in \mathbb{R} \mid b \leq a \; \forall a \in A\} \\ &= \{b \in \mathbb{R} \mid -b \geq -a \; \forall a \in A\} \\ &= \{b \in \mathbb{R} \mid -b \geq a \; \forall a \in -A\} \end{aligned}$$

implies that $-B = \{b \in \mathbb{R} \mid b \geq a \; \forall a \in -A\}$ is the set of all upper bounds of A. So, $-\inf A$ is the least upper bound of $-A$, i.e. $-\inf A = \sup(-A)$.

Exercise 1.41 (Zorich 2.2.21)

Show that the set $\mathbb{Q}(\sqrt{n})$ of numbers of the form $a + b\sqrt{n}$ where $a, b \in \mathbb{Q}$, n is a fixed natural number that is not the square of any integer, is an ordered set satisfying the principle of Archimedes but not the axiom of completeness.

Solution. The order on $\mathbb{Q}(\sqrt{n})$ can be embedded from the ordering on the reals by defining the canonical injection map $i : \mathbb{Q}(\sqrt{n}) \longrightarrow \mathbb{R}$ and defining for any $x, y \in \mathbb{Q}(\sqrt{n})$,

$$x \leq_{\mathbb{Q}(\sqrt{n})} y \iff i(x) \leq_{\mathbb{R}} i(y) \tag{169}$$

Now, let $h > 0$ be any fixed real number, and $x = (a, b) = a + b\sqrt{n}$. By the Archimidean principle, we can find a $k \in \mathbb{Z}$ such that

$$(k-1)h \leq x \leq kh \text{ for some } x \in \mathbb{Q}(\sqrt{n}) \subset \mathbb{R} \tag{170}$$

We now show that $\mathbb{Q}(\sqrt{n})$ is not complete since it doesn't satisfy the LUB property. Since there are infinite prime numbers in $\mathbb{N}$, choose a prime number p that is not a factor of n. Then, we are guaranteed that pn is not a perfect square, and can define the set

$$X = \{x \in \mathbb{Q}(\sqrt{n}) \mid x < \sqrt{pn}\} \subset \mathbb{Q}(\sqrt{n}) \tag{171}$$

and assume that $c = c_1 + c_2\sqrt{n} = \sup X$ exists ($c_1, c_2 \in \mathbb{Q}$). Clearly, $c \neq \sqrt{pn} \notin \mathbb{Q}(\sqrt{n})$.

1. Assume $c < \sqrt{pn} \iff 0 < \sqrt{pn} - c \in \mathbb{R}$. By the Archimidean principle, there exists a $k \in \mathbb{N}$ such that $0 < \frac{1}{k} < \sqrt{pn} - c$. Then, we can verify that $c + \frac{1}{k} = (c_1 + \frac{1}{k}) + c_2\sqrt{n} \in \mathbb{Q}(\sqrt{n})$ and
$$c + \frac{1}{k} < c + \sqrt{pn} - c = \sqrt{pn} \implies c + \frac{1}{k} \in X \tag{172}$$
implies that c is not an upper bound. So we must turn to case 2.
2. Assume $c > \sqrt{pn} \iff c - \sqrt{pn} > 0$. By AP, there exists a $k \in \mathbb{N}$ such that $0 < \frac{1}{k} < c - \sqrt{pn}$. Then, we can verify that $c - \frac{1}{k} \in \mathbb{Q}(\sqrt{n})$ and
$$c - \frac{1}{k} > c - c + \sqrt{pn} = \sqrt{pn} \tag{173}$$
implies that $c - \frac{1}{k}$ is an upper bound of X, so c is not least.

Therefore, by contradiction, c does not exist.

Exercise 1.42 (Zorich 2.2.22)

Let $n \in \mathbb{N}$ and $n > 1$. In the set $E_n = \{0, 1, \ldots, n-1\}$, we define the sum and product of two elements as the remainders when the usual sum and product in $\mathbb{R}$ are divided by n. With these operations on it, the set E_n is denoted $\mathbb{Z}_n$.

1. Show that if n is not a prime number, then there are nonzero numbers $m, k \in \mathbb{Z}_n$ such that $m \cdot k = 0$, i.e. there exist nonzero zero divisors.
2. Show that if p is prime, then there are no zero divisors in $\mathbb{Z}_p$ and $\mathbb{Z}_p$ is a field.
3. Show that, no matter what the prime p, $\mathbb{Z}_p$ cannot be ordered in a way consistent with the arithmetic operations on it.

Solution. Listed.

1. n is composite implies that there exist $1 < m, k < n$ such that $n = mk$. These factors m, k are precisely the zero divisors of $\mathbb{Z}_n$ since $mk = n \equiv 0 \pmod{n}$.
2. With p prime, assume that there are nontrivial zero divisors $1 < m, k < p$ in $\mathbb{Z}_p$. Then, $mk \equiv 0 \pmod{n} \implies mk = lp$ for some $l \in \mathbb{N}$. But this implies that m or k must divide p, which is impossible since $1 < m, k < p$. Then prove field axioms.
3. For any field, we must have $0 \leq 1$, because if not, then

$$0 > 1 \implies 0 < 1^{-1} \cdot 1 = 1^{-1} \implies 0 \cdot 0 < 1^{-1} \cdot 1^{-1} = 1 \tag{174}$$

So, $0 \leq 1$ implies that $0 \leq 1 \leq 2 \leq \ldots \leq p - 1$. But

$$0 + 1 \leq (p - 1) + 1 = 0 \tag{175}$$

is false, so any ordering is impossible.

Exercise 1.43 (Zorich 2.2.23)

Show that if $\mathbb{R}$ and $\mathbb{R}'$ are two models of the set of real numbers and $f : \mathbb{R} \longrightarrow \mathbb{R}'$ (with $f \not\equiv 0'$) is a mapping such that $f(x + y) = f(x) + f(y)$ and $f(x \cdot y) = f(x) \cdot f(y)$ for any $x, y \in \mathbb{R}$. Prove that f is an order-preserving isomorphism.

Solution. Let $0, 0'$ be the additive identity of $\mathbb{R}, \mathbb{R}'$, respectively, and $1, 1'$ the multiplicative identity. We claim that $f(0) = 0'$ since

$$\begin{aligned} f(0) &= f(0 + 0) && \text{(definition of additive identity)} \\ &= f(0) + f(0) && \text{(homomorphism over } +) \end{aligned}$$

which implies that $f(0) + f(0) = f(0) = 0' + f(0)$. Since $f(0)$ lives in field $\mathbb{R}'$, its additive identity $-f(0)$ is well defined, and we get $f(0) = f(0) + f(0) + (-f(0)) = 0' + f(0) + (-f(0)) = 0'$. We also claim that $f(1) = 1'$ since

$$\begin{aligned} f(1) &= f(1 \cdot 1) && \text{(definition of multiplicative identity)} \\ &= f(1) \cdot f(1) && \text{(homomorphism over } \cdot) \end{aligned}$$

which implies that $f(1) \cdot f(1) = 1' \cdot f(1)$. Since $f(1)$ lives in field $\mathbb{R}'$, its multiplicative identity $f(1)^{-1}$ is well defined, and we get $f(1) = f(1) \cdot f(1) \cdot f(1)^{-1} = 1' \cdot f(1) \cdot f(1)^{-1} = 1'$. Now that we have proved mapping of identities, this implies the mapping of inverses.

$$\begin{aligned} 0' = f(0) = f(x - x) = f(x) + f(-x) &\implies f(-x) = -f(x) \\ 1' = f(1) = f(x \cdot x^{-1}) = f(x) \cdot f(x^{-1}) &\implies f(x^{-1}) = f(x)^{-1} \end{aligned}$$

With these conditions, we have proved that f is a homomorphism of fields. Now we prove that f is a bijection, but first, we claim that $f(x) = 0' \implies x = 0$. Assume that there exists a nonzero $x \in \mathbb{R}$ such that $f(x) = 0'$. Then, x^{-1} is well defined, and

$$f(x) \cdot f(x^{-1}) = f(x) \cdot f(x)^{-1} = 0'$$
$$f(x) \cdot f(x^{-1}) = f(x \cdot x^{-1}) = f(1) = 1'$$

which implies that $0' = 1'$. So, $f(1) = 1' = 0'$, and so for all $k \in \mathbb{R}$, $f(k) = f(k \cdot 1) = f(k) \cdot f(1) = f(k) \cdot 0' = 0' \implies f \equiv 0'$, leading to a contradiction of the assumption that $f' \not\equiv 0'$.

1. (f injective). Assume f is not injective, i.e. there exists distinct $x_1, x_2 \in \mathbb{R}$ s.t. $f(x_1) = f(x_2)$. Then, using that fact $f(x) = 0 \implies x = 0$,

$$0 = f(x_1) - f(x_2) = f(x_1 - x_2) \implies x_1 - x_2 = 0 \implies x_1 = x_2 \tag{176}$$

2. (f surjective). Let y be any nonzero element in $\mathbb{R}'$ (clearly if $y = 0'$ then its preimage is 0) and y^{-1} its multiplicative inverse. Assume there exists no $x \in \mathbb{R}$ satisfying $f(x) = y$, meaning that there exist no x satisfying

$$f(x) \cdot = y \cdot y^{-1} = 1' \tag{177}$$

But since f maps inverses to inverses, we can choose $x = (y^{-1})^{-1}$, which leads to

$$f(x) \cdot y^{-1} = (\tag{178}$$

Finally, we prove that f is order preserving. Assume that $x \leq y \iff 0 \leq y - x$, we wish to prove that

$$f(x) \leq f(y) \iff 0 \leq f(y) - f(x) = f(y - x) \tag{179}$$

Therefore, since this preservation of ordering is really the statement $0 \leq y - x \implies 0 \leq f(y - x)$, it suffices to prove that $0 \leq x \implies 0 \leq f(x)$. Now, assume that we have a x such that $f(x) < 0'$. Adding it with the equation $f(1) = 1'$ gives us

$$f(x + 1) < 1' \tag{180}$$

It is easy to prove that $0 \leq x \iff 0 \leq x^{-1}$. Now assume that $0 > f(x)$. **INCOMPLETE**

Exercise 1.44 (Density of Rationals in $\mathbb{R}$)

Prove that for any two distinct $a < b \in \mathbb{R}$, there exists an infinite number of rational numbers between a and b.

Solution. Since $a < b$, then $b - a > 0$ and by the Archimidean principle, there exists a $k \in \mathbb{N}$ such that

$$0 < \frac{1}{k} < b - a \implies 1 < kb - ka \tag{181}$$

which implies that the length of $[ka, kb)$ greater than 1. By the inductive property of $\mathbb{Z}$, there must be an integer $p \in [ka, kb)$. If there were not, then this would imply that $[ka+1, kb+1)$ and $[ka-1, kb-1)$ had no integers and repeating would mean that there were no integers in $\mathbb{R}$. Therefore,

$$ka \leq p < kb \implies a \leq \frac{p}{k} < b \tag{182}$$

for all $a, b \in \mathbb{R}$, with $p/k \in \mathbb{Q}$. If a is irrational we can replace the $\leq$ to $<$, leaving $a \leq \frac{p}{k} < b$, and if a is rational, we can construct another rational $a + \frac{1}{k} \in (a, b)$.

Exercise 1.45 (Nested Interval Lemma)

With the fact that $\mathbb{R}$ is complete, prove the following.

1. For a sequence of closed nested intervals $I_1 \supset I_2 \supset \ldots$ of $\mathbb{R}$, there exists a point $c \in \mathbb{R}$ belonging to all these intervals.
2. Furthermore, if the hypothesis also satisfies the fact that for any $\epsilon > 0$, there exists a $k \in \mathbb{N}$ such that $|I_k| < \epsilon$ (i.e. the length of the intervals decreases to 0), then the point c common to all sets is unique.

Solution. Listed.

1. Let $I_n = [a_n, b_n]$, with $a_n < b_n$ finite for all $n \in \mathbb{N}$. For all $n \in \mathbb{N}$, we have $I_n = [a_n, b_n]$ and can take the two subsets $X_n = (-\infty, a_n)$ and $Y_n = (b_n, \infty)$, where $x \leq y$ for every $x \in X_n, y \in Y_n$. We also have the fact that $\mathbb{R} = X_n \cup I_n \cup Y_n$. Since $\mathbb{R}$ is complete, there exists a c such that $x \leq c \leq y$ for all $x \in X, y \in Y$. But $x \leq c \iff c \notin X_n$ and $c \leq y \iff c \notin Y_n$, so for all $n \in \mathbb{N}$, c must be in I_n.
2. Since we have proved (a), it now suffices to prove uniqueness of c. Let there be two distinct points $c_1, c_2 \in \mathbb{R}$ belonging to these intervals. Without loss of generality, assume $c_1 - c_2 > 0$, and choose
$$\epsilon = \frac{c_1 - c_2}{3} \tag{183}$$
Then, there should exist a $k \in \mathbb{N}$ such that $|I_k| < \epsilon$. Since I_k must contain c_1, it must be a subset of $[c_1 - \epsilon, c_1 + \epsilon]$ (should be able to see why) and similarly for c_2.
$$\begin{aligned} I_k &\subset \bigg[c_1 - \frac{c_1 - c_2}{3}, c_1 + \frac{c_1 - c_2}{3}\bigg] = \bigg[\frac{2c_1 + c_2}{3}, \frac{4c_1 - c_2}{3}\bigg] = L \\ I_k &\subset \bigg[c_2 - \frac{c_1 - c_2}{3}, c_2 + \frac{c_1 - c_2}{3}\bigg] = \bigg[\frac{-c_1 + 4c_2}{3}, \frac{c_1 + 2c_2}{3}\bigg] = M \end{aligned}$$
But since $c_1 > c_2 \implies \frac{c_1+2c_2}{3} < \frac{2c_1+c_2}{3}$, L and M are disjoint $\implies I_k$, as a subset of both, leaves us with $I_k = \emptyset$, contradicting that it is a closed interval.

Exercise 1.46

Compactness of Closed Interval in $\mathbb{R}$ Prove that any system of open intervals covering (i.e. an open cover of) a closed interval contains a finite subsystem that covers the closed interval. Another way to state this is by saying that every closed interval of $\mathbb{R}$ is compact.

Solution. A closed interval with a finite open covering is trivially compact since any subcovering is also finite. We only need to deal with when a closed interval $I = [a, b]$ has an infinite open covering $\{U_\alpha\}_{\alpha \in A}$, which means that the set of indices A is infinite. Assume that there exists no finite covering of I. Then, we divide I into two halves
$$I_1 = \bigg[a, \frac{a+b}{2}\bigg], \;\; I_2 = \bigg[\frac{a+b}{2}, b\bigg] \tag{184}$$
and define a subcovering for each of them. That is, we can define $A_1 \subset A$ and $A_2 \subset A$ such that $\{U_\alpha\}_{\alpha \in A_1} \subset \{U_\alpha\}_{\alpha \in A}$ is a covering of I_1 and $\{U_\alpha\}_{\alpha \in A_2} \subset \{U_\alpha\}_{\alpha \in A}$ is a covering of I_2. At least one of A_1 or A_2 must be infinite, since if they were both finite, then we can define a finite covering $\{U_\alpha\}_{\alpha \in A_1 \cup A_2}$ of I. Choose the interval with the infinite covering and repeat this procedure, which will result in a nested interval that decreases in length by a half.
$$I \supset I_1 \supset I_2 \supset \ldots \tag{185}$$

By the nested interval lemma, there exists a unique point c common to all these intervals. But since $c \in [a, b]$, the open cover $\{U\}$ should contain an open interval $(c - \delta_1, c + \delta_2)$ containing c. We wish to prove that this interval is a superset of some I_k in the sequence above, contradicting the fact that I_k has an infinite cover. Since the length of each I_i decreases arbitrarily (i.e. we can choose any $\epsilon > 0$ and find a I_k with length less than ϵ), we choose $\epsilon = \frac{1}{2} \min\{\delta_1, \delta_2\}$, and we should be able to find some I_k that is a subinterval of $[c - \epsilon, c + \epsilon]$, which itself is a subinterval of $(c - \delta_1, c + \delta_2)$.

$$I_k \subset \left[c - \frac{1}{2} \min\{\delta_1, \delta_2\}, c + \frac{1}{2} \min\{\delta_1, \delta_2\}\right] \subset (c - \delta_1, c + \delta_2) \tag{186}$$

Therefore, $(c - \delta_1, c + \delta_2)$ is a finite cover of I_k, contradicting the fact that all I_k's have infinite covers.

Exercise 1.47 (Bolzano-Weierstrass Theorem)

Prove that every bounded infinite set of real numbers has at least one limit point. (A limit point p of set X is a point such that every open neighborhood of p contains an infinite number of elements of X).

Solution. Let the set of points be denoted X, and let a be the lower bound and b be the upper bound. Then, $X \subset [a, b] = I$. Now divide $[a, b]$ into halves $[a, \frac{a+b}{2}] \cup [\frac{a+b}{2}, b]$. At least one of the halves must have an infinite number of points; choose the interval with infinite points as I_1 and doing this repeatedly gives the nested sequence

$$I \supset I_1 \supset I_2 \supset \ldots \tag{187}$$

By the nested interval lemma, there exists at least one point $c \in \mathbb{R}$ that is in all these intervals. Furthermore, since $|I_i| = \frac{1}{2^i}(b - a)$ decreases to 0, we can choose a $\epsilon > 0$ and find an interval I_k with $|I_k| < \epsilon$. We claim that c is a limit point of X. Given an ϵ, we wish to prove that there are an infinite number of points within the ϵ-neighborhood $(c - \epsilon, c + \epsilon)$ of c. Since we can find some I_k with $|I_k| < \epsilon$, we can see that

$$I_k \subset (c - \epsilon, c + \epsilon) \tag{188}$$

and therefore the ϵ-neighborhood of c contains I_k, which contains an infinite number of points in X.
We can construct another proof that is dependent on the compactness lemma. This construction will be useful for problem 2.3.4. Let X be a given subset of $\mathbb{R}$, and it follows from the definition of boundedness that X is contained in some closed interval $I \subset \mathbb{R}$. We show that at least one point of I is a limit point of X. Assume that it is not. Then each point $x \in I$ would have a neighborhood $U(x)$ containing at most a finite number of points from X. The totality of such neighborhoods $\{U(x)\}$ constructed for the points $x \in I$ forms an open covering of X. Since I is closed, it is compact and therefore we can find a finite subcovering $\{U_i(x)\}_i$ of open intervals that cover I and therefore cover X. This open cover $\{U_i(x)\}_i$ of X is a finite union of sets that each contain at most a finite number of points from X, so the covering of X contains a finite number of points from X, a contradiction that X contains infinite points.

Exercise 1.48 (Zorich 2.3.1)

Show that

1. if I is any system of nested closed intervals, then

$$\sup\{a \in \mathbb{R} \mid [a, b] \in I\} = \alpha \leq \beta = \inf\{b \in \mathbb{R} \mid [a, b] \in I\}$$

   and

$$[\alpha, \beta] = \bigcap_{[a,b] \in I} [a, b]$$

2. if I is a system of nested open intervals (a, b), the intersection

$$\bigcap_{(a,b) \in I} (a, b)$$

may happen to be empty.

Solution. Listed.

1. (May be tempted to say that $a_1 \leq a_2 \leq \ldots$, but this assumes that the indexing set I is countable). We claim that for any two intervals $[a_n, b_n]$ and $[a_m, b_m]$ in I,

$$a_n \leq b_m$$

Assume that $a_n > b_m$. Then $b_n \geq a_n > b_m \geq a_m$ implies that $[a_n, b_n]$ and $[a_m, b_m]$ are disjoint, contradicting the fact that they are nested. Now given that X is the set of a_n's and Y is the set of b_n's, we have $x \leq y$ for all $x \in X, y \in Y$. So by 2.2.16.b, we have $\sup X \geq \inf Y$.
To prove the second statement, we show that trying to "expand" the interval $[\alpha, \beta]$ will lead to a contradiction. Since α is the LUB, given any $\epsilon > 0$, there exists a $(a_l, b_l) \in X$ such that $\alpha - \epsilon < a_l < \alpha$, which implies that $[\alpha, \beta] \subset [a_l, \beta] \subset [\alpha - \epsilon, \beta]$. Assuming that this extended interval is the intersection, we should be able to choose any point in $[\alpha - \epsilon, \beta]$ and find that it is in every element of I. We choose a point in $[\alpha - \epsilon, a_l)$, which is not in the interval (a_l, b_l). We do the same for $\beta \mapsto \beta + \epsilon$. We also check that "shrinking" the interval $[\alpha, \beta] \mapsto [\alpha + \epsilon, \beta]$ is no good, since we can find an element in $[\alpha, \alpha + \epsilon)$ that is in every interval in I.
2. Take the system of nested open intervals

$$(0, 1) \supset (0, \frac{1}{2}) \supset (0, \frac{1}{3}) \ldots (0, \frac{1}{n}) \supset \ldots$$

Take their infinite intersection, denote it S, and assume that some $\epsilon \in (0, 1)$ is in S. Since ϵ is a real number, by the Archimidean principle there exists a $k \in \mathbb{N}$ such that $\frac{1}{k} < \epsilon$. Therefore, $\epsilon \notin (0, \frac{1}{k}) \implies \epsilon \notin S$.

Exercise 1.49 (Zorich 2.3.2)

Show that

1. from a system of closed intervals covering a closed interval it is not always possible to choose a finite subsystem covering the interval.
2. from a system of open intervals covering a open interval it is not always possible to choose a finite subsystem covering the interval.
3. from a system of closed intervals covering a open interval it is not always possible to choose a finite subsystem covering the interval.

Solution. We show with the interval $(0, 1)$ or $[0, 1]$. Using linear transformations it is easy to generalize this to any other interval (a, b) or $[a, b]$.

1. Consider the infinite covering

$$[0, 1] = [0, \frac{1}{2}] \cup [\frac{1}{2}, \frac{3}{4}] \cup [\frac{3}{4}, \frac{7}{8}] \cup \ldots$$

2. Consider the infinite covering

$$(0, 1) = (0, \frac{1}{2}) \cup (\frac{1}{2}, \frac{3}{4}) \cup (\frac{3}{4}, \frac{7}{8}) \cup \ldots$$

3. Consider the infinite covering

$$(0,1) = \big[0, \frac{1}{2}\big] \cup \big[\frac{1}{2}, \frac{3}{4}\big] \cup \big[\frac{3}{4}, \frac{7}{8}\big] \cup \ldots$$

Exercise 1.50 (Zorich 2.3.3)

Show that if we only take the set $\mathbb{Q}$ of rational numbers instead of the complete set $\mathbb{R}$ of real numbers, with the definitions of closed, open, and neighborhood of a point $r \in \mathbb{Q}$ to mean respectively the corresponding subsets of $\mathbb{Q}$, then none of the three lemmas is true.

Solution. We prove only for the nested interval lemma. We choose the series of nested intervals

$$\left(\sqrt{2} - \frac{1}{n}, \sqrt{2} + \frac{1}{n}\right)$$

with $n \in \mathbb{N}$. Assume that there is a $r \in \mathbb{Q}$ such that

$$r \in \left(\sqrt{2} - \frac{1}{n}, \sqrt{2} + \frac{1}{n}\right) \text{ for all } n \in \mathbb{N}$$

which is equivalent to saying that $\left|r - \sqrt{2}\right| < \frac{1}{n}$ for all $n \in \mathbb{N}$. Clearly, $r \neq \sqrt{2}$, and by the Archimidean principle, there exists a $k \in \mathbb{N}$ such that

$$0 < \frac{1}{k} < |r - \sqrt{2}|$$

which contradicts the above.

Exercise 1.51 (Zorich 2.3.4)

Show that the three lemmas above are equivalent to the axiom of completeness.

Solution. Note that from the proofs, completeness implies nested interval lemma, which implies compactness of closed intervals, which implies the Bolzano-Weierstrass theorem. So, it is sufficient to prove that Bolzano-Weierstrass theorem implies completeness to determine equivalence. There are not a lot of direct proofs, so we prove that Weierstrass implies nested interval, which implies completeness.

1. (Weierstrass $\implies$ Nested) Assume that we have $\mathbb{R}$ with the Bolzano-Weierstrass theorem. Take the series of nested closed intervals

$$I = [a, b] \supset I_1 = [a_1, b_1] \supset I_2 = [a_2, b_2] \supset \ldots$$

We see that $a \leq a_i \leq b$, so the infinite sequence of monotonically nondecreasing values a_i is bounded. Therefore, it must have a limit point, which we will denote as c. We claim that $a_i \leq c$ for all a_i. Since if it were not, then $c < a_i$ for some i, and choosing $\epsilon = 0.5(a_i - c)$, the ϵ-neighborhood of c will not contain a_j for $j \geq i$ since

$$c < a_i \implies 0.5c < 0.5a_i \implies c + \epsilon = 0.5c + 0.5a_i < a_i < a_{i+1} < \ldots$$

. With similar reasoning, we can conclude that $b_i \geq c$ for all b_i. This implies that $a_i \leq c \leq b_i$ for all i which is equivalent to saying that $c \in [a_i, b_i] = I_i$ for all $i \in \mathbb{N}$.

2. (Nested $\implies$ LUB Principle) Let $X \subset \mathbb{R}$ be a set that is bounded above, with b_1 any upper bound. Since X is nonempty, there exists $a_1 \in X$ that is not an upper bound (otherwise, X would be a singleton set and it trivially has a least upper bound). Consider the well-defined

interval $[a_0, b_0]$. Take the mean $m_0 = 0.5(a_0 + b_0)$, and if m_0 is an upper bound, set it to b_1 (with $a_1 = a_0$) and a_1 if else (with $b_1 = b_0$). Then, we have a sequence of nested intervals

$$[a_0, b_0] \supset [a_1, b_1] \supset [a_2, b_2] \supset \ldots$$

of decreasing lengths $|I_k| = \frac{1}{2^k}(b - a)$. All of them must contain a unique common point $c \in \mathbb{R}$ by the nested intervals lemma, which implies that

$$a_0 \leq a_1 \leq a_2 \leq \ldots \leq c \leq \ldots \leq b_2 \leq b_1 \leq b_0$$

I claim two things:

(a) c is an upper bound for X. Suppose it were not, then there exists some $x \in X$ such that $c < x$, and let the distance between them be $\epsilon = x - c > 0$. By AP, we can choose $k \in \mathbb{N}$ such that $\frac{1}{k} < \epsilon$. All the b_n are upper bounds of X, so we have $x \leq b_n$. Subtracting c on both sides gives

$$0 < x - c = \epsilon \leq b_n - c \leq |I_n| = \frac{1}{2^n}(b_0 - a_0)$$

where the last inequality follows from $c \in I_n = [a_n, b_n]$, so the maximum distance it can be from the endpoint b_n is $|I_n|$. The inequality above holds for all $n \in \mathbb{N}$, so increasing n arbitrarily should decrease $\frac{1}{2^n}(b_0 - a_0)$ past ϵ. To formalize this, we use the inequality

$$\frac{1}{2^n} < \frac{1}{n} \text{ for all } n \in \mathbb{N}$$

and so we have

$$\epsilon \leq b_n - c < \frac{1}{n}(b_0 - a_0)$$

We choose the natural number $n = \lceil \frac{2(b_0 - a_0)}{\epsilon} \rceil$, which does not satisfy the inequality above since

$$\epsilon < \frac{1}{n}(b_0 - a_0) = \frac{1}{\lceil 2(b_0 - a_0)/\epsilon \rceil}(b_0 - a_0) \leq \frac{\epsilon}{2(b_0 - a_0)}(b_0 - a_0) = \frac{\epsilon}{2}$$

This leads to a contradiction.

(b) We now prove that c is least. Assume that c is not least $\implies$ there exists an upper bound B such that $B < c$ and $x \leq B$ for all $x \in X$. **INCOMPLETE**

Exercise 1.52 (Zorich 2.4.1)

Show that the set of real numbers has the same cardinality as the points of the interval $(-1, 1)$.

Solution. We define the bijective map $\rho : (-1, 1) \longrightarrow \mathbb{R}$

$$p(x) = \begin{cases} 0 & \text{if } x = 0 \\ \frac{1}{x} & \text{if } x \neq 0 \end{cases}$$

Exercise 1.53 (Zorich 2.4.2)

Give an explicit one-to-one correspondence between

1. the points of two open intervals
2. the points of two closed intervals
3. the point of a closed interval and an open interval
4. the points of the closed interval $[0, 1]$ and $\mathbb{R}$

Solution. Listed.

1. $\rho : (a, b) \longrightarrow (c, d)$ defined
$$\rho(x) = \frac{d-c}{b-a}(x-a) + c$$
2. the extension of ρ defined on (a) to $[a, b]$
3. From (a) and (b), it suffices to prove a bijection from $(0, 1)$ to $[0, 1]$. We extract a countably infinite sequence from $(0, 1)$, say
$$x_1 = \frac{1}{3},\ x_2 = \frac{1}{4}, \ldots, x_i = \frac{1}{i+2}$$
Then, we define bijection $\rho : (0, 1) \longrightarrow [0, 1]$ as
$$\rho(x) = \begin{cases} x & \text{if } x \not\in \{x_i\} \\ 0 & \text{if } x = x_1 = \frac{1}{2} \\ 1 & \text{if } x = x_2 = \frac{1}{3} \\ x_{i-2} & \text{if } x = x_i \text{ for } i > 2 \end{cases}$$
Colloquially, we extract a copy of $\mathbb{N}$ from $(0, 1)$ and use the bijection $\mathbb{N} \simeq \mathbb{N} \cup \{0, -1\}$ to "shift" the terms.
4. We compose the bijections $\rho_1 : [0, 1] \longrightarrow (0, 1)$ and $\rho_2 : (0, 1) \longrightarrow \mathbb{R}$.

Exercise 1.54 (Zorich 2.4.3)

Show that

1. every infinite set contains a countable subset
2. the set of even integers has the same cardinality as the set of all natural numbers
3. the union of an infinite set and an at most countable set has the same cardinality as the original infinite set.
4. the set of irrational numbers has the cardinality of the continuum
5. the set of transcendental numbers has the cardinality of the continuum

Solution. Listed.

1. Let A be an infinite set. By axiom of choice, choose $a_0 \in A$. Then, $A \setminus \{a_0\} \neq \emptyset$ since A is infinite. By induction, assume you have chosen $a_0, a_1, \ldots, a_k \in A$. Then, since A is infinite, $A \setminus \{a_0, a_1, \ldots, a_k\} \neq \emptyset$, so we can choose $a_{k+1} \in A \setminus \{a_0, \ldots, a_k\}$. Thus, we have constructed a countable subset $\{a_k\}_{k \in \mathbb{N}}$ of A.
2. Given the quotient ring $2\mathbb{Z}$, define the bijection $\rho : 2\mathbb{Z} \longrightarrow \mathbb{N}$ as
$$p(x) = \begin{cases} x + 2 & \text{if } x \geq 0 \\ -x - 1 & \text{if } x < 0 \end{cases}$$
3. From (a), we can extract a countable set from original set A, call it X. Since the product of countable sets is countable ($\mathbb{N} \cup \mathbb{N}$ is countable), we can define a bijection $\tilde{\rho} : X \longrightarrow X \cup B$. Therefore, we can define a bijection $\rho : A \longrightarrow A \cup B$ as
$$\rho(x) = \begin{cases} x & \text{if } x \in A \setminus X \\ \tilde{\rho}(x) & \text{if } x \in X \end{cases}$$
4. $\mathbb{Q}$ is countable and $\mathbb{R}$ is uncountable. So, $\mathbb{R} \setminus \mathbb{Q}$ must be uncountable since if it were countable, then the union of the rationals and irrationals, which is $\mathbb{R}$, would be countable.

5. It suffices to prove that the set of algebraic numbers (numbers that are possible roots of a polynomial with integer coefficients with leading coefficient nonzero) is countable, since we can apply (d) right after. The set of all kth degree polynomials with integer coefficients is isomorphic to $\mathbb{Z}^k$ through the map

$$a_k x^k + a_{k-1}x^{k-1} + \ldots + a_2 x^2 + a_1 x^1 + a_0 \mapsto (a^{k-1}, a^{k-2}, \ldots, a_1, a_0)$$

and the union of these countable sets (minus the 0 map)

$$P = \bigg(\bigcup_{k=1}^{\infty} \mathbb{Z}^k \bigg) \setminus \{0\} = \big(\mathbb{Z} \setminus \{0\}\big) \cup \mathbb{Z}^2 \cup \ldots$$

is countable. For any element in $\mathbb{Z}^k$, there are at most k real roots, and so we can define the set of roots of an element $z \in \mathbb{Z}^k \subset P$ as a j-tuple of algebraic numbers, which can have at most $j = k$ roots.

$$r(z) = \underbrace{(r_{1z}, r_{2z}, \ldots, r_{jz})}_{j \leq k}$$

Therefore, the union of all these j-tuples for all $z \in P$

$$\bigcup_{z \in P} r(z) = \bigcup_{k=1}^{\infty} \bigcup_{z \in \mathbb{Z}^k} r(z)$$

is a countable union of a countable union of finite sets, making it countable.

Exercise 1.55 (Zorich 2.4.4)

Show that
1. the set of increasing sequences of natural numbers has the same cardinality as the set of fractions of the form $0.\alpha_1\alpha_2\ldots$
2. the set of all subsets of countable set has cardinality of the continuum

Solution. Listed.
1. Given a sequence of increasing naturals $S = (n_1, n_2, \ldots)$, we can define a binary expansion $0.\alpha_1\alpha_2\ldots$ where $\alpha_i = 1$ if and only if $i \in \mathbb{N}$ is in S and $\alpha_i = 0$ if not. This is clearly a bijection.
2. The set of all segments of increasing natural is equipotent with $2^{\mathbb{N}}$, since the elements of each sequence define a subset of $\mathbb{N}$. Cantor's diagonalization argument proves that the set of infinite binary expansions is uncountable, and by (a), this proves that $2^{\mathbb{N}}$ is uncountable.

This is very interesting since $\mathbb{N} \simeq \mathbb{R}$, but $2^{\mathbb{N}} \simeq \mathbb{R}$, and the set of all infinite q-ary expansions is equipotent to $\mathbb{R}$ too.

Exercise 1.56 (Zorich 2.4.5)

Show that
1. the set $\mathcal{P}(X)$ of subsets of a set X has the same cardinality as the set of all functions $f : X \longrightarrow \{0, 1\}$.
2. for a finite set X of n elements, $\operatorname{card} \mathcal{P}(X) = 2^n$
3. one can write $\operatorname{card} \mathcal{P}(X) = 2^{\operatorname{card} X}$, which implies $\operatorname{card} \mathcal{P}(\mathbb{N}) = 2^{\operatorname{card} \mathbb{N}} = \operatorname{card} \mathbb{R}$
4. for any set X, $\operatorname{card} X < 2^{\operatorname{card} X}$

Solution. Listed.

1. An element $Y \in \mathcal{P}(X)$ is a subset of X by definition. Letting
$$f_Y(x) = \begin{cases} 0 & \text{if } x \notin Y \\ 1 & \text{if } x \in Y \end{cases}$$
we can construct the bijective map $Y \mapsto f_Y$.
2. We can prove this using the identity (which can be proved using induction)
$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$
3. Let $F(X;\{0,1\})$ be the set of all binary valued functions from X to $\{0,1\}$. From (a), $\operatorname{card} \mathcal{P}(X) \simeq F(X;\{0,1\})$. Each binary-valued function f is determined by the assignment $f(x)$ for each $x \in X$. Since $f(x)$ has two possible values, the assignment of $f(x)$ for all $x \in X$ has $\{0,1\}^{\operatorname{card} X}$ possible choices. This gives another bijection $F(X;\{0,1\}) \simeq \{0,1\}^{\operatorname{card} X}$, so
$$\mathcal{P}(X) \simeq \{0,1\}^{\operatorname{card} X} \implies \operatorname{card} \mathcal{P}(x) = \operatorname{card}(\{0,1\}^{\operatorname{card} X}) = 2^{\operatorname{card} X}$$
4. If X is finite, then letting $n = \operatorname{card} X$, we can simply prove $n < 2^n$ by induction (which we will not do here). If X is countable, then $\mathcal{P}(X)$ is uncountable (from 2.4.4) and so using (c),
$$\operatorname{card} X = \operatorname{card} \mathbb{N} < \operatorname{card} \mathbb{R} = \operatorname{card} \mathcal{P}(X) = 2^{\operatorname{card} X}$$
For uncountable sets (and for the two cases mentioned above), we can use Cantor's theorem, which states that $\operatorname{card} X < \operatorname{card} \mathcal{P}(X)$, and so using (c), we have $\operatorname{card} X < \operatorname{card} \mathcal{P}(X) = 2^{\operatorname{card} X}$.

Exercise 1.57 (Zorich 2.4.6)

Let $X_1, \ldots, X_m$ be a finite system of finite sets. Show that
$$\begin{aligned}\operatorname{card}\left(\bigcup_{i=1}^{m} X_i\right) &= \sum_{i_1} \operatorname{card} X_{i_1} - \sum_{i_1 < i_2} \operatorname{card}(X_{i_1} \cap X_{i_2}) + \ldots \\ &\sum_{i_1 < i_2 < i_3} \operatorname{card}(X_{i_1} \cap X_{i_2} \cap X_{i_3}) - \ldots + (-1)^{m-1} \operatorname{card}(X_1 \cap \ldots \cap X_m) \\ &= \sum_{k=1}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \operatorname{card}\left(\bigcap_{j=1}^{k} X_{i_j}\right)\end{aligned}$$

Solution. Ignoring Russell's paradox (defining the universe set of all sets), we can use the commutative, associative, and distributive properties of $\cup, \cap$ on the algebra of sets. We prove using induction on m. For $m = 1$, we trivially have $\operatorname{card} X_1 = \operatorname{card} X_1$, and for $m = 2$, we claim
$$\operatorname{card}(X_1 \cup X_2) = \operatorname{card}(X_1) + \operatorname{card}(X_2) - \operatorname{card}(X_1 \cap X_2)$$
X_1 and $X_2 \setminus X_1$ are clearly exclusive sets by definition, with $X_1 \cup X_2 = X_1 \cup (X_2 \setminus X_1)$, so
$$\operatorname{card}(X_1 \cup X_2) = \operatorname{card}\big(X_1 \cup (X_2 \setminus X_1)\big) = \operatorname{card}(X_1) + \operatorname{card}(X_2 \setminus X_1) \tag{2}$$
By definition, the set $X_2 \setminus X_1$ and $X_1 \cap X_2$ are disjoint and satisfies $X_2 = (X_2 \setminus X_1) \cup (X_1 \cap X_2)$ (also by definition), so
$$\operatorname{card}(X_2) = \operatorname{card}(X_2 \setminus X_1) + \operatorname{card}(X_1 \cap X_2) \tag{3}$$

and substituting (3) into (2) gives the claim for $m = 2$. Assuming that the claim is satisfied for some m, we have

$$\begin{aligned}
\operatorname{card}\left(\bigcup_{i=1}^{m+1} X_i\right) &= \operatorname{card}\left(\left[\bigcup_{i=1}^{m} X_i\right] \cup X_{m+1}\right) \\
&= \operatorname{card}\left(\bigcup_{i=1}^{k} X_i\right) + \operatorname{card}(X_{m+1}) - \operatorname{card}\left(\left[\bigcup_{i=1}^{m} X_i\right] \cap X_{m+1}\right) \qquad \text{(claim for } m = 2\text{)} \\
&= \operatorname{card}\left(\bigcup_{i=1}^{k} X_i\right) + \operatorname{card}(X_{m+1}) - \operatorname{card}\left(\bigcup_{i=1}^{m} (X_i \cap X_{m+1})\right) \qquad \text{(distributive prop.)} \\
&= \sum_{k=1}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \operatorname{card}\left(\bigcap_{j=1}^{k} X_{i_j}\right) + \operatorname{card}(X_{m+1}) \\
&\quad - \sum_{k=1}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \operatorname{card}\left(\bigcap_{j=1}^{k} (X_{i_j} \cap X_{m+1})\right)
\end{aligned}$$

With a bit of thought, we can see that the kth term of the second summation contributes to adding another term to the $k+1$th summation term of the first. Therefore, we must try to shift the summation over by 1 index. Let us simplify this by taking the summations and extracting the first and last term, respectively. We have

$$\begin{aligned}
\sum_{k=1}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \operatorname{card}\left(\bigcap_{j=1}^{k} X_{i_j}\right) &= \sum_{1 \leq i_1 \ldots i_k \leq m} \operatorname{card}(X_{i_1}) \\
&\quad + \sum_{k=2}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \operatorname{card}\left(\bigcap_{j=1}^{k} X_{i_j}\right)
\end{aligned}$$

and

$$\begin{aligned}
&\sum_{k=1}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \operatorname{card}\left(\bigcap_{j=1}^{k} (X_{i_j} \cap X_{m+1})\right) \\
&\qquad = \sum_{k=1}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \operatorname{card}\left(\left[\bigcap_{j=1}^{k} X_{i_j}\right] \cap X_{m+1}\right) \\
&\qquad = \sum_{k=1}^{m-1} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \operatorname{card}\left(\bigcap_{j=1}^{k} (X_{i_j} \cap X_{m+1})\right) + (-1)^{m-1} \operatorname{card}\left(\bigcap_{j=1}^{m+1} X_j\right) \\
&\qquad = \sum_{k=2}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-2} \operatorname{card}\left(\bigcap_{j=1}^{k-1} (X_{i_j} \cap X_{m+1})\right) + (-1)^{m-1} \operatorname{card}\left(\bigcap_{j=1}^{m+1} X_j\right)
\end{aligned}$$

So subtracting the summations gives

$$\begin{aligned}
&\sum_{k=1}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \operatorname{card}\left(\bigcap_{j=1}^{k} X_{i_j}\right) - \sum_{k=1}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \operatorname{card}\left(\bigcap_{j=1}^{k} (X_{i_j} \cap X_{m+1})\right) + |X_{m+1}| \\
&\quad = \sum_{1 \leq i_1 \ldots i_k \leq m} \operatorname{card}(x_i) + \sum_{k=2}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \operatorname{card}\left(\bigcap_{j=1}^{k} X_{i_j}\right) + \operatorname{card}(X_{m+1}) \\
&\qquad + \sum_{k=2}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \operatorname{card}\left(\bigcap_{j=1}^{k-1} (X_{i_j} \cap X_{m+1})\right) + (-1)^m \operatorname{card}\left(\bigcap_{j=1}^{m+1} X_j\right) \\
&\quad = \sum_{1 \leq i_1 \ldots i_k \leq m+1} \operatorname{card}(X_i) + \sum_{k=2}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \Bigg[\operatorname{card}\left(\bigcap_{j=1}^{k} X_{i_j}\right) \\
&\quad + \operatorname{card}\left(\left[\bigcap_{j=1}^{k-1} X_{i_j}\right] \cap X_{m+1}\right)\Bigg] + (-1)^m \operatorname{card}\left(\bigcap_{j=1}^{m+1} X_j\right)
\end{aligned}$$

and since the set of sequences of k terms bounded by $m+1$ (of form $1 \leq i_1 \ldots i_k \leq m+1$) is the set of sequences of k terms bounded by m (of form $1 \leq i_1 \ldots i_k \leq m$) unioned with the set of sequences of k terms with max element $m+1$ (of form $1 \leq i_1 \ldots i_k = m+1$), we have

$$\sum_{1 \leq i_1 \ldots i_k \leq m} (-1)^{k-1} \left[\operatorname{card}\left(\bigcap_{j=1}^{k} X_{i_j}\right) + \operatorname{card}\left(\left[\bigcap_{j=1}^{k-1} X_{i_j}\right] \cap X_{m+1}\right)\right] = \sum_{1 \leq i_1 \ldots i_k \leq m+1} \operatorname{card}\left(\bigcap_{j=1}^{k} X_{i_j}\right)$$

and therefore, substituting the above and observing that the independent terms are the first and last terms of the summation gives

$$\begin{aligned}
\operatorname{card}\left(\bigcup_{i=1}^{m+1} X_i\right) &= \sum_{1 \leq i_1 \ldots i_k \leq m+1} \operatorname{card}(X_i) + \sum_{k=2}^{m} \sum_{1 \leq i_1 \ldots i_k \leq m+1} (-1)^{k-1} \operatorname{card}\left(\bigcap_{j=1}^{k} X_{i_j}\right) \\
&\quad + \ldots + (-1)^m \operatorname{card}\left(\bigcap_{j=1}^{m+1} X_j\right) \\
&= \sum_{k=1}^{m+1} \sum_{1 \leq i_1 \ldots i_k \leq m+1} (-1)^{k-1} \operatorname{card}\left(\bigcap_{j=1}^{k} X_{i_j}\right)
\end{aligned}$$

Exercise 1.58 (Zorich 2.4.7)

On the closed interval $[0,1] \subset \mathbb{R}$, describe the sets of numbers $x \in [0,1]$ whose ternary representation $x = 0.\alpha_1\alpha_2\ldots$, $\alpha_i \in \{0,1,2\}$ has the property.

1. $\alpha_1 \neq 1$
2. $\alpha_1 \neq 1$ and $\alpha_2 \neq 1$
3. For all $i \in \mathbb{N}$, $\alpha_i \neq 1$ (the Cantor set)

Solution. Listed.

1. $[0, \frac{1}{3}) \cup [\frac{2}{3}, 1)$
2. $[0, \frac{1}{9}) \cup [\frac{2}{9}, \frac{3}{9}) \cup [\frac{6}{9}, \frac{7}{9}) \cup [\frac{8}{9}, 1)$
3. Made by recursively removing the middle third of every partitioned intervals.

Exercise 1.59 (Zorich 2.4.8)

Show that

1. the set of numbers $x \in [0, 1]$ whose ternary representation does not contain 1 has the same cardinality as the set of all numbers whose binary representation has the form $0.\beta_1\beta_2\ldots$
2. the Cantor set has the same cardinality as the closed interval $[0, 1]$

Solution. Listed.

1. We can define a bijection $0.\alpha_1\alpha_2\ldots \mapsto 0.\beta_1\beta_2\ldots$ as $\alpha_i = 0 \iff \beta_i = 0$ and $\alpha_i = 1 \iff \beta_i = 2$.
2. The map above defines a bijection between the Cantor set and the set of all infinite binary expansions in $[0, 1]$, which is uncountable by Cantor's diagonalization theorem.

2 Euclidean Topology

With the construction of the real line and the real space, the extra properties of completeness, norm, and order (for the real line) allows us to restate these topological properties in terms of these "higher-order" properties. It also proves much more results than for general topological spaces. Therefore, the next few sections will focus on reiterating the topological properties of $\mathbb{R}$ and $\mathbb{R}^n$ (this can be done slightly more generally for metric spaces, but we talk about this in point-set topology).

Theorem 2.1 (Euclidean Topology)

The **Euclidean topology** $\mathscr{T}$ on $\mathbb{R}^n$ is defined in the equivalent ways:
1. If $n = 1$, it is the order topology generated by the basis of open intervals (a, b) for $a < b$.
2. It is the metric topology generated by the basis of open balls under the L^p metric for $p \geq 1$.
3. It is the n-fold product topology of the Euclidean topology on $\mathbb{R}$.

Proof. Let's call these sets $\mathscr{T}_1, \mathscr{T}_2, \mathscr{T}_3$. We first show that these are indeed topologies.
1. This is proven in the definition of the order topology.
2. This is proven for the L^2 metric in the definition of the metric topology, and then we prove equivalence to L^p metrics.
3. This is proven in the definition of the product topology.

Now it remains to show that these topologies are all the same. For $n = 1$, it suffices to show $\mathscr{T}_1 = \mathscr{T}_2$. We do this by comparing their bases; let $\mathscr{B}_k$ be the basis of $\mathscr{T}_k$.
1. $\mathscr{T}_1 \subset \mathscr{T}_2$. Given any open interval $(a, b) \in \mathscr{B}_1$, (a, b) is the open ball with center $\frac{a+b}{2}$ and radius $\frac{b-a}{2}$ and so it is in $\mathscr{B}_2$.
2. $\mathscr{T}_2 \subset \mathscr{T}_1$. Given an open interval with center x and radius r, the corresponding interval $(x-r, x+r)$ is in $\mathscr{B}_1$.

Now we show that $\mathscr{T}_2 = \mathscr{T}_3$ for $\mathbb{R}^n$.
1. We show that the product topology is the same as the metric topology under the L^∞ norm.
2. The equivalence of all L^p norms is already proven here.

Example 2.1 (Examples of Open and Not Open Sets)

Here are some examples of sets which are open and not open.
1. $U = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \neq 1\}$ is open since for every point $x \in U$, we just need to find a radius $\epsilon > 0$ that is smaller than its distance to the unit circle.
2. $(a, b) \times (c, d) \subset \mathbb{R}^2$ is open since given a point x, we can take the minimum of its distance between the two sides of the rectangle and construct an open ball.
3. $S = \{(x, y) \in \mathbb{R}^2 : xy \neq 0\}$ is open since given a point $x \in S$, we can take the minimum of the distance between it and the x and y axes.
4. The set of all complex z such that $|z| \leq 1$ is not open since we cannot construct open balls at the boundary points that are fully contained in the set.
5. The set $S = \{1/n\}_{n\in\mathbb{N}}$ is not open since given any point $x = 1/n$, we can construct an open ball with radius $\epsilon < 1/(n + 1)$, which contains irrationals that are not in S.

Corollary 2.2

Every closed bounded interval in $\mathbb{R}$ is compact.

Proof. Let $I = [a, b]$. Then if $x, y \in I$, $|x - y| \leq b - a = \delta$. Now by contradiction, suppose that there exists an open cover $\{U_\alpha\}$ of I which contains no finite subcover of I. Then letting $c = (a + b)/2$, at least one of the two intervals $[a, c], [c, b]$ cannot have a finite subcovering (otherwise their finite union can be covered). WLOG let it be $[a, c]$. We keep subdividing and get the sequence of nested intervals.

$$I \supset I_1 \supset I_2 \supset \dots \tag{189}$$

We know that I_n is not covered by any finite subcollection of $\{U_\alpha\}$ and if $x, y \in I_n$, then $|x-y| < 2^{-n}\delta$. From the nested intervals theorem, there exists a point z lying in every I_n. There must then be an open neighborhood U_z in the open cover, and by definition of openness there exists a $\epsilon > 0$ s.t. $z \in B_\epsilon(z) \subset U_z$. By the Archimidean property, we can set n so large that $2^{-n}\delta < \epsilon$ and this means that $B_\epsilon(z) \supset I_n$, which contradicts the fact that I_n is not covered by a finite subcollection. Therefore I is compact.

Now here are some useful properties.

Theorem 2.3 (Countability)

$\mathbb{R}^n$ is 2nd countable.

Theorem 2.4 (Separability)

$\mathbb{R}^n$ is Hausdorff.

Proof. It is a metric topology.

Theorem 2.5 (Connectedness)

$\mathbb{R}^n$ is
1. connected,
2. path connected,
3. locally connected,
4. locally path connected

Theorem 2.6 (Compactness)

$\mathbb{R}^n$ is
1. not compact,
2. locally compact.

Theorem 2.7 (Convexity)

An open ball is convex in a normed vector space. What happens if we weaken it to a metric?

Proof. The normed part is important here, as the properties of the metric is not sufficient. Given $B_r(p)$, $x, y \in B_r(p)$ implies that $||x-p|| < r$ and $||y-p|| < r$. Therefore,

$$\begin{aligned} ||tx + (1-t)y - p|| &= ||tx - tp + (1-t)y - (1-t)p|| \qquad (190) \\ &\leq t||x-p|| + (1-t)||y-p|| \qquad (191) \\ &= tr + (1-t)r = r \qquad (192) \end{aligned}$$

Theorem 2.8

Let E be a nonempty set of real numbers which is bounded above. Let $y = \sup E$. Then $y \in \overline{E}$. Hence $y \in E$ if E is closed.

Proof. Assume that y is not a limit point of E. Then, there exists some $\epsilon > 0$ s.t. $(y - \epsilon, y + \epsilon)$ does not intersect with E. This means that $y - \epsilon$ is an upper bound of E, and so y is not the supremum.

The general notion of compactness for topological spaces is not needed for analysis. Rather, we make use of the following theorem which allows us to focus on the compactness of subsets in Euclidean spaces $\mathbb{R}^n$.

Theorem 2.9 (Heine-Borel)

Let $E \subset \mathbb{R}^n$. The following are equivalent.

1. E is closed and bounded.
2. E is compact.
3. E is sequentially compact.
4. E is limit point compact.

Proof. To prove the equivalence of the last three, note that X is a metric space, so it is Hausdorff. Therefore, all three forms of compactness is the same.

Example 2.2 (Open Sets in Real Plane are Not Compact)

An open set in $\mathbb{R}^2$ is not compact. Take the open rectangle $R = (0,1)^2 \subset \mathbb{R}^2$. There exists an infinite cover of R

$$R = \bigcup_{n=0}^{\infty} (0,1) \times \left(0, \frac{2^{n+1} - 1}{2^{n+1}}\right) \tag{193}$$

that does not have a finite subcover.

Corollary 2.10 (Bolzano-Weierstrass Theorem)

Every bounded sequence in $\mathbb{R}^n$ has a limit point, i.e. contains a convergent subsequence.

Proof. The fact that the infinite sequence is bounded means that there exists some closed subset $I \in \mathbb{R}^n$ that contains all point of the sequence. By definition I is compact, so by the Heine-Borel theorem, every cover of I has a finite subcover.
Now, assume that there exists an infinite sequence in I that is not convergent, i.e. has no limit point. Then, each point $x_i \in I$ would have a neighborhood $U(x_i)$ containing at most a finite number of points in the sequence. We can define I such that the union of the neighborhoods is a cover of I. That is,

$$I \subset \bigcup_{i=1}^{\infty} U(x_i) \tag{194}$$

However, since every $U(x_i)$ contains at most a finite number of points, we must have an infinite open neighborhoods to cover $I \implies$ we cannot have a finite subcover. This contradicts the fact that I is compact.

Proof. It suffices to prove that there exists a monotonic sequence within a bounded sequence (x_n).

Theorem 2.11 (Connectedness)

A subset E of the real line $\mathbb{R}$ is connected if and only if it has the following property: if $x \in E, y \in E$ and $x < z < y$, then $z \in E$.

Proof.

Definition 2.1 (Perfect Sets)

A set P is perfect if it is closed and all of its points are limit points of P. In other words, the limit points of P and P itself coincide.

$$P' = P \tag{195}$$

Theorem 2.12

Let P be a nonempty perfect set in $\mathbb{R}^k$. Then P is uncountable.

2.1 Exercises

Exercise 2.1 (Math 531 Spring 2025, PS5.1)

We know what it means for a metric space (X, d) to be compact. We say that it is sequentially compact if every sequence in (X, d) has a convergent subsequence. Prove that a metric space is compact if and only if it is sequentially compact.

Solution. We prove bidirectionally.

1. $(\rightarrow)$. Assume that X is compact and let (x_n) be a sequence in X. For any $\epsilon > 0$, let

$$\mathscr{C}_\epsilon \coloneqq \{B_\epsilon(x) \mid x \in X\} \tag{196}$$

   be an open cover of open balls. Then there exists a finite subcover $\mathcal{F}_\epsilon \subset \mathscr{C}_\epsilon$. There is a countable sequence (x_n), with each point in at least one open set in $\mathscr{C}$. By the pigeonhole principle, at least one open set must be hit infinitely many times, call this $B(x^*, \epsilon)$. Now consider for $\epsilon = \frac{1}{n}$ for $n \in \mathbb{N}$.
2. $(\leftarrow)$.

Exercise 2.2 (Math 531 Spring 2025, PS5.2)

Give an example of a sequence of real numbers x_n for which

$$|x_{n+1} - x_n| \to 0 \tag{197}$$

as $n \to \infty$, but x_n is not convergent.

Solution. Consider the sequence

$$x_n = \sum_{i=1}^{n} \frac{1}{i} \tag{198}$$

It is the case that $x_{n+1} - x_n = 1/(n+1)$ which tends to 0, but this is a harmonic series which is not convergent.

Exercise 2.3 (Math 531 Spring 2025, PS5.3)

Let $\{x_n\}_{n=0}^{\infty}$ be a sequence of real numbers. Assume that

$$|x_{n+1} - x_n| \leq c|x_n - x_{n-1}| \tag{199}$$

for all $n \geq 1$, for some fixed $c < 1$. Prove that x_n is convergent. Hint: you may want to use the formula that you proved in a previous homework for $1 + c + c^2 + \cdots + c^N$.

Solution.

Exercise 2.4 (Math 531 Spring 2025, PS5.4)

Let x_n be a sequence of rational numbers defined recursively by:

$$x_0 = 1 \tag{200}$$

$$x_{n+1} = \frac{1}{x_n + 2} \tag{201}$$

when $n \geq 0$. The first few terms of this sequence are $1, \frac{1}{3}, \frac{3}{7}, \ldots$ Prove that the sequence is convergent and find its limit.

Solution.

Exercise 2.5 (Math 531 Spring 2025, PS5.5)

As we know, every bounded sequence of real numbers has a convergence subsequence.

1. Let's say we have two sequences a_n and b_n that are bounded. Find a single sequence of indices $\{n_k\}$ so that *both* a_{n_k} and b_{n_k} are convergent. This is called a common convergent subsequence for a_n and b_n.
2. Show that for any finite number of bounded sequences of real numbers, we can find a common convergent subsequence.
3. Now suppose have a sequence of bounded sequences. Find a common convergent subsequence. What does this remind you of?

Solution.

Exercise 2.6 (Math 531 Spring 2025, PS5.6)

Given a sequence of real numbers x_n, we can define the sequence of its means by:

$$x_1, \frac{x_1 + x_2}{2}, \frac{x_1 + x_2 + x_3}{3}, \ldots \tag{202}$$

Call the sequence of means y_n. Prove that if $x_n \to x$, then $y_n \to x$. Discuss the examples $x_n = (-1)^n$, $x_n = n$, $x_n = (-1)^n n$, and $x_n = (-1)^n \sqrt{n}$. The fourth example shows that it is possible to average out chaotic behavior (so long as it isn't focused in one direction). The third example shows that this is impossible if the system becomes too chaotic.

Solution.

Exercise 2.7 (Math 531 Spring 2025, PS4.1)

Let (X, d) be a metric space. Assume that K is compact and F is closed in (X, d). Assume $K \cap F = \emptyset$. Prove that

$$\inf_{x \in F, y \in K} d(x, y) > 0. \tag{203}$$

Show by an example that this number could be zero if K is only assumed to be closed (rather than compact).

Solution. $K \cap F = \emptyset \implies K \subset F^c$ with F^c open. This means that for every $x \in K \subset F^c$, there exists a $r_x > 0$ s.t. $B(x, r_x) \subset F^c \iff B(x, r_x) \cap F = \emptyset$.
Now we take the covering $\{B(x, \frac{r_x}{2}) \mid x \in K\}$, and since K is compact there must be a finite subcovering, which we denote $C = \{B(x_i, \frac{r_i}{2}) \mid i = 1, \ldots n\}$. Denote $r^* = \min\{r_i\}$, which is positive since we take the minimum of a finite number of positive elements.
Now for any $x \in K$ and $y \in F$, x must be in some $B(x_i, \frac{r_i}{2}) \iff d(x, x_i) < \frac{r_i}{2} \iff -d(x, x_i) > -\frac{r_i}{2}$. With the same i, since $B(x_i, r_i)$ is disjoint from F, we have $d(x_i, y) \geq r_i$. Therefore, by the triangle inequality,

$$d(x, y) \geq d(x_i, y) - d(x_i, x) = r_i - \frac{r_i}{2} = \frac{r_i}{2} \geq \frac{r^*}{2} \tag{204}$$

and thus we have found a nontrivial lower bound.

Exercise 2.8 (Math 531 Spring 2025, PS4.2)

Consider $\mathbb{R}$ with the usual metric. Find an open cover of $\mathbb{Q}$ that does not cover $\mathbb{R}$.

Solution. $\mathbb{Q} = (-\infty, \sqrt{2}) \cup (\sqrt{2}, +\infty)$.

Exercise 2.9 (Math 531 Spring 2025, PS4.3)

$\mathbb{R}$ is not compact with the usual metric since it is not bounded. Let us, however, define the following metric on $\mathbb{R}$:

$$d_*(x, y) = \frac{|x - y|}{(1 + |x|)(1 + |y|)}. \tag{205}$$

Verify that $(\mathbb{R}, d_*)$ is a metric space. Prove that all subsets of $(\mathbb{R}, d_*)$ are bounded. Show that $\mathbb{R}$ still isn't compact with this metric. What is the problem?

Solution. We first verify metric.

1. Since $|x - y| \geq 0, 1 + |x| \geq 1, 1 + |y| \geq 1$, $d_*(x, y) \geq 0$. We also see that

$$d_*(x, y) = 0 \iff |x - y| = 0 \iff x = y \tag{206}$$

2. It is symmetric since

$$d_*(x, y) = \frac{|x - y|}{(1 + |x|)(1 + |y|)} = \frac{|y - x|}{(1 + |y|)(1 + |x|)} = d_*(y, x) \tag{207}$$

3. It satisfies triangle inequality since

$$\begin{aligned}
d_*(x,y) + d_*(y,z) &= \frac{|x-y|}{(1+|x|)(1+|y|)} + \frac{|y-z|}{(1+|y|)(1+|z|)} && (208)\\
&= \frac{|x-y|(1+|z|) + |y-z|(1+|x|)}{(1+|x|)(1+|y|)(1+|z|)} && (209)\\
&= \frac{|x-y| + |x-y|\cdot|z| + |y-z| + |y-z|\cdot|x|}{(1+|x|)(1+|y|)(1+|z|)} && (210)\\
&\geq \frac{|x-z| + (|x|-|y|)\cdot|z| + (|y|-|z|)\cdot|x|}{(1+|x|)(1+|y|)(1+|z|)} && (211)\\
&= \frac{|x-z| + |x||y| - |y||z|}{(1+|x|)(1+|y|)(1+|z|)} && (212)\\
&= \frac{(1+|y|)|x-z|}{(1+|x|)(1+|y|)(1+|z|)} && (213)\\
&= \frac{|x-z|}{(1+|x|)(1+|z|)} && (214)\\
&= d_*(x,z) && (215)
\end{aligned}$$

where the inequality comes from $|x-y| + |y-z| \geq |x-z|$, $|x-y| \geq ||x|-|y|| \geq |x|-|y|$, and $|y-z| \geq ||y|-|z|| \geq |y|-|z|$.

This is bounded since for any $x, y \in \mathbb{R}$, we can show that the numerator is bounded by the denominator. Since both are positive from (1), it suffices to prove $|x-y|^2 \leq (1+|x|)^2(1+|y|)^2$.

$$\begin{aligned}
(1+|x|)^2(1+|y|)^2 &= (1+2|x|+|x|^2)(1+2|y|+|y|^2) && (216)\\
&= 1+2|x|+2|y|+4|x||y|+|x|^2+|y|^2+2|x||y|^2+2|y||x|^2+|x|^2|y|^2 && (217)\\
&\geq |x|^2+|y|^2+2|x||y| && (218)\\
&\geq |x|^2+|y|^2-2xy && (219)\\
&= |x-y|^2 && (220)
\end{aligned}$$

where the first inequality holds since all the terms in the expansion are nonnegative and the second holds since $|x||y| \geq xy$. Therefore, $d_*(x,y) \leq 1$. $\mathbb{R}$ is still not compact since we can construct the set of open balls $B_r(0)$ around 0 w.r.t. d_*. Consider the cover

$$\mathscr{C} = \{B_{1-\frac{1}{n}}(0)\}_{n \in \mathbb{N}, n \geq 2} \tag{221}$$

Now assume that there is a finite subcover. Then there must be a maximum index $N \in \mathbb{N}$ in this cover. I claim that this does not cover $\mathbb{R}$. Consider the element $y = N-1 \in \mathbb{R}$. The distance is

$$d_*(x,y) = \frac{|0-(N-1)|}{(1+|0|)(1+|N-1|)} = \frac{N-1}{N} = 1 - \frac{1}{N} \tag{222}$$

and so $y \notin \mathscr{C}$. Hence $\mathscr{C}$ is not a cover of $\mathbb{R}$.

Exercise 2.10 (Math 531 Spring 2025, PS4.4)

Consider the set $X = \mathbb{R} \cup \{Gandalf\}$. Define a metric d_* on X by:

$$d_*(x,y) = \frac{|x-y|}{(1+|x|)(1+|y|)} \tag{223}$$

for $x, y \in \mathbb{R}$, while

$$d_*(Gandalf, x) = \frac{1}{1+|x|}, \tag{224}$$

for all $x \in \mathbb{R}$. Verify that d_* is a metric on X (you don't have to do much for this, since you already did part of it in the previous problem). Prove that (X, d_*) is a compact metric space.

Solution.

Exercise 2.11 (Math 531 Spring 2025, PS4.5)

Let X be any set and endow it with the metric $d(x, y) = 1$ if $x \neq y$ and $d(x, x) = 0$. Check that this is a metric. Find all compact sets in (X, d).

Solution. It trivially satisfies nonnegativity since it's either 0 or 1, and $d(x, x) = 0$. It is symmetric as well. As for triangle inequality this is trivial. All compact sets are finite sets.

Exercise 2.12 (Math 531 Spring 2025, PS3.1)

Determine for each of the following sets, whether or not it is countable. Justify your answers

1. The set of all functions $f : \{0, 1\} \to \mathbb{N}$.
2. The set B_n of all functions $f : \{1, ..., n\} \to \mathbb{N}$
3. The set $C = \cup_{n\in\mathbb{N}} B_n$
4. The set of all functions $f : \mathbb{N} \to \{0, 1\}$.
5. The set of all functions $f : \mathbb{N} \to \{0, 1\}$ that are "eventually zero" (We say that f is eventually zero if there exists some $N \geq 1$ so that $f(n) = 0$ for all $n \geq N$.)
6. G the set of all functions $f : \mathbb{N} \to \mathbb{N}$ that are eventually constant.

Solution. Listed.

1. Countable since bijective to $\mathbb{N} \times \mathbb{N}$. We define the bijection as
$$(a_0, a_1) \in \mathbb{N} \times \mathbb{N} \mapsto f(i) = a_i \tag{225}$$
2. Countable since bijective to $\mathbb{N}^n$. We define the bijection as
$$(a_1, \ldots, a_n) \in \mathbb{N}^n \mapsto f(i) = a_i \tag{226}$$
3. Countable since we proved that B_n is countable, and a countable union of countable sets are countable from Rudin Theorem 2.12.
4. Uncountable since we can create a bijection from the set of all sequences (a_i) of 0 or 1, which from Rudin Theorem 2.14 is uncountable.
$$(a_i)_{i\in\mathbb{N}} \mapsto f(i) = a_i \tag{227}$$
5. Countable. Call this set B, and call the set of functions f that have their final 1 at index k to be A_k. Then,
$$B = \cup_{k=1}^{\infty} A_k \tag{228}$$
where $A_0 = A_1 = 1$, and $|A_k| = 2^{k-1}$ for $k \geq 2$. Since B is the countable union of at most countable sets, B must be countable.
6. Countable. Call this set B. Let A_k be the set of functions that are eventually constant to value k. Let A_{ki} be the set of functions that are always k starting from index i (where i is the smallest element). Since everything is determined to be k at i and beyond, A_{ki} can be divided up into the

first $i-1$ elements of any natural number, followed by a sequence of k's. Therefore $|A_{ki}| \approx \mathbb{N}^{i-1}$, where $\approx$ means equipotent, and so A_{ki} is countable. Therefore since countable unions of countable sets are countable,

$$A_k = \bigcup_{i=1}^{\infty} A_{ki} \text{ is countable } \implies B = \sum_{k=1}^{\infty} A_k \text{ is countable} \tag{229}$$

Exercise 2.13 (Math 531 Spring 2025, PS3.2)

Tell if the following subsets $A \subset \mathbb{R}$ (with the usual metric $d(x, y) = |x - y|$) are open or closed. Also, find (i) the limit points of A, (ii) the interior of A, (iii) $\bar{A}$.

1. $A = \mathbb{Q}$
2. $A = (0, 1]$
3. $A = \{1, \frac{1}{2}, \frac{1}{4}, ...\}$
4. $A = \{0, 1, \frac{1}{2}, \frac{1}{4}, ...\}$
5. $A = \mathbb{Z}$

Solution. Listed. We denote A' as the limit points of A and the interior as A^o.

1. Not open nor closed. $A' = \mathbb{R}$, $A^o = \emptyset$. $\bar{A} = \mathbb{R}$.
2. Not open nor closed. $A' = [0, 1]$, $A^o = (0, 1)$. $\bar{A} = [0, 1]$.
3. Not open nor closed. $A' = \{0\}$, $A^o = \emptyset$. $\bar{A} = \{0\} \cup A$.
4. Closed. $A' = \{0\}$, $A^o = \emptyset$. $\bar{A} = A$.
5. Closed. $A' = \emptyset$. $A^o = \emptyset$. $\bar{A} = A$.

Exercise 2.14 (Math 531 Spring 2025, PS3.3)

Prove the following statements subsets A, B of a general metric space (X, d).

- $\overline{A \cup B} = \bar{A} \cup \bar{B}$.
- Show by example that $\overline{A \cap B} \neq \bar{A} \cap \bar{B}$.

Solution. For the first part, we show bidirectionally.

1. $\overline{A \cup B} \subset \overline{A} \cup \overline{B}$. Let $x \in \overline{A \cup B}$. If $x \in A \cup B$, then it must be the case that either $x \in A \subset (A \cup A') = \overline{A}$ or $x \in B \subset (B \cup B') = \overline{B}$, which means $x \in \overline{A} \cup \overline{B}$. Now assume not. Then $x \in (A \cup B)'$. Therefore, for any $r > 0$, we know that $B(x, r) \cap (A \cup B) \neq \emptyset$. Now let us take a sequence $(r_n = \frac{1}{n})_{n \in \mathbb{N}}$, and for each r_n we have some element $x_n \in (A \cup B)$. Given that we have a countably infinite sequence of x_n, each which may be in A or B, by the pigeonhole principle either A or B must be hit infinitely many times. If $x_n \in A$ infinitely many times, then $x \in \overline{A}$, and analogous for B.
2. $\overline{A} \cup \overline{B} \subset \overline{A \cup B}$. WLOG let $x \in \overline{A}$. If $x \in A$, then $x \in (A \cup B) \subset \overline{A \cup B}$. If $x \notin A$, then $x \in A'$. Therefore for every $r > 0$, $B(x, r) \cap A \neq \emptyset$. But this means

$$\emptyset \neq (B(x, r) \cap A) \cup (B(x, r) \cap B) = B(x, r) \cap (A \cup B) \implies x \in (A \cup B)' \subset \overline{A \cup B} \tag{230}$$

For a counterexample, consider the sequences

$$A = (x_n) = \frac{1}{n} \qquad B = (y_n) = -\frac{1}{n} \tag{231}$$

for $n \in \mathbb{N}$. $\overline{A} = A \cup \{0\}, \overline{B} = B \cup \{0\}$, and so $\overline{A} \cap \overline{B} = \{0\}$. However, $A \cap B = \emptyset \implies \overline{A \cap B} = \emptyset$.

Exercise 2.15 (Math 531 Spring 2025, PS3.4)

Consider the set of rationals in canonical form (such that numerator and denominator are relatively prime) with potential distance:

$$d_1(\frac{p_1}{q_1}, \frac{p_2}{q_2}) = |q_1 - q_2|. \tag{232}$$

Is this a metric? Prove that the following defines a metric

$$d_2(\frac{p_1}{q_1}, \frac{p_2}{q_2}) = |p_1 - p_2| + |q_1 - q_2|. \tag{233}$$

Solution. This is not a metric since

$$d_1\left(\frac{2}{1}, \frac{3}{1}\right) = 1 - 1 = 0 \tag{234}$$

when $2/1 \neq 3/1$. For d_2, we show that it satisfies the three properties.

1. *Nonnegativity.* Since it is the sum of 2 absolute values which are norms and therefore nonnegative, it must be nonnegative by ordered field properties. We see that

$$\begin{aligned} \frac{p_1}{q_1} = \frac{p_2}{q_2} &\iff p_1 = p_2 \text{ and } q_1 = q_2 && (235) \\ &\iff |p_1 - p_2| = |q_1 - q_2| = 0 && (236) \\ &\iff |p_1 - p_2| + |q_1 - q_2| = 0 && (237) \end{aligned}$$

2. For symmetricity, note that

$$d_2\left(\frac{p_1}{q_1}, \frac{p_2}{q_2}\right) = |p_1 - p_2| + |q_1 - q_2| = |p_2 - p_1| + |q_2 - q_1| = d_2\left(\frac{p_1}{q_1}, \frac{p_2}{q_2}\right) \tag{238}$$

3. For triangle inequality, we see that for any $p_1/q_1, p_2/q_2, p_3/q_3$,

$$\begin{aligned} d_2\left(\frac{p_1}{q_1}, \frac{p_3}{q_3}\right) &= |p_1 - p_3| + |q_1 - q_3| && (239) \\ &= |(p_1 - p_2) + (p_2 - p_3)| + |(q_1 - q_2) + (q_2 - q_3)| && (240) \\ &\leq |p_1 - p_2| + |p_2 - p_3| + |q_1 - q_2| + |q_2 - q_3| && \text{(subadditivity of norm)} \\ &= d_2\left(\frac{p_1}{q_1}, \frac{p_2}{q_2}\right) + d_2\left(\frac{p_2}{q_2}, \frac{p_3}{q_3}\right) && (241) \end{aligned}$$

Exercise 2.16 (Math 531 Spring 2025, PS3.5)

Let $M = \{x_1, ..., x_3\}$ be a set with three points. Describe the set of all metrics on M. What if M has four points?

Solution. If M has 3 points call them x_1, x_2, x_3, then the metric is completely defined by the three values

$$\begin{aligned} d(x_1, x_2) &= d(x_2, x_1) && (242) \\ d(x_2, x_3) &= d(x_3, x_2) && (243) \\ d(x_3, x_1) &= d(x_3, x_1) && (244) \end{aligned}$$

where $d(x, x) = 0$. We must make sure that the triangle inequality satisfies for these 3 numbers. Therefore we can think of this as the set of all triangles in $\mathbb{R}^2$ (that are equivalent under translation

and rotation, but not permutation of points).
Similarly for 4 points, we can visualize the metrics as the set of all tetarhedra in $\mathbb{R}^3$ (since each face is a triangle, and therefore for any three points the triangle inequality is guaranteed to be satisfied), equivalent under translation and rotation, but not permutation of the 4 points.

Exercise 2.17 (Math 531 Spring 2025, PS3.6)

Let P be a polynomial of degree $n \geq 1$. Prove that if $P(0) = 0$, then $P(x) = xQ(x)$, for some polynomial Q of degree $n - 1$. Deduce that if $P(a) = 0$, then we can write $P(x) = (x - a)Q(x)$ for some Q of degree $n - 1$.

Solution. A nth degree polynomial will have the form

$$p(x) = \sum_{i=0}^{n} c_i x^i \tag{245}$$

Since $p(0) = c_0 = 0 \implies c_0 = 0$. This means that

$$p(x) = \sum_{i=1}^{n} c_i x^i = x \sum_{i=0}^{n-1} c_{i+1} x^i \text{ where } Q(x) = \sum_{i=0}^{n-1} c_{i+1} x^i \tag{246}$$

If $p(a) = 0$, we can construct $f(x) = p(x + a)$, where f is a polynomial since the expansion does not increase its degree. Since $f(x) = p(a) = 0$, by above f can be factorized $f(x) = xg(x)$ for some $(n-1)$th degree polynomial g, and by substitution this means that $p(x) = f(x - a) = (x - a)g(x - a)$.

Exercise 2.18 (Math 531 Spring 2025, PS3.7)

Consider all polynomials $P : \mathbb{R} \to \mathbb{R}$ of degree less than or equal to n. Call this set $\mathcal{P}_n$. Let's define potential distances on $\mathcal{P}_n$.

$$d_1(p, q) = |p(0) - q(0)|. \tag{247}$$

Show this defines a distance on $\mathcal{P}_0$ but not on $\mathcal{P}_n$ for $n \geq 1$. Now consider

$$d_N(p, q) = \sum_{j=0}^{N} |p(j) - q(j)| \tag{248}$$

Show that this defines a distance on $\mathcal{P}_n$, for every $n \leq N$. What does the solution say about polynomials of degree N?

Solution. If $n = 0$, $\mathcal{P}_n$ is a set of constant functions P, where each constant function P is determined completely by its value at any point, e.g. 0. We check the properties.

1. $d_1(p, q) \geq 0$ since we take the norm at the end. We can see that

$$\begin{aligned} d_1(p, q) = 0 &\iff |p(0) - q(0)| && (249)\\ &\iff p(0) = q(0) && (250)\\ &\iff p = q && (251) \end{aligned}$$

2. It is clearly symmetric.

$$d_1(p, q) = |p(0) - q(0)| = |q(0) - p(0)| = d_1(q, p) \tag{252}$$

3. It satisfies the triangle inequality by subadditivity of the norm.

$$\begin{aligned} d_1(p,r) &= |p(0) - r(0)| && (253)\\ &= |(p(0) - q(0)) + (q(0) - r(0))| && (254)\\ &\leq |p(0) - q(0)| + |q(0) - r(0)| && (255)\\ &= d_1(p,q) + d_1(q,r) && (256) \end{aligned}$$

It doesn't satisfy for P_n because consider $p(x) = x$ and $q(x) = x^2$. They are not the same function but $d_1(p,q) = |p(0) - q(0)| = 0$. For d_N defined on $\mathcal{P}_n$ for $n \leq N$, we verify the properties.

1. This is the sum of norms, so it must be nonnegative. Now we see that if $p = q$, then $p(x) = q(x) \implies |p(x) - q(x)| = 0 \implies d_N(p,q) = 0$. For the other way around, suppose $d_N(p,q) = 0$. Then from problem 3.8, we are solving the linear equation $0 = Vb - Vc$, where b, c are the vectors representing the coefficients of p, q, and V is the Vandermonde matrix with $a_i = i$. By linearity, this is equivalent to solving $0 = V(b - c)$, and since we showed that V is invertible (since a_i's are distinct), V has a trivial kernel and therefore $b - c = 0 \iff b = c \implies p = q$.
2. Symmetricity is trivial.

$$d_N(p,q) = \sum_{j=0}^{N} |p(j) - q(j)| = \sum_{j=0}^{N} |q(j) - p(j)| = d_N(q,p) \tag{257}$$

3. For triangle inequality,

$$\begin{aligned} d_N(p,r) &= \sum_{j=0}^{N} |p(j) - r(j)| && (258)\\ &= \sum_{j=0}^{N} |(p(j) - q(j)) + (q(j) - r(j))| && (259)\\ &\leq \sum_{j=0}^{N} |p(j) - q(j)| + |q(j) - r(j)| && (260)\\ &= \sum_{j=0}^{N} |p(j) - q(j)| + \sum_{j=0}^{N} |q(j) - r(j)| && (261)\\ &= d_N(p,q) + d_N(q,r) && (262) \end{aligned}$$

This shows that we need to "sample" more points from higher-degree polynomials to get the metric as they are higher-dimensional.

Exercise 2.19 (Math 531 Spring 2025, PS3.8)

Given distinct numbers $a_0, ..., a_N$ and numbers $b_0, ..., b_N$, prove that there exists a polynomial P of degree N with the property that

$$P(a_i) = b_i, \tag{263}$$

for $0 \leq i \leq N$. The most direct way to solve this problem, in my view, is to write the system equations you are trying to solve as a linear system for the coefficients of P. This will give you some matrix M that depends on the numbers $a_0, ..., a_N$. The key is to show that the determinant of this matrix is non-zero. It turns out that the determinant of this matrix is equal to

$$\prod_{0 \leq i < j \leq N} (a_i - a_j), \tag{264}$$

up to a potential $-$ sign depending on how you defined M. Prove this and deduce the result.

Solution. We can write the system of equations using the Vandermonde matrix $V \in \mathbb{R}^{(N+1)\times(N+1)}$ and c is the vector of coefficients of P.

$$b = Vc \iff \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_N \end{bmatrix} = \begin{bmatrix} 1 & a_0 & a_0^2 & \ldots & a_0^N \\ 1 & a_1 & a_1^2 & \ldots & a_1^N \\ \vdots & \vdots & \ddots & \vdots & \\ 1 & a_N & a_N^2 & \ldots & a_N^N \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_N \end{bmatrix} \tag{265}$$

To calculate the determinant of V, we prove using induction. Clearly for $N = 1$ we have

$$\det \begin{pmatrix} 1 & a_0 \\ 1 & a_1 \end{pmatrix} = a_1 - a_0 \tag{266}$$

Now assume that this formula holds for some $N - 1 \in \mathbb{N}$. Then for N, we can take V and subtract a_0 times the ith column from the $(i + 1)$st column. This gives us

$$V = \begin{bmatrix} 1 & 0 & 0 & \ldots & 0 \\ 1 & a_1 - a_0 & a_1^2 - a_0 a_1 & \ldots & a_1^N - a_0 a_1^{N-1} \\ 1 & a_2 - a_0 & a_2^2 - a_0 a_2 & \ldots & a_2^N - a_0 a_2^{N-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_N - a_0 & a_N^2 - a_0 a_N & \ldots & a_N^N - a_0 a_N^{N-1} \end{bmatrix} \tag{267}$$

When calculating the determinant, we can perform the cofactor expansion by the first row, and then for each ith row factor out $(a_i - a_0)$ to get

$$\det V = \prod_{j=1}^N (a_j - a_0) \det \begin{bmatrix} 1 & a_1 & \ldots & a_1^{N-1} \\ 1 & a_2 & \ldots & a_2^{N-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_N & \ldots & a_N^{N-1} \end{bmatrix} \tag{268}$$

which is the $(N-1) \times (N-1)$ Vandermonde matrix. Therefore, we can apply our inductive hypothesis to get

$$\det V = \prod_{j=1}^N (a_j - a_0) \prod_{1 \leq i < j \leq N} (a_j - a_i) = \prod_{0 \leq i \leq j \leq N} (a_j - a_i) \tag{269}$$

Note that this has a 0 determinant iff $a_i = a_j$ for some $i \neq j$. Therefore sicne a_i's are distinct, it must be nonzero. Therefore, this matrix is nonsingular, i.e. invertible, and we can solve the matrix equation to get

$$c = V^{-1} b \tag{270}$$

which from linear algebra is guaranteed to exist and is unique.

Exercise 2.20 (Rudin 2.1)

Prove that the empty set is a subset of every set.

Solution. It must suffice that if $x \in \emptyset$, then $x \in A$ for any arbitrary set A. This is vacuously true, since the initial condition is never met.

Exercise 2.21

Show that the empty function $f : \emptyset \to X$, where X is an arbitrary set, is always injective. If $X = \emptyset$, then f is bijective.

Solution. Given distinct $x, y \in \emptyset$, $f(x) \neq f(y)$ is vacuously true, but if $X \neq \emptyset$, then there exists a $w \in X$ with no preimage. If $X = \emptyset$, then the statement for all $w \in X$, there exists an $x \in \emptyset$ s.t. $f(x) = w$ is vacuously true.

Exercise 2.22 (Rudin 2.2)

A complex number z is said to be algebraic if there are integers $a_0, a_1, \ldots, a_n$, not all zero, such that

$$a_0 z^n + a_1 z^{n-1} + \ldots + a_{n-1} z + a_n = 0. \tag{271}$$

Prove that the set of all algebraic complex numbers is countable. Hint: For every positive integer N there are only finitely many equations with

$$n + |a_0| + |a_1| + \ldots + |a_n| = N \tag{272}$$

Solution. Consider all polynomials s.t. $n + \sum_{i=0}^n |a_i| = N$. There is only a finite number of them, and each polynomial has at most n distinct complex roots. So this set is finite, an unioning over all $N \in \mathbb{N}$ gives an at most countable set of roots.

Exercise 2.23 (Rudin 2.3)

Prove there exists real numbers which are not algebraic.

Solution. From the previous exercise, if there were no no real numbers which are not algebraic, then every real number is algebraic. This contradicts the fact that the set of all complex numbers is countable.

Exercise 2.24 (Rudin 2.4)

Is the set of all irrational real numbers countable?

Solution. No. Assume that it is countable. We have $\mathbb{Q}$ countable. Then, by assumption, we must have $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c$ be the union of countable sets, which must be countable, contradicting the fact that it is uncountable.

Exercise 2.25 (Rudin 2.5)

Construct a bounded set of real numbers which exactly 3 limit points.

Solution. We can construct the union of 3 sequences that converge onto the limit points $0, 1, 2$.

$$\Big\{\frac{1}{n}\Big\}_{n\in\mathbb{N}} \cup \Big\{\frac{1}{n} + 1\Big\}_{n\in\mathbb{N}} \cup \Big\{\frac{1}{n} + 2\Big\}_{n\in\mathbb{N}} \tag{273}$$

Exercise 2.26

Prove that the union of the limit points of sets is equal to the limit points of the union of the sets.

$$\bigcup_{k=1}^{m} A_k' = \left(\bigcup_{k=1}^{m} A_k \right)' \tag{274}$$

Solution.

Exercise 2.27 (Rudin 2.6)

Let E' be the set of all limit points of a set E. Prove that E' is closed. Prove that E and $\overline{E}$ have the same limit points. (Recall that $\overline{E} = E \cup E'$). Do E and E' always have the same limit points?

Solution. Listed.

1. Let x be a limit point of E'. Then, for every $\epsilon > 0$, $U = B_\epsilon(x) \cap E' \neq \emptyset$. Take a $y \in U$. Since $y \in B_\epsilon(x)$, which is open, we can construct an open ball $B_\delta(y) \subset B_\epsilon(x)$. Since $y \in E'$, $B_\delta(y)$ must contain elements of E, which means that $B_\epsilon(x)$ must also contain elements of E, and so x is a limit point of $E \implies x \in E'$ and E' is closed.
2. To prove that $E' \subset \overline{E}'$, we know that if $x \in E'$, then for every $\epsilon > 0$, there exists a $B_\epsilon^\circ(x)$ that has a nontrivial intersection with E which means that it has a nontrivial intersection with $E \cup E'$. To prove that $\overline{E}' \subset E'$, we know that if $y \in \overline{E}'$, then for every $\delta > 0$ there exists a $B_\delta(x)$ that has a nontrivial intersection with $\overline{E}$. If $B_\delta(x)$ intersects E then we are done. If $B_\delta(x)$ intersects E', then we can find a $y \in E' \cap B_\delta(x)$. Since $B_\delta(x)$ is open, we can construct $B_\varepsilon(y) \subset B_\delta(x)$ and since $y \in E'$, we know that $B_\varepsilon(y)$ contains an element of E, which means that $B_\delta(x)$ contains an element of E. Therefore, $E' = \overline{E}'$.
3. No. Consider the set $E = \{1/n\}_{n \in \mathbb{N}}$. $E' = \{0\}$, but $E'' = \emptyset$.

Exercise 2.28 (Rudin 2.7)

Let $A_1, A_2, \ldots$ be subsets of a metric space.

1. If $B_n = \cup_{i=1}^n A_i$, prove that $\bar{B}_n = \cup_{i=1}^n \bar{A}_i$ for $n = 1, 2, 3, \ldots$
2. If $B = \cup_{i=1}^\infty A_i$, prove that $\bar{B} \supset \cup_{i=1}^\infty \bar{A}_i$.

Solution. Listed.

1. We will prove that $\overline{B_n} \subseteq \cup_{i=1}^n \overline{A_i}$ and $\cup_{i=1}^n \overline{A_i} \subseteq \overline{B_n}$. If $x \in B_n$, then $x \in \cup_{i=1}^n A_i$. Therefore, assume that $x \in B_n'$. Then for every $\epsilon > 0$, there exists a $B_\epsilon^\circ(x)$ s.t.

$$B_\epsilon^\circ(x) \cap B_n \neq \emptyset \implies B_\epsilon^\circ(x) \cap \left(\bigcup_{i=1}^n A_n \right) \neq \emptyset \tag{275}$$

This means that there exists some $i = i(\epsilon)$, a function of ϵ, s.t. $B_\epsilon^\circ(x) \cap A_i \neq \emptyset$. However, this i may change if we unfix ϵ. We have so far proved that just for one $\epsilon > 0$ there exists an i. Now if we take a sequence of $\epsilon = 1, \frac{1}{2}, \frac{1}{3}, \ldots$, we have a sequence of $i(\epsilon)$'s living in $\{1, \ldots, n\}$. By the pigeonhole principle, there must be at least one i that is hit infinitely many times, and so we can choose this i, that works for all $\epsilon > 0 \implies x \in A_i' \subseteq \cup_{i=1}^n \overline{A_i}$. If $x \in \cup_{i=1}^n \overline{A_i}$, then there exists an $\overline{A_i}$ s.t. $x \in \overline{A_i}$. If $x \in A_i$, then we are done. If $x \in A_i'$, then for every $\epsilon > 0$, there exists a $B_\epsilon^\circ(x)$ s.t.

$$B_\epsilon^\circ(x) \cap A_i \neq \emptyset \implies B_\epsilon^\circ(x) \cap \left(\bigcup_{i=1}^n A_i \right) \neq \emptyset \tag{276}$$

and so $x \in B_n' \subset \overline{B_n}$.

2. $x \in \cup_{i=1}^\infty \overline{A_i} \implies x \in \overline{A_i}$ for some i. If $x \in A_i$, then $x \in B$ and we are done. If $x \in A_i'$, then for every $\epsilon > 0$ there exists $B_\epsilon(x)$ s.t.

$$B_\epsilon^\circ(x) \cap A_i \neq \emptyset \implies B_\epsilon^\circ(x) \cap \left(\bigcup_{i=1}^\infty A_i \right) \neq \emptyset \tag{277}$$

and so $B_\epsilon^\circ(x) \cap B \neq \emptyset \implies x \in B' \subset \overline{B}$.

Exercise 2.29 (Rudin 2.8)

Is every point of every open set $E \subset \mathbb{R}^2$ a limit point of E? Answer the same question for closed sets in $\mathbb{R}^2$.

Solution. Yes for open. Given any $x \in U$ open, there always exists an $\epsilon > 0$ s.t.

$$B_\epsilon^\circ(x) \subset B_\epsilon(x) \subset U \tag{278}$$

and so $B_\epsilon^\circ(x)$ has a nontrivial intersection with U. If U is closed, then no. Note that for closed U, we have that every limit point is in U, but not every point in U is a limit point. Consider the isolated point $U = \{x\}$. x is not a limit point of U.

Exercise 2.30 (Rudin 2.9)

Let E° denote the set of all interior points of E in X. Prove the following:

(a) E° is always open.
(b) E is open if and only if $E^\circ = E$.
(c) If $G \subseteq E$ and G is open, then $G \subset E^\circ$.
(d) Prove that the complement of E° is the closure of the complement of E.
(e) Do E and $\bar{E}$ always have the same interiors?
(f) Do E and E° always have the same closures?

Solution. Listed.

1. We assume that E° is not open (this does not mean that E° is necessarily closed!). That is, there exists an $x \in E^\circ$ s.t. we can't construct an open ball $B_\epsilon(x) \subseteq E^\circ$. Since $x \in E^\circ \subset E$, by definition of an interior point we can construct a $B_\epsilon(x) \subset E$. But from our assumption $B_\epsilon(x) \not\subset E^\circ$. We choose a $y \in B_\epsilon(x) \setminus E^\circ$. Since $B_\epsilon(x)$ is open, there exists a $\delta > 0$ s.t.

$$B_\delta(y) \subset B_\epsilon(x) \subset E \tag{279}$$

But the fact that we can construct an open ball around y means that $y \in E^\circ$, leading to a contradiction.

2. If E is open, then by definition $E \subset E^\circ$. Now $E^\circ \subset E$ holds for all sets since E° must be composed of points from E. If $E = E^\circ$, then for every $x \in E$, $x \in E^\circ$, so by definition there exists an $\epsilon > 0$ s.t. $B_\epsilon(x) \subset E$, which means that E is open.
3. Let $x \in G$ open. Then there exists an $\epsilon > 0$ s.t. $B_\epsilon(x) \subset G$, and so $B_\epsilon(x) \subset E$. Since we can always construct an open ball around x contained within E, $x \in E^\circ$ and $G \subset E^\circ$.
4. ($(E^\circ)^c \subset \overline{E^c}$) If $x \in (E^\circ)^c$, then there exists no $\epsilon > 0$ s.t. $B_\epsilon(x) \subset E$. Then, for any $\epsilon > 0$, $B_\epsilon(x) \not\subset E \implies B_\epsilon(x) \cap E^c \neq \emptyset \implies x \in E^c \subset \overline{E^c}$. ($\overline{E^c} \subset (E^\circ)^c$) If $x \in \overline{E^c}$, then $x \in E^c$ or $x \in E^{c\prime}$. If $x \in E^c$, note $E^\circ \subset E \implies (E^\circ)^c \supset E^c \implies x \in (E^\circ)^c$. If $x \in E^{c\prime}$, then for all $\epsilon > 0$ $B_\epsilon(x) \cap E^c \neq \emptyset \implies B_\epsilon(x) \not\subset E \implies x \in E^\circ$.

5. No. Consider the rationals $\mathbb{Q} \subset \mathbb{R}$. $\mathbb{Q}^\circ = \emptyset$ but $\overline{\mathbb{Q}}^\circ = \mathbb{R}^\circ = \mathbb{R}$. It is true and straightforward to prove that $E^\circ \subset \overline{E}^\circ$. Let $x \in E^\circ$. Then there exists an $\epsilon > 0$ s.t. $B_\epsilon(x) \subset E \implies B_\epsilon(x) \subset \overline{E} \implies x \in \overline{E}^\circ$.
6. No. Consider $\mathbb{Q} \subset \mathbb{R}$. Then $\overline{\mathbb{Q}} = \mathbb{R}$ and $\overline{\mathbb{Q}^\circ} = \overline{\emptyset} = \emptyset$.

Exercise 2.31 (Rudin 2.10)

Let X be an infinite set. For $p \in X$ and $q \in X$, define

$$d(p,q) = \begin{cases} 1 & \text{if } p \neq q \\ 0 & \text{if } p = q \end{cases} \tag{280}$$

Prove that this is a metric. Which subsets of the resulting metric space are open? Which are closed? Which are compact?

Solution. This is a metric since clearly it satisfies symmetry and the fact that $d(p,p) = 0$. The triangle inequality

$$d(p,r) \leq d(p,q) + d(q,r) \tag{281}$$

is trivially satisfied if $p = r$, and if $p \neq r$, then either $p \neq q$ or $q \neq r$, and so the RHS ≥ 1. An open ϵ-ball around $x \in X$ is either X, when $\epsilon > 1$, or $\{x\}$ when $\epsilon \leq 1$. Therefore

Exercise 2.32 (Rudin 2.11)

For $x \in \mathbb{R}$ and $y \in \mathbb{R}$, define

$$\begin{aligned} d_1(x,y) &= (x-y)^2 \\ d_2(x,y) &= \sqrt{|x-y|} \\ d_3(x,y) &= |x^2 - y^2| \\ d_4(x,y) &= |x - 2y| \\ d_5(x,y) &= \frac{|x-y|}{1+|x-y|} \end{aligned}$$

Determine, for each of these, whether it is a metric or not.

Solution. Listed. Positive semidefiniteness and symmetry are easy to check.

1. The triangle inequality gives

$$\begin{aligned} d_1(x,z) \leq d_1(x,y) + d_1(y,z) &\iff (x-z)^2 \leq (x-y)^2 + (y-z)^2 && (282) \\ &\iff 0 \leq (x-y)(y-z) && (283) \end{aligned}$$

which is not satisfied if $x < y < z$, so this is not a valid metric.
2. The triangle inequality gives $\sqrt{|x-z|} \leq \sqrt{|x-y|} + \sqrt{|y-z|}$, and since both sides are positive this inequality is equivalent to squaring both sides to get

$$|x-z| \leq |x-y| + |y-z| + 2\sqrt{|x-y||y-z|} \tag{284}$$

which is true since $|x-z| \leq |x-y| + |y-z|$ of the Euclidean distance satisfies the triangle inequality and $0 \leq \sqrt{|x-y||y-z|}$.
3. This does not satisfy triangle inequality, as taking $0, 1, 2$ gives

$$d_3(0,2) = 4 > 1 + 1 = d_3(0,1) + d_3(1,2) \tag{285}$$

4. This does not satisfy symmetry.
5. For simplicity, let us set $A = |x-y|, B = |y-z|, C = |x-z|$. Then, we get

$$\frac{C}{1+C} \leq \frac{A}{1+A} + \frac{B}{1+B} \iff C \leq A + B + 2AB + ABC \tag{286}$$

where $C \leq A + B$ is true by triangle inequality of Euclidean distance, $0 \leq AB$, and $0 \leq ABC$. Intuitively, we want a metric that doesn't "blow up" the distance between x and y. More precisely, we want a valid metric $d(x,y)$ to be $O(|x-y|)$. Having something like a quadratic growth rate $(x-y)^2$ will blow the distance $d(x,z)$ up too much overpowering the individual $d(x,y) + d(y,z)$.

Exercise 2.33 (Rudin 2.12)

Let $K \subset \mathbb{R}$ consist of 0 and the numbers $1/n$ for $n = 1, 2, 3, \ldots$. Prove that K is compact directly from the definition (without using the Heine-Borel theorem).

Solution. Every open cover of K must have an open set G s.t. $0 \in G$. Since G is open, there exists an open neighborhood $B_\epsilon(0) \subset G$ around 0. By the Archimidean principle, there exists an $N \in \mathbb{N}$ s.t.

$$\epsilon N > 1 \implies \epsilon > \frac{1}{N} \tag{287}$$

and so, $B_\epsilon(0)$ contains all points $\{1/n\}$ for $n > N$. For the rest of the points $1, 1/2, \ldots, 1/N$, we can simply construct a finite cover over each of them, hence getting a finite cover.

Exercise 2.34 (Rudin 2.13)

Construct a compact set of real numbers whose limit points form a countable set.

Solution. Consider the set

$$E = \left\{ \left(\frac{1}{10}\right)^n + \left(\frac{1}{10}\right)^{n+k} \;:\; n \in \{0\} \cup \mathbb{N}, k \in \mathbb{N} \right\} \cup \{0\} \tag{288}$$

This is clearly bounded by 0 and 1.1. Let us represent the elements of this set by (n,k). We can show that

$$(n_1, k_1) > (n_2, k_2) \tag{289}$$

if $n_1 < n_2$ or $n_1 = n_2$ and $k_1 < k_2$. Therefore, to prove closedness, we must prove that every limit point is a point in E. We can do this by proving that a point not in E cannot be a limit point. Choose any $x \notin E$. Then, due to the ordering, we can see that there exists a (n,k) s.t.

$$A = \left(\frac{1}{10}\right)^n + \left(\frac{1}{10}\right)^{n+k} < k < \left(\frac{1}{10}\right)^n + \left(\frac{1}{10}\right)^{n+k+1} = B \tag{290}$$

and so we can take $\epsilon = \min\{k - A, B - k\}$ and show that $B_\epsilon(x)$ does not contain A nor B, and so has an empty intersection with E. Therefore, it cannot be a limit point of E and is closed. Since E is bounded and closed in $\mathbb{R}$, it is compact. Its limit points contain $1, 0.1, 0.01, \ldots, 0$ (simply by fixing n and letting $k \to \infty$, and so E' is infinite. We have just shown that since E is closed, $E' \subset E$. But E is countable, so E' is countable.

Exercise 2.35 (Rudin 2.14)

Given an example of an open cover of the segment $(0, 1)$ which has no finite subcover.

Solution. Consider

$$(0, 1/2) \cup \left(\bigcup_{i=1}^{\infty} \left[1 - \frac{1}{2^i}, 1 - \frac{1}{2^{i+1}}\right) \right) \tag{291}$$

Exercise 2.36 (Rudin 2.15)

Solution.

Exercise 2.37 (Rudin 2.16)

Regard $\mathbb{Q}$, the set of all rational numbers, as a metric space, with $d(p, q) = |p - q|$. Let E be the set of all $p \in \mathbb{Q}$ s.t. $2 < p^2 < 3$. Show that E is closed and bounded in $\mathbb{Q}$ but is not compact. Is E open in $\mathbb{Q}$?

Solution. E is clearly bounded by 0 and 2 since $0^2 < 2 < p^2 < 3 < 2^2$. It is closed and we can show this by showing that E^c is open. Let $x \in E^c$. Then, $x^2 < 2$ or $x^2 > 3$.

1. $x^2 < 2 \iff -\sqrt{2} < x < \sqrt{2}$. Now let $\epsilon = \min\{\sqrt{2} - x, x + \sqrt{2}\} > 0$. Then by the Archimidean property there exists a $n \in \mathbb{N}$ s.t. $0 < \frac{1}{n} < \epsilon$. Therefore, the image of $B_{1/n}(x) \subset \mathbb{Q}$ will map onto $(0, 2)$.
2. $x^2 > 3 \iff x > \sqrt{3}$ or $x < -\sqrt{3}$. If $x > \sqrt{3}$, then by AP there exists a $n \in \mathbb{N}$ s.t. $x - \frac{1}{n} > \sqrt{3} \implies (x - \frac{1}{n})^2 > 3$. If $x < -\sqrt{3}$, then by AP there exist $n \in \mathbb{N}$ s.t. $x + \frac{1}{n} < -\sqrt{3} \implies (x + \frac{1}{n})^2 > 3$. Either way, the image of $B_{1/n}(x)$ will map within E^c.

It is not compact because E is not closed in $\mathbb{R}$. The limit points of E in $\mathbb{R}$ is $[\sqrt{2}, \sqrt{3}] \cup [-\sqrt{3}, -\sqrt{2}]$, which contains irrationals and is clearly not a subset of E. Since it is not closed in $\mathbb{R}$, it is not compact in $\mathbb{R}$, and it is not compact in $\mathbb{Q} \subset \mathbb{R}$. It is open because

$$E = \big((\sqrt{2}, \sqrt{3}) \cup (-\sqrt{3}, \sqrt{2})\big) \cup \mathbb{Q} \subset \mathbb{R} \tag{292}$$

which is the union of open $\big(\sqrt{2}, \sqrt{3}) \cup (-\sqrt{3}, \sqrt{2})\big)$ and subset $\mathbb{Q} \subset \mathbb{R}$, and so it is open.

Exercise 2.38 (Rudin 2.17)

Let E be the set of all $x \in [0, 1]$ whose decimal expansion consists of only the digits 4 and 7. Is E countable? Is E dense in $[0, 1]$? Is E compact? Is E perfect?

Solution.

Exercise 2.39 (Rudin 2.18)

Is there a nonempty perfect set in $\mathbb{R}$ which contains no rational number?

Solution.

Exercise 2.40 (Rudin 2.19)

Listed.

1. If A and B are disjoint closed sets in some metric space X, prove that they are separated.
2. Prove the same for disjoint open sets.
3. Fix $p \in X$, $\delta > 0$, define A to be the set of all $q \in X$ for which $d(p,q) < \delta$. Define B similarly, with $>$ in place of $<$. Prove that A and B are separated.
4. Prove that every connected metric space with at least two points is uncountable.

Solution. Listed.

1. This is trivial with the fact that the closure of the closure of A is the closure of A.
2. Let A, B be open. We wish to show that if $x \in A'$, then $x \notin B$. Assume $x \in B$. Then there exists $\epsilon > 0$ s.t. $B_\epsilon(x) \subset B$. But $B \cap A = \emptyset \implies B_\epsilon(x) \cap A = \emptyset$ and so $x \notin A'$, which is a contradiction.
3. Clearly, $A \cap B = \emptyset$. Not let $x \in A \implies$ there exists $\epsilon > 0$ s.t. $B_\epsilon(x) \subset A \implies B_\epsilon(x) \cap B = \emptyset \implies x \in B'$. We can prove similarly to show that $x \in B \implies x \notin A'$.
4. Assume X is countable (solutionis very similar for finite). Then, we can enumerate a $X = \{x_i\}_{i=1}^\infty$. We wish to show that X can be decomposed into the union of an open ball and the interior of its complement as shown in (3). We fix $p \in X$. Then, we take the set $D = \{d(p,x)\}_{x \neq p} \subset \mathbb{R}$. Since D is a countable subset of $\mathbb{R}$, there must exist some $\alpha > 0$ s.t. $\alpha \notin D$. This α partitions the distances into two sets, and we can define
$$X = \{q \in X \mid d(p,q) < \alpha\} \cup \{q \in X \mid d(p,q) > \alpha\} \tag{293}$$
and by (3), these two sets are separated, which means that X is not connected, leading to a contradiction.

Exercise 2.41 (Rudin 2.20)

Are closures and interiors of connected sets always connected? Look at subsets of $\mathbb{R}^2$.

Solution. The interiors are not always connected. Consider the two closed balls $\overline{B_1((1,0))}$ and $\overline{B_1((-1,0))}$ as subsets of $\mathbb{R}^2$. They are connected but their interiors, which are the two open balls, are not connected.
As for closures, they are always connected. Let W be connected. Then for any partition $A \cup B = W$, $\overline{A} \cap B \neq \emptyset$ WLOG. Consider $\overline{W} = W \cup W'$ and take any partition $\overline{W} = C \cup D$. Then, label $A = C \cap W, A^* = C \cap W', B = D \cap W, B^* = D \cap W'$. This implies that $C = A \cup A^*, D = B \cup B^*$, and $A \cup B = W$ (which is connected). Then, we can show that
$$\begin{aligned} \overline{C} \cap D = (\overline{A \cup A^*} \cap D) &= (\overline{A} \cup \overline{A^*}) \cap D = (\overline{A} \cap D) \cup (\overline{A^*} \cap D) \\ &= (\overline{A} \cap B) \cup (\overline{A} \cap B^*) \cup (\overline{A^*} \cap D) \end{aligned}$$
which cannot be empty since by connectedness of W, $\overline{A} \cap B \neq \emptyset$. Therefore, $\overline{W}$ is connected.

Exercise 2.42 (Rudin 2.21)

Let A and B be separated subsets of some $\mathbb{R}^k$. Suppose $a \in A, b \in B$ and define
$$p(t) = (1-t)a + tb \tag{294}$$
for $t \in \mathbb{R}$. Put $A_0 = p^{-1}(A), B_0 = p^{-1}(B)$.

1. Prove that A_0 and B_0 are separated subsets of $\mathbb{R}$.
2. Prove that there exists a $t_0 \in (0,1)$ s.t. $p(t_0) \notin A \cup B$.

3. Prove that every convex subset of $\mathbb{R}^k$ is connected.

Solution.

Exercise 2.43 (Rudin 2.22)

A metric space is called *separable* if it contains a countable dense subset. Show that $\mathbb{R}^k$ is separable.

Solution. Consider the set $\mathbb{Q}^k \subset \mathbb{R}^k$. It is a finite Cartesian product (and hence, a countable union) of countable $\mathbb{Q}$, and so it is countable. $\mathbb{Q}^k$ is dense in $\mathbb{R}^k$ since given any $x \in \mathbb{R}^k$, we claim x is a limit point of $\mathbb{Q}^k$. Given any $\epsilon > 0$, we can construct $B^\circ_\epsilon(x)$. For each coordinate x_i, by density of rationals in $\mathbb{R}$ we can choose a $q_i \in \mathbb{Q}$ s.t. $0 < d(x_i, q_i) < \epsilon/k$. Then, using triangle inequality, we can take the distances between each coordinate changed from x_i to q_i. Let q^k be the vector x with the components $x_1, \ldots, x_k$ changed to $q_1, \ldots, q_k$, respectively.

$$d(x, q) = d(x, q^1) + d(q^1, q^2) + \ldots + d(q^{k-1}, q_k) < \frac{\epsilon}{k} + \ldots + \frac{\epsilon}{k} = \epsilon \tag{295}$$

and so $q \in B^\circ_\epsilon(x)$. Hence the intersection of $\mathbb{Q}^k$ and $B^\circ_\epsilon(x)$ for any $\epsilon > 0$ is nontrivial, so x is a limit point of $\mathbb{Q}^k$.

Exercise 2.44 (Rudin 2.23)

A collection $\{V_\alpha\}$ of open subsets of X is said to be a *base* for X if the following is true: For every $x \in X$ and every open set $G \subset X$ such that $x \in G$, we have $x \in V_\alpha \subset G$ for some α. In other words, every open set in X is the union of a subcollection of $\{V_\alpha\}$. Prove that every separable metric space has a countable base.

Solution. Since X is separable it contains a countable dense subset, call it S. Then for every $x \in S$, we can look at the set of all open balls with center x and rational radii, call it $\mathcal{B}$. Then $\mathcal{B}$ is countable. Now consider an open set U. By definition, for every $x \in U$, there exists an $\epsilon > 0$ s.t. $B_\epsilon(x) \subset U$. By AP, we can find a $n \in \mathbb{N}$ s.t. $0 < \frac{1}{n} < \epsilon$, and therefore we can find an open ball $B \in \mathcal{B}$ s.t. $B(x) \subset U$. We claim that

$$W \coloneqq \bigcup_{x \in U} B(x) = U \tag{296}$$

If $x \in U$, then by construction it is contained in $B(x) \subset \cup_{x \in U} B(x)$, and so $U \subset W$. If $x \in W$, then it is in $B(x)$, which is fully contained in U and so $W \subset U$. Therefore every open set can be constructed by a countable union of open balls in countable $\mathcal{B}$.

Exercise 2.45 (Rudin 2.24)

Let X be a metric space in which every infinite subset has a limit point. Prove that X is separable.

Solution. We fix $\delta > 0$. Choose $x_1 \in X$. Then choose $x_2 \in X$ s.t. $d(x_1, x_2) \geq \delta$, and keep doing this until we choose $x_{j+1} \in X$ s.t. $d(x_{j+1}, x_i) \geq \delta$ for all $i \in 1, \ldots, j$.

1. We claim that this must stop after a finite number of steps. Assume it doesn't. Then by assumption $V = \{x_i\}_{i=1}^\infty$ should have a limit point in X, denote it x. Choose $\frac{\delta}{2} > 0$. Then, $B^\circ_{\delta/2}(x) \cap V \neq \emptyset$. This intersection can only have one point since if it had two x', x'', then since

both are in $B_{\delta/2}(x)$, then

$$d(x', x'') \leq d(x', x) + d(x, x'') \leq \frac{\delta}{2} + \frac{\delta}{2} = \delta \tag{297}$$

and since they are both in V, then $d(x', x'') \geq \delta$, which is a contradiction. Since there is a finite number of points in $B_{\delta/2}(x)$ of V, x cannot be a limit point. So this must terminate at some finite $J < \infty$.

2. Denote $W = \{x_i\}_{i=1}^J$. Then, $\mathscr{B}_\delta = \{B_\delta(x) \mid x \in W\}$ must cover X, since if it didn't, there would exist a $y \in X$ s.t. $d(y, x) \geq \delta$ for all $x \in W$, and we can add another element in W.
3. Consider $\delta = 1, 1/2, 1/3, \ldots$ and construct the same cover

$$\mathscr{B}_k = \{B_{1/k}(x_{ki}) \mid i = 1, \ldots, J_k\} \tag{298}$$

which is finite. Therefore, $\mathscr{B} = \cup_{k=1}^\infty \mathscr{B}_k$ must be countable.

4. We claim that countable $\{x_{ki}\}_{k,i}$ is dense. Consider any $x \in X$. For every $\epsilon > 0$, we can find an arbitrarily large $n \in \mathbb{N}$ s.t. $0 < \frac{1}{n} < \epsilon$. Since $\mathscr{B}_n$ is an open cover, there must exist some x_{ni} s.t. $x \in B_{1/n}(x_{ni})$, which by symmetry implies that $x_{ni} \in B_{1/n}(x) \subset B_\epsilon(x)$. Therefore, there always exists an x_{ni} in every $B_\epsilon(x)$, and so $B_\epsilon(x) \cap \{x_{ki}\} \neq \emptyset \implies x$ is a limit point of $\{x_{ki}\}$ and so it is dense.

Exercise 2.46 (Rudin 2.25)

Prove that every compact metric space K has a countable base, and that K is therefore separable.

Solution. For every $n \in \mathbb{N}$, let us consider an open covering $\mathscr{F}_n := \{B_{1/n}(x_n) \mid x_n \in K\}$. Since K is compact, it has a finite subcovering

$$\mathscr{G}_n := \{B_{1/n}(x_{ni}) \mid i = 1, \ldots, k(n)\} \tag{299}$$

Now consider the union $\mathscr{G} = \cup_{i=1}^n \mathscr{G}_n$, which is countable. We claim that $\mathscr{G}$ is a base. Consider any open set U. Then for every $x \in U$, we want to show that x is contained in a $B_{1/n}(x_{ni}) \subset U$. Since U is open , there exists a $\epsilon > 0$ s.t. $B_\epsilon(x) \subset U$. Now by AP, there exists a $n \in \mathbb{N}$ s.t. $0 < \frac{1}{n} < \frac{\epsilon}{2}$. Therefore $B_{1/n}(x) \subset B_\epsilon(x)$. Since $\mathscr{G}$ is an open covering, there must exist some $B_{1/n}(x_{ni})$ that contains x. Now we wish to show that $B_{1/n}(x_{ni})$ is fully contained in U. Let $y \in B_{1/n}(x_{ni})$. Then, by triangle inequality,

$$d(y, x) = d(y, x_{ni}) + d(x_{ni}, x) < \frac{1}{n} + \frac{1}{n} < \epsilon \tag{300}$$

and therefore $x \in B_{1/n}(x_{ni}) \subset B_\epsilon(x)$. Therefore, for every $x \in U$, we can construct an open ball of $\mathscr{G}$ containing x and contained in U, proving that this is a base.
We claim that the set of all $\mathscr{P} = \{x_{ni}\}_{n,i}$ forms a countable dense subset. This is clearly countable since $\mathscr{G}$ is countable. We must prove that the closure of $\mathscr{P} = K$. Let $x \in K$. Given any $\epsilon > 0$, we wish to show that $B_\epsilon(x) \cap \mathscr{P} \neq \emptyset$. Since $B_\epsilon(x)$ is open, it can be covered by a subcollection of $\mathscr{G}$, and so their centers must be in $B_\epsilon(x)$, proving that $B_\epsilon(x) \cap \mathscr{P} \neq \emptyset$. Therefore, x is a limit point of $\mathscr{P}$.

Exercise 2.47 (Rudin 2.26)

Solution.

Exercise 2.48 (Rudin 2.27)

Solution.

Exercise 2.49 (Rudin 2.28)

Solution.

Exercise 2.50 (Rudin 2.29)

Prove that every open set in $\mathbb{R}$ is the union of an at most countable collection of disjoint segments.

Solution. Let $U \subset \mathbb{R}$ be open. Then for all $x \in U$ there exists $\epsilon > 0$ s.t. $(x - \epsilon, x + \epsilon) \subset U$. Now since $\mathbb{R}$ is separable (by exercise Rudin 2.22), it has a countable dense subset $\mathbb{Q}$. Consider all segments of rational radius and rational centers

$$\mathscr{B} = \{(q - p, q + p) \subset \mathbb{R} \mid q, p \in \mathbb{Q}\} \tag{301}$$

This is clearly countable. We claim that every open U can be expressed as the union of a subset of $\mathscr{B}$. Now by AP, there exists $n \in \mathbb{N}$ s.t. $0 < \frac{1}{n} < \frac{\epsilon}{2}$, so for all $x \in U$, there exists $n \in \mathbb{N}$ s.t. $(x - \frac{1}{n}, x + \frac{1}{n}) \subset U$. Now since $\mathbb{Q}$ is dense in $\mathbb{R}$, $x \in \mathbb{Q}' \implies (x - \frac{1}{n}, x + \frac{1}{n}) \cap \mathbb{Q} \neq \emptyset$. Say r is in this intersection. Then, by symmetry of metric, $x \in (r - \frac{1}{n}, r + \frac{1}{n})$. Therefore, for all $x \in U$, we have found an open ball in $\mathscr{B}$ that contains x. Now, we must show that this actually is fully contained in U. This is easy, since if $y \in B_{1/n}(r)$, then

$$d(y, x) \leq d(y, r) + d(r, x) \leq \frac{1}{n} + \frac{1}{n} < \epsilon \tag{302}$$

and so $B_{1/n}(r)$ is complete contained in the ϵ-ball around x, which is a subset of U. So for all $x \in U$, we found an open set $U_x \in \mathscr{B}$ covering x and fully contained in U, which means that $\cup_{x \in U} U_x = U$. Now for some intervals $B_1, B_2 \in \mathscr{B}$, if $B_1 \cap B_2 \neq \emptyset$, take their union, which is another segment, and keep doing this until $B_i \cap B_j \neq \emptyset$ for all i, j. The cardinality of this new pruned set will be less than or equal to $\mathscr{B}$, which is countable, and so this must be at most countable.

3 Sequences in Euclidean Space

We assume that you know what sequences are, and what it means for them to converge in $\mathbb{R}^n$. Let's primarily focus on sequences in $\mathbb{R}$. Note that there are many way in which a sequence can be divergent.

1. Increasing/decreasing indefinitely
2. Oscillating between two constant values
3. Oscillating between a value tending to $+\infty$ and a value tending to $-\infty$
4. Many other classes of divergence

Definition 3.1 (Sequence Tending to Infinity)

The sequence $\{x_n\}$ **tends to positive infinity** if for each number c there exists $N \in \mathbb{N}$ such that $x_n > c$ for all $n > N$. It is denoted

$$x_n \to +\infty \text{ or } \lim_{n\to\infty} x_n = +\infty \tag{303}$$

We define sequences that **tend to negative infinity** similarly. And $\{x_n\}$ **tends to infinity** if for each c there exists $N \in \mathbb{N}$ such that $|x_n| > c$ for all $n > N$, which is written

$$x_n \to \infty \tag{304}$$

Note that

$$x_n \to +\infty \text{ or } x_n \to -\infty \implies x_n \to \infty \tag{305}$$

but the converse is not necessarily true. The simple example is the sequence $x_n = (-1)^n n$. Also, it is important to know that a sequence may be unbounded and yet not tend to $+\infty$, $-\infty$, or ∞.

Example 3.1 (Unbounded Sequence that Doesn't tend to ∞)

The sequence $x_n = n^{(-1)^n}$ is divergent yet does not tend to positive infinity, negative infinity, nor infinity.

Definition 3.2 (Monotonic Sequences)

Let X be an ordered set. (x_n) is
1. **strictly increasing** if $x_{n+1} > x_n$ for all n.
2. **strictly decreasing** if $x_{n+1} < x_n$ for all n.
3. **increasing (nondecreasing)** if $x_{n+1} \geq x_n$ for all n.
4. **decreasing (nonincreasing)** if $x_{n+1} \leq x_n$ for all n.

Sequences of these types are called **monotonic**.

The properties of a general metric space give us some general conditions to determine whether a sequence converges in $\mathbb{R}^n$. To add to our toolbox in determining convergence, we will focus on sufficient conditions for convergence in $\mathbb{R}$, which has the additional properties of being an ordered field. These additional structures unlock a whole new suite of theorems in convergence. Why do we want to focus on just real-valued, i.e. numerical, sequences? First is that since $\mathbb{R}^n$ is constructed as the product topology of $\mathbb{R}$, we can prove a lot about continuity of functions in $\mathbb{R}^n$ by proving limits in $\mathbb{R}$, and letting the construction of the product topology do the rest. Second, the codomain of many natural structures such as inner products, norms, and measures lie in the reals, and we often need to prove convergence of these values.

3.1 Convergence Theorems from Completeness and Order

So far, in order to show that a sequence is convergent, we must identify a real number first and then show using the ϵ-δ definition that it converges. This might be overkill in a case where we just want to prove that a sequence converges, but we don't care what it converges to. The following theorems are direct consequences of the Cauchy-completeness and the least upper bound property of the reals.

Theorem 3.1 (Cauchy-Convergence Criterion)

A cauchy sequence in $\mathbb{R}$ converges.

Proof. This is by definition true, and to see how other forms of completeness implies Cauchy convergence, see above.

The second result is an immediate consequence of Dedekind completeness, which is equivalent to Cauchy completeness in the reals.

Theorem 3.2 (Convergence Criterion for Monotonic Sequences)

Let $(x_n)_n$ be a sequence.

1. If (x_n) is monotonically increasing/decreasing and bounded above/below in $\mathbb{R}$, then it converges.
2. If (x_n) is monotonically increasing/decreasing in $\mathbb{R} \cup \{\pm\infty\}$, then it converges.

Proof. It satisfies to prove the first case, as the second case can be done similarly without much difficulty. Let $x_n \leq x_{n+1}$. Then the set $\{x_n\}$ is bounded above in $\mathbb{R}$, which has the least upper bound property, and so there exists a least upper bound x. We claim that the sequence converges to x. For every $\epsilon > 0$, since it is least, there exists at least one $x_N \in (x - \epsilon, x)$. By monotonicity, this means that $x_n \in (x - \epsilon, x)$ for all $n \geq N$, and so the sequence converges to x.

We are able to see how both Cauchy and Dedekind completeness of the reals define convergence in $\mathbb{R}$. Now let's squeeze a bit more out of the total ordering to gain some properties of convergence and divergence.

Theorem 3.3 (Preservation of Ordering Between Sequences and Limits)

Given convergent sequences (x_n) and (y_n), if

$$\lim_{n\to\infty} x_n < \lim_{n\to\infty} y_n \tag{306}$$

then there exists an index $N \in \mathbb{N}$ such that $x_n < y_n$ for all $n > N$.

Proof. Given $x_n \to x, y_n \to y$ and $x < y (\in \mathbb{R})$, then for every $\epsilon > 0$, there exists $N_1, N_2 \in \mathbb{N}$ s.t. $d(x, x_n) < \epsilon$ for all $n > N_1$ and $d(y, y_n) < \epsilon$ for all $n > N_2$. Setting $N = \max\{N_1, N_2\}$, we can say the same for all $n > N$. We choose $\epsilon = \frac{y-x}{2} > 0$. Then, there exists $N \in \mathbb{N}$ s.t. $x_n \in (x - \epsilon, x + \epsilon)$ and $y_n \in (y - \epsilon, y + \epsilon)$ for all $n > N$. Therefore, if $a \in (x - \epsilon, x + \epsilon)$ and $b \in (y - \epsilon, y + \epsilon)$, then

$$a < \sup B_\epsilon(x) = x + \epsilon = y - \epsilon = \inf B_\epsilon(y) < b \tag{307}$$

which implies that $x_n < y_n$ for all $n > N$.

Theorem 3.4 (Squeeze Theorem for Sequences)

Given sequences $(x_n), (y_n), (z_n)$ such that

$$x_n \leq y_n \leq z_n \tag{308}$$

for all $n > N$, if $\{x_n\}$ and $\{z_n\}$ both converge to the same limit, then the sequence $\{y_n\}$ also converges to that limit. That is,

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} z_n = A \implies \lim_{n\to\infty} y_n = A \tag{309}$$

Proof. We first prove that if there exists a $N \in \mathbb{N}$ s.t. $a_n \leq b_n$ for all $n > N$, then $\lim_{n\to\infty} a_n = a \leq b = \lim_{n\to\infty} b_n$. Assume this weren't true, that $a > b$. Then for $\epsilon = \frac{a-b}{2} > 0$, there must exist $M \in \mathbb{N}$ s.t. $a_n \in (a-\epsilon, a+\epsilon)$ and $b_n \in (b-\epsilon, b+\epsilon)$ for all $n > M$. But

$$b_n < \sup(b-\epsilon, b+\epsilon) = b+\epsilon = a-\epsilon = \inf(a-\epsilon, a+\epsilon) < a_n \tag{310}$$

which contradicts $a_n \leq b_n$. Therefore, $a \leq b$. Therefore, we can use this to get

$$A = \lim_{n\to\infty} x_n \leq \lim_{n\to\infty} y_n \leq \lim_{n\to\infty} z_n = A \implies \lim_{n\to\infty} y_n = A \tag{311}$$

Theorem 3.5 (Unbounded Sequence has Subsequence Tending to Infinity)

If (x_n) is not bounded above then it has a subsequence $x_{n_k} \to +\infty$.

Proof. We can construct such a subsequence.

Therefore, we can construct a subsequential limit to $\pm\infty$ if (x_n) is not bounded. If it is bounded, then by the Bolzano-Weierstrass theorem it contains a convergent subsequence. Therefore, we have the following.

Corollary 3.6

From each sequence of real numbers there exists either a convergent subsequence or a subsequence tending to infinity.

Proof. If it's bounded, then Bolzano-Weierstrass. If it's not bounded, then use previous theorem.

Example 3.2

We claim that

$$\lim_{n\to\infty} \sqrt[n]{n} = 1 \tag{312}$$

3.2 Arithmetic

Theorem 3.7 (Arithmetic on Limits)

Given that $(x_n), (y_n)$ are numerical sequences with $y_n \neq 0$ for all n, and let

$$\lim_{n\to\infty} x_n = x, \qquad \lim_{n\to\infty} y_n = y \neq 0 \tag{313}$$

then,

$$\lim_{n\to\infty}(x_n + y_n) = x + y \tag{314}$$
$$\lim_{n\to\infty}(cx_n) = cx \tag{315}$$
$$\lim_{n\to\infty}(x_n \cdot y_n) = x \cdot y \tag{316}$$
$$\lim_{n\to\infty}\frac{x_n}{y_n} = \frac{x}{y} \tag{317}$$

It immediately follows that the set of all convergent sequences in $\mathbb{R}^{\mathbb{N}}$ is a subspace of $\mathbb{R}^{\mathbb{N}}$.

Proof. For every $\epsilon > 0$, there exists $N_1, N_2 \in \mathbb{N}$ such that

$$|x_n - x| < \epsilon \text{ for all } n > N_1 \tag{318}$$
$$|y_n - y| < \epsilon \text{ for all } n > N_2 \tag{319}$$

Therefore, for a given ϵ, we wish to prove that there exists a N such that for all $n > N$,

$$1.|(x_n + y_n) - (x + y)| < \epsilon \tag{320}$$
$$2.|cx_n - cx| < \epsilon \tag{321}$$
$$3.|(x_n y_n) - (xy)| < \epsilon \tag{322}$$
$$4.\left|\frac{x_n}{y_n} - \frac{x}{y}\right| < \epsilon \tag{323}$$

1. yy the triangle inequality, we can see that

   $$|(x_n + y_n) - (x + y)| = |x_n - x| + |y_n - y| \tag{324}$$

   Since we can choose the error between x_n and x for $n > N_1$, and y_n and y for $n > N_2$ as small as we want, we set it to $\epsilon/2$. Then, we have

   $$|(x_n + y_n) - (x + y)| = |x_n - x| + |y_n - y| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \tag{325}$$

   for all $n > N = \max\{N_1, N_2\}$. Therefore, for a given ϵ, there exists an N such that

   $$|(x_n + y_n) - (x + y)| < \epsilon \text{ for all } n > N \tag{326}$$

2. This proof is easy. For a given ϵ, we choose the error to be $\frac{\epsilon}{c}$.

   $$|x_n - x| < \frac{\epsilon}{c} \text{ for all } n > N_1 \tag{327}$$

   Then, there exists natural number N_1 such that

   $$|cx_n - cx| < c|x_n - x| = c\frac{\epsilon}{c} = \epsilon \text{ for all } n > N_1 \tag{328}$$

3. We first observe that since the limit of $\{y_n\}$ exists, it must be bounded by a value, say y. That is,

   $$|y_n| < Y \text{ for all } n \in \mathbb{N} \tag{329}$$

   Then, we see that

   $$\begin{aligned} |x_n y_n - xy| &= |(x_n y_n - x y_n) + (x y_n - xy)| && (330) \\ &< |x_n y_n - x y_n| + |x y_n - xy| && (331) \\ &= |y_n||x_n - x| + |x||y_n - y| && (332) \end{aligned}$$

Suppose $\epsilon > 0$ is given. Then, we can set the error bounds freely; there exists $N_1, N_2 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\epsilon}{2Y} \text{ for all } n > N_1 \tag{333}$$
$$|y_n - y| < \frac{\epsilon}{2|x|} \text{ for all } n > N_2 \tag{334}$$

Then, we can see that

$$|x_n y_n - xy| \leq |y_n||x_n - x| + |x||y_n - y| < Y \cdot \frac{\epsilon}{2Y} + |x|\frac{\epsilon}{2|x|} = \epsilon \tag{335}$$

for all $n > N = \max\{N_1, N_2\}$.

4. We use the estimate

$$\left|\frac{x}{y} - \frac{x_n}{y_n}\right| = \frac{|x_n||y_n - y| + |y_n||x_n - x|}{y_n^2} \cdot \frac{1}{1 - \delta(y_n)}, \quad \delta(y_n) = \frac{|y_n - y|}{|y_n|} \tag{336}$$

For a given $\epsilon > 0$, we find natural numbers N_1, N_2 such that

$$|x_n - x| < \min\left\{1, \frac{\epsilon|y|}{8}\right\} \text{ for all } n > N_1 \tag{337}$$
$$|y_n - y| < \min\left\{\frac{|y|}{4}, \frac{\epsilon y^2}{16(|x|+1)}\right\} \text{ for all } n > N_2 \tag{338}$$

From this we can deduce that

$$|x_n| = |x_n - x + x| < |x_n - x| + |x| < |x| + 1 \tag{339}$$

and

$$|y| = |y_n + y - y_n| < |y_n| + |y - y_n| \tag{340}$$
$$\implies |y_n| > |y| - |y_n - y| > |y| - \frac{|y|}{4} > \frac{|y|}{2} \tag{341}$$
$$\implies \frac{1}{|y_n|} < \frac{2}{|y|} \tag{342}$$
$$\implies 0 < \delta(y_n) = \frac{|y_n - y|}{|y_n|} < \frac{|y|/4}{|y|/2} = \frac{1}{2} \tag{343}$$
$$\implies 1 - \delta(y_n) > \frac{1}{2} \tag{344}$$
$$\implies 0 < \frac{1}{1 - \delta(y_n)} < 2 \tag{345}$$

So, we can substitute

$$|x_n| \cdot \frac{1}{y_n^2} \cdot |y_n - y| < (|x| + 1) \cdot \frac{4}{y^2} \cdot \frac{\epsilon \cdot y^2}{16(|x|+1)} = \frac{\epsilon}{4} \tag{346}$$
$$\left|\frac{1}{y_n}\right| \cdot |x_n - x| < \frac{2}{|y|} \cdot \frac{\epsilon|y|}{8} = \frac{\epsilon}{4} \tag{347}$$

into the final equation to get

$$\left|\frac{x}{y} - \frac{x_n}{y_n}\right| < \epsilon \text{ for all } n > N = \max\{N_1, N_2\} \tag{348}$$

Example 3.3

We claim that

$$\lim_{n\to\infty} \frac{n}{q^n} = 0 \text{ if } q > 1 \tag{349}$$

Since $x_n = \frac{n}{q^n} \implies x_{n+1} = \frac{n+1}{nq} x_n$ for $n \in \mathbb{N}$. Since

$$\lim_{n\to\infty} \frac{n+1}{nq} = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)\frac{1}{q} = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right) \cdot \lim_{n\to\infty} \frac{1}{q} = 1 \cdot \frac{1}{q} = \frac{1}{q} < 1 \tag{350}$$

there exists an index N such that $\frac{n+1}{nq} < 1$ for $n > N$. Thus, we have

$$x_n > x_{n+1} = x_n \cdot \frac{n+1}{nq} \text{ for } n > N \tag{351}$$

which means that the sequence will be monotonically decreasing from index N on. The terms of the sequence

$$x_{N+1} > x_{N+2} > x_{N+3} > \dots \tag{352}$$

are positive (bounded below) and are monotonically decreasing, so it must have a limit.
Finding the actual limit is easy. Let $x = \lim_{n\to\infty} x_n$. It follows from the relation $x_{n+1} = \frac{n+1}{nq} x_n$ that

$$x = \lim_{n\to\infty} \big(x_{n+1}\big) = \lim_{n\to\infty} \left(\frac{n+1}{nq} x_n\right) = \lim_{n\to\infty} \frac{n+1}{nq} \cdot \lim_{n\to\infty} x_n = \frac{1}{q} x \tag{353}$$

which implies that $\left(1 - \frac{1}{q}\right) = 0 \implies x = 0$.

Definition 3.3 (Cauchy Product)

The Cauchy Product is the direct convolution of two sequences.

Definition 3.4 (Recursive Sequence)

Sometimes, a sequence may be defined **recursively**, where the nth term contains a combination of the $n-1$ terms before it.

3.3 Limsup and Liminf

The superior and inferior limits represent some sort of "bound" on the sequence in the long run. That is, on the long run, the terms of the sequence (x_n) cannot be greater than its superior limit and cannot be less than its inferior limit. With this interpretation, the following definition should be clear.

Definition 3.5 (Inferior, Superior Limits)

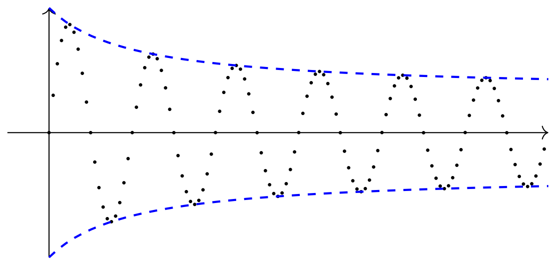
The **superior/inferior limit** of a sequence (x_n) is defined in the equivalent ways.

1. Given that E is the set of all partial limits, the limsup/liminf is the supremum/infimum of E.

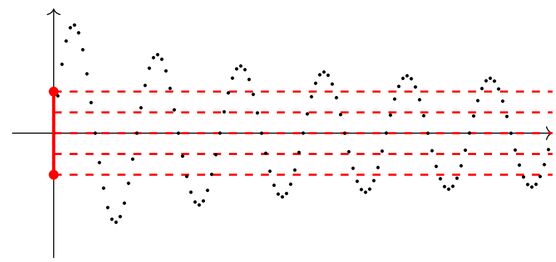
$$\limsup_{n\to\infty} x_n \coloneqq \sup\{E\} \qquad \liminf_{n\to\infty} x_n \coloneqq \inf\{E\} \tag{354}$$

2. The limsup/liminf is the limit of the sequence of supremums/infimums of the elements up to k.

$$\limsup_{n\to\infty} x_n \coloneqq \lim_{n\to\infty} \sup_{k\geq n} x_k \qquad \limsup_{n\to\infty} x_n \coloneqq \lim_{n\to\infty} \inf_{k\geq n} x_k \tag{355}$$



(a) In order to find the limsup, we first look the whole sequence in $\mathbb{N}$ and find the supremum. We now "decrease" our domain from $\mathbb{N}$ to $\{2, \ldots\}$, then $\{3, \ldots\}$, then $\{4, \ldots\}$ and so on, continuing to label the supremum of the sequence. The limit of this sequence of supremums is the limsup.



(b) The 5 red lines marked in the middle (along with infinitely many others) are viable partial limits because one can choose a subsequence such that all of its points after a certain n lie in some ϵ-neighborhood of the limit. Therefore, we claim that the limsup/inf is the supremum of this set E.

Figure 1: Two ways to visualize the superior and inferior limits of the divergent sequence $x_n = \big(\frac{1}{x+1} + 0.5\big)\sin(2\pi x)$. The left is the limit of the supremum, and the right is the supremum of the closed set of subsequential limits.

Example 3.4 (Computing Limsup and Liminf)

We give some basic examples.

1. Let $x_n = (-1)^n$. Then $E = \{-1, +1\}$ and
$$\limsup_{n\to\infty} x_n = 1, \qquad \liminf_{n\to\infty} x_n = -1 \tag{356}$$
2. Let $x_n = (-1)^n/[1 + (1/n)]$. Then
$$\limsup_{n\to\infty} x_n = 1, \qquad \liminf_{n\to\infty} x_n = -1 \tag{357}$$

Let's give two warnings. First, limsup and liminfs do *not* behave like limits under addition and multiplication. That is,
$$\limsup x_n + \limsup y_n \neq \limsup x_n + y_n \tag{358}$$

Example 3.5 (Counterexamples of Arithmetic Consistency of Limit superior)

Consider $(x_n) = (-1)^n$ and $y_n = (-1)^{n+1}$. Then
$$\limsup x_n = \limsup y_n = 1, \qquad \liminf x_n = \liminf y_n = -1 \tag{359}$$
But $(x_n + y_n) = 0$, so
$$\limsup x_n + y_n = \liminf x_n + y_n = 0 \tag{360}$$

Second, note that even though we are talking about subsequential limits, the limsup and liminf are *not* subsequential limits! It is the supremum of subsequential limits E, which may or may not be in E.

Example 3.6 (Limsup that is not attained by any subsequential limit)

This should be a sequence not in $\mathbb{R}$.

However, in $\mathbb{R}$, it turns out that the limsup and liminf are both contained in E, so we are fine.

Lemma 3.8

If (x_n) is a sequence in $\mathbb{R}$, then

1. the limsup is indeed a subsequential limit, i.e. $\limsup x_n \in E$.
2. If $x > \limsup x_n$, then $\exists N \in \mathbb{N}$ s.t. $n \geq N \implies x_n < x$.

Proof. For the first claim, there are two cases to consider. If (x_n) is unbounded from above, then $\exists (x_{n_k})$ such that $x_{n_k} \to +\infty \implies \infty = \limsup x_n \in E$. If (x_n) is bounded from above, then the subsequential limits of (x_n) are either in (x_n) or they are limit points of x_n. This implies that the set E consists of points either in $\{x_n\}$ or are limit points of the set $\{x_n\} \implies \sup E$ is in E since it's a limit point.
For the second claim, if there are infinitely many terms of the sequence larger than x, then we could find a subsequence (x_{n_k}) with $x_{n_k} > x$ for all k. Therefore (x_n) has a subsequential limit which must be $\geq x$. Every subsequential limit of (x_{n_k}) is also a subsequential limit of (x_n). This contradicts $\limsup x_n = \sup E$.

Theorem 3.9 (Requirements of Partial Limits for Limit to Exist)

Here are two results in which we can use partial limits to determine if a sequence has a limit or not.

1. A sequence has a limit or tends to $\pm\infty$ if and only if its inferior and superior limits are the same.
$$\limsup x_n = \liminf x_n = x \implies \lim_{n \to +\infty} x_n = x \tag{361}$$
2. A sequence converges if and only if every subsequence of it converges.

Proof. For (1), we pick $x + \epsilon > x$. Then every term past some N_1 must be less than $x + \epsilon$. By the same logic, we have N_2 for $x - \epsilon < x$. So take $N = \max\{N_1, N_2\}$, which is contained in the ϵ-ball around x.

Theorem 3.10 (Ordering on Subsequential Limits)

If $s_n \leq t_n$ for $n \geq N$, where N is fixed, then
$$\liminf_{n \to \infty} s_n \leq \liminf_{n \to \infty} t_n$$
$$\limsup_{n \to \infty} s_n \leq \liminf_{n \to \infty} t_n$$

Example 3.7

We claim
$$\lim_{n \to \infty} n^{1/n} = 1 \tag{362}$$
We can consider $x_n = n^{1/n} - 1$ and want to show that $x_n \to 0$. We have $x_n \geq 0$. If $n > 1$, then $n = (x_n + 1)^n \geq x_n^2 \cdot \frac{n(n-1)}{2}$ from the binomial theorem. This means that
$$x_n^2 \leq \frac{2}{n-1} \implies 0 \leq x_n \leq \sqrt{\frac{2}{n-1}} \to 0 \tag{363}$$
And so by the squeeze theorem, $x_n \to 0$.

Example 3.8

If $x > 1, \alpha \in \mathbb{R}$, then

$$\lim_{n \to +\infty} \frac{n^\alpha}{x^n} = 0 \tag{364}$$

3.4 Convergence Tests for Real Series

Definition 3.6 (Series over $\mathbb{R}$)

Given a sequence of real numbers (x_n), the **series (of partial sums)** is the sequence

$$(s_n) = \sum_{k=1}^{n} x_k \tag{365}$$

The **sum of the series** is the limit of (s_n). Usually we define (s_n) implicitly and use the summation notation.

$$\sum_{n=1}^{\infty} x_n \coloneqq \lim_{n \to \infty} s_n \tag{366}$$

1. If the sequence (s_n) converges to s, the series is **convergent**, written

$$\sum x_n < +\infty \tag{367}$$

2. If the sequence does not converge, it is **divergent**.
3. If the series of partial sums of $(|x_n|)$ converges, then it is said to be **absolutely convergent**.[a]

$$\sum_{n=1}^{\infty} |x_n| \tag{368}$$

[a] Clearly, every absolutely convergent series is convergent because $\left|\sum_{n=1}^{\infty} a_n\right| \leq \sum_{n=1}^{\infty} |a_n|$. This is sort of the infinite analogue of the triangle inequality.

We must reiterate a few warnings here. Note that the series $\sum x_n$ is simply notation and should *not* be treated as an "infinite sum." Such a thing does not exist for algebraic structures which have finary operations. More specifically, given a series, we cannot in general split nor combine series, and we cannot reindex nor rearrange (an infinite number of) terms. However, we can manipulate each term for a fixed index.

Example 3.9 (Disasters of Reindexing and Rearranging)

Let us take the series $\sum 0$. We clearly know that the corresponding sequence of partial sums $0, 0, \ldots$ is convergent to 0. But if we do this series of steps.

$$\begin{aligned}
\sum_{n=1}^\infty 0 &= \sum_{n=1}^\infty n - n && \text{(Can manipulate terms)} \\
&= \sum_{n=1}^\infty n - \sum_{n=1}^\infty n && \text{(Cannot split series)} \\
&= 1 + \sum_{n=2}^\infty n - \sum_{n=1}^\infty n && \text{(Can take 1st term out)} \\
&= 1 + \sum_{n=1}^\infty (n+1) - \sum_{n=1}^\infty n && \text{(Cannot reindexing)} \\
&= 1 + \sum_{n=1}^\infty (n+1) - n && \text{(Cannot combine series)} \\
&= 1 + \sum_{n=1}^\infty 1 && \text{(Can manipulate terms)} \\
&= 1 + \infty = +\infty && (369)
\end{aligned}$$

The wrong steps show that the series is divergent.

We have seen the consequences of these mistakes that beginners make and are often on popular media. However, note that we can always do splitting, combining, reindexing, and rearranging for *finite sums*, which are algebraically defined. Later on, we will show that some of these operations are allows for series that we know are convergent.

Lemma 3.11 (All Series in Extended Positive Reals Converge)

All series in $\mathbb{R}_{\geq 0} \cup \{+\infty\}$ converge.

$$\sum_{n=1}^\infty x_n \in [0, +\infty] \tag{370}$$

Proof.

Since the convergence of a series is equivalent to convergence of its sequence of partial sums, applying the Cauchy convergence criterion to the sequence $\{s_n\}$ leads to the following theorem.

Theorem 3.12 (Cauchy Convergence Criterion for Series)

The series $a_1 + \ldots + a_n + \ldots$ converges if and only if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $m \geq n > N$,

$$|a_n + \ldots + a_m| < \epsilon \tag{371}$$

Corollary 3.13 (nth Term Test)

A necessary (but not sufficient) condition for convergence of the series $a_1 + \ldots a_n + \ldots$ is that the terms tend to 0 as $n \to \infty$. That is, it is necessary that

$$\lim_{n\to\infty} a_n = 0 \tag{372}$$

Proof. It suffices to set $m = n$ in the Cauchy convergence criterion. This would mean that for every $\epsilon > 0$ there exists a $N \in \mathbb{N}$ such that

$$|a_n| = |a_n - 0| < \epsilon \text{ for all } n > N \tag{373}$$

which, by definition, means that $\{a_n\}$ converges to 0.

Nothing so far is really suprising here. The Cauchy convergence criterion really just follows from the definition of Cauchy completeness, and the nth term test is pretty trivial. The way that we will build up convergence tests is by proving some special cases of convergence and then using the direct comparison test to then classify further series.

Example 3.10 (Telescoping Series)

A **telescoping series** is a series in which the partial sums can cancel out. An example is the series of partial sums of the sequence $(x_n) = \frac{1}{n(n+1)}$. In here, the series term is

$$s_n = \sum_{k=1}^{n} \frac{1}{k(k+1)} \tag{374}$$

$$= \sum_{k=1}^{n} \frac{1}{k} - \frac{1}{k+1} \tag{375}$$

$$= \sum_{k=1}^{n} \frac{1}{k} - \sum_{k=1}^{n} \frac{1}{k+1} \tag{376}$$

$$= \sum_{k=1}^{n} \frac{1}{k} - \sum_{k=2}^{n+1} \frac{1}{k} \tag{377}$$

$$= \frac{1}{1} + \left(\sum_{k=2}^{n} \frac{1}{k} \right) - \left(\sum_{k=2}^{n} \frac{1}{k} \right) - \frac{1}{n+1} \tag{378}$$

$$= 1 + \left(\sum_{k=2}^{n} \frac{1}{k} - \frac{1}{k} \right) - \frac{1}{n+1} \tag{379}$$

$$= 1 - \left(\sum_{k=2}^{n} 0 \right) - \frac{1}{n+1} \tag{380}$$

$$= 1 - \frac{1}{n+1} \tag{381}$$

Note that all of the examples that we have done here are for finite sums, so they are all legal.

Example 3.11 (Geometric Series)

The series $\sum_{n=0}^{\infty} q^n$ is called a **geometric series**.

$$1 + q + q^2 + \ldots + q^n + \ldots \tag{382}$$

is called the **geometric series**. We can see that

1. $|q| \geq 1 \iff \sum q^n$ is divergent. $|q| \geq 1 \implies |q|^n \geq 1$, and so the terms q^n does not converge to 0, and the nth term test is not met.

2. $|q| < 1 \iff \sum q^n$ is convergent. We can use the identity

$$s_n = 1 + q + \ldots + q^{n-1} = \frac{1-q^n}{1-q} \implies \lim_{n\to\infty} \frac{1-q^n}{1-q} = \frac{1}{1-q} \tag{383}$$

since $\lim_{n\to\infty} q^n = 0$ if $|q| < 1$.

The Cauchy convergence criterion can be used to prove the direct comparison test.

Theorem 3.14 (Direct Comparison Test)

For some fixed N, if

1. If $|x_n| \leq y_n$ for all $n \geq N$ and $\sum y_n$ converges, then $\sum x_n$ converges.
2. If $x_n \geq y_n \geq 0$ for all $n \geq N$ and $\sum y_n$ diverges, then $\sum x_n$ diverges.

Example 3.12 (Comparison with Telescoping Series)

We can prove the special case a geometric series with the direct comparison test. We claim that $\sum_{n=1}^\infty \frac{1}{n^2}$ is finite. We can see that

$$\frac{1}{n^2} \leq \frac{2}{n(n+1)} \tag{384}$$

where the series of the terms in the RHS is telescoping and therefore converges. So by the direct comparison test, $\sum \frac{1}{n^2}$ converges.

Now we prove another corollary of the Cauchy convergence criterion.

Theorem 3.15 (Cauchy Condensation Test)

If $a_1 \geq a_2 \geq \ldots \geq 0$, the series $\sum_{n=1}^\infty a_n$ converges if and only if the series

$$\sum_{k=0}^\infty 2^k a_{2^k} = a_1 + 2a_2 + 4a_4 + 8a_8 + \ldots \tag{385}$$

converges.

Proof. Letting $A_k = a_1 + a_2 + \ldots + a_k$ and $S_n = a_1 + 2a_2 + \ldots + 2^n a_{2^n}$, it is clear that by adding up the inequalities

$$\begin{aligned}
&a_2 \leq a_2 \leq a_1 \\
&2a_4 \leq a_3 + a_4 \leq 2a_2 \\
&4a_8 \leq a_5 + a_6 + a_7 + a_8 \leq 4a_4 \\
&\ldots \\
&2^n a_{2^{n+1}} \leq a_{2^n+1} + \ldots + a_{2^{n+1}} \leq 2^n a_{2^n},
\end{aligned}$$

we get

$$\frac{1}{2}(S_{n+1} - a_1) \leq A_{2^{n+1}} - a_1 \leq S_n \tag{386}$$

Since the sequences $\{A_k\}$ and $\{S_k\}$ are nondecreasing, and hence from the inequalities we can conclude that they are either both bounded above (which means that they are both convergent since it is a bounded, nondecreasing series) or both unbounded above (which means that they are both divergent since they are nondecreasing and unbounded).

Corollary 3.16 (p-series Test)

The series

$$\sum_{n=1}^\infty \frac{1}{n^p} \tag{387}$$

converges for $p > 1$ and diverges for $p \leq 1$.[a]

[a] This sort of reminds you of u-substitution. For example, look at $\int_1^\infty f(t)\,dt = \int_0^\infty e^u f(e^u)\,du$, where the convergence of LHS $\iff$ convergence of RHS.

Proof. Suppose $p \geq 0$. By the previous theorem, the series converges or diverges simultaneously with the series

$$\sum_{k=0}^\infty 2^k \frac{1}{(2^k)^p} = \sum_{k=0}^\infty (2^{1-p})^k \tag{388}$$

which is really just a geometric series. A necessary and sufficient condition for the convergence of this series is that $2^{1-p} < 1$, that is, $p > 1$.
Now suppose $p \leq 0$. The series is then clearly divergent since all of the terms are larger than 1.

Example 3.13 (Harmonic Series)

The **harmonic series**

$$1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n} + \ldots \tag{389}$$

seems at first glance to be converging since the terms converge to 0. However, it does not pass the Cauchy condensation test since

$$\sum_{n=1}^\infty 2^n x_n = \sum_{n=1}^\infty 2^n \frac{1}{2^n} = \sum_{n=1}^\infty 1 = +\infty \tag{390}$$

As you can see, this increases logarithmically, so in early calculators it was hard to numerically detect divergence (you would have to double the number of series terms to get a linear increase).

3.5 Ratio and Root Tests

Now we introduce the root and ratio tests, which are derived by the comparison test with a geometric series. The ratio test is used more day-to-day, but not as decisive as the root test. Both tests have a similar flavor.

Theorem 3.17 (Ratio Test)

Suppose the limit $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \alpha$ exists for the series $\sum_{n=1}^\infty a_n$. Then,

1. $\alpha < 1 \implies \sum a_n$ converges absolutely.
2. $\alpha > 1 \implies \sum a_n$ diverges.
3. $\alpha = 1 \implies \sum a_n$ is inconclusive.

Alternatively, if

1. $\limsup |a_{n+1}/a_n| = \alpha < 1$, then $\sum a_n$ converges
2. If $\exists N$ s.t. $|a_{n+1}/a_n| \geq 1$ for all $n \geq N$, then $\sum a_n$ diverges.

Proof. Since $\limsup \left|\frac{a_{n+1}}{a_n}\right| = \alpha < 1$, fix any $\alpha < \beta < 1$. Then $\exists N$ s.t. if $n > N$, $|a_{n+1}/a_n| < \beta$. So $|a_{N+1}| < \beta |a_N| \implies |a_{N+2}| < \beta^2 |a_N|$. So letting $C = |a_N|$, for all $m \geq N$,

$$|a_m| \leq \frac{C}{\beta^N} \beta^m \implies |a_m| \leq \tilde{C} \beta^m \text{ for all } m \geq N \tag{391}$$

So $\sum a_n$ converges by comparison test since $\sum \beta^m < \infty$ when $\beta < 1$.

Theorem 3.18 (Root Test)

Let $\sum_{n=1}^\infty a_n$ be a given series and

$$\alpha = \limsup_{n\to\infty} \sqrt[n]{|a_n|} \tag{392}$$

Then,
1. $\alpha < 1 \implies \sum a_n$ converges.
2. $\alpha > 1 \implies \sum a_n$ diverges.
3. $\alpha = 1 \implies \sum a_n$ is inconclusive.

Proof. Listed.
1. If $\limsup \sqrt[n]{|a_n|} = \alpha < 1$, take any $\alpha < \beta < 1$. Then $\exists N \in \mathbb{N}$ s.t. if $n \geq N$, then $|a_n|^{1/n} < \beta \iff |a_n| < \beta^n$. Since $\beta < 1$, $\sum \beta^n < \infty$, and by comparison test, $\sum a_n$ converges.
2. Suppose $\alpha > 1$. Then $\limsup |a_n|^{1/n} = \alpha > 1$. So there exists a subsequence (a_{n_k}) s.t. $(|a_{n_k}|^{1/n_k}) \to \alpha > 1$. This means $\exists N$ s.t. for $n \geq N$, $|a_{n_k}|^{1/n_k} > 1 \implies |a_{n_k}| > 1$. But this fails the nth term test.
3. We do not claim anything and so there's nothing to prove.

Example 3.14 (Root Test Inconclusive Results)

Consider $\sum \frac{1}{n} = +\infty$, but from the root test

$$\sqrt[n]{\frac{1}{n}} \to 1, \text{ so } \alpha = 1 \tag{393}$$

Consider $\sum \frac{1}{n^2} < +\infty$, but from from the root test

$$\sqrt[n]{\frac{1}{n^2}} = \left(\frac{1}{n^{1/n}}\right)^2 \to 1, \text{ so } \alpha = 1 \tag{394}$$

Example 3.15

The sequence $\sum \frac{c^n}{n!}$ always converges for $c \in \mathbb{R}$.

Theorem 3.19 (Weierstrass M-test for Absolute Convergence)

Let $\sum_{n=1}^\infty a_n$ and $\sum_{n=1}^\infty b_n$ be series. Suppose there exists an index $N \in \mathbb{N}$ such that $|a_n| \leq b_n$ for all $n > N$. Then,

$$\sum_{n=1}^\infty b_n \text{ converges} \implies \sum_{n=1}^\infty a_n \text{ converges absolutely} \tag{395}$$

We finally conclude by giving a theorem about the convergence of some special sequences.

Theorem 3.20 (Special Sequences)

Some special sequences:
1. If $p > 0$, then $\lim_{n\to\infty} \frac{1}{n^p} = 0$.
2. If $p > 0$, then $\lim_{n\to\infty} \sqrt[n]{p} = 1$.
3. $\lim_{n\to\infty} \sqrt[n]{n} = 1$.
4. If $p > 0$ and α is real, then $\lim_{n\to\infty} \frac{n^\alpha}{(1+p)^n} = 0$.
5. If $|x| < 1$, then $\lim_{n\to\infty} x^n = 0$.

3.6 Euler's Number and Trigonometric Functions

Definition 3.7 (Euler's Number)

We define **Euler's number** in the equivalent forms:
1. As a series
$$e \coloneqq \sum_{n=0}^\infty \frac{1}{n!} \tag{396}$$
2. As a limit
$$\lim_{n\to+\infty} \left(1 + \frac{1}{n}\right)^n = e \tag{397}$$

Proof. The first thing we should do is show that it converges, this is a one-liner.
$$\sum_{n=0}^\infty \frac{1}{n!} = 1 + 1 + \sum_{n=2}^\infty \frac{1}{n!} \leq 2 + \sum_{n=2}^\infty \frac{1}{n(n-1)} \tag{398}$$
Since it is a bounded and monotonically increasing function, we are done.
To prove equivalence. Let us define the sequence
$$t_n = \sum_{k=0}^n \frac{1}{k!}, \qquad s_n = \left(1 + \frac{1}{n}\right)^n \tag{399}$$
We know that $t_n \to e$, and we want to show that $s_n \to e$. We do this with the squeeze theorem.
1. We can see that
$$s_n = \left(1 + \frac{1}{n}\right)^n \tag{400}$$
$$= \sum_{k=0}^n \binom{n}{k} 1^{n-k} \left(\frac{1}{n}\right)^k \tag{401}$$
$$= 1 + 1 + \frac{n(n-1)}{2!}\frac{1}{n^2} + \frac{n(n-1)(n-2)}{3!}\frac{1}{n^3} + \ldots \tag{402}$$
$$= 1 + 1 + \frac{1}{2!}(1)\left(1 - \frac{1}{n}\right) + \frac{1}{3!}(1)\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) + \ldots + \frac{1}{n!}(1)\prod_{k=1}^{n-1}\left(1 - \frac{k}{n}\right) \tag{403}$$
$$\leq \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \ldots + \frac{1}{n!} = t_n \tag{404}$$
and so $s_n < t_n \implies \limsup s_n \leq \limsup t_n = e$.
2. Let $m \leq n$ be fixed. Then,
$$s_n \geq 1 + 1 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) + \ldots + \frac{1}{m!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\ldots\left(1 - \frac{m-1}{n}\right) \tag{405}$$

since we are just taking the first m positive terms of the element. Therefore, letting $n \to +\infty$ and keeping m fixed, we get

$$\liminf_{n\to+\infty} s_n \geq 1 + 1 + \frac{1}{2!} + \ldots + \frac{1}{n!} \text{ for all } m \in \mathbb{N} \tag{406}$$

which implies $\liminf s_n \geq t_m$ for all $m \in \mathbb{N}$, and now letting $m \to +\infty$, we have $\liminf s_n \geq \liminf t_m = e$.

Now we prove the irrationality of e. It is usually extremely difficult to prove that an arbitrary number is irrational, e.g. π^e or π^{e^e}.

Theorem 3.21 (e is Irrational)

e is irrational.

Proof. Letting $t_n = \sum_{k=0}^{n} \frac{1}{k!}$, we have

$$\begin{aligned}
e - t_n &= \sum_{k=n+1}^{\infty} \frac{1}{k!} && (407)\\
&= \frac{1}{(n+1)!}\left(1 + \frac{1}{n+2} + \frac{1}{(n+3)(n+2)} + \ldots\right) && (408)\\
&< \frac{1}{(n+1)!}\underbrace{\left(1 + \frac{1}{n+2} + \frac{1}{(n+2)^2} + \ldots\right)}_{\text{geometric}} && (409)\\
&= \frac{1}{(n+1)!}\left(\frac{1}{1-(1/(n+2)!)}\right) && (410)\\
&= \frac{1}{n!n} \cdot \underbrace{\frac{(n+2)n}{(n+1)^2}}_{<1} && (411)\\
&= \frac{1}{n!n} && (412)
\end{aligned}$$

Note that we can combine and split sums since we know that e is convergent. Now suppose that $e = p/q$. Then,

$$0 < q!(e - t_q) < \frac{1}{q} \tag{413}$$

But $q!e$ is an integer and $q!t_q$ is also an integer. So we have $q! \cdot \frac{p}{q}$, an integer, between 0 and 1, which is a contradiction.

Since we have defined some number $e \in \mathbb{R}$, we know that exponential exist, and therefore we the function $x \mapsto e^x$ is well-defined. In fact, it is so important that we have a separate name for it.

Definition 3.8 (Exponential Function)

The **exponential function** is generally referred to as the function $x \mapsto e^x$.

There is a nice series representation.

Theorem 3.22 (Exponential Function as a Series)

We have

$$e^x = \sum_{n=0}^\infty \frac{x^n}{n!} \tag{414}$$

Proof.

Now that this is done, we can define the trigonometric functions formally as such.

Definition 3.9 (Trigonometric Functions)

We have

$$\sin x = \sum_{n=0}^\infty \frac{(-1)^n}{(2n+1)!} x^{2n+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \tag{415}$$

$$\tag{416}$$

$$\cos x = \sum_{n=0}^\infty \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \tag{417}$$

3.7 Exercises

Exercise 3.1 (Math 531 Spring 2025, PS4.6)

Consider the set of all bounded sequences of real numbers. That is, we consider sequences $\{x_n\}$ for which

$$\sup_{n \in \mathbb{N}} |x_n| \tag{418}$$

exists. For example, the sequence $\{1, 2, 3, \ldots\}$ does not belong to the set, but the sequence $\{1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \ldots\}$ does. Call this set X. Endow it with a metric:

$$d(\{x_n\}, \{y_n\}) = \sup_{n \in \mathbb{N}} |x_n - y_n|. \tag{419}$$

Explain why this is a metric. Make sure to explain why the supremum on the right hand side exists.

Solution.

Exercise 3.2 (Math 531 Spring 2025, PS4.7)

Consider the metric space (X, d) from the previous problem. Is $\overline{B_1(\{0\})}$ a compact set? Here, $\{0\}$ is just the sequence of zeros: $\{0, 0, 0, 0, \ldots\}$.

Solution.

Exercise 3.3 (Math 531 Spring 2025, PS6.1)

Give a direct proof that

$$\sum_{n=1}^{\infty} \frac{1}{k^2+k} \tag{420}$$

converges and find the exact value of the series.

Solution. This is a telescoping series.

$$\sum_{n=1}^{\infty} \frac{1}{k^2+k} = \sum_{n=1}^{\infty} \frac{1}{k} - \frac{1}{k-1} = 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \ldots \tag{421}$$

and so the nth partial sum is $1 - \frac{1}{n} \to 1$ as $n \to \infty$.

Exercise 3.4 (Math 531 Spring 2025, PS6.2)

Suppose that a_n is a sequence of non-negative real numbers and suppose that $\sum_{n=1}^{\infty} a_n$ converges. Prove that there exists a sequence b_n with $\lim_{n\to\infty} b_n = +\infty$ so that $\sum_{n=1}^{\infty} a_n b_n$ is still convergent.

Solution. Let us take $r_n = \sum_{m=n}^{\infty} a_n$. Since a_n is nonnegative, $r_n = 0 \iff a_m = 0$ for all $m \geq n$. In this case set

$$(b_m) := \begin{cases} 1 & \text{if } m < n \\ m & \text{if } m \geq n \end{cases} \tag{422}$$

and

$$\sum_{n=1}^{\infty} a_n = \sum_{m=1}^{n} a_m = \sum_{m=1}^{n} a_m b_m = \sum_{m=1}^{\infty} a_m b_m < +\infty \tag{423}$$

So we may assume $r_n > 0$. Now define $(b_n) = \big(\frac{1}{\sqrt{r_n}}\big)$. By the Cauchy criterion, $r_n \to 0$ as $n \to +\infty$, so $b_n \to +\infty$. But we claim that $\frac{a_n}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}})$ since

$$\frac{a_n}{\sqrt{r_n}}(\sqrt{r_n} + \sqrt{r_{n+1}}) = a_n + a_n \frac{\sqrt{r_{n+1}}}{\sqrt{r_n}} < 2a_n = 2(r_n - r_{n+1}) \tag{424}$$

Since the series $\sum(\sqrt{r_n} - \sqrt{r_{n+1}})$ converges to $\sqrt{r_1}$, it follows by the comparison test that $\sum \frac{a_n}{\sqrt{r_n}}$ also converges.

Exercise 3.5 (Math 531 Spring 2025, PS6.3)

Prove that $\sum_{n=1}^{\infty} \frac{\sin(n)}{n}$ converges[a]. Hint: Use summation by parts and the fact that $\sin(x) = \frac{1}{2i}(\exp(ix) - \exp(-ix))$ where $i = \sqrt{-1}$.

[a] Here, we are going to assume that we know a little bit about trigonometric functions!

Solution. We can see that

$$\left|\mathrm{Im} \sum_{n=1}^{N} e^{in}\right| = \left|\sum_{n=1}^{N} \sin(n)\right| \leq \left|e^i \frac{1-e^{in}}{1-e^i}\right| \leq \frac{2}{|1-e^i|} < +\infty \tag{425}$$

and so the partial sums form a bounded sequence, implying that $\sum \frac{\sin(n)}{n}$ is convergent.

Exercise 3.6 (Math 531 Spring 2025, PS6.4)

Fix $c \in (0, \infty)$.

- Assume x_n is a sequence of numbers with $x_n \to 0$. Prove that
$$c^{x_n} \to 1. \tag{426}$$
- Deduce that, if $x_n \to x$, then
$$c^{x_n} \to c^x, \tag{427}$$
for any $x \in \mathbb{R}$.

Solution. $x_n \to 0$ as $n \to \infty$ means that $\forall \delta > 0$, $\exists N \in \mathbb{N}$ s.t. $|x_n| < \delta$ for all $n \geq N$. Now take $\epsilon > 0$. Then we wish to show that $\exists M \in \mathbb{N}$ s.t. $|1 - c^{x_m}| < \epsilon$ for all $n \geq M$. We choose $\delta = \min\{-\log_c(1-\epsilon), \log_c(1+\epsilon)\}$. Then $\exists M$ s.t.

$$x_n \in \big(-\log_c(1-\epsilon), \log_c(1+\epsilon)\big) \iff |x_n| < \delta \tag{428}$$

for $n \geq M$ by convergence $x_n \to 0$. Since $f(x) = c^x$ is monotonically increasing, this is equivalent to

$$f(x_n) = c^{x_n} \tag{429}$$

Exercise 3.7 (Math 531 Spring 2025, PS6.5)

For every $z \in \mathbb{C}$, verify that the series

$$E(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!} \tag{430}$$

converges. Prove that for every $z, w \in \mathbb{C}$ we have that $E(z)E(w) = E(z+w)$. Deduce that, for $q \in \mathbb{Q}$, we have that $E(q) = e^q$. Deduce that $E(x) = e^x$, for all $x \in \mathbb{R}$.

Solution.

Exercise 3.8 (Rudin 3.1)

Prove that convergence of $\{x_n\}$ implies convergence of $\{|x_n|\}$. Is the converse true?

Solution. If $\{x_n\}$ converges to x, then for all $\epsilon > 0$, there exists a $N \in \mathbb{N}$ s.t. $|x_n - x| < \epsilon$ if $n > N$. We use the inequality $\big||x_n| - |x|\big| \leq |x_n - x|$ to show that then for every $\epsilon > 0$ there exists a $N \in \mathbb{N}$ s.t.

$$\big||x_n| - |x|\big| \leq |x_n - x| \leq \epsilon$$

and so $\{|x_n|\}$ converges to $|x|$.

Exercise 3.9 (Rudin 3.2)

Calculate

$$\lim_{n \to \infty} \sqrt{n^2 + n} - n$$

Solution. We can compute

$$\lim_{n\to\infty} \sqrt{n^2+n} - n = \lim_{n\to\infty} (\sqrt{n^2+n} - n) \cdot \frac{\sqrt{n^2+n}+n}{\sqrt{n^2+n}+n} = \lim_{n\to\infty} \frac{n}{\sqrt{n^2+n}+n}$$

where

$$A_n = \frac{n}{\sqrt{n^2+2n+1}+n} \leq \frac{n}{2n+1} \leq \frac{n}{\sqrt{n^2+n}+n} \leq \frac{n}{\sqrt{n^2}+n} = \frac{n}{2n} = \frac{1}{2} = C_n$$

C_n is ultimately constant. It suffices to prove that A_n limits to $\frac{1}{2}$ by showing that

$$\frac{n}{2n+1} = \frac{n/n}{(2n+1)/n} = \frac{1}{2+\frac{1}{n}}$$

where $\{\frac{1}{n}\}$ is infinitesimal.

Exercise 3.10 (Rudin 3.3)

If $s_1 = \sqrt{2}$ and

$$s_{n+1} = \sqrt{2+\sqrt{s_n}}$$

for $n = 1, 2, \ldots$, prove that $\{s_n\}$ converges and that $s_n < 2$ for $n = 1, 2, \ldots$.

Solution. We can show that $s_n < 2$ by induction. $s_1 = \sqrt{2} < 2$, so the base case is proved. Now, given that $s_n < 2$, $\sqrt{s_n} < 2 \implies 2 + \sqrt{s_n} < 2 + \sqrt{2} < 4 \implies s_{n+1} = \sqrt{2+\sqrt{s_n}} < 2$ and we are done.

Exercise 3.11 (Rudin 3.4)

Find the upper and lower limits of the sequence $\{s_n\}$ defined by

$$s_1 = 0; \quad s_{2m} = \frac{s_{2m-1}}{2}; \quad s_{2m+1} = \frac{1}{2} + s_{2m}.$$

Solution.

Exercise 3.12 (Rudin 3.5)

For any two real sequences $\{a_n\}$, $\{b_n\}$, prove that

$$\limsup_{n\to\infty}(a_n + b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n,$$

provided the sum on the right is not of the form $\infty - \infty$.

Solution.

Exercise 3.13 (Rudin 3.6)

Investigate the behavior (convergence or divergence) of Σa_n if

(a) $a_n = \sqrt{n+1} - \sqrt{n}$;

(b) $a_n = (\sqrt{n+1} - \sqrt{n})/n$;

(c) $a_n = (\sqrt[n]{n} - 1)^n$;
(d) $a_n = \frac{1}{1+z^n}$, for complex values of z.

Solution.

Exercise 3.14 (Rudin 3.7)

Prove that the convergence of Σa_n implies the convergence of

$$\sum \frac{\sqrt{a_n}}{n},$$

if $a_n \geq 0$.

Solution.

Exercise 3.15 (Rudin 3.8)

If Σa_n converges, and if $\{b_n\}$ is monotonic and bounded, prove that $\Sigma a_n b_n$ converges.

Solution.

Exercise 3.16 (Rudin 3.9)

Find the radius of convergence of each of the following power series:
(a) $\sum n^3 z^n$,
(b) $\sum \frac{2^n}{n!} z^n$,
(c) $\sum \frac{2^n}{n^2} z^n$,
(d) $\sum \frac{n^3}{3^n} z^n$.

Solution.

Exercise 3.17 (Rudin 3.10)

Suppose that the coefficients of the power series $\sum a_n z^n$ are integers, infinitely many of which are distinct from zero. Prove that the radius of convergence is at most 1.

Solution.

Exercise 3.18 (Rudin 3.11)

Suppose $a_n > 0$, $s_n = a_1 + \cdots + a_n$, and Σa_n diverges.
(a) Prove that $\sum \frac{a_n}{1+a_n}$ diverges.
(b) Prove that

$$\frac{a_{N+1}}{s_{N+1}} + \cdots + \frac{a_{N+k}}{s_{N+k}} \geq 1 - \frac{s_N}{s_{N+k}}$$

and deduce that $\sum \frac{a_n}{s_n}$ diverges.

(c) Prove that

$$\frac{a_n}{s_n^2} \leq \frac{1}{s_{n-1}} - \frac{1}{s_n}$$

and deduce that $\sum \frac{a_n}{s_n^2}$ converges.

(d) What can be said about

$$\sum \frac{a_n}{1 + na_n} \text{ and } \sum \frac{a_n}{1 + n^2 a_n}?$$

Solution.

Exercise 3.19 (Rudin 3.12)

Suppose $a_n > 0$ and Σa_n converges. Put

$$r_n = \sum_{m=n}^{\infty} a_m.$$

(a) Prove that

$$\frac{a_m}{r_m} + \cdots + \frac{a_n}{r_n} > 1 - \frac{r_n}{r_m}$$

if $m < n$, and deduce that $\sum \frac{a_n}{r_n}$ diverges.

(b) Prove that

$$\frac{a_n}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}})$$

and deduce that $\sum \frac{a_n}{\sqrt{r_n}}$ converges.

Solution.

Exercise 3.20 (Rudin 3.13)

Prove that the Cauchy product of two absolutely convergent series converges absolutely.

Solution.

Exercise 3.21 (Rudin 3.14)

If $\{s_n\}$ is a complex sequence, define its arithmetic means σ_n by

$$\sigma_n = \frac{s_0 + s_1 + \cdots + s_n}{n+1} \quad (n = 0, 1, 2, \ldots).$$

(a) If $\lim s_n = s$, prove that $\lim \sigma_n = s$.

(b) Construct a sequence $\{s_n\}$ which does not converge, although $\lim \sigma_n = 0$.

(c) Can it happen that $s_n > 0$ for all n and that $\limsup s_n = \infty$, although $\lim \sigma_n = 0$?

(d) Put $a_n = s_n - s_{n-1}$, for $n \geq 1$. Show that

$$s_n - \sigma_n = \frac{1}{n+1} \sum_{k=1}^{n} k a_k.$$

Assume that $\lim(na_n) = 0$ and that $\{\sigma_n\}$ converges. Prove that $\{s_n\}$ converges. [This gives a converse of (a), but under the additional assumption that $na_n \to 0$.]

(e) Derive the last conclusion from a weaker hypothesis: Assume $M < \infty$, $|na_n| \leq M$ for all n, and $\lim \sigma_n = \sigma$. Prove that $\lim s_n = \sigma$, by completing the following outline:
If $m < n$, then

$$s_n - \sigma_n = \frac{m+1}{n-m}(\sigma_n - \sigma_m) + \frac{1}{n-m} \sum_{i=m+1}^{n} (s_n - s_i).$$

For these i,

$$|s_n - s_i| \leq \frac{(n-i)M}{i+1} \leq \frac{(n-m-1)M}{m+2}.$$

Fix $\varepsilon > 0$ and associate with each n the integer m that satisfies

$$m \leq \frac{n-\varepsilon}{1+\varepsilon} < m+1.$$

Then $(m+1)/(n-m) \leq 1/\varepsilon$ and $|s_n - s_i| < M\varepsilon$. Hence

$$\limsup_{n\to\infty} |s_n - \sigma| \leq M\varepsilon.$$

Since ε was arbitrary, $\lim s_n = \sigma$.

Solution.

Exercise 3.22 (Rudin 3.15)

Definition 3.21 can be extended to the case in which the a_n lie in some fixed $\mathbb{R}^k$. Absolute convergence is defined as convergence of $\Sigma|\mathbf{a}_n|$. Show that Theorems 3.22, 3.23, 3.25(a), 3.33, 3.34, 3.42, 3.45, 3.47, and 3.55 are true in this more general setting. (Only slight modifications are required in any of the proofs.)

Solution.

Exercise 3.23 (Rudin 3.16)

Fix a positive number α. Choose $x_1 > \sqrt{\alpha}$, and define $x_2, x_3, x_4, \ldots$, by the recursion formula

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{\alpha}{x_n}\right).$$

(a) Prove that $\{x_n\}$ decreases monotonically and that $\lim x_n = \sqrt{\alpha}$.

(b) Put $\varepsilon_n = x_n - \sqrt{\alpha}$, and show that

$$\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$$

so that, setting $\beta = 2\sqrt{\alpha}$,

$$\varepsilon_{n+1} < \beta \left(\frac{\varepsilon_1}{\beta}\right)^{2^n} \qquad (n = 1, 2, 3, \ldots).$$

(c) This is a good algorithm for computing square roots, since the recursion formula is simple and the convergence is extremely rapid. For example, if $\alpha = 3$ and $x_1 = 2$, show that $\varepsilon_1/\beta < \frac{1}{10}$ and that therefore

$$\varepsilon_5 < 4 \cdot 10^{-16}, \quad \varepsilon_6 < 4 \cdot 10^{-32}.$$

Solution.

Exercise 3.24 (Rudin 3.17)

Fix $\alpha > 1$. Take $x_1 > \sqrt{\alpha}$, and define

$$x_{n+1} = \frac{\alpha + x_n}{1 + x_n} = x_n + \frac{\alpha - x_n^2}{1 + x_n}.$$

(a) Prove that $x_1 > x_3 > x_5 > \cdots$.
(b) Prove that $x_2 < x_4 < x_6 < \cdots$.
(c) Prove that $\lim x_n = \sqrt{\alpha}$.
(d) Compare the rapidity of convergence of this process with the one described in Exercise 16.

Solution.

Exercise 3.25 (Rudin 3.18)

Replace the recursion formula of Exercise 16 by

$$x_{n+1} = \frac{p-1}{p} x_n + \frac{\alpha}{p} x_n^{-p+1}$$

where p is a fixed positive integer, and describe the behavior of the resulting sequences $\{x_n\}$.

Solution.

Exercise 3.26 (Rudin 3.19)

Associate to each sequence $a = \{\alpha_n\}$, in which α_n is 0 or 2, the real number

$$x(a) = \sum_{n=1}^{\infty} \frac{\alpha_n}{3^n}.$$

Prove that the set of all $x(a)$ is precisely the Cantor set described in Sec. 2.44.

Solution.

Exercise 3.27 (Rudin 3.20)

Suppose $\{p_n\}$ is a Cauchy sequence in a metric space X, and some subsequence $\{p_{n_i}\}$ converges to a point $p \in X$. Prove that the full sequence $\{p_n\}$ converges to p.

Solution.

Exercise 3.28 (Rudin 3.21)

Prove the following analogue of Theorem 3.10(b): If $\{E_n\}$ is a sequence of closed nonempty and bounded sets in a complete metric space X, if $E_n \supset E_{n+1}$, and if

$$\lim_{n\to\infty} \text{diam } E_n = 0,$$

then $\bigcap_1^\infty E_n$ consists of exactly one point.

Solution.

Exercise 3.29 (Rudin 3.22)

Suppose X is a nonempty complete metric space, and $\{G_n\}$ is a sequence of dense open subsets of X. Prove Baire's theorem, namely, that $\bigcap_1^\infty G_n$ is not empty. (In fact, it is dense in X.) Hint: Find a shrinking sequence of neighborhoods E_n such that $E_n \subset G_n$, and apply Exercise 21.

Solution.

Exercise 3.30 (Rudin 3.23)

Suppose $\{p_n\}$ and $\{q_n\}$ are Cauchy sequences in a metric space X. Show that the sequence $\{d(p_n, q_n)\}$ converges. Hint: For any m, n,

$$d(p_n, q_n) \leq d(p_n, p_m) + d(p_m, q_m) + d(q_m, q_n);$$

it follows that

$$|d(p_n, q_n) - d(p_m, q_m)|$$

is small if m and n are large.

Solution.

Exercise 3.31 (Rudin 3.24)

Let X be a metric space.

(a) Call two Cauchy sequences $\{p_n\}$, $\{q_n\}$ in X equivalent if

$$\lim_{n\to\infty} d(p_n, q_n) = 0.$$

Prove that this is an equivalence relation.

(b) Let X^* be the set of all equivalence classes so obtained. If $P \in X^*, Q \in X^*$, $\{p_n\} \in P, \{q_n\} \in Q$, define

$$\Delta(P, Q) = \lim_{n \to \infty} d(p_n, q_n);$$

by Exercise 23, this limit exists. Show that the number $\Delta(P, Q)$ is unchanged if $\{p_n\}$ and $\{q_n\}$ are replaced by equivalent sequences, and hence that Δ is a distance function in X^*.

(c) Prove that the resulting metric space X^* is complete.

(d) For each $p \in X$, there is a Cauchy sequence all of whose terms are p; let P_p be the element of X^* which contains this sequence. Prove that

$$\Delta(P_p, P_q) = d(p, q)$$

for all $p, q \in X$. In other words, the mapping φ defined by $\varphi(p) = P_p$ is an isometry (i.e., a distance-preserving mapping) of X into X^*.

(e) Prove that $\varphi(X)$ is dense in X^*, and that $\varphi(X) = X^*$ if X is complete. By (d), we may identify X and $\varphi(X)$ and thus regard X as embedded in the complete metric space X^*. We call X^* the completion of X.

Solution.

Exercise 3.32 (Rudin 3.25)

Let X be the metric space whose points are the rational numbers, with the metric $d(x, y) = |x - y|$. What is the completion of this space? (Compare Exercise 24.)

Solution.

4 Limits and Continuity

Just like we did with our notes in point set topology, we will continue our discussions of convergence by talking about limits of functions, followed by continuity. In the real line, we have extra structure and should consider slight variations of the limit of a function. I think it's important to distinguish two-sided limits, which can be defined only for interior points (which can have sequences converging from both left and right), one-sided limits, which can be defined on boundaries and require less assumptions to compute, and limits, which is the most general and attempts to unify the two.

4.1 Limits of Functions

Note that this definition is really just a special case of the definition of the limits of functions between topological spaces and that of metric spaces.

Definition 4.1 (Limit of Real-Valued Functions)

Let $f : E \subset \mathbb{R} \to \mathbb{R}$ and $p \in \mathbb{R}$ be a limit point of E. We say $f(x) \to q$ as $x \to p$, i.e. the **limit**

$$\lim_{x \to p} f(x) = q \tag{431}$$

exists, if it meets the following equivalent conditions.

1. *ϵ-δ Definition.* For all $\epsilon > 0$, there exists a $\delta > 0$ s.t.

$$x \in E, 0 < |x - p| < \delta \implies |f(x) - q| < \epsilon \tag{432}$$

2. *Sequential Definition.* If for all sequences $(x_n) \subset E$,

$$x_n \to p \implies f(x_n) \to q \tag{433}$$

Furthermore, if E is open, then we call the limit a **two-sided limit**.

Note that the strictly inequality $0 < d_X(x, p)$ is important to ensure that $x \neq p$, since functions can jump at p. Another useful tool to define is the one-sided limit, which is specific to the reals as an ordered set. This allows us to define weaker concepts that will be useful later on (e.g. to define CDF functions in probability which are right-continuous).

Definition 4.2 (One-Sided Limit of a Real-Valued Function)

Let $f : E \subset \mathbb{R} \to \mathbb{R}$.

1. *Right-Hand limit.* If $p \in \mathbb{R}$ is a limit point of $E \cap [p, +\infty)$, then we say the **right-hand limit** exists

$$\lim_{x \to p^+} f(x) = q \tag{434}$$

   if it meets the following equivalent conditions.

   (a) *ϵ-δ Definition.* For all $\epsilon > 0$, there exists a $\delta > 0$ s.t.

$$p < x < p + \delta \implies |f(x) - q| < \epsilon \tag{435}$$

   (b) *Sequential Definition.* If for all sequences $(x_n) \subset E \cap [p, +\infty)$,

$$x_n \to p \implies f(x_n) \to q \tag{436}$$

2. *Left-Hand limit.* If $p \in \mathbb{R}$ is a limit point of $E \cap [p, +\infty)$, then we say the **left-hand limit** exists

$$\lim_{x \to p^-} f(x) = q \tag{437}$$

if it meets the following equivalent conditions.

(a) *ϵ-δ Definition.* For all $\epsilon > 0$, there exists a $\delta > 0$ s.t.

$$p - \delta < x < p \implies |f(x) - q| < \epsilon \tag{438}$$

(b) *Sequential Definition.* If for all sequences $(x_n) \subset E \cap (-\infty, p]$,

$$x_n \to p \implies f(x_n) \to q \tag{439}$$

Note that the one-sided limit deals with a smaller class of sequences, so it one-sided limits may exist while two sided limits do not.

Example 4.1 (Limit of the Signum Function)

The function $\operatorname{sgn} : \mathbb{R} \longrightarrow \mathbb{R}$ defined

$$\operatorname{sgn} x = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases} \tag{440}$$

has no limit as $x \to 0$.

First, it is ludicrous that the limit would be any number that is not $\{-1, 0, 1\}$. If we assume that $A \notin \{-1, 0, 1\}$, then we can choose any arbitrarily small ϵ-neighborhood of A that does not include the three numbers. Clearly, there doesn't exist any $\delta > 0$ such that the deleted δ-neighborhood of 0 maps to a set completely contained in the ϵ-neighborhood of A. That is,

$$\operatorname{sgn}\big(\mathring{U}_\delta(0)\big) = \{-1, 1\} \not\subset U_\epsilon(A) \tag{441}$$

It doesn't even intersect the ϵ-neighborhood at all.

1. If $A = 1$, we can construct a ϵ-neighborhood V_A for $\epsilon = \frac{1}{2}$. Clearly, there exists no open neighborhood U_0 of 0 that is entirely mapped to V, since U_0 contains both negative numbers and 0 and hence must be mapped to $0, -1$.
2. Similarly, given the $(\epsilon = \frac{1}{2})$-neighborhood of $A = -1$, there exists no open neighborhood U_0 of 0 that is entirely mapped to it, since U_0 contains both positive numbers and 0 and hence must be mapped to $0, 1$.
3. Finally, given the $(\epsilon = \frac{1}{2})$-neighborhood of $A = 0$, there exists no open neighborhood U_0 of 0 that is entirely mapped to it, since U_0 contains both positive and negative numbers and hence must be mapped to ± 1.

Therefore, the limit does not exist.

Example 4.2 (Limit of Absolute Value of Signum Function)

We will show that

$$\lim_{x \to 0} |\operatorname{sgn} x| = 1 \tag{442}$$

We construct a ϵ-neighborhood $U_\epsilon(1)$ around 1. Given this neighborhood, we can imagine choosing the deleted δ-neighborhood $\mathring{U}_\delta(0)$ around 0. Since every element in $\mathring{U}_\delta(0)$ maps to 1, it is clearly in U_ϵ. In fact, for arbitrarily small $\epsilon > 0$, we can choose **any** $\delta > 0$ since everything in $\mathbb{R} \setminus 0$ maps to 1. We can visualize this in $\mathbb{R}^2$ as

Theorem 4.1 (Two-Sided vs One-Sided Limits)

Let $f : E \subset \mathbb{R} \to \mathbb{R}$. If both of the one-sided limits of real-valued functions exist at p and

$$\lim_{x\to p^-} f(x) = \lim_{x\to p^+} f(x) = q \tag{443}$$

Then the two-sided limit of f exists at p, with

$$\lim_{x\to p} f(x) = q \tag{444}$$

Proof.

Note that on boundaries, the limit can exist, but one of the one-sided limits may not exist, so the converse isn't true.

Example 4.3 (Limit and Right-Hand Limit Exists, but Not Left-Hand Limit)

Consider the function

$$f : (0, +\infty) \to \mathbb{R}, f(x) = x \tag{445}$$

The limit and the right-hand limit exists, but not the left-hand limit since $(-\infty, 0] \cap (0, +\infty) = \emptyset$, and so there exist no limit points. It doesn't even matter if we change the domain to $[0, +\infty)$, since the intersection will be $\{0\}$, which also has no limit points.

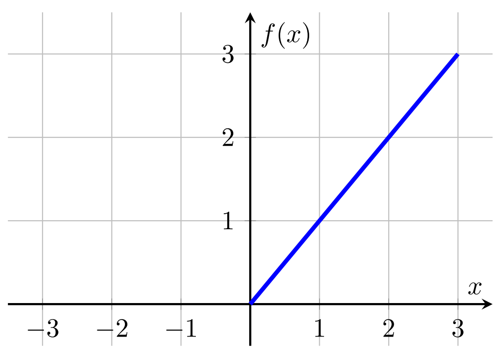


Figure 2

The general rule of thumb is to always try to find the two-sided limit if applicable, but if not, then you can find the one-sided limit.

Theorem 4.2 (Arithmetic on Limits of Functions)

Given two numerical valued functions $f, g : E \subset \mathbb{R} \longrightarrow \mathbb{R}$ with a common domain where $g(x) \neq 0$ for all $x \in E$, let

$$\lim_{x\to a} f(x) = A, \qquad \lim_{x\to a} g(x) = B \tag{446}$$

then,

$$\begin{aligned}
&\lim_{x\to a}(f+g)(x) = A + B \\
&\lim_{x\to a}(cf)(x) = cA \\
&\lim_{x\to a}(f\cdot g)(x) = A\cdot B \\
&\lim_{x\to a}\left(\frac{f}{g}\right)(x) = \frac{A}{B}
\end{aligned}$$

Proof. Cauchy sequence criterion for a limit immediately proves this.

Finally, we reiterate some limit theorems already stated for sequences, but now corresponding to functions. Interpreting the function limit as the Cauchy sequence definition of limits renders the proofs of these theorems trivial.

Theorem 4.3 (Bounds on Limits of Functions)

If the functions $f, g : E \to \mathbb{R}$ are such that

$$\lim_{x\to a} f(x) < \lim_{x\to a} g(x) \tag{447}$$

then there exists a deleted neighborhood $U_\delta(a)$ in E at each point of which $f(x) < g(x)$.

Theorem 4.4 (Squeeze Theorem for Limits of Functions)

Given the functions $f, g, h : E \subset \mathbb{R} \longrightarrow \mathbb{R}$ such that

$$f(x) \leq g(x) \leq h(x) \text{ for all } x \in E \tag{448}$$

then,

$$\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = q \implies \lim_{x\to a} g(x) = q \tag{449}$$

4.2 Continuous Functions

Definition 4.3 (Continuity of a Function)

A function f is **continuous at point** a if for any neighborhood $V\big(f(a)\big)$ of $f(a)$, there is a neighborhood $U(a)$ of a whose image under the mapping f is contained in $V\big(f(a)\big)$.
Generalizing this, we say that a function is **(globally) continuous** if the preimage of every neighborhood in its codomain is an open set in its domain.

Lemma 4.5 (Existence of Limits of Continuous Functions)

$f : E \longrightarrow \mathbb{R}$ is continuous at $a \in E$, where a is a limit point of E if and only if

$$\lim_{x\to a} f(x) = f(a) \tag{450}$$

Proof. The limit equaling $f(a)$ means that, by definition, for any arbitrarily small deleted neighborhood of $f(a)$, denoted $U_{f(a)} \setminus f(a)$, its preimage will be an open neighborhood of a, which itself will contain an open set.

This also means that we can use the Cauchy limit definition to defined continuity of a function at a point. That is, for any sequence $\{a_n\}$ of point in codomain E which converges to point a, the function f is continuous at a if the corresponding sequence $\{f(a_n)\}$ converges to $f(a)$.

Theorem 4.6

This means that the continuous functions commute with the operation of passing to the limit at a point.

$$\lim_{x \to a} f(x) = f\Big(\lim_{x \to a} x\Big) \tag{451}$$

Lemma 4.7 (Properties of Continuous Functions)

Let $f : \mathbb{R}^n \longrightarrow \mathbb{R}^m,\ g : \mathbb{R}^m \longrightarrow \mathbb{R}^p$ with $c \in \mathbb{R}$.

1. f continuous at $x_0 \implies cf$ continous at x_0.
2. f, g continuous at $x_0 \implies f + g$ continuous at x_0.
3. Let $m = 1$. f, g continuous at $x_0 \implies fg$ continuous at x_0.
4. f continuous at x_0 and $f(x) \neq 0 \forall x \in \mathbb{R}^n \implies 1/f$ continuous at x_0.
5. If $f(x) = \big(f_1(x), f_2(x), ..., f_n(x)\big)$ coordinate-wise, then

$$f \text{ continuous at } x_0 \iff f_1, f_2, ..., f_m \text{ continuous at } x_0 \tag{452}$$

6. f continuous at x_0 and g continuous at $y_0 = f(x_0) \implies g \circ f$ continuous at x_0.

Proof. This is an immediate result of the equivalence of a function being continuous at point a and its limit at point a existing.

Theorem 4.8 (Local Properties of Continuous Functions)

Let $f : E \longrightarrow \mathbb{R}$ be a function that is continuous at the point $a \in E$. Then,

1. f is bounded in some neighborhood $U(a)$.
2. If $f(a) \neq 0$, then in some neighborhood $U(a)$ all the values of the function have the same sign as $f(a)$.
3. If the function $g : U(a) \subset E \longrightarrow \mathbb{R}$ is defined in some neighborhood of a and is continuous at a, then the following functions

$$(f + g)(x) \tag{453}$$
$$(f \cdot g)(x) \tag{454}$$
$$\Big(\frac{f}{g}\Big)(x) \text{ where } g(a) \neq 0 \tag{455}$$

are also defined in $U(a)$ and continuous at a.

4. If the function $g : Y \longrightarrow \mathbb{R}$ is continuous at a point $b \in Y$ and f is such that $f : E \longrightarrow Y$, $f(a) = b$, and f is continuous at a, then the composite function

$$g \circ f : E \longrightarrow \mathbb{R} \tag{456}$$

is defined on E and continuous at a. This is easy to see because given the open neighborhood of $g(b)$, we know for a fact that $U_\delta(a)$ maps completely into $U_\epsilon(b)$, and that $U_\epsilon(b)$ maps completely into $U_\kappa(g(b))$ and so the composition of these mappings must mean that $U_\delta(a)$ maps completely into $U_\kappa(g(b))$.

Example 4.4

An algebraic polynomial

$$P(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \ldots + a_{n-1} x + a_n \tag{457}$$

is a continuous function on $\mathbb{R}$. Since $f(x) = x$ and $f(x) = c$ are continuous functions, by induction on x, we can multiply them together to find that $f(x) = x^n$ is continuous, which implies that ax^n is continuous, which implies that the sums of these functions are also continuous.

4.2.1 Intermediate and Extreme Value Theorem

Unlike local properties, the global property of a function is a property involving the entire domain of definition of the function.

Theorem 4.9 (Compact Sets to Compact)

If $f : X \to Y$ is continuous and $K \subset X$ is compact, then $f(K)$ is compact in Y.

Proof.

Corollary 4.10 (Extreme Value Theorem)

A continuous real-valued function over a compact set attains its maximum and minimum.

Theorem 4.11 (Intermediate Value Theorem)

If a function f is continuous on an open interval and assumes values $f(a) = A, f(b) = B$, then for any number $C \in (A, B)$, there is a point c between a and b such that $f(c) = C$.

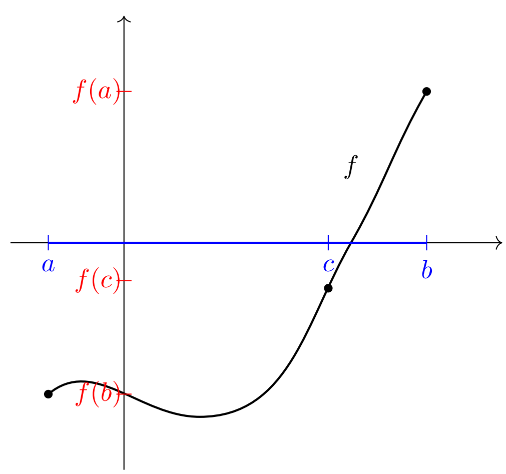


Figure 3: Illustration of a continuous function with a root in the interval $[a, b]$

Proof.

This following proof provides a very simple algorithm for finding the zero of the equation $f(x) = 0$ on an interval whose endpoints has values with opposite signs. Note that the colloquial description of the intermediate value theorem, that is is impossible to pass continuously from positive to negative values without assuming the value 0 along the way), assumes more than they state. That is, this theorem is actually dependent on the domain of definition: that is is a closed interval, or more generally, that it is **connected**.

4.2.2 Inverse Function Theorem

We begin by introducing this intuitive lemma.

Lemma 4.12

A continuous mapping $f : E \longrightarrow \mathbb{R}$ of a closed interval $E = [a, b]$ into $\mathbb{R}$ is injective if and only if the function f is strictly monotonic on $[a, b]$.
Furthermore, every strictly monotonic function $f : X \subset \mathbb{R} \longrightarrow \mathbb{R}$ (for arbitrary X) has an inverse

$$f^{-1} : f(X) \subset \mathbb{R} \longrightarrow \mathbb{R} \tag{458}$$

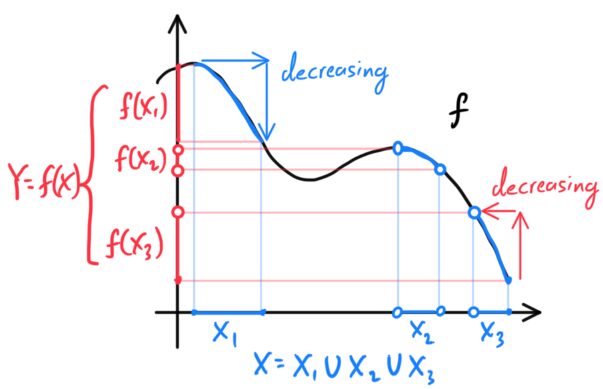
with the same kind of monotonicity on $f(X)$ that f has on X.

Lemma 4.13 (Criterion for Continuity of a Monotonic Function)

A monotonic function $f : E \longrightarrow \mathbb{R}$ defined on a closed interval $E = [a, b]$ is continuous if and only if its set of values $f(E)$ is the closed interval with endpoints $f(a)$ and $f(b)$.
Note that both conditions imply that there are no points of discontinuities in the graph of f.

Theorem 4.14 (Inverse Function Theorem)

A function $f : X \longrightarrow \mathbb{R}$ that is strictly monotonic on a set $X \subset \mathbb{R}$ has an inverse $f^{-1} : Y \longrightarrow \mathbb{R}$ defined on the set $Y = f(X)$ of values of f. The function $f^{-1} : Y \longrightarrow \mathbb{R}$ is monotonic and has the same type of monotonicity on Y that f has on X.



If in addition, X is a closed interval $[a, b]$ and f is continuous on X, then the set $Y = f(X)$ is the closed interval with endpoints $f(a)$ and $f(b)$ and the function $f^{-1} : Y \longrightarrow \mathbb{R}$ is continuous on it.

Example 4.5 (Sin and Arcsin)

The function $f(x) = \sin x$ is increasing and continuous on the closed interval $\big[-\frac{\pi}{2}, \frac{\pi}{2}\big]$. Hence, the restriction to the closed interval $\big[-\frac{\pi}{2}, \frac{\pi}{2}\big]$ has an inverse $x = f^{-1}(y)$, called the **arcsin**, and denoted by $x = \arcsin y$.

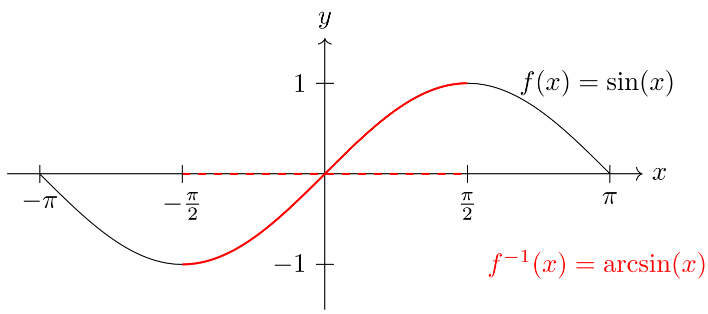


Figure 4: This function is defined on the closed interval $\big[-\sin\big(-\frac{\pi}{2}\big), \sin\big(-\frac{\pi}{2}\big)\big] = [-1, 1]$ and increases continuously from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$.

4.3 Discontinuities

If a function is not continuous at a point, then we call this a *discontinuity*. It turns out that we can nicely categorize them into three categories, based of their oscillation, raning from least to most extreme.

Definition 4.4 (Oscillation)

Given an interval I, the **oscillation** of f on I is defined

$$\operatorname*{osc}_I(f) \coloneqq \sup_I(f) - \inf_I(f) \tag{459}$$

Definition 4.5 (Removable Discontinuity)

A **removable discontinuity** is characterized by the fact that the limit $\lim_{x \to a} f(x) = A$ exists, but $A \neq f(a)$.

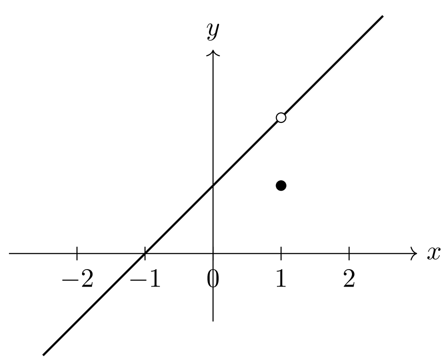


Figure 5: A function with a removable discontinuity at $x = 1$. The function is defined as $f(x) = \frac{x^2-1}{x-1}$ for $x \neq 1$ and $f(1) = 1$. The limit of the function as x approaches 1 is 2 (shown by the open circle), but the function value at $x = 1$ is 1 (shown by the filled circle).

Removable discontinuities are usually benign, and we can fix them by simply replacing the value of the function with its limit.

Definition 4.6 (Jump Discontinuity)

A **jump discontinuity** is characterized by both the left and right-hand limits

$$\lim_{x \to p^-} f(x), \quad \lim_{x \to p^+} f(x) \tag{460}$$

existing, but at least one of them is not equal to the value $f(p)$ that the function assumes at p.

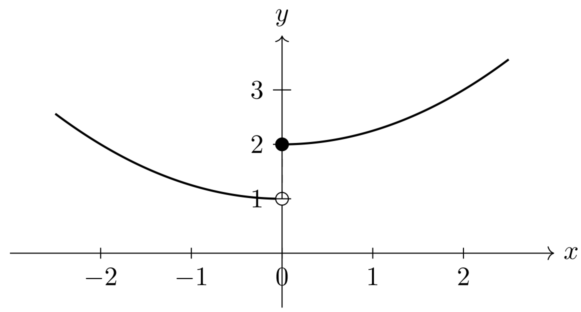


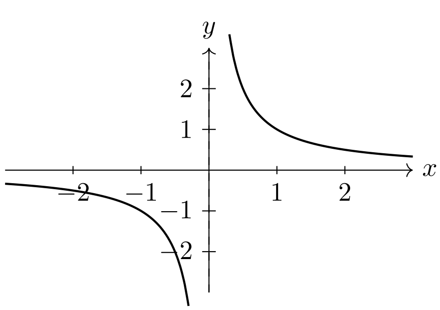
Figure 6: A function with a step discontinuity at $x = 0$. The function is defined as $f(x) = 1 + 0.25x^2$ for $x < 0$ and $f(x) = 2 + 0.25x^2$ for $x \geq 0$. The limit from the left $\lim_{x \to 0^-} f(x) = 1$ is shown by the open circle, while the function value at $x = 0$ is $f(0) = 2$ shown by the filled circle. The dashed line highlights the jump in value.

Definition 4.7 (Essential Discontinuity, of Second Kind)

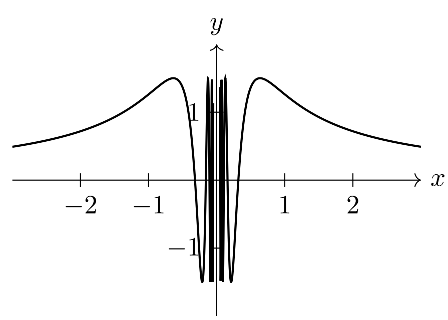
An **essential discontinuity** is characterized by at least one of the two limits

$$\lim_{x \to p^-} f(x), \quad \lim_{x \to p^+} f(x) \tag{461}$$

not existing.



(a) The function $f(x) = \frac{1}{x}$ has an infinite discontinuity at $x = 0$



(b) The function $f(x) = |1.5 \sin\left(\frac{1}{x}\right)|$ has an oscillatory discontinuity at $x = 0$

Figure 7: Examples of discontinuities of the second kind, where the limit does not exist as x approaches the point of discontinuity

Example 4.6 (Dirichlet Function)

The Dirichlet function, defined

$$\mathcal{D}(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases} \tag{462}$$

is discontinuous at every point, and obviously all of its discontinuities are of second kind, since in every interval there are both rational and irrational numbers and therefore there exists no limit at any point $a \in \mathbb{R}$.

More specifically, given any point $a \in \mathbb{R}$, assume that a is rational. We can set $\epsilon = 0.1$-neighborhood around the value 1, but no matter how small we let δ, the interval $(a - \delta, a + \delta)$ will contain both rationals and irrationals, meaning that it will map to $\{0, 1\}$ always, which is not fully contained in $(0.9, 1.1)$.

Here is a slightly more interesting example.

Example 4.7 (Riemann Function)

Let the Riemann function $\mathcal{R}$ be defined

$$\mathcal{R}(x) = \begin{cases} \frac{1}{n}, & \text{if } x = \frac{m}{n} \in \mathbb{Q}, \text{ where } \gcd(m, n) = 1 \\ 0, & \text{if } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases} \tag{463}$$

We first note that for any point $a \in \mathbb{R}$, any bounded neighborhood $U(a)$ of it, and any number $N \in \mathbb{N}$, the neighborhood $U(a)$ contains only a finite number of rational numbers $\geqslant n$, where $n < N$. By shrinking the neighborhood, we can assume that the denominators of all rational numbers in the neighborhood are larger than N, since rationals with larger denominators have smaller gaps between them. Thus, at any point $x \in U(a) \setminus a$, we have

$$\big|\mathcal{R}(x)\big| < \frac{1}{N} \tag{464}$$

and therefore $\lim_{x \to a} \mathcal{R}(x) = 0$ at any point $a \in \mathbb{R} \setminus \mathbb{Q}$. Hence, the Riemann function is continuous at any irrational number.

4.4 Uniform Continuity

Roughly speaking, a function f is uniformly continuous if it is possible to guarantee that $f(x)$ and $f(y)$ be as close to each other as we please by requiring only that x and y be sufficiently close to each other. Intuitively, uniform continuity says that given any two points x, y in the domain where their distance is arbitrarily small (δ apart), we can guarantee that the distance between $f(x), f(y)$ is at maximum some arbitrarily small ϵ.

Definition 4.8 (Uniform Continuity)

A function $f : E \longrightarrow \mathbb{R}$ is **uniformly continuous** on a set $E \subset \mathbb{R}$ if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$\big|f(x_1) - f(x_2)\big| < \epsilon \tag{465}$$

for all points $x_1, x_2 \in E$ such that $|x_1 - x_2| < \delta$.

Example 4.8 (Uniformly Continuous)

The following visual shows the radical function $f(x) = \sqrt{x}$ defined on $\mathbb{R}^+$. We can see that it satisfies uniform continuity because the graph does not escape the top and/or bottom of the $\epsilon \times \delta$ window, no matter where the box is located on the graph. More strictly speaking, no matter what we set the ϵ (how long the box is), uniform continuity says that we can choose a sufficient δ (width of the box) such that the graph does not escape the top/bottom of the window no matter where the window is.

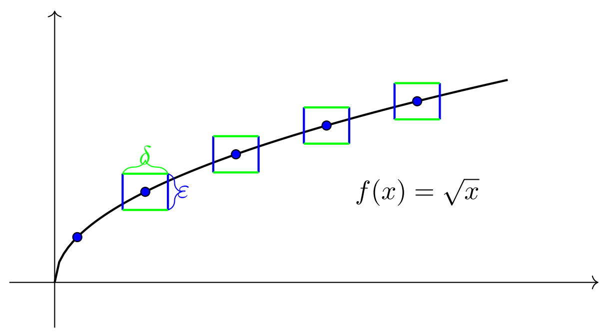


Figure 8: Graph of $f(x) = \sqrt{x}$ with ε-δ rectangles at various points

Example 4.9 (Not Uniformly Continuous)

We can clearly see that the function $f(x) = 1/x$ is not uniformly continuous, since the graph escapes the $\epsilon \times \delta$ window at some point (marked in red). More strictly speaking, given any length ϵ of the window, we cannot create a thin-enough δ box that will contain the graph, since as $x \to 1$, the function becomes unbounded. That is, arbitrarily thin boxes don't help when the slope is arbitrarily steep.

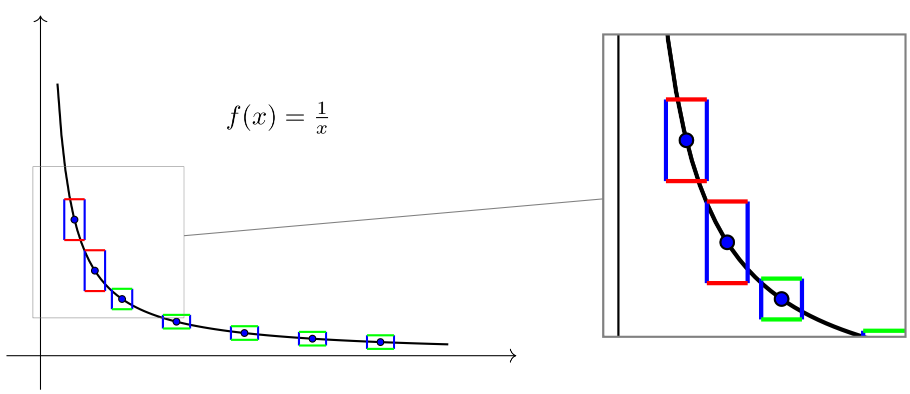
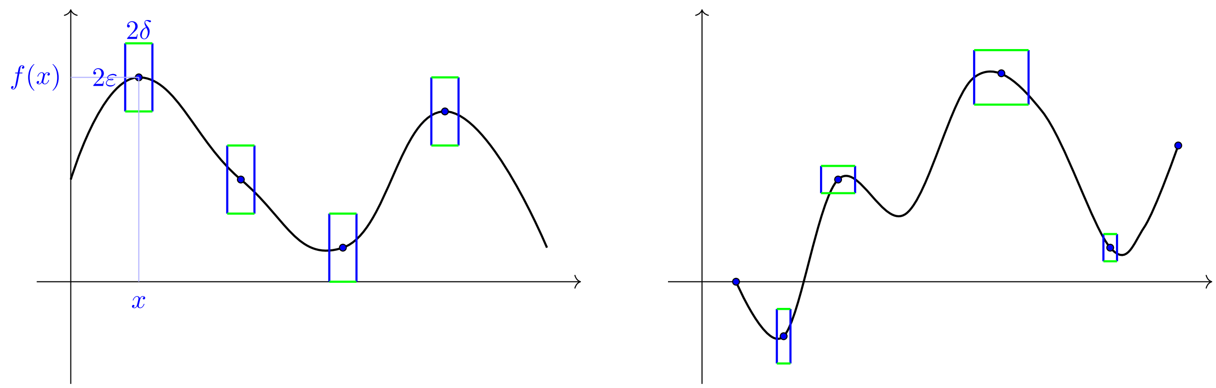


Figure 9: Graph of $f(x) = \frac{1}{x}$ with epsilon-delta boxes and magnified view

To compare uniform continuity with regular continuity, we can adapt this alternate (yet equivalent interpretation): Let there exist function $f : E \longrightarrow \mathbb{R}$. Given any $\epsilon > 0$, we can choose a $\delta > 0$ such that given any point $x \in E$ and $f(x)$, as long as a second point y is δ away from x, then $f(y)$ is ϵ away from $f(x)$. This visualization would lead to there being a $2\epsilon \times 2\delta$ window around point x. Therefore, given a certain $\epsilon > 0$, the way we choose δ is only dependent on ϵ, and so it must be a function of ϵ:

$$\delta = \delta(\epsilon) \tag{466}$$

However, in continuity, there just has to exist *some* δ-neighborhood of x such that its image is contained in the ϵ-neighborhood of $f(x)$.



(a) Uniform continuity means that the box above does not change dimensions no matter where the point is (hence, the name uniform).

(b) In continuity, there are no restrictions on the dimensions of this box. It just has to exist for every point, through a function of ϵ.

Figure 10: Uniform continuity vs continuity.

With this intuition, it is easy to see the result below.

Lemma 4.15 (Uniform Continuity Implies Continuity)

If f is uniformly continuous on the set E, it is continuous at each point of that set. However, the converse is not generally true.

Example 4.10

Let $f : \mathbb{R} \longrightarrow \mathbb{R},\ f(x) = 3x + 7$. Then f is uniformly continuous. Choose $\epsilon > 0$. Let $\delta = \epsilon/3$. Choose $x, y \in \mathbb{R}$ and assume $|x - y| < \delta$. Then,

$$|f(x) - f(y)| = |3x + 7 - 3y - 7| = 3|x - y| < 3\delta = \epsilon \tag{467}$$

Example 4.11

Let $f : (0, 4) \subset \mathbb{R} \longrightarrow \mathbb{R},\ f(x) = x^2$. Then f is uniformly continuous on $(0, 4)$. Choose $\epsilon > 0$. Let $\delta = \epsilon/8$. Choose $x, y \in (0, 4)$ and assume $|x - y| < \delta$. Then,

$$|f(x) - f(y)| = |x^2 - y^2| = (x + y)|x - y| < (4 + 4)|x - y| = 8\delta = \epsilon \tag{468}$$

A natural question one might ask is: under what assumptions is the converse true?

Theorem 4.16 (Cantor's Theorem on Uniform Continuity)

A function that is continuous on a compact set is uniformly continuous on that set.

Proof. Here is a proof I derived myself. The general idea is that with continuity, we we can control the δ with an ϵ *plus* some point z. To decouple this dependence on z, we can restrict our scope to each z that are the center of the balls in the finite cover of x, and then taking the minimum (most restrictive) over them.

We wish to show that $\forall \epsilon > 0$, $\exists \delta > 0$ s.t.

$$d(x, x') < \delta \implies d\big(f(x), f(x')\big) < \epsilon \quad \forall x, x' \in X \tag{469}$$

We begin by constructing an open cover. Fix $\epsilon > 0$. $\forall z \in X$, $\exists \delta_z > 0$ s.t.

$$d(z, x) < \delta_z \implies d(f(z), f(x)) < \epsilon \tag{470}$$

This gives us an open cover $\{B(\frac{\delta_z}{2}, z)\}_{z \in X}$ of X, and by compactness, let's take a finite cover $\{B(\frac{\delta_{z_i}}{2}, z_i)\}_{i=1}^n$.
Now set $\delta = \min_i \{\delta_{z_i}/2\}$. We hope that when we choose $x, y \in X$ s.t. $d(x, y) < \delta$, we can trivially assume that $x \in B_i$ for some i. By setting the radius to be half, we can also assume that y is also in the same ball as x since

$$d(z_i, y) \leq d(z_i, x) + d(x, y) \leq \frac{\delta_{z_i}}{2} + \min_i \{\delta_{z_i}/2\} \leq \delta_{z_i} \tag{471}$$

This pretty much means that we can use the continuity properties locally within B_i, and so we can bound

$$d(f(x), f(y)) \leq d(f(x), f(z_i)) + d(f(z_i), f(y)) \leq 2\epsilon \tag{472}$$

Proof. Here is another proof recommended by Prof. Kiselev, who mentioned that a sequential proof is what he would go with intuitively. Let's prove by contradiction.[a] Assume f is not uniformly continuous. Then $\exists \epsilon_0 > 0$ s.t. $\forall \delta = \frac{1}{n} > 0$, there exists $x_n, y_n \in X$ s.t.

$$d(x, y) < \frac{1}{n} \text{ but } d(f(x_n), f(y_n)) \geq \epsilon_0 \tag{473}$$

Since X is compact, we can take a convergent subsequence (x_{n_k}) of (x_n), which converges to say z. We can take the same subindex to define the corresponding (y_{n_k}). We first show that these two subsequences converge to the same value. By triangle inequality, we see that

$$d(z, y_{n_k}) \leq d(z, x_{n_k}) + d(x_{n_k}, y_{n_k}) \tag{474}$$

As we take the limit as $k \to \infty$, we see that $d(z, x_{n_k}) \to 0$ by construction of our subsequence, and $d(x_{n_k}, y_{n_k}) < \frac{1}{n_k}$ be construction. Therefore, $d(z, y_{n_k}) \to 0$ and we establish convergence. By continuity of f, we can see that

$$\lim_{k \to +\infty} f(x_k) = f(z) = \lim_{k \to +\infty} f(y_k) \tag{475}$$

However, we have picked x_n, y_n—and thus x_{n_k}, y_{n_k}—such that $d(f(x_{n_k}), f(y_{n_k})) \geq \epsilon_0$, which is a contradiction to the equality above.

[a] Thanks to Kiselev. This is also Exercise 4.10 in Baby Rudin.

Example 4.12 (Wrong Proof Attempt to Reflect on My Actions)

Here's a wrong proof I had in my first attempt. Fix $\epsilon > 0$, and consider any two points $x, x' \in X$. Take $r = d(x, x') + 1$, and take the finite subcover of the open cover of all r-balls of X, giving us

$$\{B_r(z_i)\}_{i=1}^n \tag{476}$$

Let's focus on one ball, which we can assume contains both x, x' since we can just add it. Since we know that f is continuous on z_i, we know that $\exists \delta_i' > 0$ s.t.

$$d(x, z_i) < \delta_i \implies d(f(x, z_i)) < \frac{\epsilon}{2} \tag{477}$$

$$d(x', z_i) < \delta_i \implies d(f(x', z_i)) < \frac{\epsilon}{2} \tag{478}$$

Therefore, if we unfix x, x', we can always find a ball $B_r(z_i)$ containing both of them and find a $\delta_i = 2\delta_i'$ such that

$$d\big(f(x), f(x')\big) \leq d\big(f(x), f(z_i)\big) + d\big(f(z_i), f(x')\big) < \epsilon \tag{479}$$

Theorem 4.17 (Uniformly Continuous Function is Linearly Bounded)

If $f : \mathbb{R} \to \mathbb{R}$ is uniformly continuous, then there exists constants, $a, b \in \mathbb{R}$ such that $|f(x)| \leq a + b|x|$ for all $x \in \mathbb{R}$.

Proof. Since f is uniformly continuous, for every $\epsilon > 0$ there exists a $\delta > 0$ s.t.

$$|x - y| < \delta \implies |f(x) - f(y)| < \epsilon \tag{480}$$

for all $x, y \in \mathbb{R}$. Then, setting $y = 0$ and $\epsilon = 1$, we have some $\delta > 0$ s.t.

$$|x| < \delta \implies |f(0) - f(x)| < 1 \tag{481}$$

Now for any $k \in \mathbb{N}$, take $|x| < k\delta$. Then we can construct a sequence $(x_i = i\frac{|x|}{k})_{i=0}^k$, where $x_0 = 0$ and $x_k = |x|$, and from the triangle inequality followed by uniform convergence, we have

$$|x_i - x_{i+1}| = \frac{|x|}{k} < \frac{k\delta}{k} = \delta \implies |f(0) - f(x)| \leq \sum_{i=0}^{k-1} |f(x_i) - f(x_{i+1})| < \sum_{i=0}^{k-1} 1 = k \tag{482}$$

and so we come to the result

$$|x| < k\delta \implies |f(0) - f(x)| < k \tag{483}$$

Now set $m(x) = \min\{k \in \mathbb{N} \mid |x| < k\delta\}$. It must be nonempty by the Archimedean property, so it's lower bounded by 1.

$$|x| < \delta m(x) \implies \frac{|x|}{\delta} < m(x) \implies m(x) \leq \frac{|x|}{\delta} + 1 \tag{484}$$

where the final implication comes from $m(x)$ being minimum. Therefore,

$$|f(x)| \leq |f(0)| + |f(x) - f(0)| \leq |f(0)| + m(x) \leq |f(0)| + \frac{|x|}{\delta} + 1 \tag{485}$$

and we have found such $a = |f(0)| + 1, b = \frac{1}{\delta}$.

4.5 Holder Continuity

Definition 4.9 (Holder Continuity)

Given $f : E \subset \mathbb{R} \to \mathbb{R}$, f is α-**Holder continuous** if there exists a $C > 0$ such that for all $x, y \in E$,

$$|f(x) - f(y)| \leq M|x - y|^\alpha \tag{486}$$

4.6 Lipshitz Continuity

In both examples, the function satisfied an inequality of form

$$|f(x_1) - f(x_2)| \leq M|x_1 - x_2| \tag{487}$$

this is called the *Lipshitz inequality.* Lipshitz continuity is a strong form of uniform continuity for functions. Intuitively, a Lipshitz continuous function is limited in how fast it can change (by the Lipshitz constant).

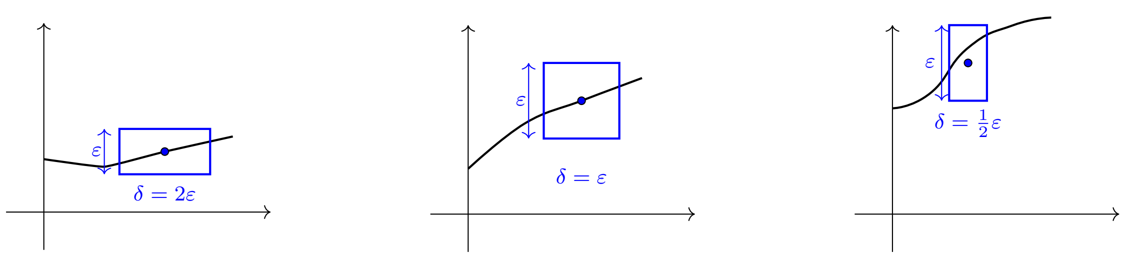
Definition 4.10 (Lipshitz Continuous Function)

Given $f : E \subset \mathbb{R} \longrightarrow \mathbb{R}$, f is **Lipshitz continuous** if there exists a $M > 0$—called the *Lipschitz constant*—such that for all $x, y \in E$,

$$|f(x) - f(y)| \leq M|x - y| \tag{488}$$

Note that Lipshitz continuity pops up as a very natural extension of uniform continuity. The inequality above just means that given an ϵ, we can choose a δ such that a linear multiple of δ is always greater than ϵ. This means that Lipshitz continuity is just uniform continuity such that the δ function is linear:

$$\delta = \delta(\epsilon) = \frac{1}{M}\epsilon \tag{489}$$



(a) $M = \frac{1}{2}$ (b) $M = 1$ (c) $M = 2$

Figure 11: Relationship between slope M and the ratio of δ to ε

Definition 4.11 (Bi-Lipshitz Continuity)

A function $f : E \subset \mathbb{R}$ is **Bi-Lipshitz continuous** if there exists constant $M \geq 1$ such that for all real $x, y \in E$,

$$\frac{1}{M}|x - y| \leq |f(x) - f(y)| \leq M|x - y| \tag{490}$$

It immediately follows that for $x \neq y$, $|f(x) - f(y)|$ cannot equal 0, which means that a bilipshitz map is injective. A bilipshitz map is really just Lipshitz map with its inverse also being Lipshitz.

Theorem 4.18

A bilipshitz map f is a homeomorphism onto its image.

4.7 Asymptotic Analysis

Definition 4.12 (Little-O Notation)

The function $f : E \longrightarrow \mathbb{R}$ is said to be **infinitesimal compared with the function** $g : E \longrightarrow \mathbb{R}$ as $x \to a$, written (by abuse of notation) $f = o(g)$ as $x \to a$, if

$$\lim_{x \to a} \frac{f(x)}{g(x)} = 1 \tag{491}$$

or in other words, if f/g is an infinitesimal function as $x \to a$. Therefore, $f = o(1)$ as $x \to a$ means that f is infinitesimal as $x \to a$. [a]

[a]Note that writing $f = o(g)$ is again, an abuse of notation. $f = o(g)$ is really a shorthand way of writing that f is in the class of functions that is infinitesimal compared with the function g.

Intuitively, $f = o(g)$ means that the ratio between $f(x)$ and $g(x)$ will tend to infinity as $x \to a$ (this does not mean that f will be infinitely greater than g, however!).

Example 4.13 (Linear vs Quadratic)

For example, looking at the two functions $f(x) = x^2$ and $g(x) = x$, we have

1. $x^2 = o(x)$ as $x \to 0$ (since $\frac{x^2}{x} = x$ is infinitesimal as $x \to 0$)
2. $x = o(x^2)$ as $x \to \infty$ (since $\frac{x}{x^2} = \frac{1}{x}$ is infinitesimal as $x \to \infty$)

We can visualize $g/f(x)$ tending to infinity within a neighborhood of 0 and $f/g(x)$ tending to infinity within a neighborhood of ∞.

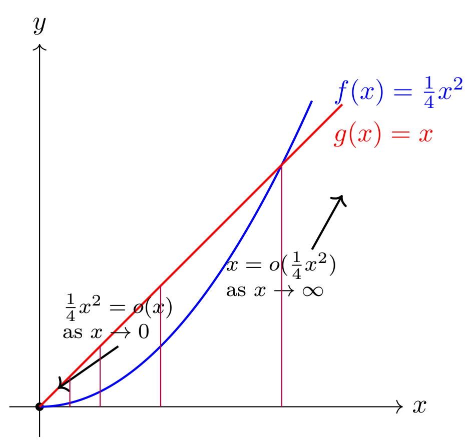


Figure 12

Definition 4.13 (Orders of Infinitesimals, Infinities)

If $f = o(g)$ and g is infinitesimal as $x \to a$, then f is an **infinitesimal of higher order than** g **as** $x \to a$. Furthermore, if f and g are infinite functions as $x \to a$ and $f = o(g)$ as $x \to a$, then g is a **higher order infinity than** f **as** $x \to a$.

Definition 4.14 (Big-O Notation)

By abuse of notation, $f = O(g)$ as $x \to a$ means that

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \infty \tag{492}$$

or in other words, f/g is ultimately bounded as $x \to a$. In particular, $f = O(1)$ as $x \to a$ means that f is bounded within a certain neighborhood $U(a)$ of a.

Definition 4.15 (Functions of Same Order)

The functions f and g are of the same over as $x \to a$, written

$$f \asymp g \text{ as } x \to a \tag{493}$$

if $f = O(g)$ and $g = O(f)$ as $x \to a$. Intuitively, this means that the ratio between f and g within some deleted neighborhood of a is finite.
Note that the condition that f and g be of the same order as $x \to a$ is (by definition of ultimately bounded functions) equivalent to the condition that there exist $c_1, c_2 > 0$ and an open neighborhood $U(a)$ such that the relations

$$c_1|g(x)| \leq |f(x)| \leq c_2|g(x)| \tag{494}$$

is true for $x \in U(a)$.

Definition 4.16 (Asymptotic Equivalence of Functions)

For functions f and g, if

$$\lim_{x \to a} \frac{f(x)}{g(x)} = 1 \tag{495}$$

we say that f **behaves asymptotically like** g **as** $x \to a$, or that f **is equivalent to** g **as** $x \to a$, written

$$f \sim g \text{ as } x \to a \tag{496}$$

Moreover, $\sim$ is an equivalence relation, which means that

1. $f \sim f$ as $x \to a$
2. $f \sim g$ as $x \to a \implies g \sim f$ as $x \to a$
3. $f \sim g$ and $g \sim h$ as $x \to a \implies f \sim h$ as $x \to a$

We list a few examples in order to develop some sort of visual intuition for when two functions are asymptotically equivalent.

Example 4.14 (Both Converges at Finite Value to Nonzero Finite Value)

If $f(a) = g(a) \neq 0$, then $f \sim g$ trivially since the ratio of f and g converges to 1 within a neighborhood of a.

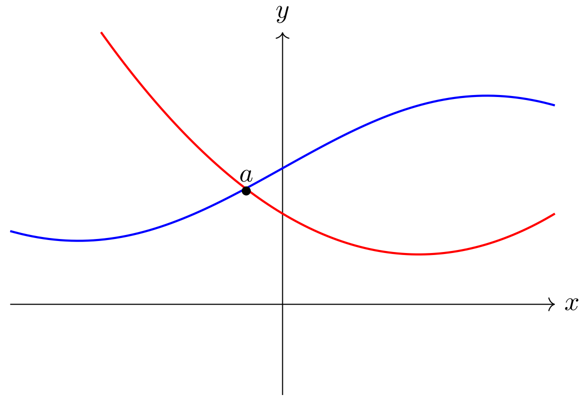
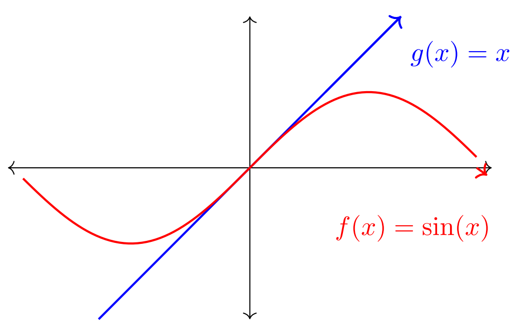


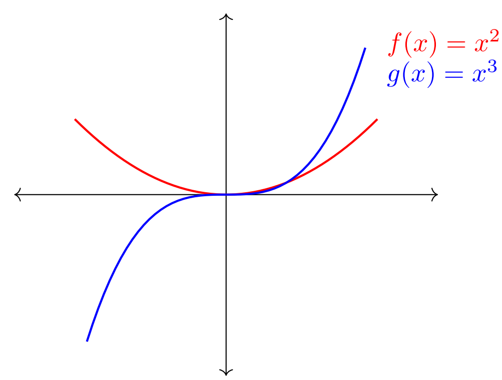
Figure 13

Example 4.15 (Both Converges at Finite Value to 0)

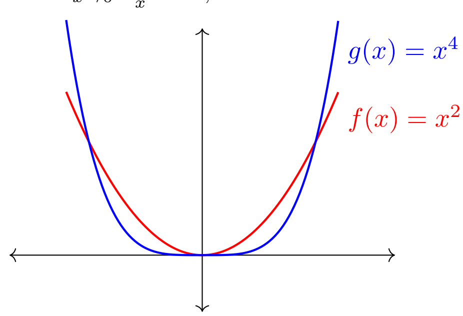
When $f(a) = g(a) = 0$, it may be f may be equivalent to g or one function may be infinitesimally smaller than the other.



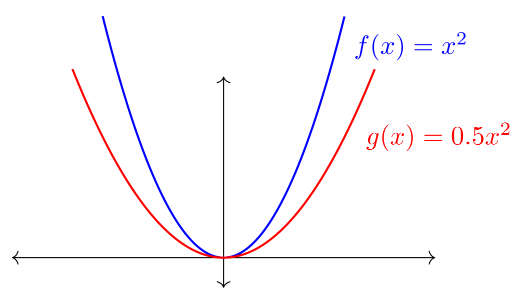
(a) When $f(x) = \sin x$ and $g(x) = x$, then $f \sim g$ since we see that $\lim_{x \to 0} \frac{\sin x}{x} = 1$, and so $\sin x \sim x$ as $x \to 1$.



(b) When $f(x) = x^2$ and $g(x) = x^3$, then $\lim_{x \to 0} \frac{x^3}{x^2} = 0$, and so $x^3 \not\sim x^2$. In fact, $x^3 = o(x^2)$.



(c) When $f(x) = x^2$ and $g(x) = x^4$, then $\lim_{x \to 0} \frac{x^4}{x^2} = 0$, and so $x^4 \not\sim x^2$. In fact, $x^4 = o(x^2)$.

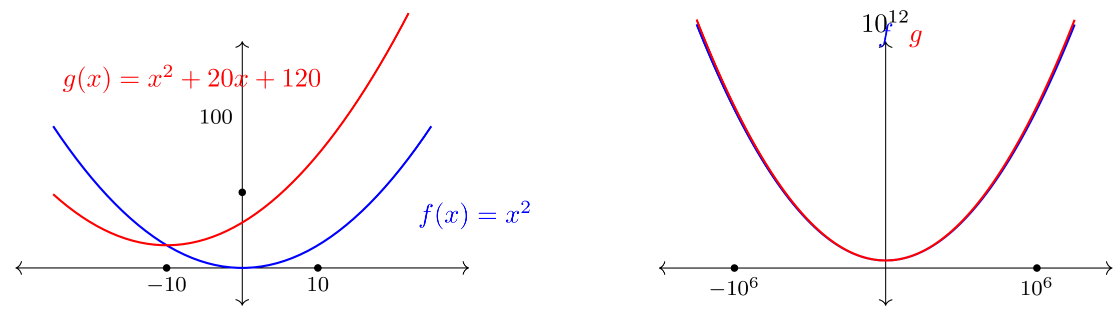


(d) When $f(x), g(x) = x^2, 0.5x^2$, then $\lim_{x \to 0} \frac{0.5x^2}{x^2} = \frac{1}{2}$. So $0.5x^2 \not\sim x^2$.

Figure 14: Examples of different scenarios.

Example 4.16 (Analyzing at Infinity)

When analyzing the behavior of functions as $x \to \infty$, we can picture the two graphs of f and g on the plane and "zoom out" to see if the ratio of the values converge to 1. This would mean that as $x \to \infty$, we should see the graphs overlapping more and more.



(a) Comparison of $f(x) = x^2$ and $g(x) = x^2 + 10x + 100$ (b) Asymptotic behavior with $g/f \approx 1$

Figure 15: taking $f(x) = x^2$ and $g(x) = x^2 + 10x + 100$, we can see that the discrepancy is high around a neighborhood of $x = 0$. But as $x \to +\infty$, we get $\lim_{x\to+\infty} \frac{x^2+10x+100}{x^2} = 1$, and so the graphs look like they are overlapping. Notice that even though the absolute difference $|(x^2 + 10x + 100) - x^2| = |10x + 100|$ tends to infinity, this difference increases infinitesimally compared to f and g.

From this, we can see that if $f \sim g$ as $x \to a$, then their difference

$$f - g = o(g) = o(f) \tag{497}$$

That is, $(f - g)(x)$ is infinitesimal compared to g or f (doesn't matter which one we compare it to). This leads to our next section, where we formalize this concept with absolute and relative errors.

It is useful to note that since the relation $\lim_{x\to a} \gamma(x) = 1$ is equivalent to

$$\gamma(x) = 1 + \alpha(x), \text{ where } \lim_{x\to a} \alpha(x) = 0 \tag{498}$$

the relation $f \sim g$ as $x \to a$ is equivalent to saying that

$$\frac{f(x)}{g(x)} = \gamma(x), \text{ where } \lim_{x\to a} \gamma(x) = 1 \tag{499}$$

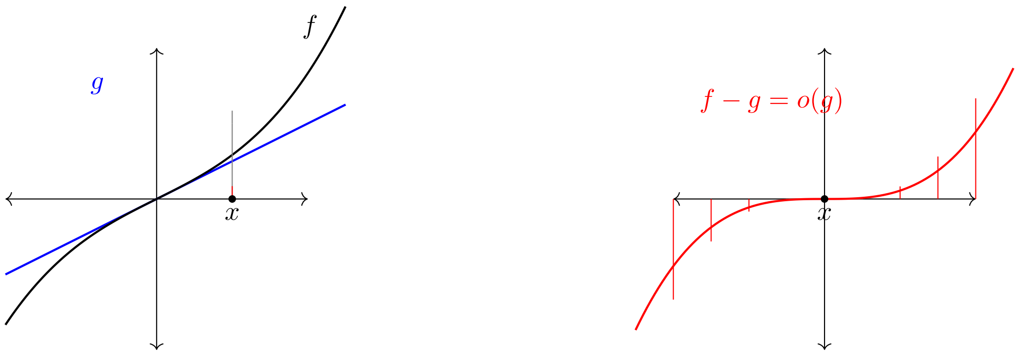
which implies

$$f(x) = g(x) + \alpha(x)g(x) = g(x) + o(g(x)) \text{ as } x \to a \tag{500}$$

or, symmetrically,

$$g(x) = f(x) + \alpha(x)f(x) = f(x) + o(f(x)) \text{ as } x \to a \tag{501}$$

This means that f can be exactly represented by another function g, plus another (error) function $o(g(x))$ that is infinitesimal compared to g.



(a) Functions f, g, and their difference

(b) Behavior of $f - g$ (little-o of g)

Figure 16: Visualization of asymptotic behavior where $f - g = o(g)$

Note that it is not a sufficient condition that the error function be infinitesimal! The error function $f - g$ must be infinitesimal *compared to* g! This tells us that not only does the error function decrease infinitesimally, but also is infinitesimal compared to the approximation function we already have, which is in general a much stronger claim. This representation of certain types functions will provide the foundation for differential calculus when we talk about "good" approximations for a function.

Definition 4.17 (Relative Error)

Since $f \sim g$ as $x \to a$ means that

$$f(x) = g(x) + \alpha(x)g(x) = g(x) + o(g(x)) \tag{502}$$

we can define the **relative error** of g as an approximation of f to be

$$|\alpha(x)| = \left| \frac{f(x) - g(x)}{g(x)} \right| \tag{503}$$

Clearly, since $f \sim g$, the relative error must be infinitesimal as $x \to a$.

We use the following lemma to check whether two functions are asymptotically equivalent.

Lemma 4.19

$f \sim g$ as $x \to a$ if and only if the relative error of g is infinitesimal as $x \to a$.

Example 4.17

We claim that

$$x^2 + x = \left(1 + \frac{1}{x}\right)x^2 \sim x^2 \text{ as } x \to \infty \tag{504}$$

We see that the absolute error of this approximation $|(x^2 + x) - x^2| = |x|$ tends to infinity, but the relative error $\frac{|x|}{x^2} = \frac{1}{|x|} \to 0$ as $x \to \infty$.

Theorem 4.20 (Prime Number Theorem)

Let $\pi(x)$ be the number of prime numbers strictly less than x. Then $\pi \sim \frac{x}{\ln x}$ as $x \to +\infty$, or more precisely,

$$\pi(x) = \frac{x}{\ln x} + o\left(\frac{x}{\ln x}\right) \text{ as } x \to +\infty \tag{505}$$

Example 4.18

It is a fact that $\lim_{x \to 0} \frac{sinx}{x} = 1$, so we have $\sin x \sim x$ as $x \to 0$. So,

$$\sin x = x + o(x) \text{ as } x \to 0 \tag{506}$$

The following theorem proves useful when computing limits.

Theorem 4.21

If $f \sim \tilde{f}$ as $x \to a$, then

$$\lim_{x \to a} f(x)g(x) = \lim_{x \to a} \tilde{f}(x)g(x) \tag{507}$$

provided one of these limits exist.

Theorem 4.22 (Properties of $o(g)$ and $O(g)$ Functions)

For $x \to a$,

1. $o(f) + o(f) = o(f)$
2. $o(f)$ is also $O(f)$
3. $o(f) + O(f) = O(f)$
4. $O(f) + O(f) = O(f)$
5. If $g(x) \neq 0$, then

$$\frac{o(f(x))}{g(x)} = o\left(\frac{f(x)}{g(x)}\right), \text{ and } \frac{O(f(x))}{g(x)} = O\left(\frac{f(x)}{g(x)}\right) \tag{508}$$

4.8 Exercises

Exercise 4.1 (Rudin 4.1)

Suppose f is a real function defined on R^1 which satisfies

$$\lim_{h \to 0} [f(x+h) - f(x-h)] = 0 \tag{509}$$

for every $x \in R^1$. Does this imply that f is continuous?

Solution. No, and the point is the realize that under this assumption, the value of $f(x_0)$ would be irrelevant, and so you can change this value to be any value you want—creating a point discontinuity—without affecting the limit at all. For example, consider the function

$$f(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{else} \end{cases} \tag{510}$$

In this case, $f(h) = f(-h)$, so the limit is trivially 0. But f is not continuous at $x = 0$.

At first glance, this may seem to be true since we are considering “sort of” an open neighborhood. Let’s attempt to prove this. We wish to show that for every $\epsilon > 0$, $\exists \delta > 0$ s.t.

$$0 < |y - x| < \delta \implies |f(y) - f(x)| < \epsilon \tag{511}$$

I tried to prove this by fixing $\epsilon > 0$ and $x \in \mathbb{R}$. By assumption, $\exists \delta > 0$ s.t. $|h| < \delta \implies |f(x+h) - f(x-h)| < \epsilon$. But in other words, this means that

$$|\underbrace{(h + x)}_{=y} - x| < \delta \implies |f(x+h) - f(x) + f(x) - f(x-h)| < \epsilon \tag{512}$$

But the triangle inequality is an upper bound on the RHS, which doesn’t lead us anywhere.

Exercise 4.2 (Rudin 4.2)

If f is a continuous mapping of a metric space X into a metric space Y, prove that

$$f(\overline{E}) \subset \overline{f(E)} \tag{513}$$

for every set $E \subset X$. ($\overline{E}$ denotes the closure of E.) Show, by an example, that $f(\overline{E})$ can be a proper subset of $\overline{f(E)}$.

Solution.

Exercise 4.3 (Rudin 4.3)

Let f be a continuous real function on a metric space X. Let $Z(f)$ (the *zero set* of f) be the set of all $p \in X$ at which $f(p) = 0$. Prove that $Z(f)$ is closed.

Solution.

Exercise 4.4 (Rudin 4.4)

Let f and g be continuous mappings of a metric space X into a metric space Y, and let E be a dense subset of X. Prove that $f(E)$ is dense in $f(X)$. If $g(p) = f(p)$ for all $p \in E$, prove that $g(p) = f(p)$ for all $p \in X$. (In other words, a continuous mapping is determined by its values on a dense subset of its domain.)

Solution.

Exercise 4.5 (Rudin 4.5)

If f is a real continuous function defined on a closed set $E \subset R^1$, prove that there exist continuous real functions g on R^1 such that $g(x) = f(x)$ for all $x \in E$. (Such functions g are called *continuous extensions* of f from E to R^1.) Show that the result becomes false if the word "closed" is omitted. Extend the result to vector-valued functions. *Hint:* Let the graph of g be a straight line on each of the segments which constitute the complement of E (compare Exercise 29, Chap. 2). The result remains true if R^1 is replaced by any metric space, but the proof is not so simple.

Solution.

Exercise 4.6 (Rudin 4.6)

If f is defined on E, the *graph* of f is the set of points $(x, f(x))$, for $x \in E$. In particular, if E is a set of real numbers, and f is real-valued, the graph of f is a subset of the plane.
Suppose E is compact, and prove that f is continuous on E if and only if its graph is compact.

Solution.

Exercise 4.7 (Rudin 4.7)

If $E \subset X$ and if f is a function defined on X, the *restriction* of f to E is the function g whose domain of definition is E, such that $g(p) = f(p)$ for $p \in E$. Define f and g on R^2 by: $f(0,0) = g(0,0) = 0$, $f(x,y) = xy^2/(x^2+y^4)$, $g(x,y) = xy^2/(x^2+y^6)$ if $(x,y) \neq (0,0)$. Prove that f is bounded on R^2, that g is unbounded in every neighborhood of $(0,0)$, and that f is not continuous at $(0,0)$; nevertheless, the restrictions of both f and g to every straight line in R^2 are continuous!

Solution.

Exercise 4.8 (Rudin 4.8)

Let f be a real uniformly continuous function on the bounded set E in R^1. Prove that f is bounded on E.
Show that the conclusion is false if boundedness of E is omitted from the hypothesis.

Solution.

Exercise 4.9 (Rudin 4.9)

Show that the requirement in the definition of uniform continuity can be rephrased as follows, in terms of diameters of sets: To every $\epsilon > 0$ there exists a $\delta > 0$ such that diam $f(E) < \epsilon$ for all $E \subset X$ with diam $E < \delta$.

Solution.

Exercise 4.10 (Rudin 4.10)

Complete the details of the following alternative proof of Theorem 4.19: If f is not uniformly continuous, then for some $\epsilon > 0$ there are sequences $\{p_n\}, \{q_n\}$ in X such that $d_X(p_n, q_n) \to 0$ but $d_Y(f(p_n), f(q_n)) > \epsilon$. Use Theorem 2.37 to obtain a contradiction.

Solution.

Exercise 4.11 (Rudin 4.11)

Suppose f is a uniformly continuous mapping of a metric space X into a metric space Y and prove that $\{f(x_n)\}$ is a Cauchy sequence in Y for every Cauchy sequence $\{x_n\}$ in X. Use this result to give an alternative proof of the theorem stated in Exercise 13.

Solution.

Exercise 4.12 (Rudin 4.12)

A uniformly continuous function of a uniformly continuous function is uniformly continuous.
State this more precisely and prove it.

Solution.

Exercise 4.13 (Rudin 4.13)

Let E be a dense subset of a metric space X, and let f be a uniformly continuous real function defined on E. Prove that f has a continuous extension from E to X (see Exercise 5 for terminology). (Uniqueness follows from Exercise 4.) *Hint:* For each $p \in X$ and each positive integer n, let $V_n(p)$ be the set of all $q \in E$ with $d(p, q) < 1/n$. Use Exercise 9 to show that the intersection of the closures of the sets $f(V_1(p)), f(V_2(p)), \ldots$, consists of a single point, say $g(p)$, of R^1. Prove that the function g so defined on X is the desired extension of f.
Could the range space R^1 be replaced by R^k? By any compact metric space? By any complete metric space? By any metric space?

Solution.

Exercise 4.14 (Rudin 4.14)

Let $I = [0, 1]$ be the closed unit interval. Suppose f is a continuous mapping of I into I. Prove that $f(x) = x$ for at least one $x \in I$.

Solution.

Exercise 4.15 (Rudin 4.15)

Call a mapping of X into Y *open* if $f(V)$ is an open set in Y whenever V is an open set in X.
Prove that every continuous open mapping of R^1 into R^1 is monotonic.

Solution.

Exercise 4.16 (Rudin 4.16)

Let $[x]$ denote the largest integer contained in x, that is, $[x]$ is the integer such that $x - 1 < [x] \leq x$; and let $(x) = x - [x]$ denote the fractional part of x. What discontinuities do the functions $[x]$ and (x) have?

Solution.

Exercise 4.17 (Rudin 4.17)

Let f be a real function defined on (a, b). Prove that the set of points at which f has a simple discontinuity is at most countable. *Hint:* Let E be the set on which $f(x-) < f(x+)$. With each point x of E, associate a triple (p, q, r) of rational numbers such that

(a) $f(x-) < p < f(x+)$,
(b) $a < q < t < x$ implies $f(t) < p$,
(c) $x < t < r < b$ implies $f(t) > p$.

The set of all such triples is countable. Show that each triple is associated with at most one point of E. Deal similarly with the other possible types of simple discontinuities.

Solution.

Exercise 4.18 (Rudin 4.18)

Every rational x can be written in the form $x = m/n$, where $n > 0$, and m and n are integers without any common divisors. When $x = 0$, we take $n = 1$. Consider the function f defined on R^1 by

$$f(x) = \begin{cases} 0 & (x \text{ irrational}), \\ \frac{1}{n} & (x = \frac{m}{n}). \end{cases} \tag{514}$$

Prove that f is continuous at every irrational point, and that f has a simple discontinuity at every rational point.

Solution.

Exercise 4.19 (Rudin 4.19)

Suppose f is a real function with domain R^1 which has the intermediate value property: If $f(a) < c < f(b)$, then $f(x) = c$ for some x between a and b.
Suppose also, for every rational r, that the set of all x with $f(x) = r$ is closed. Prove that f is continuous.
Hint: If $x_n \to x_0$ but $f(x_n) > r > f(x_0)$ for some r and all n, then $f(t_n) = r$ for some t_n between x_0 and x_n; thus $t_n \to x_0$. Find a contradiction. (N. J. Fine, *Amer. Math. Monthly*, vol. 73, 1966, p. 782.)

Solution.

Exercise 4.20 (Rudin 4.20)

If E is a nonempty subset of a metric space X, define the distance from $x \in X$ to E by

$$\rho_E(x) = \inf_{z \in E} d(x, z). \tag{515}$$

(a) Prove that $\rho_E(x) = 0$ if and only if $x \in \overline{E}$.
(b) Prove that ρ_E is a uniformly continuous function on X, by showing that

$$|\rho_E(x) - \rho_E(y)| \leq d(x, y) \tag{516}$$

for all $x \in X, y \in X$.
Hint: $\rho_E(x) \leq d(x,z) \leq d(x,y) + d(y,z)$, so that $\rho_E(x) \leq d(x,y) + \rho_E(y)$.

Solution.

Exercise 4.21 (Rudin 4.21)

Suppose K and F are disjoint sets in a metric space X, K is compact, F is closed. Prove that there exists $\delta > 0$ such that $d(p,q) > \delta$ if $p \in K, q \in F$. *Hint:* ρ_F is a continuous positive function on K. Show that the conclusion may fail for two disjoint closed sets if neither is compact.

Solution.

Exercise 4.22 (Rudin 4.22)

Let A and B be disjoint nonempty closed sets in a metric space X, and define

$$f(p) = \frac{\rho_A(p)}{\rho_A(p) + \rho_B(p)} \quad (p \in X). \tag{517}$$

Show that f is a continuous function on X whose range lies in $[0,1]$, that $f(p) = 0$ precisely on A and $f(p) = 1$ precisely on B. This establishes a converse of Exercise 3: Every closed set $A \subset X$ is $Z(f)$ for some continuous real f on X. Setting

$$V = f^{-1}([0,1/2)), \quad W = f^{-1}((1/2,1]), \tag{518}$$

show that V and W are open and disjoint, and that $A \subset V, B \subset W$. (Thus pairs of disjoint closed sets in a metric space can be covered by pairs of disjoint open sets. This property of metric spaces is called *normality*.)

Solution.

Exercise 4.23 (Rudin 4.23)

A real-valued function f defined in (a,b) is said to be *convex* if

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y) \tag{519}$$

whenever $a < x < b, a < y < b, 0 < \lambda < 1$. Prove that every convex function is continuous. Prove that every increasing convex function of a convex function is convex. (For example, if f is convex, so is e^f.) If f is convex in (a,b) and if $a < s < t < u < b$, show that

$$\frac{f(t)-f(s)}{t-s} \leq \frac{f(u)-f(s)}{u-s} \leq \frac{f(u)-f(t)}{u-t}. \tag{520}$$

Solution.

Exercise 4.24 (Rudin 4.24)

Assume that f is a continuous real function defined in (a, b) such that

$$f\left(\frac{x+y}{2}\right) \leq \frac{f(x)+f(y)}{2} \tag{521}$$

for all $x, y \in (a, b)$. Prove that f is convex.

Solution.

Exercise 4.25 (Rudin 4.25)

If $A \subset R^k$ and $B \subset R^k$, define $A + B$ to be the set of all sums $\mathbf{x} + \mathbf{y}$ with $\mathbf{x} \in A, \mathbf{y} \in B$.

(a) If K is compact and C is closed in R^k, prove that $K + C$ is closed.
Hint: Take $\mathbf{z} \notin K + C$, put $F = \mathbf{z} - C$, the set of all $\mathbf{z} - \mathbf{y}$ with $\mathbf{y} \in C$. Then K and F are disjoint. Choose δ as in Exercise 21. Show that the open ball with center $\mathbf{z}$ and radius δ does not intersect $K + C$.

(b) Let α be an irrational real number. Let C_1 be the set of all integers, let C_2 be the set of all $n\alpha$ with $n \in C_1$. Show that C_1 and C_2 are closed subsets of R^1 whose sum $C_1 + C_2$ is not closed, by showing that $C_1 + C_2$ is a countable dense subset of R^1.

Solution.

Exercise 4.26 (Rudin 4.26)

Suppose X, Y, Z are metric spaces, and Y is compact. Let f map X into Y, let g be a continuous one-to-one mapping of Y into Z, and put $h(x) = g(f(x))$ for $x \in X$.
Prove that f is uniformly continuous if h is uniformly continuous.
Hint: g^{-1} has compact domain $g(Y)$, and $f(x) = g^{-1}(h(x))$.
Prove also that f is continuous if h is continuous.
Show (by modifying Example 4.21, or by finding a different example) that the compactness of Y cannot be omitted from the hypotheses, even when X and Z are compact.

Solution.

Exercise 4.27 (Math 531 Spring 2025, PS6.6)

Let X be a complete metric space. Suppose that $f : X \to X$ is such that $d(f(x), f(y)) \leq \frac{1}{2} d(x, y)$ for all $x, y \in X$.

(a) Prove that f is continuous

(b) Pick any $x_0 \in X$. Define a sequence of points $\{x_n\}$ in X by:

$$x_{n+1} = f(x_n). \tag{522}$$

Prove that $\{x_n\}$ converges. Hint: use a previous homework assignment.

(c) Denote the limit of $\{x_n\}$ by x. Prove that $f(x) = x$.

(d) Prove that if $f(y) = y$ and $f(x) = x$ then $x = y$.

Solution.

Exercise 4.28 (Math 531 Spring 2025, PS6.7)

Does there exist a continuous function $f : [0, 1) \to \mathbb{R}$ that is onto? If so, construct one from scratch. If not, prove such a function cannot exist.

Solution.

Exercise 4.29 (Math 531 Spring 2025, PS7.1)

If a function $f : [0, 1] \to \mathbb{R}$ is continuous, it is uniformly continuous. This means that, given $\epsilon > 0$, there exists $\delta(\epsilon) > 0$ so that

$$d(x, y) < \delta(\epsilon) \Rightarrow d(f(x), f(y)) < \epsilon. \tag{523}$$

- Prove that if $f(x) = 1$ for all x, we can take $\delta(\epsilon) = +\infty$.
- Prove that if $f(x) = Mx$, we can take $\delta(\epsilon) = \frac{\epsilon}{M}$.
- Prove that if $f(x) = \sqrt{x}$, we can take $\delta(\epsilon) = \frac{\epsilon^2}{4}$.
- Prove that if $f(x) = \frac{x}{x+1}$, we can take $\delta(\epsilon) = \epsilon$.
- Prove that if $f(x) = x^N$, for some $N \in \mathbb{N}$, then we can take $\delta(\epsilon) = \frac{\epsilon}{N}$.

You cannot use differentiation for Problem 1. Do everything from scratch.

Solution.

Exercise 4.30 (Math 531 Spring 2025, PS7.2)

Let X be a general metric space. A function $f : X \to Y$ is called compact if the image of every closed ball $B_r(x)$ is compact in Y. Prove that any continuous $f : \mathbb{R}^n \to \mathbb{R}^n$ is compact. Give an example of a discontinuous function $f : \mathbb{R} \to \mathbb{R}$ that is compact.

Solution.

Exercise 4.31 (Math 531 Spring 2025, PS7.3)

Prove that if $f : \mathbb{R} \to \mathbb{R}$ sends connected sets to connected sets and compact sets to compact sets, then it is continuous. Hint: Assume that $f : \mathbb{R} \to \mathbb{R}$ sends every connected set to a connected set. Assume also that f is discontinuous at some $x \in \mathbb{R}$. Find a compact set K so that $f(K)$ is not compact.

Solution.

Exercise 4.32 (Math 531 Spring 2025, PS7.4)

- Let X be a general metric space and assume $K \subset X$ is compact. Let $f : K \to K$ and assume that
$$d(f(x), f(y)) < d(x, y) \tag{524}$$
for all $x \neq y \in K$. Show that there exists $x \in K$ with $f(x) = x$.
- Find a function $f : [0, \infty) \to [0, \infty)$ with the property that $|f(x) - f(y)| < |x - y|$, for all non-equal $x, y \in [0, \infty)$, but for which there is no point $x \in [0, \infty)$ for which $f(x) = x$.

Hint for the first part: define the function $g : K \to \mathbb{R}$ by $g(x) = d(f(x), x)$. First prove that g is continuous. Then deduce that g must attain its minimum at some x^*. Then show $g(f(x^*)) < g(x^*)$ unless $f(x^*) = x^*$. Conclude that $f(x^*) = x^*$.

Solution.

Exercise 4.33 (Math 531 Spring 2025, PS7.6)

A function $f : X \to X$ is said to be Hölder continuous of degree α for some $\alpha \in (0, \infty)$ if there exists $M > 0$ so that

$$d(f(x), f(y)) \leq C \cdot d(x, y)^{\alpha}. \tag{525}$$

Prove that if f is Hölder continuous of degree $\alpha > 0$, then f is continuous. Give an example of a function on $\mathbb{R}$ that is Hölder continuous of degree $\frac{1}{2}$ but not of degree 1. Prove that continuously differentiable functions $[0, 1]$ are Hölder continuous of degree 1. Prove that the only functions on $\mathbb{R}$ that are Hölder continuous of degree larger than 1 are constants.

Solution.

5 Differentiation

In general, there are two ways that we define a derivative: as a limit and with infinitesimals. In a standard analysis course, we introduce the derivative as the limit of the difference quotient $\mathrm{Diff}_h f(x) = \frac{f(x+h)-f(x)}{h}$. However, there is a second definition that emphasizes the existence of a best linear approximation. This is an equivalent definition and has two advantages. First, a lot of physicists tend to use this formulation, and second, it frames the derivative of one-variable functions well so that it is easy to generalize into Frechet derivatives for multivariate functions. Therefore, I think it is best to introduce them concurrently. Furthermore, these two formulations each naturally lead to the Taylor's theorems with Lagrange and Peano forms of the remainder.

Definition 5.1 (Derivative)

Let $f : E \subset \mathbb{R} \to \mathbb{R}$ and $x \in \mathbb{R}$ be a limit point of E. f is said to be **differentiable at** x if it satisfies either of the equivalent definitions.

1. *Limit of Difference Quotients.* If the following limit exists in $\mathbb{R}$.[a]

$$f'(x) \coloneqq \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{y \to x} \frac{f(y) - f(x)}{y - x} \tag{526}$$

2. *Fundamental Increment Lemma.* If there exists a number $f'(x) \in \mathbb{R}$ s.t.

$$f(x+h) - f(x) = f'(x)h + o(h) \tag{527}$$

We call the linear mapping $df_x : \mathbb{R} \to \mathbb{R}$ defined $df_x(h) = f'(x)h$ as the **differential** of f at x. If f is differentiable at x, then we call $f'(x) \in \mathbb{R}$ the **derivative of** f **at** x. If f is differentiable for all $x \in [a, b]$, we say f is **differentiable on** $[a, b]$.

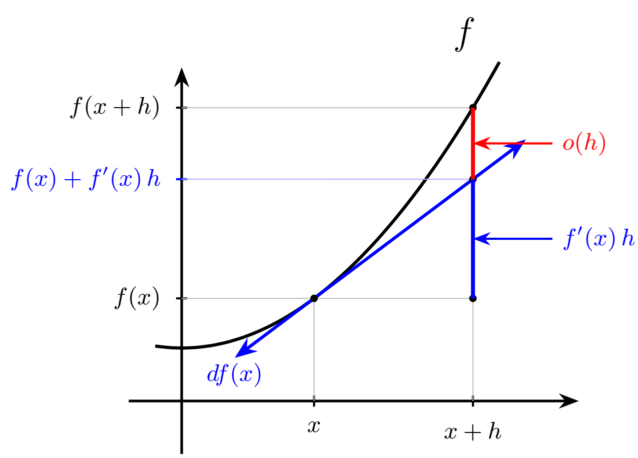


Figure 17: The graph of the differential $df(x) : h \mapsto f'(x)h$ is the best linear approximation of f around x, i.e. the tangent line. Note that the difference (in red) is the function $\varphi(h) = f(x+h) - f(x) - f'(x)h \in o(h)$.

[a] Some authors write the difference quotient as $\phi(y) = \frac{f(x+h)-f(x)}{h}$ while others write $\gamma(h) = \frac{f(y)-f(x)}{y-x}$. These two are equivalent definitions since they are related in the sense that $\phi(y) = \gamma(y - x)$. So the following two limits exist or fail to exist simultaneously. If they do both exist, then $\lim_{y\to x} \phi(y) = \lim_{y \to x} \gamma(y - x) = \lim_{y \to 0} \gamma(y) = \lim_{h \to 0} \gamma(h)$ where the only nontrivial equality is the second equality, which is true (should be shown).

Proof. We prove bidirectionally.

1. ($\rightarrow$). Let us define $\varphi(h) = f(x+h) - f(x) - f'(x)h$. We claim that this is $o(h)$, since

$$\lim_{h \to 0} \frac{\varphi(h)}{h} = \lim_{h \to 0} \left(\frac{f(x+h) - f(x) - f'(x)h}{h} \right) \tag{528}$$
$$= \lim_{h \to 0} \left(\frac{f(x+h) - f(x)}{h} - f'(x) \right) \tag{529}$$
$$= \lim_{h \to 0} \left(\frac{f(x+h) - f(x)}{h} \right) - f'(x) \tag{530}$$
$$= f'(x) - f'(x) = 0 \tag{531}$$

Therefore, we can verify that

$$f'(x)h + \varphi(h) = f'(x)h + f(x+h) - f(x) - f'(x)h = f(x+h) - f(x) \tag{532}$$

2. ($\leftarrow$). Assume such a $f'(x)$ and $\varphi(h)$ exist. Then, we can see that by dividing both sides by h.

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \frac{\varphi(h)}{h} \tag{533}$$

We must throw away the case where $h = 0$ since we divided by h, but it doesn't matter since we are taking the limit as $h \to 0$.

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \left(f'(x) + \frac{\varphi(h)}{h} \right) \tag{534}$$
$$= f'(x) + \lim_{h \to 0} \frac{\varphi(h)}{h} \tag{535}$$
$$= f'(x) + 0 = f'(x) \tag{536}$$

Therefore, we call the derivative also the *best linear approximation* of the function f. With this, we can establish the fun fact of the origins of *Leibniz's notation* of the derivative. Consider the identity function $\iota : x \mapsto x$, and denote its differential $dx(h) = d\iota(h) = h$. Then, for any differentiable function, we can do

$$df(x)(h) = f'(x)\, h = f'(x)\, dx(h) \tag{537}$$

where the first equality follows from the definition and the second equality comes from substituting $dx(h) = h$ in it. Now removing the input parameter h, we have

$$df(x) = f'(x)dx \implies \frac{df(x)}{dx} = f'(x) \tag{538}$$

Therefore, the "fraction" on the left hand side is really just a fraction of two functions of h! It turns out that since $df(x)$ is linear, the h's in the numerator and denominator cancels out, and the fraction reduces to a constant.

Note that I've been very careful with this definition. A lot of authors use the fact that f must be defined on an open set U and choose $x \in U$. However, this might be restrictive in cases where we want to look at "derivatives" on boundaries. Our definition is a generalization of that since if x is indeed in the interior of E, then Equation 526 by definition of the limit of a function becomes a two-sided limit. If x is on the boundary of E, then this is still well-defined as a one-sided derivative. We also adopt the convention from limits that a function may be right or left differentiable, which is useful as I've seen it pop up a few times here and there.

Definition 5.2 (One-Sided Derivative)

Let $f : E \subset \mathbb{R} \to \mathbb{R}$.

1. If x is a limit point of $E \cap [x, +\infty)$, then the following limit—if it exists—is called the **right derivative of f at x**.
$$f'_+(x) = \lim_{y \to x^+} \frac{f(y) - f(x)}{y - x} \tag{539}$$
2. If x is a limit point of $E \cap (-\infty, x]$, then the following limit—if it exists—is called the **left derivative of f at x**.
$$f'_-(x) = \lim_{y \to x^-} \frac{f(y) - f(x)}{y - x} \tag{540}$$

Lemma 5.1 (Differentiability Implies Continuity)

If f is differentiable at x, it is continuous at x.

Proof. If f is differentiable at x, then the derivative

$$f'(x) = \lim_{y \to x} \frac{f(x) - f(y)}{x - y} \tag{541}$$

exists. Therefore,

$$0 = f'(x) \cdot 0 = f'(x)\big(\lim_{y \to x} x - y\big) \tag{542}$$
$$= \lim_{y \to x} \frac{f(x) - f(y)}{x - y} \cdot \lim_{y \to x} x - y \tag{543}$$
$$= \lim_{y \to x} \frac{f(x) - f(y)}{x - y}(x - y) \tag{544}$$
$$= \lim_{y \to x} f(x) - f(y) \tag{545}$$

which implies that $f(x) = \lim_{y \to x} f(y)$, and hence f is continuous at x.

Therefore, we can see that the set of differentiable functions is a subset of the set of continuous functions.

5.1 Rules of Differentiation

Lemma 5.2 (Linearity of Differentiation)

If $f, g : E \to \mathbb{R}$ are differentiable at x, then $f + g$ is differentiable at x with

$$(f + g)'(x) = f'(x) + g'(x) \tag{546}$$

Proof. Trivial.

Lemma 5.3 (Product Rule)

If $f, g : E \to \mathbb{R}$ are differentiable at x, then fg is differentiable at x with

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \tag{547}$$

Proof. For products, let's not take the quotient just yet.

$$(fg)(x) - (fg)(y) = f(x)g(x) - f(y)g(y) \tag{548}$$

We know something about $f(x) - f(y)$ and $g(x) - g(y)$, so try to put it into this form.

$$\big(f(x) - f(y)\big)g(x) + f(y)\big(g(x) - g(y)\big) \tag{549}$$

Therefore,

$$\frac{(fg)(x) - (fg)(y)}{x - y} = \underbrace{\frac{f(x) - f(y)}{x - y}}_{\text{exists}} g(x) + f(y) \underbrace{\frac{g(x) - g(y)}{x - y}}_{\text{exists}} \tag{550}$$

So by taking limits, f is continuous so $f(y) \to x$ as $y \to x$, and we finally have

$$f'(x)g(x) + f(x)g'(x) \tag{551}$$

Lemma 5.4 (Quotient Rule)

If $f, g : E \to \mathbb{R}$ are differentiable at x and $g(x) \neq 0$ in a neighborhood of x, then f/g is differentiable at x with

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2} \tag{552}$$

Proof. We will prove that $(1/g)'(x) = -\frac{g'(x)}{g(x)^2}$. Using the limit rules,

$$\begin{aligned}
\left(\frac{1}{g}\right)' &= \lim_{y \to x} \frac{(1/g)(x) - (1/g)(y)}{x - y} && (553)\\
&= \lim_{y \to x} \frac{g(y) - g(x)}{(x - y)g(x)g(y)} && (554)\\
&= \left(\lim_{y \to x} \frac{g(y) - g(x)}{x - y}\right)\left(\lim_{y \to x} \frac{1}{g(x)g(y)}\right) && (555)\\
&= -g'(x) \cdot \frac{1}{g(x)^2} && (556)
\end{aligned}$$

Therefore,

$$\begin{aligned}
\left(\frac{f}{g}\right)'(x) = f(x)\left(\frac{1}{g}\right)'(x) + f'(x)\left(\frac{1}{g}\right)(x) && (557)\\
-\frac{f(x)g'(x)}{(g(x))^2} + \frac{f'(x)}{g(x)} && (558)\\
= \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} && (559)
\end{aligned}$$

It turns out that proving all the familiar rules is quite a pain to do from first principles.

Theorem 5.5 (Power Rule of Differentiation for Naturals)

For $\alpha \in \mathbb{R}$,

$$\frac{d}{dx} x^\alpha = \alpha x^{\alpha - 1} \tag{560}$$

Proof. By the binomial theorem, we have

$$\begin{aligned}
\frac{d}{dx} x^n &= \lim_{h \to 0} \frac{(x+h)^n - x_n}{h} && (561)\\
&= \lim_{h \to 0} \frac{1}{h} \bigg\{ \bigg(\sum_{k=0}^n \binom{n}{k} x^k h^{n-k} \bigg) - x^n \bigg\} && (562)\\
&= \lim_{h \to 0} \frac{1}{h} \sum_{k=1}^n \binom{n}{k} x^k h^{n-k} && (563)\\
&= \lim_{h \to 0} \sum_{k=1}^n \binom{n}{k} x^k h^{n-k-1} && (564)\\
&= \lim_{h \to 0} \bigg\{ n x^{n-1} + \binom{n}{2} x^{n-2} h + \ldots \bigg\} && (565)\\
&= n x^{n-1} && (566)
\end{aligned}$$

It is clear that now, we can differentiate all polynomials.

Theorem 5.6 (Chain Rule)

Let $f : [a, b] \subset \mathbb{R} \to \mathbb{R}$ be a differentiable function, fix $x \in [a, b]$, and assume differentiable $g : I \to \mathbb{R}$ where $f(x) \in I$. Then, $h = g \circ f$ is differentiable at x with the derivative

$$h'(x) = g'(f(x)) f'(x) \tag{567}$$

Proof. We have

$$g\big(f(x)\big) - g\big(f(y)\big) = \big(f(x) - f(y)\big)\big(g'(f(x)) + E_{f(y) \to f(x)}\big) \tag{568}$$

Now we divide by $x - y$.

$$\frac{g(f(x)) - g(f(y))}{x - y} - \frac{f(x) - f(y)}{x - y} \cdot \big(g'(f(x)) + E_{f(y) \to f(x)}\big) \tag{569}$$

Now if $x \to y$, then f is continuous, which implies $f(y) \to f(x)$ and so $E \to 0$, and so by taking this limit, the above evaluates to

$$f'(x) \cdot g'(f(x)) \tag{570}$$

We could have also done

$$\frac{g(f(x)) - g(f(y))}{x - y} = \frac{g(f(x)) - g(f(y))}{f(x) - f(y)} \cdot \frac{f(x) - f(y)}{x - y} \to g'(f(x)) \cdot f'(x) \tag{571}$$

Theorem 5.7 (Inverse Function Theorem)

Let $E_1, E_2 \subset \mathbb{R}$ and $f : E_1 \to E_2$ be a homeomorphism. Let $x_0 \in E_1$ be a limit point of E_1, and let $y_0 = f(x_0)$.

1. If f is differentiable at x_0 and $f'(x_0) \neq 0$, then f^{-1} is differentiable at y_0 (which is a limit point of E_2),
2. Furthermore,

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)} \iff df^{-1}(y_0) = \big(df(x_0)\big)^{-1} \tag{572}$$

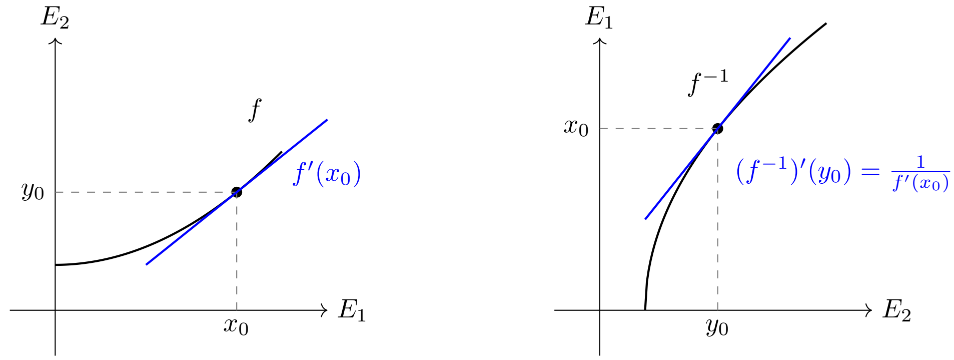


Figure 18: Geometric relationship between the derivative of f at x_0 and f^{-1} at y_0.

Proof. Let $y_0 = f(x_0)$. Since f^{-1} is continuous at y_0, for any sequence $y \to y_0$ in $E_2 \setminus \{y_0\}$, the sequence $x = f^{-1}(y)$ must converge to $x_0 = f^{-1}(y_0)$ in E_1. Because f is injective, $y \neq y_0 \implies x \neq x_0$. The difference quotient for f^{-1} at y_0 is:

$$\frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \frac{x - x_0}{f(x) - f(x_0)} \tag{573}$$

Since f is differentiable at x_0 and $f'(x_0) \neq 0$, we apply the limit:

$$\lim_{y \to y_0} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \lim_{x \to x_0} \frac{1}{\frac{f(x)-f(x_0)}{x-x_0}} = \frac{1}{f'(x_0)} \tag{574}$$

This shows that f^{-1} is differentiable at y_0 with $(f^{-1})'(y_0) = 1/f'(x_0)$.

Note that if we knew in advance that f^{-1} was differentiable at y_0 (which is a stronger hypothesis), we can find immediately by the identity

$$(f^{-1} \circ f)(x) = x \tag{575}$$

and the theorem on the differentiation of a composite function that

$$(f^{-1})'(y_0) \cdot f'(x_0) = 1 \tag{576}$$

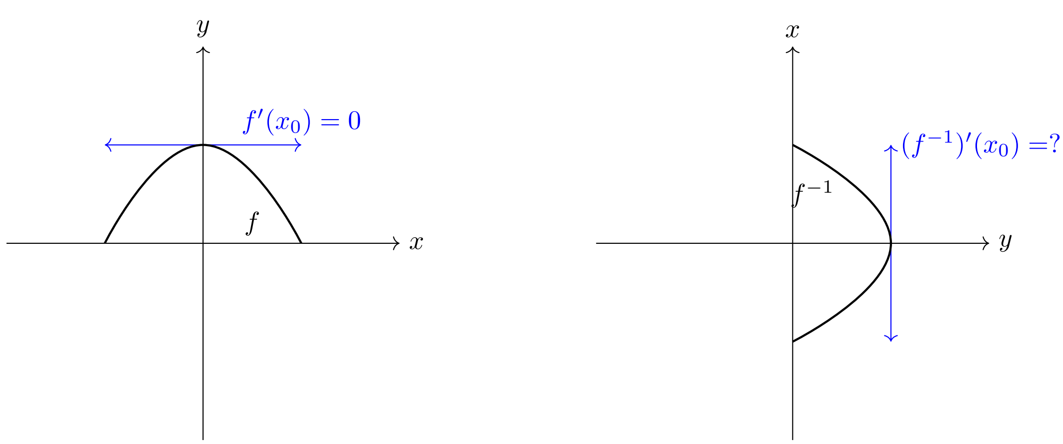


Figure 19: Note that if the hypothesis was satisfied, but $f'(x_0) = 0$, then f^{-1} would not be differentiable since it would have an undefined differential.

Theorem 5.8 (Differentiation of Exponential and Logarithmic Functions)

Let $a > 0$,

1. *Exponential.* The exponential function $x \mapsto a^x$ is differentiable over $\mathbb{R}$, and the derivative is
$$\frac{d}{dx} a^x = a^x \ln(a) \tag{577}$$
2. *Logarithmic.* The logarithmic function $x \mapsto \log_a x$ is differentiable over $(0, +\infty)$, and the derivative is
$$\frac{d}{dx} \log_a(x) = \frac{1}{x \ln(a)} \tag{578}$$

In particular,
$$\frac{d}{dx} e^x = e^x, \qquad \frac{d}{dx} \ln(x) = \frac{1}{x} \tag{579}$$

Proof. Listed.

1. *Exponential.* We can write
$$\frac{d}{dx} a^x = \lim_{h \to 0} \frac{a^{x+h} - a^x}{h} = \lim_{h \to 0} \frac{a^x (a^h - 1)}{h} = a^x \left(\lim_{h \to 0} \frac{a^h - 1}{h} \right) \tag{580}$$
Therefore, it suffices to prove that the limit term equals $\ln(a)$. We claim that for $x < 1$,
$$1 + x \leq e^x \leq \frac{1}{1 - x} \tag{581}$$
The left hand inequality comes from Bernoulli's inequality that is taken as the limit $n \to +\infty$. For the second inequality, note that be the previous inequality, $e^{-x} \geq 1 - x$. Since $x < 1$, $1 - x > 0$, and so inverting the inequality gives $e^x = \frac{1}{e^{-x}} \leq \frac{1}{1-x}$. Since
$$b = e^{\ln(b)} \implies b^h = (e^{\ln(b)})^h = e^{h \ln(b)} \tag{582}$$
We can substitute this into equation 581 to get
$$1 + h \ln(b) \leq e^{h \ln(b)} \leq \frac{1}{1 - h \ln(b)} \tag{583}$$
Now subtracting 1 and dividing by h from all sides, we get
$$\ln(b) \leq \frac{e^{h \ln(b)} - 1}{h} \leq \frac{\ln(b)}{1 - h \ln(b)} \tag{584}$$
By taking the limit as $h \to 0$, and invoking the squeeze theorem, we are done.
2. *Logarithmic.* Since we have defined the logarithm as the inverse of the exponential, by the inverse function theorem, $\ln(x)$ is differentiable. Therefore, by the chain rule, we have
$$1 = \frac{d}{dx} x = \frac{d}{dx} e^{\ln(x)} = e^{\ln(x)} \frac{d}{dx} \ln(x) = x \frac{d}{dx} \ln(x) \implies \frac{d}{dx} \ln(x) = \frac{1}{x} \tag{585}$$

Theorem 5.9 (Power Rule of Differentiation for Reals)

For $\alpha \in \mathbb{R}$, the power function $x \mapsto x^\alpha$ is differentiable over the *interior* of its domain[a] and the derivative is

$$\frac{d}{dx}x^\alpha = \alpha x^{\alpha-1} \tag{586}$$

[a]Depending on α and x, the domain may be different. Consider x^2 which has domain $\mathbb{R}$ versus $\sqrt{x}$ which has domain $[0, +\infty)$ and has derivative defined over $(0, +\infty)$. More generally, when α is irrational or is a rational with even denominator, the x^α is not even well-defined for $x < 0$.

Proof. When $x > 0$, we use the trick $x^\alpha = e^{\alpha \ln(x)}$. By the chain rule,

$$f'(x) = e^{\alpha \ln(x)} \cdot \alpha \frac{1}{x} = x^\alpha \alpha \frac{1}{x} = \alpha x^{\alpha-1} \tag{587}$$

When $x < 0$, we do the same thing but with $x^r = (-1)^r(-x)^r$.

Theorem 5.10 (Differentiation of Trigonometric Functions)

The following are true for trigonometric functions.

1. *Sine.* The sine function $x \mapsto \sin(x)$ is differentiable over $\mathbb{R}$, and the derivative is

$$\frac{d}{dx}\sin(x) = \cos(x) \tag{588}$$

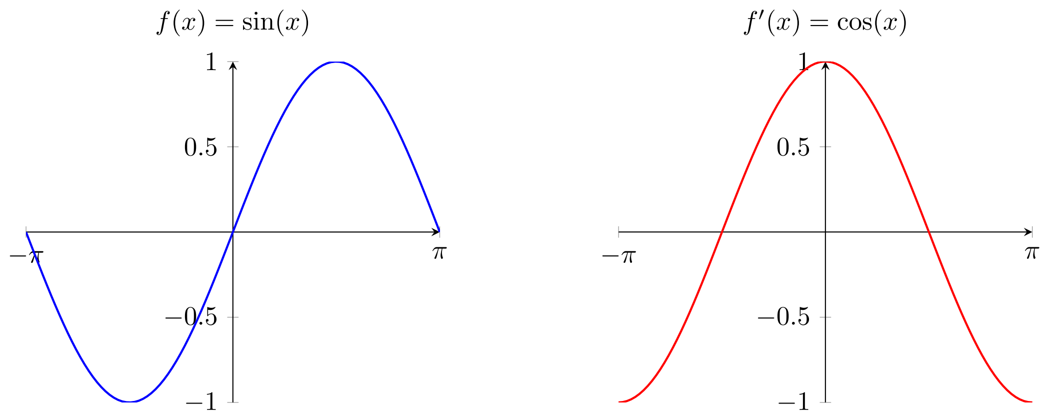


Figure 20: Sine and its derivative

2. *Cosine.* The cosine function $x \mapsto \cos(x)$ is differentiable over $\mathbb{R}$, and the derivative is

$$\frac{d}{dx}\cos(x) = -\sin(x) \tag{589}$$

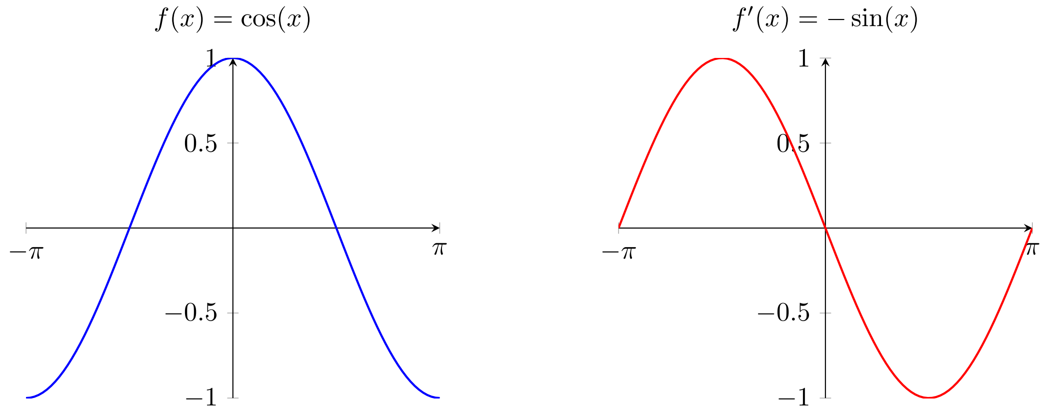


Figure 21: Cosine and its derivative

3. *Tangent.* The tangent function $x \mapsto \tan(x)$ is differentiable over $\mathbb{R} \setminus \{\frac{(2z+1)}{2}\pi\}_{z \in \mathbb{Z}}$, and the derivative is

$$\frac{d}{dx} \tan(x) = \sec^2(x) \tag{590}$$

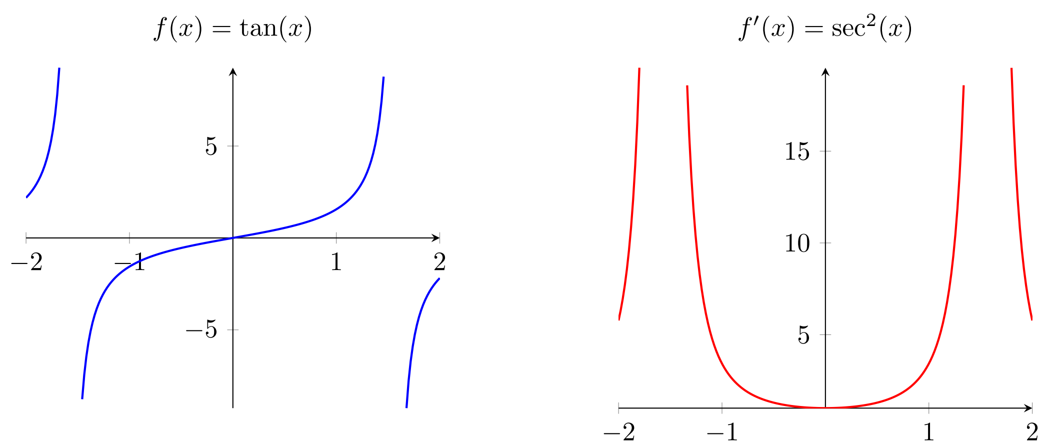


Figure 22: Tangent and its derivative

4. *Arcsine.* The inverse sine function $x \mapsto \arcsin(x)$ is differentiable on $(-1, 1)$, and the derivative is

$$\frac{d}{dx} \arcsin(x) = \frac{1}{\sqrt{1 - x^2}} \tag{591}$$

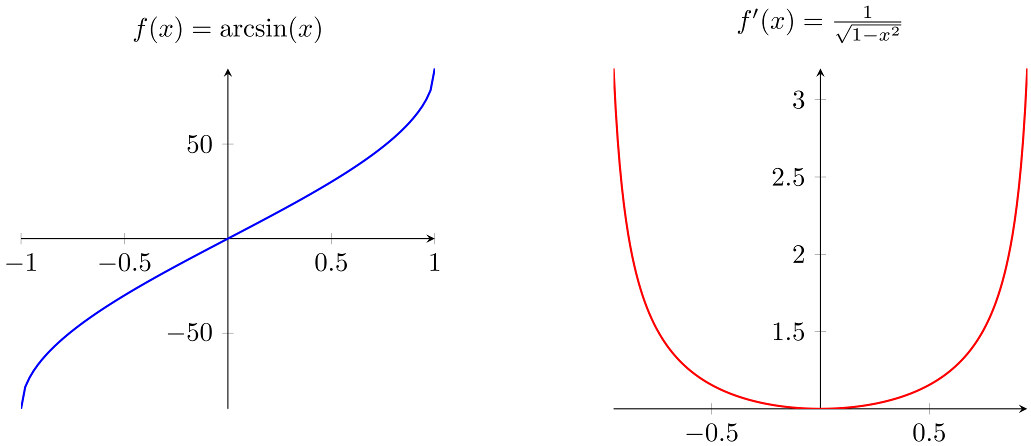


Figure 23: Arcsine and its derivative

5. *Arccosine.* The inverse cosine function $x \mapsto \arccos(x)$ is differentiable on $(-1, 1)$, and the derivative is

$$\frac{d}{dx} \arccos(x) = -\frac{1}{\sqrt{1-x^2}} \tag{592}$$

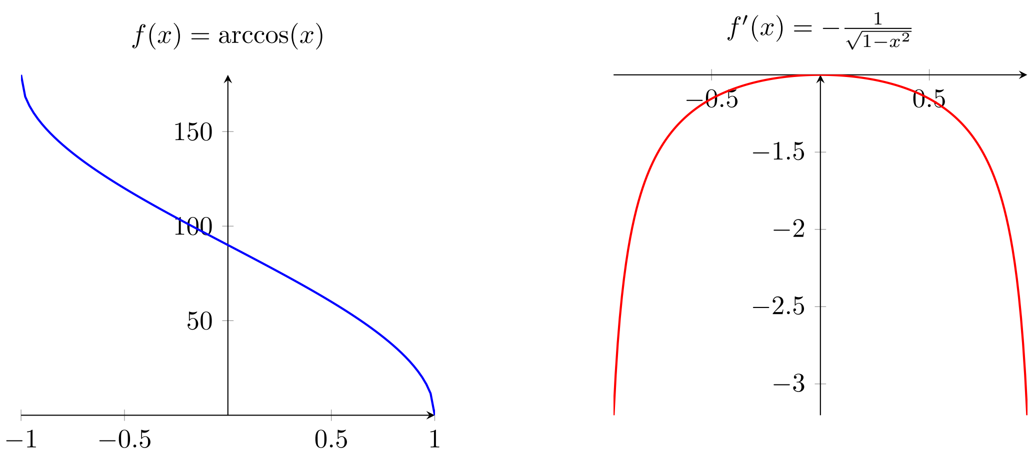


Figure 24: Arccosine and its derivative

6. *Arctangent.* The inverse tangent function $x \mapsto \arctan(x)$ is differentiable over $\mathbb{R}$, and the derivative is

$$\frac{d}{dx} \arctan(x) = \frac{1}{1+x^2} \tag{593}$$

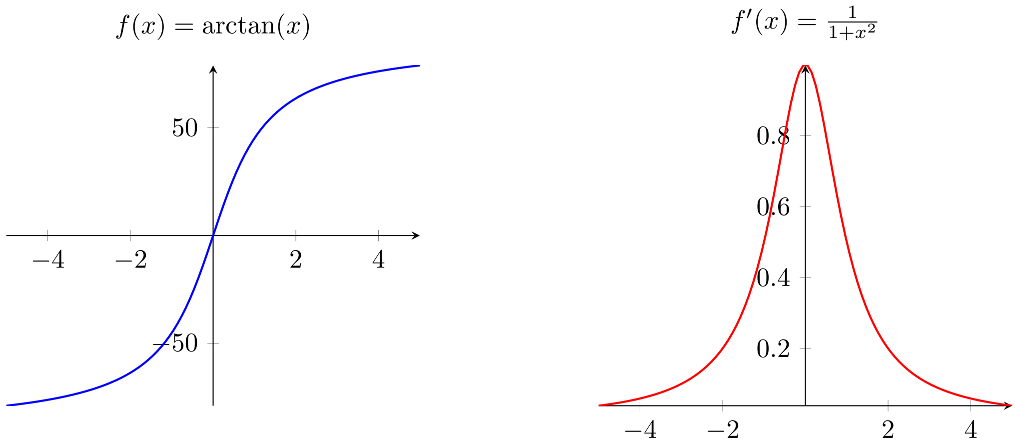


Figure 25: Arctangent and its derivative

Proof. Listed.

1. *Sine.* Let $f(x) = \sin x$.

$$\lim_{h \to 0} \frac{\sin(x+h) - \sin(x)}{h} = \lim_{h \to 0} \frac{2 \sin\left(\frac{h}{2}\right) \cos\left(x + \frac{h}{2}\right)}{h} \tag{594}$$

$$= \lim_{h \to 0} \cos\left(x + \frac{h}{2}\right) \cdot \lim_{h \to 0} \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)} = \cos(x) \tag{595}$$

Here, we have used the theorem on the limit of a product, the continuity of the function $\cos(x)$, the equivalence $\sin t \sim t$ as $t \to 0$, and the theorem on the limit of a composite function.

2. *Cosine.* We have

$$\lim_{h \to 0} \frac{\cos(x+h) - \cos(x)}{h} = \lim_{h \to 0} \frac{-2 \sin\left(\frac{h}{2}\right) \sin\left(x + \frac{h}{2}\right)}{h} \tag{596}$$

$$= -\lim_{h \to 0} \sin\left(x + \frac{h}{2}\right) \cdot \lim_{h \to 0} \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)} = -\sin(x) \tag{597}$$

3. *Tangent.* We simply use the quotient rule.

$$\frac{d}{dx} \tan(x) = \frac{d}{dx} \frac{\sin(x)}{\cos(x)} = \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)} = \sec^2(x) \tag{598}$$

4. *Arcsine.* Since the arcsine is the inverse of the sine function, by the inverse function theorem, arcsin is differentiable. So, using the chain rule, we have

$$1 = \frac{d}{dx} x = \frac{d}{dx} \sin(\arcsin(x)) = \cos(\arcsin(x)) \frac{d}{dx} \arcsin(x) \tag{599}$$

But since $\cos(x) = \sqrt{1 - \sin^2(x)}$, we have

$$\frac{d}{dx} \arcsin(x) = \frac{1}{\cos(\arcsin(x))} = \frac{1}{\sqrt{1 - \sin^2(\arcsin(x))}} = \frac{1}{\sqrt{1 - x^2}} \tag{600}$$

5. *Arccosine.* Since arccosine is the inverse of the cosine function, by the inverse function theorem, arccos is differentiable. So, using the chain rule, we have

$$1 = \frac{d}{dx} x = \frac{d}{dx} \cos(\arccos(x)) = -\sin(\arccos(x)) \frac{d}{dx} \arccos(x) \tag{601}$$

But since $\sin(x) = \sqrt{1 - \cos^2(x)}$, we have

$$\frac{d}{dx} \arccos(x) = -\frac{1}{\sin(\arccos(x))} = -\frac{1}{\sqrt{1 - \cos^2(\arccos(x))}} = -\frac{1}{\sqrt{1 - x^2}} \tag{602}$$

6. *Arctangent.* Since arctangent is the inverse of the tangent function, by the inverse function theorem, arctan is differentiable. So, using the chain rule, we have

$$1 = \frac{d}{dx} x = \frac{d}{dx} \tan(\arctan(x)) = \sec^2(\arctan(x)) \frac{d}{dx} \arctan(x) \tag{603}$$

But since $1 + \tan^2(x) = \sec^2(x)$, we have

$$\frac{d}{dx} \arccos(x) = \frac{1}{\sec^2(\arctan(x))} = \frac{1}{1 + \tan^2(\arctan(x))} = \frac{1}{1 + x^2} \tag{604}$$

Now let's do some computation.

Example 5.1

Consider the function

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) := \begin{cases} x^2 & \text{if } x \geq 0 \\ x^3 & \text{if } x < 0 \end{cases} \tag{605}$$

Is it differentiable? The answer is clearly yes for $x > 0$ or $x < 0$. However, at $x = 0$, we can compute the right and left hand limits and ensure that they are equal, proving differentiability.

$$f'_-(0) = \lim_{h \to 0^-} \frac{f(h) - f(0)}{h} = \lim_{h \to 0^-} \frac{h^3}{h} = \lim_{h \to 0^-} h^2 = 0 \tag{606}$$

$$f'_+(0) = \lim_{h \to 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \to 0^-} \frac{h^2}{h} = \lim_{h \to 0^+} h = 0 \tag{607}$$

Since the right and left hand limits coincide, the limit exists and thus by definition $f'(0) = 0$.
Note that this is the same as when we look at the restrictions of f to $g = f|_{[0,+\infty)}$ and $h = f|_{(-\infty,0)}$. Their derivatives at 0 (which is indeed a limit point of each of their domains) is well defined, which by the power rule (as an application of the left and right hand limits) is

$$g'(0) = 0, \quad h'(0) = 0 \tag{608}$$

Example 5.2 (Differentiability at Endpoints)

Consider the function

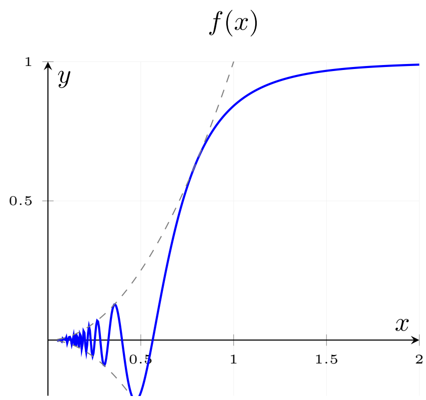
$$f : [0, 1] \to \mathbb{R}, \quad f(x) := \begin{cases} x^2 \sin(1/x^2) & \text{if } x \in (0, 1] \\ 0 & \text{if } x = 0 \end{cases} \tag{609}$$

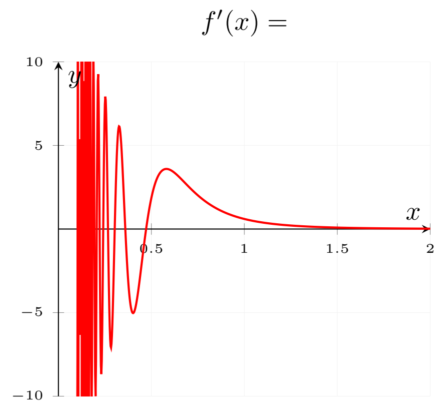
On $(0, 1]$, f as the composition and product of two differentiable functions is differentiable, and so we can compute f' over $(0, 1]$. However, $f(0) = 0$, so we can't use product rule. We must just use the limit definition.

$$f'(0) = \lim_{h \to 0} \frac{f(h) - f(0)}{h} = \lim_{h \to 0^+} \frac{h^2 \sin(1/h^2)}{h} = \lim_{h \to 0^+} h \sin(1/h^2) = 0 \tag{610}$$

Therefore, we solved

$$f' : [0,1] \to \mathbb{R}, \quad f'(x) \coloneqq \begin{cases} 2x\sin(1/x) - \frac{2}{x}\cos\frac{1}{x^2} & \text{if } x \in (0,1] \\ 0 & \text{if } x = 0 \end{cases} \tag{611}$$





5.2 Higher Order Derivatives and Continuously Differentiablity

Differentiability is nice, but if we have the additional property that the derivative is continuous as well, we can unlock some better properties. These continuously differentiable functions—called *smooth functions*—are denoted $f \in C^1([a,b])$.[8]

Definition 5.3 (C^k Functions)

A function $f : E \subset \mathbb{R} \to \mathbb{R}$ is said to be
1. C^0 if it is continuous.
2. C^1, or **continuously differentiable**, if its derivative is continuous.
3. C^k, or **k-times continuously differentiable**, if its kth derivative $f^{(k)}$ exists on E and is continuous over E.
4. C^∞, or **smooth**, if it is infinitely differentiable with derivatives of all order that are smooth.

Example 5.3 (Continuous But Not Differentiable)

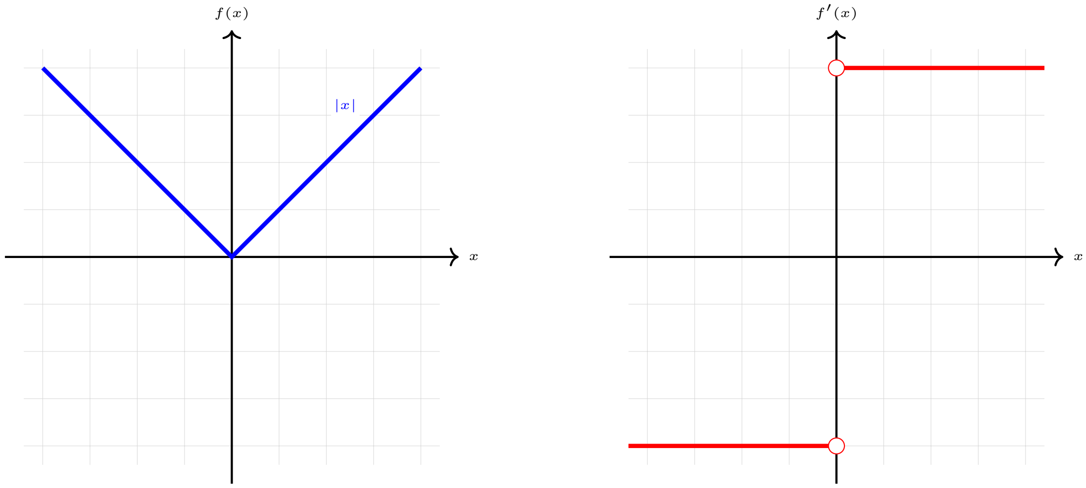
Consider the absolute value function:

$$f(x) = |x| \tag{612}$$

This function is continuous everywhere. However, its derivative is a step function that is undefined at the origin, with a jump discontinuity from -1 to 1:

$$f'(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases} \tag{613}$$

[8] Smoothness is a colloquial term that can be used differently in different contexts, so don't take this to heart too much.



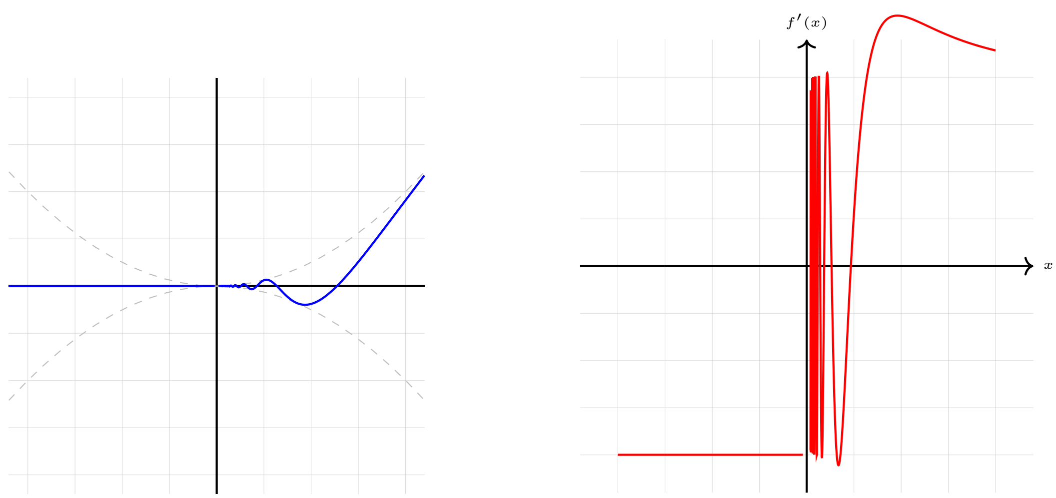
(a) $f(x) = |x|$ (b) $f'(x) = \mathrm{sgn}(x)$

Figure 26: The cusp at the origin in $f(x)$ corresponds to the jump discontinuity in $f'(x)$.

Example 5.4 (Differentiable but Not Continuously Differentiable)

Consider the oscillatory function:

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases} \tag{614}$$



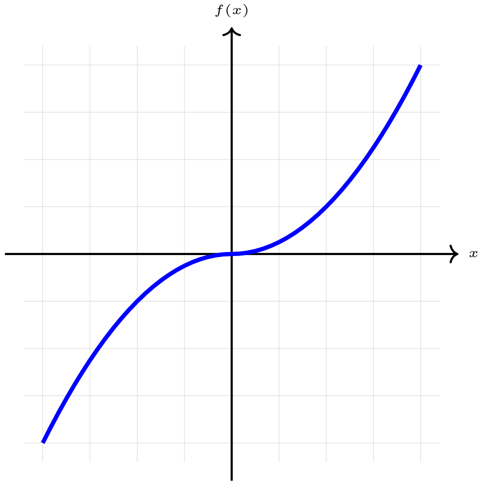
(a) $f(x)$ (Magnified at Origin) (b) $f'(x)$ oscillation

Example 5.5 (Continuously Differentiable but No Second Derivative)

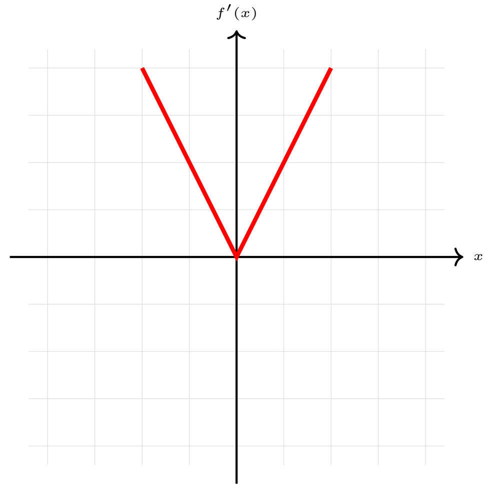
Consider the signed square function:

$$f(x) = x|x| \tag{615}$$

The derivative is $f'(x) = 2|x|$. While f' is continuous (C^0), it is not differentiable at $x = 0$, so f is not C^2.



(a) $f(x) = x|x|$



(b) $f'(x) = 2|x|$

Figure 28: A C^1 function has a continuous derivative, but that derivative may still have a cusp.

Theorem 5.11 (C^1 Implies Lipschitz)

A continuously differentiable function is Lipschitz continuous.

Proof.

Example 5.6 (Differentiable But Not Lipschitz)

Consider

$$f : (0, +\infty), f(x) = \sqrt{x}, \quad f'(x) = \frac{1}{2\sqrt{x}} \tag{616}$$

Then, the derivative is unbounded around 0.

Theorem 5.12 (L'Hopital's Rule)

Suppose f, g are continuously differentiable functions with $f(c) = g(c) = 0$ and $g'(c) \neq 0$. Then,

$$\lim_{x \to c} \frac{f(x)}{g(x)} = \lim_{x \to c} \frac{f'(x)}{g'(x)} \tag{617}$$

Proof. We have

$$\lim_{x\to c}\frac{f(x)}{g(x)} = \lim_{x\to c}\frac{f(x)-f(c)}{g(x)-g(c)} \tag{618}$$
$$= \lim_{x\to c}\frac{\frac{f(x)-f(c)}{x-c}}{\frac{g(x)-g(c)}{x-c}} \tag{619}$$
$$= \frac{\lim_{x\to c}\frac{f(x)-f(c)}{x-c}}{\lim_{x\to c}\frac{g(x)-g(c)}{x-c}} \tag{620}$$
$$= \frac{f'(c)}{g'(c)} \tag{621}$$
$$= \frac{\lim_{x\to c} f'(x)}{\lim_{x\to c} g'(x)} \tag{622}$$
$$= \lim_{x\to c}\frac{f'(x)}{g'(x)} \tag{623}$$

Example 5.7 (Limit of $sin(x)/x$)

We can see that

$$\lim_{x\to 0}\frac{\sin(x)}{x} = \lim_{x\to 0}\frac{\cos(x)}{1} = 1 \tag{624}$$

Example 5.8

Let $f(x) = \sin x$ and $g(x) = -0.5x$. Then, the function

$$h(x) = \frac{f(x)}{g(x)} = \frac{\sin x}{-0.5x} \tag{625}$$

is clearly undefined at $x = 0$. However, we can solve the limit using L'Hopital's rule to get

$$\lim_{x\to 0}\frac{\sin x}{-0.5x} = \lim_{x\to 0}\frac{\cos x}{-0.5} = -2 \tag{626}$$

Therefore, $h : \mathbb{R}\backslash 0 \longrightarrow \mathbb{R}$ can be completed to continuous function on all of $\mathbb{R}$ by defining the extension:

$$H(x) \coloneqq \begin{cases} h(x), & x \neq 0 \\ -2, & x = 0 \end{cases} \tag{627}$$

5.3 Extrema and Concavity

Similarly, we can connect the concepts of extrema and derivatives.

Theorem 5.13 (First Derivative Test)

Let $f : [a, b] \to \mathbb{R}$ and assume f has a local maximum at $c \in (a, b)$ with f differentiable at c. Then $f'(c) = 0$.

Proof. Let us pick two sequences—a left one and a right one—that converges to x from either side.

1. If $x > c$ (with x sufficiently close to c), then $f(c) \geq f(x)$ and so

$$\frac{f(c) - f(x)}{c - x} \leq 0 \implies f'(c) = \lim_{c - x} \frac{f(c) - f(x)}{c - x} \leq 0 \tag{628}$$

2. If $x < c$, then $f(c) \geq f(x)$ and so

$$\frac{f(c) - f(x)}{c - x} \geq 0 \implies f'(c) = \lim_{c - x} \frac{f(c) - f(x)}{c - x} \geq 0 \tag{629}$$

So $0 \leq f'(c) \leq 0 \implies f'(c) = 0$.

Definition 5.4 (Convex, Concave Functions)

A function $f : (a, b) \longrightarrow \mathbb{R}$ defined on an open interval $(a, b) \subset \mathbb{R}$ is **convex** if the inequality

$$f(\alpha_1 x_1 + \alpha_2 x_2) \leq \alpha_1 f(x_1) + \alpha_2 f(x_2) \tag{630}$$

holds and **concave**, or **convex upward**, if the inequality

$$f(\alpha_1 x_1 + \alpha_2 x_2) \geq \alpha_1 f(x_1) + \alpha_2 f(x_2) \tag{631}$$

holds for all pairs of points $x_1, x_2 \in (a, b)$ and any numbers $\alpha_1, \alpha_2 \geq 0$ such that $\alpha_1 + \alpha_2 = 1$. If this inequality is strict whenever $x_1 \neq x_2$ and $\alpha_1 \alpha_2 \neq 0$, the function is said to be **strictly convex** and **strictly concave**, respectively.

Note that using induction on the number of points, we get a primitive form of Jensen's inequality.

Lemma 5.14 (Jensen's Inequality)

If $f : (a, b) \longrightarrow \mathbb{R}$ is a convex function, $x_1, ..., x_n$ are points of (a, b), and $\alpha_1, ..., \alpha_n$ are nonnegative numbers such that $\alpha_1 + ... + \alpha_n = 1$, then

$$f(\alpha_1 x_1 + ... + \alpha_n x_n) \leq \alpha_1 f(x_1) + ... + \alpha_n f(x_n) \tag{632}$$

The following is also another equivalent condition for a function to be convex over (a, b).

Theorem 5.15

A function $f : (a, b) \longrightarrow \mathbb{R}$ that is differentiable on the open interval (a, b) is convex on (a, b) if and only if its graph contains no points below any tangent drawn to it.

Theorem 5.16 (Second Derivative Test)

Given a function $f : (a, b) \longrightarrow \mathbb{R}$ that is differentiable in its domain,

1. f is convex $\iff$ f' is increasing on (a, b) $\iff$ $f'' \geq 0$ on (a, b)
2. f is strictly convex $\iff$ f' is strictly increasing on (a, b) $\iff$ $f'' > 0$ on (a, b)
3. f is concave $\iff$ f' is decreasing on (a, b) $\iff$ $f'' \leq 0$ on (a, b)
4. f is strictly concave $\iff$ f' is strictly decreasing on (a, b) $\iff$ $f'' < 0$ on (a, b)

Definition 5.5 (Inflection Point)

Let $f : E \longrightarrow \mathbb{R}$ be a function defined and differentiable on a neighborhood $U(x_0)$. If the function is convex downward (resp. upward) on the set $\mathring{U}^-(x_0) = \{x \in U(x_0) \mid x < x_0\}$ and convex upward (resp. downward) on $\mathring{U}^+(x_0) = \{x \in U(x_0) \mid x > x_0\}$, then the point

$$\big(x_0, f(x_0)\big) \tag{633}$$

is called a **inflection point of the graph**.

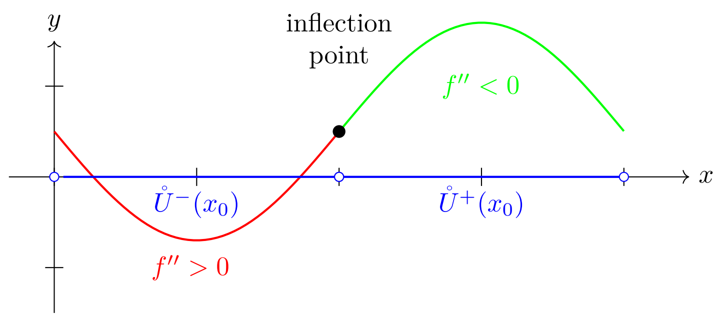


Figure 29: Curve with changing concavity and inflection point at π

5.4 Mean Value Theorem

Theorem 5.17 (Rolle's Theorem)

Suppose $f : [a, b] \to \mathbb{R}$ is differentiable on (a, b). Then, if $f(a) = f(b)$, then there exists a $c \in (a, b)$ such that $f'(c) = 0$.

Proof. Since f is continuous on $[a, b]$, by the extreme value theorem, it has to attain its global max and min values somewhere in $[a, b]$. If either is in (a, b), then the derivative is 0. If max and min are attained on $\{a, b\}$, then since $f(a) = f(b)$, this implies that $f(x) = f(a)$ for all $x \in [a, b]$, which implies $f'(x) = 0$.

Theorem 5.18 (Mean Value Theorem)

Assume $f : [a, b] \to \mathbb{R}$ is differentiable. Then there exists a $c \in (a, b)$ for which

$$f'(c) = \frac{f(b) - f(a)}{b - a} \iff f'(c)(b - a) = f(b) - f(a) \tag{634}$$

Proof. Just use Rolle's on

$$g(x) = f(x) - \left[\frac{f(b) - f(a)}{b - a}(x - a) + f(a)\right] \tag{635}$$

which satisfies $g(a) = g(b) = 0$, and so there must exist some $c \in (a, b)$ such that $g'(c) = 0$, i.e.

$$f'(c) - \frac{f(b) - f(a)}{b - a} = 0 \implies f'(c) = \frac{f(b) - f(a)}{b - a} \tag{636}$$

Geometrically, this means that there exists a tangent line somewhere at $\zeta \in (a, b)$ that is parallel the secant line connecting the two points $\big(a, f(a)\big)$ and $\big(b, f(b)\big)$. Physically, if x is interpreted as time and $f(b) - f(a)$ as the amount of displacement over the time $b - a$ of a particle moving along the line, this theorem says that the velocity $f'(x)$ of the particle at some time $\zeta \in (a, b)$ is such that if the particle had moved with constant velocity $f'(\zeta)$ over the whole time interval, it would have been displaced by the same amount $f(b) - f(a)$. We call $f'(\zeta)$ the **average velocity** over the time interval $[a, b]$.

Note that the MVT is important in that it connects the increment of a function over a finite interval with the derivative of the function on that interval. Up to now, we have characterized only the local (infinitesimal) increment of a function in terms of the derivative or differential at a given point. MVT connects the increment of a function over a **finite** interval with the derivative of the function. The MVT actually leads to multiple useful corollaries, such as the intermediate value theorem for derivatives and for monotonic functions.

Theorem 5.19 (Intermediate Value Theorem For Derivatives)

Suppose $f : [a, b] \to \mathbb{R}$ is a real and differentiable function and suppose $f'(a) < \lambda < f'(b)$. Then there exists $x \in (a, b)$ such that $f'(x) = \lambda$.[a]

[a] i.e. f doesn't have to be continuous, but it must have a middle value.

Proof. Let $g(x) = f(x) - \lambda x$. Then g is differentiable with $g'(x) = f'(x) - \lambda$. But this implies that

1. $g'(a) = f'(a) - \lambda < 0$, which implies that
$$\frac{g(t_1) - g(a)}{t_1 - a} < 0 \implies g(t_1) - g(a) < 0 \implies g(t_1) < g(a) \tag{637}$$
for some $t_1 > a$ sufficiently close to a.
2. $g'(b) = f'(b) - \lambda > 0$, which implies that
$$\frac{g(t_2) - g(b)}{t_2 - b} > 0 \implies g(t_2) - g(b) > 0 \implies g(t_2) > g(b) \tag{638}$$
for some $t_2 < b$ sufficiently close to b.

By the mean value theorem there exists $x \in (a, b)$ s.t. $g'(x) = 0 \implies f'(x) = \lambda$.

Corollary 5.20 (Derivatives Cannot Have Jump Discontinuities)

You can't have a jump discontinuity for derivatives.

That is, the derivative of a differentiable function cannot "jump," so it's like the IVT of derivatives. However, it may as well have discontinuities of the second kind.

Example 5.9 (Derivative Might Jump if Not Differentiable)

A non-example is $f(x) = |x|$. It is not differentiable over $[-1, 1]$, and so we see a jump in the derivative.

5.5 Monotone Functions

Definition 5.6 (Monotone)

A function $f : E \subset \mathbb{R} \to \mathbb{R}$ is

1. **monotonically increasing** if $x < y \implies f(x) \leq f(y)$.
2. **monotonically decreasing** if $x < y \implies f(x) \geq f(y)$.
3. **strictly monotonically increasing** if $x < y \implies f(x) < f(y)$.

4. **strictly monotonically increasing** if $x < y \implies f(x) > f(y)$.

for all $x, y \in E$. We say f is (strictly) monotone if is is (strictly) monotonically increasing or decreasing.

Often, we restrict the domain of a monotone function to be an interval.

Definition 5.7 (Interval)

An **interval** $I \subset \mathbb{R}$ is a set of the form $[a, b], [a, b), (a, b], (a, b)$.

Theorem 5.21 (Continuous Injective Function is Monotone)

Let $f : I \to \mathbb{R}$ be continuous. If f is injective, then f is strictly monotone.

Proof. We prove the contrapositive. If f is not strictly monotone, then there exists points $x, y, z \in I$, with $x < y < z$, such that $f(x) \geq f(y) \leq f(z)$ or $f(x) \leq f(y) \geq f(z)$. Let's focus on the first case: $f(x) \geq f(y) \leq f(z)$. If any of them are equal, then f is not injective, so $f(x) > f(y) < f(z)$. There are two cases.

1. $f(x) > f(z)$. Then, we have $f(y) < f(z) < f(x)$, and by the intermediate value theorem, there exists a $c \in (x, y)$ such that $f(c) = f(z)$, which implies f is not injective.
2. $f(x) < f(z)$. Then, we have $f(y) < f(x) < f(z)$, and by the intermediate value theorem, there exists a $c \in (y, z)$ such that $f(c) = f(x)$, which implies f is not injective.

We apply the same logic to the second case, and we are done.

Theorem 5.22 (Derivative of a Monotonic Function)

Given function $f : [a, b] \longrightarrow \mathbb{R}$ that is differentiable on (a, b),[a]

1. $f' > 0 \implies f$ is strictly increasing
2. $f' \geq 0 \iff f$ is increasing
3. $f' = 0 \iff f$ is constant
4. $f' \leq 0 \iff f$ is decreasing
5. $f' < 0 \implies f$ is strictly decreasing

The reverse implication is a bit weaker.

1. f is strictly increasing $\implies f'(x) \geq 0$
2. f is strictly decreasing $\implies f'(x) \leq 0$

[a]Note the one-sided direction for the strict inequalities.

Proof. If $x_1 < x_2$ are two points of the interval, then the MVT

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1) \tag{639}$$

shows that the sign of the left hand side must equal that of the right. For constant functions, the MVT equation

$$0 = f(x_2) - f(x_1) = f'(c)(x_2 - x_1) \tag{640}$$

implies that $f'(c) = 0$ for all $x_1, x_2 \in E$.

Proof. Here is another proof, where we prove the contrapositive. Let there be an $x \in E$ s.t. $f'(x) < 0$. Then,

$$\lim_{y \to x} \frac{f(y) - f(x)}{y - x} < 0 \tag{641}$$

In short, there exists a neighborhood around x where the quotient is less than 0, so in that neighborhood, we can find a y such that $f(y) - f(x) < 0$. More specifically, for every $\epsilon > 0$, there exists a $\delta > 0$ s.t.

$$|x - y| < \delta \implies \frac{f(y) - f(x)}{y - x} < 0 \tag{642}$$

If we set $x < y$, then $f(y) - f(x) < 0 \implies f(x) > f(y)$, and so f cannot be monotonically increasing. The same logic applies to monotonically decreasing functions.

Note that f strictly monotone does not imply that f' is strictly greater than or less than 0!

Example 5.10

Let $f(x) = x^3$. f is strictly increasing on $(-1, 1)$, but $f'(0) = 0$.

Corollary 5.23 (Equal Derivatives Imply Constant Difference)

Given two functions $f, g : E \subset \mathbb{R} \to \mathbb{R}$ have the same derivatives—i.e. $f' = g'$—then the difference is a constant

$$f - g = c \tag{643}$$

Proof. Since $f' - g' = (f - g)' = 0$, it must be the case that $f - g$ is constant.

5.6 Taylor's Formula

Taylor's formula is quite misunderstood when you first learn about it, and most people don't know that there are two separate theorems on it with two completely different purposes. The first one from Lagrange extends the MVT and the purpose is to find an explicit formula for the error value. The second from Peano extends the definition of differentiability to higher order terms, and the main purpose is to tell you how fast the error term vanishes. Their assumptions are also completely different, too.

Let's start with Peano's first. So far, we have constructed a linear approximation, but let's step back for a moment and try to construct successive approximations to an arbitrary function $f : E \longrightarrow \mathbb{R}$ around a given point x_0. That is, we find a function g such that

$$f(x) = g(x) + o(g) \tag{644}$$

Depending on what g is, we can construct better approximations of f.

1. *Constant Approximation.* The 0th order approximation is when g is a constant. So

$$f(x) = c_0 + o(1) \text{ as } x \to x_0 \tag{645}$$

   which basically means that we want the difference $f(x) - c_0 \in o(1)$. If f is continuous[9], the only possible case is when $c_0 = f(x)$.

2. *Linear Approximation.* We have already done the 1st order approximation, since we just set

$$f(x) = c_0 + c_1(x - x_0) + o(x - x_0) \text{ as } x \to x_0 \tag{646}$$

   If we assume f is continuous, then we must have $c_0 = f(x)$ since $c_1(x - x_0)$ will vanish. Then, we have already proven that $c_1 = f'(x_0)$.

[9]This is important!

3. *Quadratic Approximation.* We can continue this pattern to try and find a quadratic approximation of f in the form

$$f(x) = c_0 + c_1(x - x_0) + c_2(x - x_0)^2 + o\big((x - x_0)^2\big) \text{ as } x \to x_0 \tag{647}$$

By assuming continuity of f, we can derive $c_0 = f(x_0)$. Furthermore, continuity of the derivative will imply $c_1 = f'(x_0)$. What should our guess be for c_2? Well, we can rearrange the terms so that

$$\begin{aligned} c_2 &= \frac{f(x) - c_0 - c_1(x - x_0) - o((x - x_0)^2)}{(x - x_0)^2} &(648) \\ &= \frac{f(x) - c_0 - c_1(x - x_0)}{(x - x_0)^2} - o(1) &(649) \end{aligned}$$

Now for such polynomial coefficients to exist, c_2 must be constant, so the right hand ride shouldn't depend on x. We can take advantage of the $o(1)$ term by setting the limit $x \to x_0$. By using L'Hopital's rule, we have

$$c_2 = \lim_{x \to x_0} \frac{f(x) - c_0 - c_1(x - x_0)}{(x - x_0)^2} = \lim_{x \to x_0} \frac{f'(x) - c_1}{2(x - x_0)} = \frac{1}{2} f''(x_0) \tag{650}$$

We then postulate that $c_2 = \frac{1}{2} f''(x_0)$ and define the error term as

$$\varphi(x) = f(x) - f(x_0) - f'(x_0)(x - x_0) - \frac{f''(x_0)}{2}(x - x_0)^2 \tag{651}$$

We claim that it is $o((x - x_0)^2)$. Indeed, using L'Hopital's rule

$$\begin{aligned} \lim_{x \to x_0} \frac{\varphi(x)}{(x - x_0)^2} &= \lim_{x \to x_0} \left(\frac{f(x) - f(x_0) - f'(x_0)(x - x_0)}{(x - x_0)^2} - \frac{f''(x_0)}{2} \right) &(652) \\ &= \lim_{x \to x_0} \left(\frac{f'(x) - f'(x_0)}{2(x - x_0)} \right) - \frac{f''(x_0)}{2} &(653) \\ &= \frac{f''(x_0)}{2} - \frac{f''(x_0)}{2} = 0 &(654) \end{aligned}$$

and we are done.

At this point, the pattern is pretty clear.

Theorem 5.24 (Taylor's Theorem with Peano's Form of Remainder)

If $f : [a, b] \to \mathbb{R}$ is n-times differentiable at point $x_0 \in (a, b)$, then there exists an $o\big((x - x_0)^n\big)$ function such that

$$f(x) = \left(\sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \right) + o\big((x - x_0)^n\big) \tag{655}$$

Proof. We prove by induction. The cases for $n = 0, 1, 2$ are done above. Now assume that this holds for $n - 1$. We are seeking a polynomial $P_n(x; x_0) = c_0 + c_1(x - x_0) + \ldots + c_n(x - x_0)^n$ such that

$$f(x) = c_0 + c_1(x - x_0) + \ldots + c_n(x - x_0)^n + o\big((x - x_0)^n\big) \text{ as } x \to x_0 \tag{656}$$

Since $c_n(x - x_0)^n \in o((x - x_0)^{n-1})$, we also know that $c_n(x - x_0)^n + o((x - x_0)^n) \in o((x - x_0)^{n-1})$, and so such a polynomial must also satisfy

$$f(x) = c_0 + c_1(x - x_0) + \ldots + c_{n-1}(x - x_0)^{n-1} + o\big((x - x_0)^{n-1}\big) \text{ as } x \to x_0 \tag{657}$$

Since any solution $c_0, \ldots, c_n$ must satisfy the above for $c_0, \ldots, c_{n-1}$, we can invoke our induction step for $n - 1$ to see that

$$c_k = \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k, \qquad k = 0, 1, \ldots, n - 1 \tag{658}$$

To make a guess for c_n, we do some algebra to isolate

$$c_n = \frac{f(x) - \big(c_0 + c_1(x - x_0) + \ldots + c_{n-1}(x - x_0)^{n-1} + o((x - x_0)^n)\big)}{(x - x_0)^n} \tag{659}$$

$$= \frac{f(x) - \big(c_0 + c_1(x - x_0) + \ldots + c_{n-1}(x - x_0)^{n-1}\big)}{(x - x_0)^n} - o(1) \tag{660}$$

$$\tag{661}$$

and by taking the limit, with repeated application of L'Hopital's rule $n - 1$ times,

$$c_n = \lim_{x \to x_0} \frac{f(x) - (c_0 + \ldots + c_{n-1}(x - x_0)^{n-1})}{(x - x_0)^n} = \lim_{x \to x_0} \frac{f^{(n-1)}(x) - f^{(n-1)}(x_0)}{n!(x - x_0)} = \frac{1}{n!} f^{(n)}(x_0) \tag{662}$$

Now we claim that $c_n = \frac{1}{n!} f^{(n)}(x_0)$, and all that's left to do is show that the error term

$$\varphi(x) = f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \in o\big((x - x_0)^{n-1}\big) \tag{663}$$

Indeed, we have

$$\lim_{x \to x_0} \frac{\varphi(x)}{(x - x_0)^n} = \lim_{x \to x_0} \frac{f(x) - f(x_0) - f'(x_0)(x - x_0) - \frac{1}{2} f''(x_0)(x - x_0)^2 - \ldots - \frac{1}{n!} f^{(n)}(x_0)(x - x_0)^n}{(x - x_0)^n} \tag{664}$$

$$= \lim_{x \to x_0} \frac{f(x) - f(x_0) - \ldots - \frac{1}{(n-1)!} f^{(n-1)}(x_0)(x - x_0)^{n-1}}{(x - x_0)^n} - \frac{1}{n!} f^{(n)}(x_0) \tag{665}$$

$$= \lim_{x \to x_0} \frac{f^{(n-1)}(x) - f^{(n-1)}(x_0)}{n!(x - x_0)} - \frac{1}{n!} f^{(n)}(x_0) \tag{666}$$

$$= \frac{1}{n!} f^{(n)}(x_0) - \frac{1}{n!} f^{(n)}(x_0) = 0 \tag{667}$$

Given differentiable $f : [a, b] \to \mathbb{R}$, and $x_0 \in [a, b]$, say that we want to try and approximate $f(x)$ with $f(x_0)$. Using MVT on the closed subinterval $[x_0, x]$[10], we can find a $c \in (x_0, x)$ s.t. $f(x) = f(x_0) + f'(c)(x - x_0)$. This is like a first order approximation, and perhaps we can extend this if we have an additional assumption that f is twice differentiable. To do this, recall how we proved the MVT. We subtracted a linear function from $f(x)$ to get a new function $g(x)$ satisfying $g(a) = g(b) = 0$. Now we can perform this again to get higher order approximations.

Lemma 5.25 (Taylor's Formula of Order 2 with Lagrange's Form of Remainder)

Suppose $f : [a, b] \to \mathbb{R}$
1. is twice differentiable over (a, b), and
2. is C^1 over $[a, b]$.

Then, there exists a ξ between x_0 and x s.t.

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(\xi)}{2}(x - x_0)^2 \tag{668}$$

Proof. Let us define the function

$$g(x) \coloneqq f(x) - f(a) - f'(a)(x - a) - M(x - a)^2 \tag{669}$$

[10]or $[x, x_0]$ if $x < x_0$

where M was chosen such that $g(b) = 0$. Now notice that

$$g(a) = f(a) - f(a) - 0 - 0 = 0 \tag{670}$$
$$g'(a) = f'(a) - f'(a) - 0 = 0 \tag{671}$$

and so by using Rolle's theorem on g, there exists a $c_1 \in (a, b)$ s.t.

$$g'(c_1)(b-a) = g(b) - g(a) = 0 - 0 = 0 \implies g'(c_1) = 0 \tag{672}$$

therefore, we can use Rolle's theorem again on g' and claim there exists a $c \in (a, c_1)$ s.t.

$$g''(c)(c_1 - a) = g'(c_1) - g'(a) = 0 - 0 = 0 \implies g''(c) = 0 \tag{673}$$

This gives us all we need. By taking the double derivative of g, we get

$$0 = g''(c) = f''(c) - 2M \implies M = \frac{f''(c)}{2} \tag{674}$$

and substituting this in gives

$$0 = g(b) = f(b) - f(a) - f'(a)(b-a) - \frac{f''(c)}{2}(b-a)^2 \tag{675}$$

This is like a mean value theorem for the second order, where only the final term is dependent on c. We can continue this process to get a nth order approximation.

Theorem 5.26 (Taylor's Theorem with Lagrange's Form of Remainder)

Suppose $f : [a, b] \to \mathbb{R}$
1. is $(n+1)$-times differentiable over (a, b), and
2. is C^n over $[a, b]$.

Then, there exists a ξ between x_0 and x s.t.

$$f(x) = \left(\sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \right) + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1} \tag{676}$$

Proof. We do the exact same process. Let us define

$$P(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \tag{677}$$

and set

$$g(x) \coloneqq f(x) - P(x) - M(x-a)^{n+1} \tag{678}$$

where M was chosen such that $g(b) = 0$. Now notice that evaluating on g gives us

$$g(a) = g'(a) = g''(a) = \ldots = g^{(n)}(a) = 0 \tag{679}$$

So by using Rolle's theorem on g, there exists a $c_1 \in (a, b)$ s.t. $g'(c_1) = 0$. Therefore we can use Rolle's theorem on g' to show there exists a $c_2 \in (a, c_1)$. We keep doing this until we show that there exists a $c \in (a, b)$ s.t. $g^{(n)}(c) = 0$. With this, we can directly evaluating the nth derivative of g to find

$$0 = g^{(n)}(c) = f^{(n)}(c) - n!M \implies M = \frac{f^{(n)}(c)}{n!} \tag{680}$$

5.7 Exercises

Exercise 5.1 (Rudin 5.1)

Let f be defined for all real x, and suppose that

$$|f(x) - f(y)| \leq (x - y)^2 \tag{681}$$

for all real x and y. Prove that f is constant.

Solution.

Exercise 5.2 (Rudin 5.2)

Suppose $f'(x) > 0$ in (a, b). Prove that f is strictly increasing in (a, b), and let g be its inverse function. Prove that g is differentiable, and that

$$g'(f(x)) = \frac{1}{f'(x)} \quad (a < x < b) \tag{682}$$

.

Solution.

Exercise 5.3 (Rudin 5.3)

Suppose g is a real function on R^1, with bounded derivative (say $|g'| \leq M$). Fix $\epsilon > 0$, and define $f(x) = x + \epsilon g(x)$. Prove that f is one-to-one if ϵ is small enough. (A set of admissible values of ϵ can be determined which depends only on M.)

Solution.

Exercise 5.4 (Rudin 5.4)

If

$$C_0 + \frac{C_1}{2} + \cdots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0 \tag{683}$$

where $C_0, \ldots, C_n$ are real constants, prove that the equation

$$C_0 + C_1 x + \cdots + C_{n-1} x^{n-1} + C_n x^n = 0 \tag{684}$$

has at least one real root between 0 and 1.

Solution.

Exercise 5.5 (Rudin 5.5)

Suppose f is defined and differentiable for every $x > 0$, and $f'(x) \to 0$ as $x \to +\infty$. Put $g(x) = f(x+1) - f(x)$. Prove that $g(x) \to 0$ as $x \to +\infty$.

Solution.

Exercise 5.6 (Rudin 5.6)

Suppose
(a) f is continuous for $x \geq 0$,
(b) $f'(x)$ exists for $x > 0$,
(c) $f(0) = 0$,
(d) f' is monotonically increasing.
Put

$$g(x) = \frac{f(x)}{x} \quad (x > 0) \tag{685}$$

and prove that g is monotonically increasing.

Solution.

Exercise 5.7 (Rudin 5.7)

Suppose $f'(x), g'(x)$ exist, $g'(x) \neq 0$, and $f(x) = g(x) = 0$. Prove that

$$\lim_{t \to x} \frac{f(t)}{g(t)} = \frac{f'(x)}{g'(x)} \tag{686}$$

. (This holds also for complex functions.)

Solution.

Exercise 5.8 (Rudin 5.8)

Suppose f' is continuous on $[a, b]$ and $\epsilon > 0$. Prove that there exists $\delta > 0$ such that

$$\left| \frac{f(t) - f(x)}{t - x} - f'(x) \right| < \epsilon \tag{687}$$

whenever $0 < |t - x| < \delta, a \leq x \leq b, a \leq t \leq b$. (This could be expressed by saying that f is uniformly differentiable on $[a, b]$ if f' is continuous on $[a, b]$.) Does this hold for vector-valued functions too?

Solution.

Exercise 5.9 (Rudin 5.9)

Let f be a continuous real function on R^1, of which it is known that $f'(x)$ exists for all $x \neq 0$ and that $f'(x) \to 3$ as $x \to 0$. Does it follow that $f'(0)$ exists?

Solution.

Exercise 5.10 (Rudin 5.10)

Suppose f and g are complex differentiable functions on $(0, 1)$, $f(x) \to 0, g(x) \to 0, f'(x) \to A, g'(x) \to$

B as $x \to 0$, where A and B are complex numbers, $B \neq 0$. Prove that

$$\lim_{x \to 0} \frac{f(x)}{g(x)} = \frac{A}{B} \tag{688}$$

. Compare with Example 5.18. *Hint:*

$$\frac{f(x)}{g(x)} = \left\{ \frac{f(x)}{x} - A \right\} \cdot \frac{x}{g(x)} + A \cdot \frac{x}{g(x)} \tag{689}$$

. Apply Theorem 5.13 to the real and imaginary parts of $f(x)/x$ and $g(x)/x$.

Solution.

Exercise 5.11 (Rudin 5.11)

Suppose f is defined in a neighborhood of x, and suppose $f''(x)$ exists. Show that

$$\lim_{h \to 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = f''(x) \tag{690}$$

. Show by an example that the limit may exist even if $f''(x)$ does not. *Hint:* Use Theorem 5.13.

Solution.

Exercise 5.12 (Rudin 5.12)

If $f(x) = |x|^3$, compute $f'(x), f''(x)$ for all real x, and show that $f^{(3)}(0)$ does not exist.

Solution.

Exercise 5.13 (Rudin 5.13)

Suppose a and c are real numbers, $c > 0$, and f is defined on $[-1, 1]$ by

$$f(x) = \begin{cases} x^a \sin(|x|^{-c}) & (\text{if } x \neq 0), \\ 0 & (\text{if } x = 0) \end{cases} \tag{691}$$

. Prove the following statements:

(a) f is continuous if and only if $a > 0$.
(b) $f'(0)$ exists if and only if $a > 1$.
(c) f' is bounded if and only if $a \geq 1 + c$.
(d) f' is continuous if and only if $a > 1 + c$.
(e) $f''(0)$ exists if and only if $a > 2 + c$.
(f) f'' is bounded if and only if $a \geq 2 + 2c$.
(g) f'' is continuous if and only if $a > 2 + 2c$.

Solution.

Exercise 5.14 (Rudin 5.14)

Let f be a differentiable real function defined in (a, b). Prove that f is convex if and only if f' is monotonically increasing. Assume next that $f''(x)$ exists for every $x \in (a, b)$, and prove that f is convex if and only if $f''(x) \geq 0$ for all $x \in (a, b)$.

Solution.

Exercise 5.15 (Rudin 5.15)

Suppose $a \in R^1$, f is a twice-differentiable real function on (a, ∞), and M_0, M_1, M_2 are the least upper bounds of $|f(x)|, |f'(x)|, |f''(x)|$, respectively, on (a, ∞). Prove that

$$M_1^2 \leq 4M_0M_2 \tag{692}$$

. *Hint:* If $h > 0$, Taylor's theorem shows that

$$f'(x) = \frac{1}{2h}[f(x + 2h) - f(x)] - hf''(\xi) \tag{693}$$

for some $\xi \in (x, x + 2h)$. Hence

$$|f'(x)| \leq hM_2 + \frac{M_0}{h} \tag{694}$$

. To show that $M_1^2 = 4M_0M_2$ can actually happen, take $a = -1$, define

$$f(x) = \begin{cases} 2x^2 - 1 & (-1 < x < 0), \\ \frac{x^2-1}{x^2+1} & (0 \leq x < \infty) \end{cases} \tag{695}$$

and show that $M_0 = 1, M_1 = 4, M_2 = 4$. Does $M_1^2 \leq 4M_0M_2$ hold for vector-valued functions too?

Solution.

Exercise 5.16 (Rudin 5.16)

Suppose f is twice-differentiable on $(0, \infty)$, f'' is bounded on $(0, \infty)$, and $f(x) \to 0$ as $x \to \infty$. Prove that $f'(x) \to 0$ as $x \to \infty$. *Hint:* Let $a \to \infty$ in Exercise 15.

Solution.

Exercise 5.17 (Rudin 5.17)

Suppose f is a real, three times differentiable function on $[-1, 1]$, such that

$$f(-1) = 0, \quad f(0) = 0, \quad f(1) = 1, \quad f'(0) = 0 \tag{696}$$

. Prove that $f^{(3)}(x) \geq 3$ for some $x \in (-1, 1)$. Note that equality holds for $\frac{1}{2}(x^3 + x^2)$. *Hint:* Use Theorem 5.15, with $\alpha = 0$ and $\beta = \pm 1$, to show that there exist $s \in (0, 1)$ and $t \in (-1, 0)$ such that $f^{(3)}(s) + f^{(3)}(t) = 6$.

Solution.

Exercise 5.18 (Rudin 5.18)

Suppose f is a real function on $[a, b]$, n is a positive integer, and $f^{(n-1)}$ exists for every $t \in [a, b]$. Let α, β, and P be as in Taylor's theorem (5.15). Define

$$Q(t) = \frac{f(t) - f(\beta)}{t - \beta} \tag{697}$$

for $t \in [a, b], t \neq \beta$, differentiate $f(t) - f(\beta) = (t - \beta)Q(t)$ $n - 1$ times at $t = \alpha$, and derive the following version of Taylor's theorem:

$$f(\beta) = P(\beta) + \frac{Q^{(n-1)}(\alpha)}{(n-1)!}(\beta - \alpha)^n \tag{698}$$

.

Solution.

Exercise 5.19 (Rudin 5.19)

Suppose f is defined in $(-1, 1)$ and $f'(0)$ exists. Suppose $-1 < \alpha_n < \beta_n < 1$, $\alpha_n \to 0$, and $\beta_n \to 0$ as $n \to \infty$. Define the difference quotients

$$D_n = \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n} \tag{699}$$

. Prove the following statements:

(a) If $\alpha_n < 0 < \beta_n$, then $\lim D_n = f'(0)$.

(b) If $0 < \alpha_n < \beta_n$ and $\{\beta_n/(\beta_n - \alpha_n)\}$ is bounded, then $\lim D_n = f'(0)$.

(c) If f' is continuous in $(-1, 1)$, then $\lim D_n = f'(0)$.

Give an example in which f is differentiable in $(-1, 1)$ (but f' is not continuous at 0) and in which α_n, β_n tend to 0 in such a way that $\lim D_n$ exists but is different from $f'(0)$.

Solution.

Exercise 5.20 (Rudin 5.22)

Suppose f is a real function on $(-\infty, \infty)$. Call x a *fixed point* of f if $f(x) = x$.

(a) If f is differentiable and $f'(t) \neq 1$ for every real t, prove that f has at most one fixed point.

(b) Show that the function f defined by

$$f(t) = t + (1 + e^t)^{-1} \tag{700}$$

has no fixed point, although $0 < f'(t) < 1$ for all real t.

(c) However, if there is a constant $A < 1$ such that $|f'(t)| \leq A$ for all real t, prove that a fixed point x of f exists, and that $x = \lim x_n$, where x_1 is an arbitrary real number and $x_{n+1} = f(x_n)$ for $n = 1, 2, 3, \ldots$.

(d) Show that the process described in (c) can be visualized by the zig-zag path $(x_1, x_2) \to (x_2, x_2) \to (x_2, x_3) \to (x_3, x_3) \to (x_3, x_4) \to \ldots$.

Solution.

Exercise 5.21 (Rudin 5.23)

The function f defined by

$$f(x) = \frac{x^3 + 1}{3} \tag{701}$$

has three fixed points, say α, β, γ, where

$$-2 < \alpha < -1, \quad 0 < \beta < 1, \quad 1 < \gamma < 2 \tag{702}$$

. For arbitrarily chosen x_1, define $\{x_n\}$ by setting $x_{n+1} = f(x_n)$.

(a) If $x_1 < \alpha$, prove that $x_n \to -\infty$ as $n \to \infty$.
(b) If $\alpha < x_1 < \gamma$, prove that $x_n \to \beta$ as $n \to \infty$.
(c) If $\gamma < x_1$, prove that $x_n \to +\infty$ as $n \to \infty$.

Thus β can be located by this method, but α and γ cannot.

Solution.

Exercise 5.22 (Rudin 5.25)

Suppose f is twice differentiable on $[a, b]$, $f(a) < 0, f(b) > 0, f'(x) \geq \delta > 0$, and $0 \leq f''(x) \leq M$ for all $x \in [a, b]$. Let ξ be the unique point in (a, b) at which $f(\xi) = 0$. Complete the details in the following outline of *Newton's method* for computing ξ.

(a) Choose $x_1 \in (\xi, b)$, and define $\{x_n\}$ by

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{703}$$

. Interpret this geometrically, in terms of a tangent to the graph of f.

(b) Prove that $x_{n+1} < x_n$ and that $\lim x_n = \xi$.

(c) Use Taylor's theorem to show that

$$x_{n+1} - \xi = \frac{f''(t_n)}{2f'(x_n)}(x_n - \xi)^2 \tag{704}$$

for some $t_n \in (\xi, x_n)$.

(d) If $A = M/2\delta$, deduce that

$$0 \leq x_{n+1} - \xi \leq \frac{1}{A}[A(x_1 - \xi)]^{2^n} \tag{705}$$

.

(e) Show that Newton's method amounts to finding a fixed point of the function g defined by

$$g(x) = x - \frac{f(x)}{f'(x)} \tag{706}$$

. How does $g'(x)$ behave for x near ξ?

(f) Put $f(x) = x^{1/3}$ on $(-\infty, \infty)$ and try Newton's method. What happens?

Solution.

Exercise 5.23 (Rudin 5.26)

Suppose f is differentiable on $[a, b], f(a) = 0$, and there is a real number A such that $|f'(x)| \leq A|f(x)|$ on $[a, b]$. Prove that $f(x) = 0$ for all $x \in [a, b]$. *Hint:* Fix $x_0 \in [a, b]$, let

$$M_0 = \sup |f(x)|, \quad M_1 = \sup |f'(x)| \tag{707}$$

for $a \leq x \leq x_0$. For any such x, $|f(x)| \leq M_1(x_0 - a) \leq A(x_0 - a)M_0$. Hence $M_0 = 0$ if $A(x_0 - a) < 1$. That is, $f = 0$ on $[a, x_0]$. Proceed.

Solution.

Exercise 5.24 (Rudin 5.27)

Let ϕ be a real function defined on a rectangle R in the plane, given by $a \leq x \leq b, \alpha \leq y \leq \beta$. A *solution of the initial-value problem*

$$y' = \phi(x, y), \quad y(a) = c \quad (\alpha \leq c \leq \beta) \tag{708}$$

is, by definition, a differentiable function f on $[a, b]$ such that $f(a) = c, \alpha \leq f(x) \leq \beta$, and

$$f'(x) = \phi(x, f(x)) \quad (a \leq x \leq b) \tag{709}$$

. Prove that such a problem has at most one solution if there is a constant A such that

$$|\phi(x, y_2) - \phi(x, y_1)| \leq A|y_2 - y_1| \tag{710}$$

whenever $(x, y_1) \in R$ and $(x, y_2) \in R$. *Hint:* Apply Exercise 26 to the difference of two solutions. Note that this uniqueness theorem does not hold for the initial-value problem $y' = y^{1/2}, y(0) = 0$, which has two solutions: $f(x) = 0$ and $f(x) = x^2/4$. Find all other solutions.

Solution.

Exercise 5.25 (Rudin 5.28)

Formulate and prove an analogous uniqueness theorem for systems of differential equations of the form

$$y_j' = \phi_j(x, y_1, \ldots, y_k), \quad y_j(a) = c_j \quad (j = 1, \ldots, k) \tag{711}$$

. Note that this can be rewritten in the form

$$\mathbf{y}' = \mathbf{\Phi}(x, \mathbf{y}), \quad \mathbf{y}(a) = \mathbf{c} \tag{712}$$

where $\mathbf{y} = (y_1, \ldots, y_k)$ ranges over a k-cell, $\mathbf{\Phi}$ is the mapping of a $(k+1)$-cell into the Euclidean k-space whose components are the functions $\phi_1, \ldots, \phi_k$, and $\mathbf{c}$ is the vector $(c_1, \ldots, c_k)$. Use Exercise 26, for vector-valued functions.

Solution.

Exercise 5.26 (Rudin 5.29)

Specialize Exercise 28 by considering the system

$$y_j' = y_{j+1} \quad (j = 1, \dots, k-1) \tag{713}$$

,

$$y_k' = f(x) - \sum_{j=1}^{k} g_j(x) y_j \tag{714}$$

, where $f, g_1, \dots, g_k$ are continuous real functions on $[a, b]$, and derive a uniqueness theorem for solutions of the equation

$$y^{(k)} + g_k(x) y^{(k-1)} + \cdots + g_2(x) y' + g_1(x) y = f(x) \tag{715}$$

, subject to initial conditions

$$y(a) = c_1, \quad y'(a) = c_2, \quad \dots, \quad y^{(k-1)}(a) = c_k \tag{716}$$

.

Solution.

Exercise 5.27 (Math 531 Spring 2025, PS7.5)

Coming back to Problem 1 above, assume that f is differentiable on $[0, 1]$ and that $|f'(x)| \leq M$ for every $x \in [0, 1]$. Prove that we can take $\delta(\epsilon) = \frac{\epsilon}{M}$.

Solution.

Exercise 5.28 (Math 531 Spring 2025, PS7.7)

Suppose that $f : [0, 2] \to [0, 2]$ is twice differentiable. Suppose $f(0) = 0$, $f(1) = 1$, and $f(2) = 2$. Prove that there exists $c \in (0, 2)$ so that $f''(c) = 0$.

Solution.

Exercise 5.29 (Math 531 Spring 2025, PS7.8)

Suppose that f is differentiable on $[0, 1]$ satisfies:

$$f'(x) = f(x). \tag{717}$$

- Prove that f is automatically infinitely differentiable.
- Let $M = \max\{f(x) : x \in [0, 1]\}$. Why does M exist? Similarly, let $M(\epsilon) = \max\{|f(x)| : x \in [0, \epsilon]\}$.
- Prove that for all $x \in [0, \epsilon]$, we have that

$$|f(x)| \leq M(\epsilon) x + |f(0)|. \tag{718}$$

- Deduce that if $f(0) = 0$, we have that

$$M\left(\frac{1}{2}\right) = 0. \tag{719}$$

Similarly, deduce that $M(c) = 0$ for all $c \in [0, 1]$.
- What you have just proved is that if $f'(x) = f(x)$ while $f(0) = 0$, it follows that $f(x) = 0$ for all x.
- Assume that $E : \mathbb{R} \to [0, \infty)$ is differentiable and

$$|E'(t)| \leq CE(t). \tag{720}$$

Prove that if $E(0) = 0$, then $E(t) = 0$ for all t.

Solution.

Exercise 5.30 (Math 531 Spring 2025, PS8.1)

Prove that if $f : [0, 1] \to \mathbb{R}$ is differentiable and $f' > 0$ on $[0, 1]$, then f is strictly increasing on $[0, 1]$.

Solution.

Exercise 5.31 (Math 531 Spring 2025, PS8.2)

Explain why if $x(t)$ represents the position of a particle at time t, $x'(t)$ is called the velocity of the particle, $x''(t)$ is called its acceleration, and $x'''(t)$ is called its jerk.

Solution.

Exercise 5.32 (Math 531 Spring 2025, PS8.3)

A function $f : \mathbb{R} \to \mathbb{R}$ is said to be T periodic if $f(t + T) = f(t)$, for all $t \in \mathbb{R}$. Now, given a function $\tilde{f}$ defined on $[0, T)$, we can always extend $\tilde{f}$ to be T periodic in the following way. First, we can write

$$\mathbb{R} = \bigcup_{n \in \mathbb{Z}} [nT, (n + 1)T). \tag{721}$$

Then define for $t \in [nT, (n + 1)T)$:

$$f(t) = \tilde{f}(t - nT). \tag{722}$$

- Show that f defined this way is T periodic.
- Suppose $\tilde{f}$ is continuous on $[0, T)$. Does this mean that its extension f will also be continuous on $\mathbb{R}$? What condition do you have to add?
- Suppose $\tilde{f}$ is continuously differentiable on $[0, T)$. What conditions do you have to put on $\tilde{f}$ to ensure that f is continuously differentiable? How about k-times continuously differentiable?

Solution.

Exercise 5.33 (Math 531 Spring 2025, PS8.4)

Assume $f : \mathbb{R} \to \mathbb{R}$ is differentiable and $|f'(x)| \leq \frac{1}{1+x^2}$. Prove that

$$\lim_{x \to \infty} f(x) \tag{723}$$

exists. You are not allowed to use integration. Hint: How do you prove convergence without knowing what the limit is?

Solution.

Exercise 5.34 (Math 531 Spring 2025, PS8.5)

In a previous homework assignment, we defined the function

$$E(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}. \tag{724}$$

We proved it is continuous, satisfies $E(z+w) = E(z)E(w)$ for all $z, w \in \mathbb{C}$, and then deduced that for all $x \in \mathbb{R}$, we have that $E(x) = e^x$.

- Prove that $E'(0) = 1$. Hint: write $\frac{H(x)-H(0)}{x} = \frac{1}{x}\sum_{n=1}^{\infty} \frac{x^n}{n!} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{n!}$. Then prove that
$$\lim_{x \to 0} \sum_{n=1}^{\infty} \frac{x^{n-1}}{n!} = 1. \tag{725}$$
- By studying the difference quotient, prove that $E'(x) = E(x)$ for all $x \in \mathbb{R}$. This is much easier than the preceding point.
- Prove that $\lim_{x\to\infty} E(x) = \infty$ and $\lim_{x\to-\infty} E(x) = 0$.
- Prove that $E : (-\infty, \infty) \to (0, \infty)$ is 1-1 and onto.
- Let $L = E^{-1} : (0, \infty) \to (-\infty, \infty)$. Prove that
$$L'(t) = \frac{1}{t}, \tag{726}$$
for all $t \in (0, \infty)$.

Solution.

Exercise 5.35 (Math 531 Spring 2025, PS8.6)

For each $k \in \mathbb{N}$, consider $\sum_{j=1}^{N} j^k$. It turns out that this can be expressed as a polynomial of degree $k+1$ in N. For example,

$$\sum_{j=1}^{N} j^0 = N, \tag{727}$$

is a polynomial of degree 1 in N. Similarly,

$$\sum_{j=1}^{N} j = \frac{N(N+1)}{2} = \frac{N^2}{2} + \frac{N}{2}, \tag{728}$$

is a polynomial of degree 2 in N. If we write:

$$\sum_{j=1}^{N} j^k = a_0 + a_1 N + a_2 N^2 + ... + a_{k+1} N^{k+1}, \tag{729}$$

what is the value of a_{k+1}? Hint: divide by N^{k+1} and take the limit as $N \to \infty$.

Solution.

Exercise 5.36 (Math 531 Spring 2025, PS9.3)

Assume we have a twice differentiable function $f : [0, \infty) \to \mathbb{R}$. Assume that $f''(x) > 0$ for all $x \in [0, \infty)$, while $f'(0) \geq 0$. Prove that $\lim_{x \to \infty} f(x) = +\infty$.

Solution.

Exercise 5.37 (Math 531 Spring 2025, PS9.4)

We proved that the function $E : \mathbb{C} \to \mathbb{C}$ defined by:

$$E(z) = \sum_{k=0}^{\infty} \frac{z^k}{k!} \tag{730}$$

satisfies $E(z + w) = E(z)E(w)$. Let us now investigate the real and imaginary parts of $E(it)$, where $t \in \mathbb{R}$. Let us call the real part $C(t)$ and the imaginary part $S(t)$.

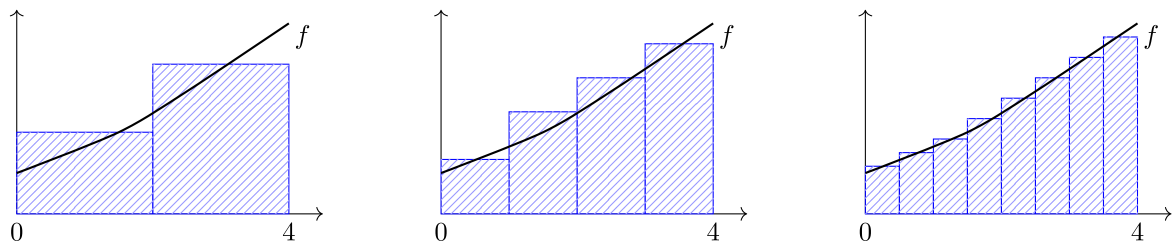
- Prove that $E(\overline{z}) = \overline{E(z)}$, for every $z \in \mathbb{C}$.
- Deduce that $|E(it)| = 1$ for all $t \in \mathbb{R}$ and thus:
$$C(t)^2 + S(t)^2 = 1, \tag{731}$$
for all $t \in \mathbb{R}$.
- Prove that $C(-t) = C(t)$ for all t and that $S(-t) = -S(t)$ for all t.
- Prove that $C(0) = 1$ and $S(0) = 0$, while $C'(t) = -S(t)$ and $S'(t) = C(t)$, for all $t \in \mathbb{R}$.
- Deduce that $C''(t) = -C(t)$ and prove that there must be some $t > 0$ for which $C(t) = 0$. (Hint: Use Problem 3)
- Prove that there is a smallest $t_* > 0$ for which $C(t_*) = 0$.
- Define $\pi = 2t_*$ so that $C(\frac{\pi}{2}) = 0$. Since S is increasing on $[0, t_*]$, deduce that $S(\frac{\pi}{2}) = 1$.
- Now use the formula $E(z + w) = E(z)E(w)$ to deduce that
$$C(t + 2\pi) = C(t), \quad S(t + 2\pi) = S(t), \tag{732}$$
for all $t \in \mathbb{R}$.
- It is now reasonable to unveil that C and S are none other but our old friends: cos and sin.

Solution.

6 Riemann Integration

Now we will construct the theory of integration, which can be informally thought of as an "summation for uncountable sets." Note that as of now, integration is a completely independent construction from the derivative. That is, we could have established integration first before differentiation. The unification of the two comes from the fundamental theorem of calculus.

If you have taken high school calculus, we can do this by defining the "Riemann sums" which is just the sum of the areas of a bunch of rectangles. By taking increasingly finer and finer rectangles, we can approximate the actual function. Note that this notion of "approximation" is not just a colloquial term. We will precisely characterize this, since we already established our theory of limits and suprema/infima. Let's begin.



(a) (b) (c)

Figure 30: Approximating an integral with increasingly fine partitions

6.1 Darboux Integration

Definition 6.1 (Partition)

Let $[a, b]$ be an interval. A **partition** P of $[a, b]$ is a finite set of $P = \{x_0, \ldots, x_n\}$ s.t.

$$a = x_0 \leq x_1 \leq \ldots \leq x_{n-1} \leq x_n = b \tag{733}$$

with $\Delta x_i = [x_{i-1}, x_i]$ for $i = 1, \ldots, n$.

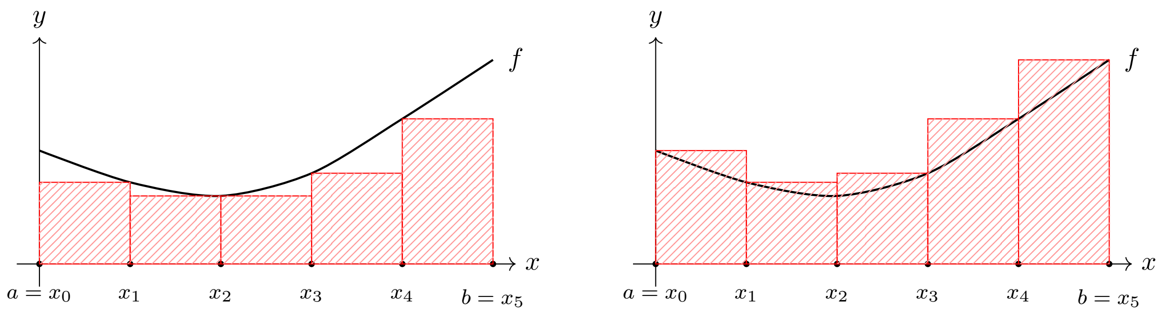
The cleanest way is to simply look at the set of all partitions along with the set of the corresponding upper and lower Riemann sums, and then hope that they behave nicely with each other. This is the approach we will take.

Definition 6.2 (Darboux Sums with Respect to Partition)

Let P be a partition of $[a, b]$ and $f : [a, b] \to \mathbb{R}$ be bounded. Then, the **upper and lower Darboux sum** is defined

$$U(P, f) = \sum_{i=1}^{n} \Big(\sup_{[x_{i-1}, x_i]} f \Big) \big[x_i - x_{i-1}\big] \tag{734}$$

$$L(P, f) = \sum_{i=1}^{n} \Big(\inf_{[x_{i-1}, x_i]} f \Big) \big[x_i - x_{i-1}\big] \tag{735}$$



(a) Lower Riemann sum.

(b) Upper Riemann sum.

Figure 31

Definition 6.3 (Darboux Integral)

Let $f : [a, b] \to \mathbb{R}$ be bounded. Then, the **upper and lower Darboux integrals** of $f(x)$ are defined to be the real numbers

$$\underline{\int_a^b} f \coloneqq \inf_P U(P, f), \qquad \overline{\int_a^b} f \coloneqq \sup_P L(P, f) \tag{736}$$

If the upper and lower Riemann integrals are equal, then f is said to be **Darboux integrable** over $[a, b]$, denoted $f \in \mathcal{R}([a, b])$, and we write the integral as

$$\int_a^b f \tag{737}$$

Proof. Note that f is bounded by assumption, so it must satisfy $m \leq f \leq M$ on $[a, b]$. Therefore, it is not hard to see that

$$m(b-a) \leq \int \underline{\int_a^b} f, \qquad \overline{\int_a^b} f \leq M(b-a) \tag{738}$$

Since every upper (lower) bounded set in $\mathbb{R}$ has a least upper (greatest lower) bound, it is Darboux integrable.

Note that we prefer to write $\int_a^b f$ rather than $\int_a^b f(x)\,dx$, which contains the variable of integration. The integral depends on a, b, f, but not x, and so the role played by x is analogous to that of the index of summation ($\sum_i c_i = \sum_j c_j$).

Just from a manipulation of definition, we would like to characterize conditions for integrability in a way that is easy to present. This culminates in the Cauchy criterion for integrability.

Definition 6.4 (Refinement)

P^* is a **refinement** of P if $P \subset P^*$. If P_1, P_2 are two partitions, then their **common refinement** $P^* = P_1 \cup P_2$.

Lemma 6.1 (Fundamental Lemma)

If P^* is a refinement of P and $f : [a, b] \to \mathbb{R}$ is bounded, then

$$L(P, f) \leq L(P^*, f) \leq U(P^*, f) \leq U(P, f) \tag{739}$$

Proof. By induction on the number of points we add to P to get P^*, we might as well assume that $P^* = P \cup \{x_*\}$. So,

$$P = \{a = x_0, x_1, \ldots, x_{n-1}, x_n\} \tag{740}$$

$$P^* = \{a = x_0, x_1, \ldots, x_{i-1}, x_i, x_*, x_{i+1}, \ldots, x_{n-1}, x_n\} \tag{741}$$

$$\tag{742}$$

Now let's compute $L(f, P^*) - L(f, P)$. Since the only intervals affected are $[x_i, x_{i+1}]$, we have

$$L(f, P^*) - L(f, P) = \inf_{[x_i, x_*]} f(x)(x_* - x_i) + \inf_{[x_*, x_{i+1}]} f(x)(x_{i+1} - x_*) - \inf_{[x_i, x_{i+1}]} f(x)(x_{i+1} - x_i) \tag{743}$$

$$= \Big(\underbrace{\inf_{[x_i, x_*]} f(x) - \inf_{[x_*, x_{i+1}]} f(x)}_{>0}\Big)(x_* - x_i) + \Big(\underbrace{\inf_{[x_*, x_{i+1}]} f(x) - \inf_{[x_i, x_{i+1}]}}_{>0}\Big)(x_{i+1} - x_*) \tag{744}$$

which is therefore greater than 0.

Theorem 6.2

We claim

$$\underline{\int_a^b} f \leq \overline{\int_a^b} f \tag{745}$$

Proof. Given P_1, P_2 partitions, let $P^* = P_1 \cup P_2$ be their common refinement. Then, from the theorem above,

$$L(P_2, f) \leq L(P^*, f) \leq U(P^*, f) \leq U(P, f) \tag{746}$$

So taking the supremum over all partitions P_2 and fixing P_1 gives

$$\underline{\int_a^b} f = \sup_{P_2} L(P_2, f) \leq \sup_{P_2} U(P_1, f) = U(P_1, f) \tag{747}$$

Then taking the infimum over all partitions P_1 gives us

$$\underline{\int_a^b} f(x) = \inf_{P_1} \underline{\int_a^b} f(x) \leq \inf_{P_1} U(P_1, f) = \overline{\int_a^b} f(x) \tag{748}$$

where we note that the infimum does not affect the terms that do not depend on P_1.

We have seen some bounds of the upper and lower integrals, and defined the Riemann integral. However, checking Riemann integrability is quite tedious, since we have to take the supremum and infimum over all possible partitions. The following theorem is extremely useful as it only requires us to find *one* partition given some ϵ. This is because that the Riemann integral, as complicated as the formula is, is still a limit of a function. That means that we can apply the Cauchy criterion to it to determine convergence.

Theorem 6.3 (Cauchy Criterion for Riemann Integrability)

$f \in \mathcal{R}$ iff $\forall \epsilon > 0$, there exists partition P such that $U(P, f) - L(P, f) < \epsilon$.

Proof. We prove bidirectionally. The reverse implication is easy, but for the forward direction you must use refinements.

1. $(\leftarrow)$. Pick any partition P. Since
$$L(f, P) \leq \underline{\int_a^b} f \leq \overline{\int_a^b} f \leq U(f, P) \tag{749}$$
This implies that
$$0 \leq \overline{\int_a^b} f - \underline{\int_a^b} f \leq U(f, P) - L(f, P) < \epsilon \tag{750}$$
and since any nonnegative number less than any positive number must be 0 (since there are no infinitesimals in $\mathbb{R}$), the LHS is 0, and the result is proven.
2. $(\rightarrow)$. f is Riemann integrable, so
$$\overline{\int_a^b} f = \underline{\int_a^b} f \iff \inf_P U(f, P) = \sup_Q L(f, Q) \tag{751}$$
for partitions P, Q. So we can find P that gets really close to the infimum and same for Q close to the supremum, i.e. there exists a P, Q, such that
$$U(f, P) < \overline{\int_a^b} f + \frac{\epsilon}{2}, \qquad L(f, Q) > \underline{\int_a^b} f - \frac{\epsilon}{2} \tag{752}$$
Now take the common refinement $P^* = P \cup Q$, and so by the fundamental lemma,
$$\underline{\int_a^b} f - \frac{\epsilon}{2} < L(f, Q) \leq L(f, P^*) \leq U(f, P^*) \leq U(f, P) < \overline{\int_a^b} f + \frac{\epsilon}{2} \tag{753}$$
which implies that $0 \leq U(f, P^*) - L(f, P^*) < \epsilon$.

Now, let's do some computation.

Example 6.1

The function
$$f : [0, 1] \rightarrow \mathbb{R}, \quad f(x) = \begin{cases} \frac{1}{x} & \text{if } x \in (0, 1] \\ \alpha & \text{if } x = 0 \end{cases} \tag{754}$$
is not Riemann integrable over $[0, 1]$ for all values of α since f is unbounded around 0.

Here is a classic example of a non-integrable function.

Example 6.2 (Non-Integrability of the Dirichlet Function)

The Dirichlet function

$$\mathcal{D}(x) := \begin{cases} 1, & \text{for } x \in \mathbb{Q} \\ 0, & \text{for } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases} \tag{755}$$

on the interval $[0, 1]$ is not integrable on that interval. For any partition P of $[0, 1]$ we can find in each interval (x_{i-1}, x_i) both a rational point ξ_i' and an irrational point ξ_i''. Then, we can see that the lower and upper Darboux sums do not necessarily converge to each other since

$$\sigma(f; P, \xi') = \sum_{i=1}^n 1 \cdot (x_i - x_{i-1}) = 1, \qquad \sigma(f; P, \xi'') = \sum_{i=1}^n 0 \cdot (x_i - x_{i-1}) = 0 \tag{756}$$

as $\lambda(P) \to 0$.

The most important properties of integrable functions is linearity, monotonicity, and additivity.

Theorem 6.4 (Axiomatic Properties of Darboux Integral)

Let f, g be real-valued bounded functions over bounded interval $[a, b]$. Then, the Darboux integral satisfies the following.

1. *Linearity.* Let $c \in \mathbb{R}$ be a constant. Then

$$\int_a^b f + \int_a^b g = \int_a^b (f + g), \qquad \int_a^b cf = c \int_a^b f \tag{757}$$

2. *Monotonicity.* If $f \leq g$ on $[a, b]$, then

$$\int_a^b f \leq \int_a^b g \tag{758}$$

3. *Additivity.* For $a < c < b$,

$$\int_a^c f + \int_c^b f = \int_a^b f \tag{759}$$

Proof. Listed.

1. *Linearity.* We prove the two properties.
   (a) If $c \in \mathbb{R}$ and $f \in \mathcal{R}$, then we wish to show that $cf \in \mathcal{R}$ and $\int cf = c \int f$.
      i. If $c > 0$, then $U(cf, P) = cU(f, P)$, and $L(cf, P) = cL(f, P)$.
      ii. If $c < 0$, then $U(cf, P) = cL(f, P)$, and $L(cf, P) = cU(f, P)$.
   So, for all $\epsilon > 0$, we can find P s.t.

$$U(f, P) - L(f, P) < \frac{\epsilon}{c} \implies U(cf, P) - L(cf, P) < \epsilon \tag{760}$$

   and so $cf \in \mathcal{R}$
   (b) If $f_1, f_2 \in \mathcal{R}$, then

$$\operatorname{osc}_E(f_1 + f_2) \leq \operatorname{osc}_E(f_1) + \operatorname{osc}_E(f_2) \text{ since } \begin{cases} \sup_E(f_1 + f_2) \leq \sup_E(f_1) + \sup_E(f_2) \\ \inf_E(f_1 + f_2) \geq \inf_E(f_1) + \inf_E(f_2) \end{cases} \tag{761}$$

   for all $E \subset [a, b]$, which implies that $f_1 + f_2 \in \mathcal{R}$.
2. *Monotonicity.*
3. *Additivity.* Let P be a partition of $[a, b]$. If $c \in P$, then we can view $P = P_1 \cup P_2$. If $c \notin P$,

consider $P \cup \{c\}$. Then we have

$$U(f, P) = U(f, P_1) + U(f, P_2) \tag{762}$$
$$U(f, P) = L(f, P_1) + L(f, P_2) \tag{763}$$

So $f \in \mathcal{R}([a, c])$, $f \in \mathcal{R}([c, b])$.

Proof. Removing the a, b for convenience, we first show that $\int f_1 + f_2 = \int f_1 + \int f_2$. Let $\epsilon > 0$. Then there exists P_i s.t.

$$U(f_i, P_i) < \int f_i + \epsilon \tag{764}$$

for $i = 1, 2$. Define $P = P_1 \cup P_2$ as the common refinement. Then

$$U(f_i, P) < \int f_i + \epsilon \tag{765}$$

and so

$$\int f_1 + f_2 \leq U(f_1 + f_2, P) \leq U(f_1, P) + U(f_2, P) \tag{766}$$
$$\leq 2\epsilon + \int f_1 + \int f_2 \tag{767}$$

which implies $\int f_1 + f_2 \leq \int f_1 + \int f_2$. To prove the other way, we see that

$$\int (-f_1) + (-f_2) \leq \int (-f_1) + \int (-f_2) \tag{768}$$

and so

$$-\int f_1 + f_2 \leq -\bigg(\int f_1 + \int f_2 \bigg) \implies \int f_1 + f_2 \geq \int f_1 + \int f_2 \tag{769}$$

For scalar multiplication, we can do similarly.

Note that by monotonicity, this immediately implies that given constants m, M such that $m \leq f(x) \leq M$ at each $x \in [a, b]$, we have

$$m \cdot (b - a) \leq \int_a^b f(x) \leq M \cdot (b - a) \tag{770}$$

In particular, if $0 \leq f(x)$ on $[a, b]$, then $0 \leq \int_a^b f(x)$.

Theorem 6.5 (Restrictions of Integrable Functions)

The restriction of f in any $[c, d] \subset [a, b]$, denoted $f\big|_{[c,d]}$, is in $\mathcal{R}[c, d]$

Proof. We can prove this by proving that the integral satisfies 6.3.

6.2 Riemann Integration

The rest of this section is not really necessary, but it's good to know for historical reasons and because some textbooks still default to the Riemann integral. Historically, a slightly earlier form of the integral invented by Riemann is properly known as the *Riemann integral*. This is constructed using a tagged partition.

Definition 6.5 (Tagged Partition)

A **tagged partition** (P, ξ) of interval $[a, b]$ is a partition P with distinguished points $\xi_i \in (x_{i-1}, x_i)$ that lands in each interval.

Definition 6.6 (Riemann Sums with Respect to Tagged Partition)

Let $f : [a, b] \to \mathbb{R}$ be bounded and (P, ξ) be a tagged partition of $[a, b]$. Then, the **Riemann sum with respect** to P is

$$S(f, P) = \sum_{i=1}^{n} f(\xi_i)(x_i - x_{i-1}) \tag{771}$$

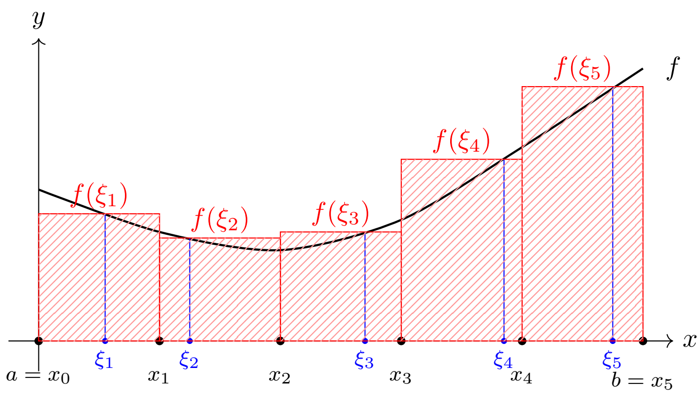


Figure 32: Riemann sum approximation using sample points ξ_i within each subinterval. This is known as a Riemann sum of a partition with distinguished points. The Riemann sum is a mapping that takes in a partition with distinguished points $p = (P, \xi)$ on the closed interval $[a, b]$ and outputs a number representing the total area of the Riemann sums.

Note that the natural way to define the Riemann integral is as the limit of the finite Riemann sums as partitions gets finer and finer. But we must be careful in saying what "finer" means. It is not simply as the number of partitions $n \to \infty$, since this may lead to multiple subsequential values of convergence by increasing the partition within different subsets of $[a, b]$.

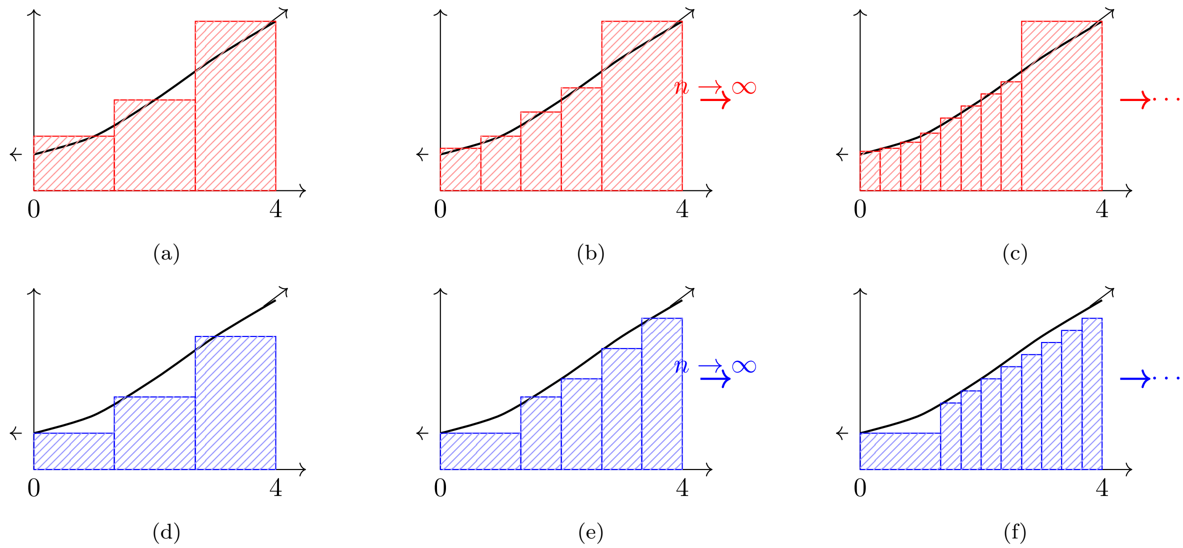


Figure 33: Upper (top row) and lower (bottom row) Riemann sums with refinement of partition. In the upper row, the rightmost rectangle remains fixed while other rectangles become thinner. In the lower row, the leftmost rectangle remains fixed while other rectangles become thinner.

Definition 6.7 (Mesh of a Partition)

Given a partition P, the **mesh** $\lambda(P)$ is the length of the largest subinterval.

An alternative way is to have the partitions all converge "uniformly" as in the maximum length of an interval in a partition must go to 0.

This allows for extra degrees of freedom for choosing points, but it is generally not needed here.

Theorem 6.6 (Equivalence of Riemann and Darboux Integrals)

Let $f : [a, b] \to \mathbb{R}$ be a bounded function. Then, f is Riemann integrable iff it is Darbox integrable. Furthermore, if f is integrable, then its Riemann and Darboux integrals are equal.

Proof.

6.3 Riemann-Stieltjes Integration

The Riemann-Stieltjes integral is a less popular but still useful integration method that generalizes the Riemann integral. The construction is exactly the same, but now we consider a *second* function φ that determines the size of these intervals. This is generally used in physics a lot, and it is a must-know if you are doing physics.

Definition 6.8 (Riemann-Stieltjes Sums with Respect to Partition)

Let P be a partition of $[a, b]$, $f : [a, b] \to \mathbb{R}$ be bounded, and φ be a monotonically increasing function on $[a, b]$. Then, the **upper and lower Riemann-Stieltjes sums** are defined

$$U(P, f, \varphi) = \sum_{i=1}^{n} \Big(\sup_{[x_{i-1}, x_i]} f \Big) \big[\varphi(x_i) - \varphi(x_{i-1}) \big] \tag{772}$$

$$L(P, f, \varphi) = \sum_{i=1}^{n} \Big(\inf_{[x_{i-1}, x_i]} f \Big) \big[\varphi(x_i) - \varphi(x_{i-1}) \big] \tag{773}$$

Definition 6.9 (Riemann-Stieltjes Integral)

Let $f : [a, b] \to \mathbb{R}$ be bounded and φ monotonically increasing on $[a, b]$. Then, the **upper and lower Riemann-Stieltjes integrals** of f are defined

$$\underline{\int_a^b} f \coloneqq \inf_P U(P, f, \varphi), \qquad \overline{\int_a^b} f \coloneqq \sup_P L(P, f, \varphi) \tag{774}$$

If the upper and lower Riemann integrals are equal, then f is said to be **Riemann-Stieltjes integrable** over $[a, b]$, denoted $f \in \mathcal{R}([a, b], \varphi)$, and we write the integral as

$$\int_a^b f \, d\varphi \tag{775}$$

We rederive the same properties again.

Lemma 6.7 (Fundamental Lemma)

If P^* is a refinement of P and $f : [a, b] \to \mathbb{R}$ is bounded, then

$$L(P, f) \leq L(P^*, F) \leq U(P^*, F) \leq U(P, f) \tag{776}$$

Proof. Ommitted, since it is exactly the same as before.

Theorem 6.8

We claim

$$\underline{\int_a^b} f \leq \overline{\int_a^b} f \tag{777}$$

Proof. Ommitted, since it is exactly the same as before.

Theorem 6.9 (Cauchy Criterion for Riemann-Stieltjes Integrability)

$f \in \mathcal{R}$ iff $\forall \epsilon > 0$, there exists partition P such that $U(P, f) - L(P, f) < \epsilon$.

Proof. Ommitted.

Theorem 6.10 (Axiomatic Properties of Riemann-Stieltjes Integral)

Let f, g be real-valued bounded functions over bounded interval $[a, b]$, and g a monotonic function over $[a, b]$. Then, the Riemann-Stieltjes satisfies the following.

1. *Linearity.* Let $c \in \mathbb{R}$ be a constant. Then,
2. *Monotonicity.*
3. *Additivity.*

Proof. Listed.

1. *Linearity.*
2. *Monotonicity.*
3. *Additivity.*

6.4 Conditions for Riemann Integrability

Note that a necessary condition of f being Riemann integrable is that f is bounded. In fact it is defined that way. You may know that a sufficient condition of integrability is that it is continuous, but we can prove something slightly weaker.

Intuitively, a function f is Riemann integrable if we can make $U(f, P) - L(f, P)$ as small as we wish. This is the case if we can find a sufficiently refined partition P such that the oscillation on f on each interval is small.

Lemma 6.11 (Functions with Vanishing Osillations are Riemann Integrable)

Let f be a bounded on a closed interval $[a, b]$. If, for every $\epsilon > 0$, there exists a partition P such that

$$\sum_{i=0}^{n-1} \operatorname*{osc}_{[x_i, x_{i+1}]} f < \epsilon \tag{778}$$

then f is Riemann integrable.

Proof. Given $\epsilon > 0$, choose $\epsilon/(b-a)$. By assumption we can find a partition P in which the total oscillation is bounded above by $\epsilon/(b-a)$. Therefore,

$$U(P, f) - L(P, f) = \sum_{i=0}^{n-1} \sup_{[x_i, x_{i+1}]} f(x) \Delta x_i - \sum_{i=0}^{n-1} \inf_{[x_i, x_{i+1}]} f(x) \Delta x_i \tag{779}$$

$$= \sum_{i=0}^{n-1} (\sup_{[x_i, x_{i+1}]} f(x) - \inf_{[x_i, x_{i+1}]} f(x)) \Delta x_i \tag{780}$$

$$< \sum_{i=0}^{n-1} \operatorname*{osc}_{[x_i, x_{i+1}]} f \Delta x_i \tag{781}$$

$$\leq \sum_{i=0}^{n-1} \frac{\epsilon}{b-a} \Delta x_i \tag{782}$$

$$= \frac{\epsilon}{b-a} \sum_{i=0}^{n-1} \Delta x_i \tag{783}$$

$$= \frac{\epsilon}{b-a} (b-a) = \epsilon \tag{784}$$

With this, we can use the uniform continuity of continuous functions over a compact set to place a bound on the oscillation of each subinterval—and thus a bound on the oscillation of the whole interval.

Theorem 6.12 (Continuous Functions are Riemann Integrable)

f continuous on $[a, b] \implies f$ is Riemann integrable on $[a, b]$.

Proof. If f is continuous, then by EVT it is bounded and uniformly continuous. Therefore we can take the evenly-partitioned intervals of $[a, b]$ and by uniform continuity, the oscillation tends to 0, and we are done.
Perhaps more explicitly, we wish to show that for all $\epsilon > 0$, there exists partition P s.t. $U(P, f) - L(P, f) < \epsilon$. Now let $\epsilon > 0$, and since it's uniformly continuous, take $\delta > 0$ s.t.

$$|x - y| < \delta \implies |f(x) - f(y)| < \frac{\epsilon}{2(b-a)} \tag{785}$$

Let $N \in \mathbb{N}$ be so large that $\frac{b-a}{N} < \delta$. Now consider the partition of $[a, b]$ given by $x_i = a + \frac{b-a}{N} i$ for $0 \leq i < N$. Intuitively, we want these subintervals to be so small that f will not deviate too widely. So it better be the case that $\frac{b-a}{N} < \delta$. So, we have

$$U(P, f) - L(P, f) = \sum_{i=1}^n \sup_{[x_{i-1}, x_i]} f(x)(x_i - x_{i-1}) - \sum_{i=1}^n \inf_{[x_{i-1}, x_i]} f(x)(x_i - x_{i-1}) \tag{786}$$

$$= \sum_{i=1}^n (\sup_{[x_{i-1}, x_i]} f(x) - \inf_{[x_{i-1}, x_i]} f(x))(x_i - x_{i-1}) \tag{787}$$

$$< \sum_{i=0}^{N-1} \frac{\epsilon}{2(b-a)}(x_i - x_{i-1}) \tag{788}$$

$$= \frac{\epsilon}{2(b-a)} \cdot (b - a) < \frac{\epsilon}{2} < \epsilon \tag{789}$$

We can actually make a stronger claim.

Corollary 6.13 (Integrability of Discontinuous Functions)

If a bounded function f on a closed interval $[a, b]$ is continuous everywhere except at a finite set of points, then $f \in \mathcal{R}[a, b]$.

Corollary 6.14 (Integrability of Monotonic Functions)

A bounded monotonic function on a closed interval is integrable on that interval.

Theorem 6.15 (Continuous Compositions of Integrable Functions are Integrable)

Let $f \in \mathcal{R}([a, b])$. Assume $\phi : \mathbb{R} \to \mathbb{R}$ is continuous. Then $\phi \circ f \in \mathcal{R}([a, b])$.

Proof. Since $f \in \mathcal{R}([a, b])$ is bounded, let $|f(x)| \leq M$ for all $x \in [a, b]$ for some $M \geq 0$. Now let $K = \sup_{t \in [-M, M]} \phi(t)$, which exists since $[-M, M]$ is compact and ϕ is continuious. ϕ is also uniformly continiuous on $[-M, M]$.
Now let $\epsilon > 0$. Then there exists a $\delta > 0$ s.t. $|t - s| < \delta \implies |\phi(t) - \phi(s)| < \epsilon$. Consequently,

$$|f(x) - f(y)| < \delta \implies |\phi(f(x)) - \phi(f(y))| < \epsilon \tag{790}$$

Since $f \in \mathcal{R}([a,b])$, we can find a partition P of $[a,b]$ s.t.

$$U(f,P) - L(f,P) < \delta^2 \implies \sum_{i=1}^{n-1} \big(\sup_{[x_i, x_{i+1}]} f - \inf_{[x_i,x_{i+1}]} f \big) \Delta x_i < \delta^2 \tag{791}$$

Let

$$A = \{ i \mid \sup_{[x_i, x_{i+1}]} f - \inf_{[x_i,x_{i+1}]} f < \delta \} \tag{792}$$

$$B = \{ i \mid \sup_{[x_i, x_{i+1}]} f - \inf_{[x_i,x_{i+1}]} f \geq \delta \} \tag{793}$$

Colloquially, we can think of A as the "good" intervals with small oscillations, and B as the "bad" intervals with larger oscillations. So,

$$\sum_{i \in B} \Delta x_i = \frac{1}{\delta} \sum_{i \in B} \delta \Delta x_i \leq \frac{1}{\delta} \sum_{i \in B} \operatorname*{osc}_{[x_i, x_{i+1}]} \Delta x_i < \frac{1}{\delta} \delta^2 = \delta \tag{794}$$

Now, compute

$$U(\phi(f), P) - L(\phi(f), P) = \sum_i \mathrm{osc}_{[x_i,x_{i+1}]}(\phi(f)) \Delta x_i \tag{795}$$

$$= \sum_{i \in A} \mathrm{osc}_{[x_i,x_{i+1}]}(\phi(f)) \Delta x_i + \sum_{i \in B} \mathrm{osc}_{[x_i,x_{i+1}]}(\phi(f)) \Delta x_i \tag{796}$$

In the good sets, if $f(x)$'s are within δ of each other, the oscillation by uniform continuity implies $\mathrm{osc}(\phi(f)) < \epsilon$. In the bad set, we have $\mathrm{osc}_{[x_i,x_{i+1}]}(\phi(f)) < 2K$, so the above can be bounded by

$$'' \leq \epsilon \sum_{i \in A} \Delta x_i + \sum_{i \in B} 2K \Delta x_i \tag{797}$$

$$\leq \epsilon(b-a) + 2K\delta \tag{798}$$

$$< \epsilon(b - a + 2K) \tag{799}$$

where the penultimate step is due to $\sum_{i \in B} \Delta x_i < \delta$.

Theorem 6.16 (Triangle Inequality for Riemann Integrals)

We have

$$\bigg| \int_a^b f \bigg| \leq \int_a^b |f| \tag{800}$$

Therefore, $|f| \in \mathcal{R}[a,b]$, implies $f \in \mathcal{R}[a,b]$.

Proof. $\phi(x) = |x|$ is continuous, so $\phi(f) \in \mathcal{R}$. Note that if $f \geq 0$, then $\int_a^b f \geq 0$. Consider $|f| - f$ and $|f| + f$, both ≥ 0. They are integrable as the image of f composed with continuous functions. So we have

$$\int |f| + f \geq 0 \implies \int |f| \geq - \int f \tag{801}$$

$$\int |f| - f \geq 0 \implies \int |f| \geq \int f \tag{802}$$

and so taking the maximum of the right hand side gives $\int |f| \geq | \int f|$.

However, contrary to intuition, f, g both integrable does not imply that $g \circ g$ is integrable. We present a counterexample.

Example 6.3 (Composition of Integrable Functions May Not be Integrable)

Consider the functions

$$|\text{sgn}|(x) \coloneqq \begin{cases} 1 & x \neq 0 \\ 0 & x = 0 \end{cases} \tag{803}$$

and the Riemann function

$$\mathcal{R}(x) \coloneqq \begin{cases} \frac{1}{n} & x = \frac{m}{n} \in \mathbb{Q}, \gcd(m, n) = 1 \\ 0 & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases} \tag{804}$$

We can see that $\mathcal{R}$ is continuous at all irrational points and discontinuous at all rational points except 0, meaning that it is integrable ($\mathbb{Q}$ has measure zero). Then, the composition of these two functions is precisely the Dirichlet function

$$\mathcal{D}(x) = |\text{sgn}| \circ \mathcal{R} \tag{805}$$

which is not integrable.

Theorem 6.17 (Products of Riemann Integrable Functions are Riemann Integrable)

If $f, g \in \mathcal{R}[a, b]$, then $fg \in \mathcal{R}[a, b]$.

Proof. A nice trick is that

$$fg = \frac{1}{4}\big((f+g)^2 - (f-g)^2\big) \tag{806}$$

which is in $\mathcal{R}([a, b])$ since the sum, difference, and squaring functions are all continuous, and hence the composition $\phi(f, g)$ is Riemann integrable.

6.5 The Fundamental Theorem of Calculus

Theorem 6.18 (Mean Value Theorem of the Integral)

Given $f \in \mathcal{R}[a, b]$, with

$$m = \inf_{x \in [a,b]} f(x), \qquad M = \sup_{x \in [a,b]} f(x) \tag{807}$$

Then

1. there exists a number $\mu \in [m, M]$ such that

$$\int_a^b f = \mu \cdot (b - a) \tag{808}$$

2. Furthermore, if $f \in C[a, b]$, it there exists a point $\xi \in [a, b]$ such that

$$\int_a^b f = f(\xi)(b - a) \tag{809}$$

Theorem 6.19 (Bonnet's Formula)

If $f, g \in \mathcal{R}[a, b]$ and g is a monotonic function on $[a, b]$, then there exists a point $\xi \in [a, b]$ such that

$$\int_a^b (f \cdot g)(x)\, dx = g(a) \int_a^\xi f(x)\, dx + g(b) \int_\xi^b f(x)\, dx \tag{810}$$

Let $f \in \mathcal{R}[a, b]$, and let us choose an $x \in [a, b]$ in order to construct the function

$$F(x) \coloneqq \int_a^x f(t)\, dt \tag{811}$$

which is called an integral with a variable upper limit. By doing this, we can "upgrade" a Riemann integrable function f to a continuous function F.

Theorem 6.20 (First Fundamental Theorem of Calculus)

Define $F : [a, b] \to \mathbb{R}$ by

$$F(x) \coloneqq \int_a^x f(t)\, dt \tag{812}$$

Then
1. F is continuous.
2. If F is continuous at x_0, then $F'(x_0) = f(x_0)$.

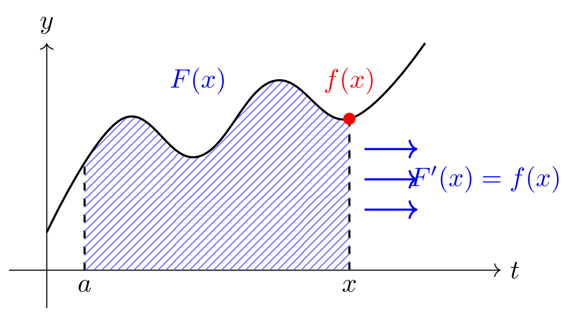


Figure 34: This theorem amazingly tells us that the rate at which the integral F is increasing at x (represented by the increasing area under the curve of f) is equal to the value of f at the point x itself!

Proof. Listed.
1. Since $f \in \mathcal{R}([a, b])$, let $M = \sup_{x \in [a,b]} |f(x)| < +\infty$. WLOG let $x, y \in [a, b]$ with $x < y$. Then, we can use the "trick" by writing the difference of F as an integral, which follows from linearity of the integral over an interval. So, we have

$$|F(x) - F(y)| = \left| \int_x^y f(t)\, dt \right| \leq \int_x^y |f(t)|\, dt \tag{813}$$

$$\leq \int_x^y M\, dt = M|y - x| \tag{814}$$

So given $\epsilon > 0$, we can take $\delta = \epsilon / M$ and F is continuous.

2. Now let's claim

$$\lim_{h \to 0} \frac{1}{h} \big(F(x_0 + h) - F(x_0) - f(x_0)h \big) = 0 \iff F'(x_0) = f(x_0) \tag{815}$$

since if the limit exists, we can add $f(x_0)$ to both sides. The term in the limit is

$$\frac{1}{h}\bigg|\int_a^{x_0+h} f(t)\,dt - \int_a^{x_0} f(t)\,dt - f(x_0)h\bigg| \leq \frac{1}{h}\bigg|\int_{x_0}^{x_0+h} f(t)\,dt - hf(x_0)\bigg| \tag{816}$$

Now we do a trick that is simple but powerful. Notice that $hf(x_0) = \int_{x_0}^{x_0+h} f(x_0)\,dt$, so we can join it with the integral.[a] So,

$$'' = \frac{1}{h}\bigg|\int_{x_0}^{x_0+h} f(t) - f(x_0)\,dt\bigg| \tag{817}$$

$$\leq \frac{1}{h}\int_{x_0}^{x_0+h} \big|f(t) - f(x_0)\big|\,dt \tag{818}$$

$$\leq \frac{1}{h}\int_{x_0}^{x_0+h} \sup_{t\in[x_0,x_0+h]} \big|f(t) - f(x_0)\big|\,dt \tag{819}$$

Note that the supremum term in the integral is just a number, so evaluating it and taking the limit as $h \to 0$ gives

$$\sup_{t\in[x_0,x_0+h]} |f(t) - f(x_0)| \to 0 \text{ as } h \to 0 \tag{820}$$

since f is continuous at x_0.

[a] Elgindi talked about how simple tricks can go a long way, e.g. the guy who was a master of Cauchy-Schwarz inequality.

Corollary 6.21

Every bounded function $f : [a, b] \longrightarrow \mathbb{R}$ on the closed interval $[a, b]$ and has only a finite number of points of discontinuity has a primitive, and every primitive of f on $[a, b]$ has the form

$$F(x) \coloneqq \int_a^x f(t)\,dt + c \tag{821}$$

where c is a constant.

Theorem 6.22 (Second Fundamental Theorem of Calculus)

Let f be a real-valued function on a closed interval $[a, b]$ with F any primitive of f on $[a, b]$. If f is Riemann-integrable (i.e. f bounded with finite points of Lebesgue measure zero) on $[a, b]$, then

$$\int_a^b f = F(b) - F(a) \tag{822}$$

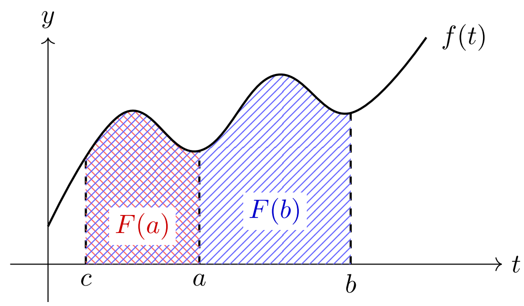


Figure 35: Graphical illustration of the Fundamental Theorem of Calculus, showing how the definite integral equals the difference of antiderivative values.

Proof. We already know that a bounded function on a closed interval having a finite number of discontinuities is integrable, and by the corollary, we are guaranteed an existence of a primitive $F(x)$ of the function f on $[a, b]$ with the form

$$F(x) \coloneqq \int_a^x f(t)\, dt + c \tag{823}$$

Setting $x = a$, we find that $c = F(a)$, and so

$$F(x) \coloneqq \int_a^x f(t)\, dt + F(a) \tag{824}$$

Evaluating F at $x = b$ gives

$$\int_a^b f(t)\, dt = F(b) - F(a) \tag{825}$$

Since we have established a bunch of derivative rules for polynomials, exponential/logarithmic, trigonometric functions, we can invoke the second fundamental theorem of calculus to derive the cheat sheet rules of integration.

Corollary 6.23 (Power Rule of Integration)

Corollary 6.24 (Integration of Exponential and Logarithmic Functions)

Corollary 6.25 (Integration of Trigonometric Functions)

Now let's do some computation.

Example 6.4 (Cannot Apply Fundamental Theorem for Nonintegrable Functions)

A common mistake that one can make is to apply the fundamental theorem for non-Riemann integrable functions. For example, one may consider evaluating

$$\int_{-1}^{1} x^{-2} = \left(-\frac{1}{x} \right) \bigg|_{-1}^{1} = -1 - 1 = -2 \tag{826}$$

But this is just incorrect. Later, we will find out how we can apply improper integration to compute this.

Theorem 6.26 (Integral Form of the Remainder)

If $f : E \longrightarrow \mathbb{R}$ has continuous derivatives up to order n on the closed interval $[a, x]$, then Taylor's formula holds

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \ldots + \frac{f^{(n-1)}(a)}{(n-1)!}(x-a)^{n-1} + r_{n-1}(a; x) \tag{827}$$

where

$$r_{n-1}(a; x) = \frac{1}{(n-1)!} \int_a^x f^{(n)}(t)(x-t)^{n-1}\, dt \tag{828}$$

This form is called **Taylor's formula with the integral form of the remainder**.

Proof. Using the 2nd fundamental theorem and the definite integration by parts formula, we can carry out the following chain of transformations, assuming continuity and differentiability when needed.

$$\begin{aligned}
f(x) - f(a) &= \int_a^x f'(t)\, dt \\
&= -\int_a^x f'(t)(x-t)'\, dt \\
&= -f'(t)(x-t)\big|_a^x + \int_a^x f''(t)(x-t)\, dt \\
&= f'(a)(x-a) - \frac{1}{2}\int_a^x f''(t)\big((x-t)^2\big)'\, dt \\
&= f'(x-a) - \frac{1}{2} f''(t)(x-t)^2\big|_a^x + \frac{1}{2}\int_a^x f'''(t)(x-t)^2\, dt \\
&= f'(a)(x-a) + \frac{1}{2} f''(a)(x-a)^2 - \frac{1}{2 \cdot 3}\int_a^x f'''(t)\big((x-t)^3\big)'\, dt \\
&= \ldots \\
&= f'(a)(x-a) + \ldots + \frac{1}{(n-1)!} f^{(n-1)}(a)(x-a)^{n-1} + r_{n-1}(a; x)
\end{aligned}$$

where $r_{n-1}(a; x)$ is given by the integral formula mentioned.

6.6 Change of Variable

We now show and prove the method what we call "u-substitution" for definite integration. We first start off with a change of variable induced by a monotonic continuous function, and then generalize it for C^1 functions.

Theorem 6.27 (Change of Variable for Monotonic Functions)

If
1. $f \in \mathcal{R}([a,b])$ and
2. φ is continuous and strictly increasing on $[a,b]$,

then
$$\int_a^b f(\varphi(x)) \cdot \varphi'(x)\,dx = \int_{\varphi(a)}^{\varphi(b)} f(u)\,du \tag{829}$$

Proof.

Theorem 6.28 (Change of Variable)

If
1. $f \in C([a,b])$[a], and
2. φ is continuously differentiable on $[a,b]$,

then
$$\int_a^b f(\varphi(x)) \cdot \varphi'(x)\,dx = \int_{\varphi(a)}^{\varphi(b)} f(u)\,du \tag{830}$$

[a]Note that we strengthen our assumption to continuous, not just Riemann integrable f!

Proof.

Note that we have made the additional assumption that continuity is needed.

Example 6.5 (Counterexample when f is Not Continuous)

If f is Riemann integrable, ...

6.7 Integration by Parts

Now a direct application of the fundamental theorem of calculus is the integration by parts. By the product rule of differentiation, we have
$$(u \cdot v)'(x) = (u' \cdot v)(x) + (u \cdot v')(x) \tag{831}$$
where by hypothesis, $u' \cdot v, u \cdot v'$ are continuous and hence integrable on $[a,b]$. Using the linearity of the integral and the 2nd fundamental theorem of calculus, we get
$$(u \cdot v)(x)\big|_a^b = \int_a^b (u' \cdot v) + \int_a^b (u \cdot v') \tag{832}$$

Theorem 6.29 (Integration by Parts)

Suppose $F, G : [a,b] \to \mathbb{R}$ are differentiable, with $F' = f, G' = g \in \mathcal{R}([a,b])$. Then
$$\int_a^b Fg = (FG)\big|_a^b - \int_a^b fG \tag{833}$$

Proof.

6.8 Improper Integrals

Note that by definition of the Riemann integral, it makes sense to talk about integrability over *bounded* functions over *closed and bounded intervals*. This is in a sense quite restrictive, since we cannot integrate over "singularities" where either the interval or the function is unbounded. We develop the tools of improper integration to deal with this problem; there are two types of improper integrals.

Definition 6.10 (Improper Integral of Unbounded Interval)

Let f be bounded on its domain.

1. If it is defined on $[a, +\infty)$ and is integrable on every closed interval $[a, b] \subset [a, +\infty)$, then we define
$$\int_a^{+\infty} f := \lim_{b \to +\infty} \int_a^b f \tag{834}$$
2. If it is defined on $(-\infty, b]$ and is integrable on every closed interval $[a, b] \subset (-\infty, b]$, then we define
$$\int_{-\infty}^b f := \lim_{a \to -\infty} \int_a^b f \tag{835}$$
3. If it is defined on $\mathbb{R}$ and is integrable on every closed interval $[a, b] \subset \mathbb{R}$, then we define[a]
$$\int_{-\infty}^{+\infty} f := \lim_{b \to +\infty} \lim_{a \to -\infty} \int_a^b f \tag{836}$$

If the limit exists, then we say that the integral **converges** and **diverges** otherwise.

[a]Small technicality is that we have to prove that this is independent of the order in which we take the limits.

Definition 6.11 (Improper Integral of Unbounded Function)

Let $f : [a, b] \to \mathbb{R}$ be unbounded at some point ω in its domain and is integrable on any closed interval $[c, d] \subset [a, b]$ not containing ω.

1. If $\omega = a$, then we define
$$\int_a^b f := \lim_{c \to a^+} \int_c^b f \tag{837}$$
2. If $\omega = b$, then we define
$$\int_a^b f := \lim_{d \to b^-} \int_a^d f \tag{838}$$
3. If $\omega \in (a, b)$, then we define
$$\int_a^b f = \lim_{d \to s^+} \int_a^d f + \lim_{c \to s^-} \int_c^b f \tag{839}$$

Example 6.6

Consider the function
$$f(x) = \frac{1}{\sqrt{x}} \tag{840}$$

1. It is not Riemann integrable over $(0, 1]$ since it is unbounded around 0. However, it is defined for all $[\epsilon, 1]$. So, we can define the improper integral
$$\int_0^1 x^{-1/2} := \lim_{\epsilon \to 0^+} \int_\epsilon^1 x^{1/2} = \lim_{\epsilon \to 0^+} (2x^{1/2})\big|_\epsilon^1 = \lim_{\epsilon \to 0^+} \big(2 - 2\sqrt{\epsilon}\big) = 2 \tag{841}$$

2. It is not Riemann integrable over $[1, +\infty)$ since the domain of integration is not bounded. However, it is defined for all $[1, h]$. So we can define the improper integral

$$\int_1^{+\infty} x^{-1/2} \coloneqq \lim_{h \to +\infty} \int_1^h x^{1/2} = \lim_{h \to +\infty} (2x^{1/2})\Big|_1^h = +\infty \tag{842}$$

So the improper Riemann integral diverges.

Example 6.7 (Vertical Asymptotes)

Consider the integral

$$\int_{-1}^{1} x^{-2} \tag{843}$$

This is not Riemann integrable since it is unbounded around 0.

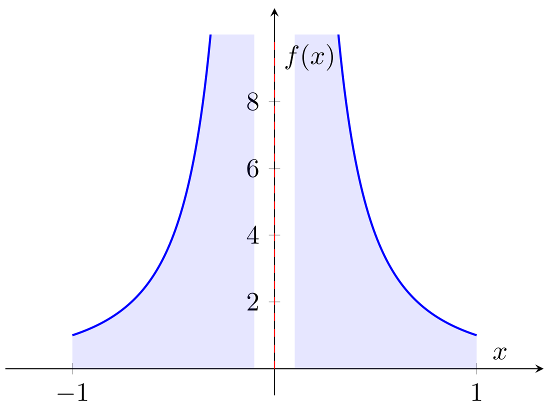


Figure 36: Plot of $f(x) = \frac{1}{x^2}$ showing the vertical asymptote at $x = 0$.

The correct strategy to compute this is to use improper integrals and apply the fundamental theorem to the restriction of the function that is Riemann integrable.

$$\int_{-1}^{1} x^{-2} = \int_{-1}^{0} x^{-2} + \int_{0}^{1} x^{-2} \tag{844}$$

$$= \lim_{\epsilon \to 0^-} \int_{-1}^{\epsilon} x^{-2} + \lim_{\epsilon \to 0^+} \int_{\epsilon}^{1} x^{-2} \tag{845}$$

$$= \lim_{\epsilon \to 0^-} \bigg(-\frac{1}{x} \bigg) \bigg|_{-1}^{\epsilon} + \lim_{\epsilon \to 0^+} \bigg(-\frac{1}{x} \bigg) \bigg|_{\epsilon}^{1} \tag{846}$$

$$= -2 + \lim_{\epsilon \to 0^-} -\frac{1}{\epsilon} + \lim_{\epsilon \to 0^+} \frac{1}{\epsilon} \tag{847}$$

$$= -2 + \infty + \infty = +\infty \tag{848}$$

Example 6.8 (Essential Discontinuities)

Consider the function and its derivative $f, f' : [0, 1] \to \mathbb{R}$ which we have derived here.

$$f(x) \coloneqq \begin{cases} x^2 \sin(1/x^2) & \text{if } x \in (0, 1] \\ 0 & \text{if } x = 0 \end{cases}, \quad f'(x) \coloneqq \begin{cases} 2x \sin(1/x) - \frac{2}{x} \cos \frac{1}{x^2} & \text{if } x \in (0, 1] \\ 0 & \text{if } x = 0 \end{cases} \tag{849}$$

Therefore, f is a primitive of f'. However, f' is not Riemann integrable since it is unbounded around 0. But for every $\epsilon > 0$, f' is continuous and bounded—and hence Riemann integrable—over $[\epsilon, 1]$, so now we can apply FTC.

$$\int_\epsilon^1 f' = f(1) - f(\epsilon) \tag{850}$$

By taking the limit as $\epsilon \to 0^+$ on both sides, we get

$$\int_0^1 f' \coloneqq \lim_{\epsilon \to 0^+} f' = f(1) - \lim_{\epsilon \to 0^+} f(\epsilon) = f(1) - f(0) \tag{851}$$

The fact that the fundamental theorem seems to work for "improper integrals" is a coincidence, and thus should be checked manually by explicitly computing the limits.

Lemma 6.30 (Properties of the Improper Integral)

Suppose f, g are functions defined on interval $[a, \omega)$ (without loss of generality, we let ω be the upper limit of integration) and integrable on every closed interval $[a, b] \subset [a, \omega)$. Suppose the improper integrals

$$\int_a^\omega f \text{ and } \int_a^\omega g \tag{852}$$

are well-defined.

1. For any $\lambda_1, \lambda_2 \in \mathbb{R}$ the function $(\lambda_1 f + \lambda_2 g)(x)$ is integrable in the improper sense on $[a, \omega)$ and

$$\int_a^\omega \lambda_1 f + \lambda_2 g = \lambda_1 \int_a^\omega f + \lambda_2 \int_a^\omega g \tag{853}$$

2. For any $c \in [a, \omega)$,

$$\int_a^\omega f = \int_a^c f + \int_c^\omega f \tag{854}$$

3. If $\varphi : [\alpha, \gamma) \longrightarrow [a, \omega)$ is a smooth strictly monotonic mapping with $\varphi(\alpha) = a$ and $\varphi(\beta) \to \omega$ as $\beta \to \gamma^-$, then the improper integral of the function $t \mapsto (f \circ \varphi)(t)\varphi'(t)$ over $[\alpha, \gamma)$ exists and

$$\int_a^\omega f(x)\, dx = \int_\alpha^\gamma (f \circ \varphi)(t)\varphi'(t)\, dt \tag{855}$$

Note that by definition, an improper integral $\int_a^\omega f \coloneqq \lim_{b \to \omega} \int_a^b f$ is a limit of the function $F(b) \coloneqq \int_a^b f$ as $b \to \omega$. This means that we can use the Cauchy criterion to determine the convergence of this limit, and hence, existence of this improper integral.

Theorem 6.31 (Cauchy Criterion for Convergence of an Improper Integral)

If the function $x \mapsto f(x)$ is defined on the interval $[a, \omega)$ and integrable on every closed interval $[a, b] \subset [a, \omega)$, then the integral

$$\int_a^\omega f \tag{856}$$

converges if and only if for every $\epsilon > 0$ there exists $B \in [a, \omega)$ such that the relation

$$\left| \int_{b_1}^{b_2} f \right| < \epsilon \tag{857}$$

holds for any $b_1, b_2 \in [a, \omega)$ satisfying $B < b_1$ and $B < b_2$.

Proof. We have

$$\int_{b_1}^{b_2} f = \int_a^{b_2} f - \int_a^{b_1} f = F(b_2) - F(b_1) \tag{858}$$

and therefore the condition is simply the Cauchy criterion for the existence of a limit for the function $\mathcal{F}(b)$ as $b \to \omega$.

Definition 6.12 (Absolute Convergence of an Improper Integral)

The improper integral

$$\int_a^{\omega} f \tag{859}$$

converges absolutely if the integral

$$\int_a^{\omega} |f| \tag{860}$$

converges. Clearly, the inequality

$$\left| \int_{b_1}^{b_2} f \right| \leq \left| \int_{b_1}^{b_2} |f| \right| \tag{861}$$

implies that if an improper integral converges absolutely, then it converges.

This study of absolute convergence reduces to the study of convergence of integrals of nonnegative functions. The following lemma is useful in determining convergence of such functions.

Lemma 6.32

Let there be a function f defined on interval $[a, \omega)$ that is also integrable over every closed interval $[a, b] \subset [a, \omega)$. If $f(x) \geq 0$ on $[a, \omega)$, then the improper integral

$$\int_a^{\omega} f \tag{862}$$

exists if and only if the function

$$\mathcal{F}(b) := \int_a^b f \tag{863}$$

is bounded on $[a, \omega)$.

Proof. It is clear that

$$\int_a^{\omega} f = \lim_{b \to \omega} \mathcal{F}(b) \tag{864}$$

If $f(x) \geq 0$, then the function $\mathcal{F}(b)$ is nondecreasing on $[a, \omega)$ and therefore has a limit as $b \to \omega$ only if it is bounded (since every monotonically increasing sequence that is bounded always converges).

This leads to the familiar integral test for convergence of a series.

Theorem 6.33 (Integral Test for Convergence of a Series)

If the function $x \mapsto f(x)$ is defined on the interval $[1, +\infty)$, nonnegative, nonincreasing, and integrable on each closed interval $[1, b] \subset [1, +\infty)$, then the series

$$\sum_{n=1}^{\infty} f(n) = f(1) + f(2) + \dots \tag{865}$$

and the integral

$$\int_a^{+\infty} f \tag{866}$$

either both converge or both diverge.

We can use the comparison test analogue to determine convergence of improper integrals.

Theorem 6.34 (Comparison Test for Convergence of Improper Integrals)

Suppose the functions $f(x), g(x)$ are defined on the interval $[a, \omega)$ and integrable on any closed interval $[a, b] \subset [a, \omega)$. If

$$0 \leq f \leq g \tag{867}$$

on $[a, \omega)$, then

$$\int_a^{\omega} g \text{ converges} \implies \int_a^{\omega} f \text{ converges} \tag{868}$$

and the inequality

$$\int_a^{\omega} f \leq \int_a^{\omega} g \tag{869}$$

holds. Also,

$$\int_a^{\omega} f \text{ diverges} \implies \int_a^{\omega} g \text{ diverges} \tag{870}$$

Example 6.9 (Sinc Function)

Consider the integral

$$\int_{-\infty}^{+\infty} \frac{\sin(x)}{x} \tag{871}$$

This is not Riemann integrable since the domain of integration is unbounded, though it is bounded. It is improperly Riemann integrable, however.

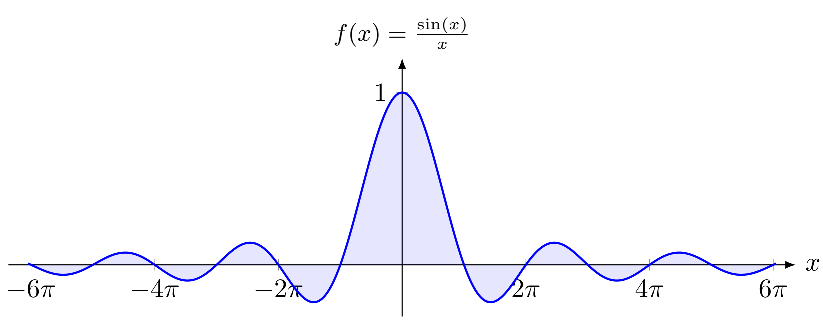


Figure 37: Plot of the function $f(x) = \frac{\sin(x)}{x}$. The integral is improper due to the unbounded domain $(-\infty, +\infty)$, despite the function itself being bounded.

We first show that $\int_0^{+\infty} \frac{\sin(x)}{x}\, dx$ converges. By the symmetry of the integrand (it is an even function), the total integral will be twice this value. For $h > 1$, we apply integration by parts with $u = \frac{1}{x}$ and $dv = \sin(x)\,dx$:

$$\int_1^h \frac{\sin(x)}{x}\,dx = \left[-\frac{\cos(x)}{x}\right]_1^h - \int_1^h \frac{\cos(x)}{x^2}\,dx \tag{872}$$

$$= \cos(1) - \frac{\cos(h)}{h} - \int_1^h \frac{\cos(x)}{x^2}\,dx \tag{873}$$

Taking the limit as $h \to \infty$, the term $\frac{\cos(h)}{h} \to 0$ because $|\cos(h)| \leq 1$. The integral $\int_1^\infty \frac{\cos(x)}{x^2}\,dx$ converges absolutely by the p-test, since $\left|\frac{\cos(x)}{x^2}\right| \leq \frac{1}{x^2}$ and $\int_1^\infty \frac{1}{x^2}\,dx$ converges. It turns out that

$$\int_{-\infty}^{+\infty} \frac{\sin(x)}{x}\,dx = \pi \tag{874}$$

6.9 Integration over Paths

Definition 6.13 (Integration For Vector Valued Functions)

A function $f : [a, b] \to R^d$ is Riemann integrable if $f = (f_1, \ldots, f_d)$ and each component $f_i : [a, b] \to \mathbb{R}$ is in $\mathcal{R}([a, b])$. The integral is defined

$$\int_a^b f = \left(\int_a^b f_1, \ldots, \int_a^b f_d\right) \tag{875}$$

Now since the codomain is $\mathbb{R}^d$, we can use the Euclidean norm $|v| \coloneqq \left(\sum_i v_i^2\right)^{1/2}$ on it.

Theorem 6.35

If $f \in \mathcal{R}([a, b], \mathbb{R}^d)$, then $|f| \in \mathcal{R}([a, b], \mathbb{R}^d)$ and

$$\left|\int f\right| \leq \int |f| \tag{876}$$

Proof. If $f \in \mathcal{R}([a, b], \mathbb{R}^d)$, then $f_i \in \mathcal{R}([a, b])$, and so

$$|f| = \sqrt{f_1^2 + \ldots f_d^2} \in \mathcal{R} \tag{877}$$

since $x \mapsto x^2$ and $x \mapsto \sqrt{x}$ are continuous. Now consider the vector $v = \int_a^b f$. Then

$$|v| = \left| \int_a^b f \right| \implies |v|^2 = \sum_{j=1}^d v_j^2 = \sum_{j=1}^d v_j \int_a^b f_j \tag{878}$$

$$= \int_a^b \sum_{j=1}^d v_j f_j \tag{879}$$

$$= \int_a^b \sum_{j=1}^d v_j f_j \tag{880}$$

$$= \int_a^b \langle v, f(t) \rangle \, dt \tag{881}$$

$$\leq \int_a^b |v| \, |f(t)| \, dt \tag{882}$$

and so

$$|v|^2 \leq |v| \cdot \int_a^b |f(t)| \, dt \implies |v| \leq \int_a^b |f(t)| \, dt \tag{883}$$

Definition 6.14 (Curve)

A **curve** is a function $\gamma : [0, 1] \to \mathbb{R}^d$.
1. If $\gamma(0) = \gamma(1)$, then it is a **closed curve**.
2. If γ is injective, then it is called a **simple curve**.

Curves are usually continuous but does not have to be.

Example 6.10

The curve can have different parameterizations and/or image. For example, the two are different curves with the image in $S^1 \subset \mathbb{R}^2$.

$$\gamma(t) = (\cos(2\pi t), \sin(2\pi t)) \tag{884}$$

$$\tilde{\gamma}(t) = (\cos(4\pi t), \sin(4\pi t)) \tag{885}$$

Definition 6.15 (Length of a Curve)

Given a curve $\gamma : [0, 1] \to \mathbb{R}^d$ and partition P of $[0, 1]$, let

$$\Lambda(\gamma, P) = \sum_{i=1}^N |\gamma(x_i) - \gamma(x_{i-1})| \tag{886}$$

i.e. the sum of the straight line distances between the curves. The **length** of the curve is defined as

$$\Lambda(\gamma) \coloneqq \sup_P \Lambda(\gamma, P) \tag{887}$$

If the length is finite, then we call this a **rectifiable curve**.

Example 6.11

Consider the curve given by

$$\gamma(t) = \left(t, t \sin \frac{1}{t}\right) \tag{888}$$

γ is continuous but $\gamma(t) < +\infty$.

For most continuous curves, this is not finite, but there is a sufficient condition for it to be finite.

Theorem 6.36 (C^1 Curves are Rectifiable)

If $\gamma : [0, 1] \to \mathbb{R}^d$ is continuously differentiable, then γ is rectifiable, and

$$\Lambda(\gamma) = \int_0^1 |\gamma'(t)| \, dt \tag{889}$$

Proof. Since $\gamma'(t)$ is continuous, then $|\gamma'(t)|$ is continuous and $|\gamma'(t)|$ is Riemann integrable. Now is P is any partition of $[0, 1]$, then

$$\begin{aligned} \Lambda(x, P) = \sum_{i=1}^n |\gamma(t_i) - \gamma(t_{i-1})| &= \sum_{i=1}^n \left| \int_{t_{i-1}}^{t_i} \gamma'(s) \, ds \right| && \text{(Fund. Thm. of Calc.)} \\ &\leq \sum_{i=1}^n \int_{t_{i-1}}^{t_i} |\gamma'(s)| \, ds && (890) \\ &= \int_{t_0}^{t_n} |\gamma'(s)| \, ds && (891) \end{aligned}$$

So we've proved one inequality. Now we prove the other. Let $\epsilon > 0$ be given. Then since $\gamma'(t)$ is continuous on compact $[0, 1]$, it must be uniformly continuous on $[0, 1]$. So $\exists \delta > 0$ s.t.

$$|s - t| < \delta \implies |\gamma'(s) - \gamma'(t)| < \epsilon \tag{892}$$

Now take a partition P of $[0, 1]$ s.t. $|t_i - t_{i-1}| < \delta$ for each $1 \leq i \leq N$. ...

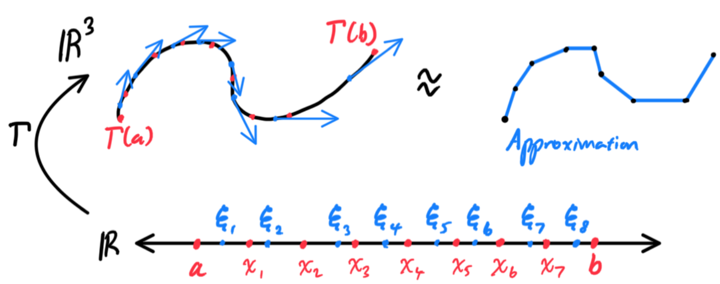


Figure 38: We can visualize this by partitioning the interval $[a, b]$ into the intervals Δ_i, each with point $\xi_i \in \Delta_i$. This would partition the path to $\Gamma(\Delta_i)$, each with points $\Gamma(\xi_i)$, and at each point $\Gamma(\xi_i)$, we can imagine the velocity vector of the curve. By taking the magnitude of this vector $\Gamma'(\xi_i)$, we multiply it by the length of the interval Δx_i to get one rectangle, creating an approximation for one partition of the path.

Corollary 6.37 (Length of the Graph of a C^1 Function)

An immediate result of this formula is the formula for the length of a graph of a function $f : [a, b] \longrightarrow \mathbb{R}$ in $\mathbb{R}^2$, by looking at the paramaterization $t \mapsto (t, f(t))$.

$$\Lambda(\gamma) = \int_a^b \sqrt{1 + (f'(t))^2}\, dt \tag{893}$$

The question on the effect of paramaterization on the integral now arises.

Definition 6.16 (Admissible Change of Parameter)

The path $\tilde{\Gamma} : [\alpha, \beta] \longrightarrow \mathbb{R}^3$ is obtained from $\Gamma : [a, b] \longrightarrow \mathbb{R}^3$ by an **admissible change of parameter** if there exists a smooth mapping

$$T : [\alpha, \beta] \longrightarrow [a, b] \tag{894}$$

such that $T(\alpha) = a, T(\beta) = b$, $T'(\tau) > 0$ (that is, the reparamaterization T is monotonic) on $[\alpha, \beta]$, and

$$\tilde{\Gamma} = \Gamma \circ T \tag{895}$$

The series of mappings can be represented with the following commutative diagram, where $I_{\alpha,\beta} = [\alpha, \beta] \subset \mathbb{R}$ and $I_{a,b} = [a, b] \subset \mathbb{R}$.

$$\begin{tikzcd} I_{\alpha,\beta} \arrow[r, "T"] \arrow[dr, "\tilde{\Gamma}"'] & I_{a,b} \arrow[d, "\Gamma"] \\ & \mathbb{R}^3 \end{tikzcd} \tag{896}$$

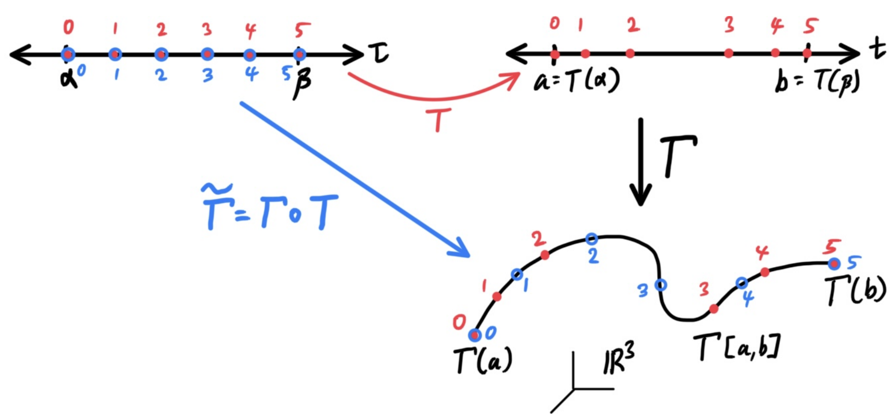


Figure 39: Note that the points are labeled $0, 1, 2, 3, 4, 5$ do not represent numerical values, but rather the order in which the points are paramaterized. We can see from this ordering that T is monotonic.

Theorem 6.38 (Invariance of Arclength Integral under Admissible Change of Parameters)

If a smooth path $\tilde{\Gamma} : [\alpha, \beta] \longrightarrow \mathbb{R}^3$ is obtained from a smooth path $\Gamma : [a, b] \longrightarrow \mathbb{R}^3$ by an admissible change of parameter, then the lengths of the two paths are equal. That is, a

$$\int_a^b |\Gamma'(t)|\, dt = \int_\alpha^\beta |\tilde{\Gamma}'(t)|\, dt := \int_\alpha^\beta |(\Gamma \circ T)'(t)|\, dt \tag{897}$$

6.10 Exercises

Example 6.12

Consider the space $X = C([a,b])$. Define $d : X \times X \to \mathbb{R}_0^+$ as

$$d(f,g) \coloneqq \int_a^b |f - g| \tag{898}$$

Then d is a metric. Note that in $\mathcal{R}([a,b])$, it is *not* a metric since $d(f,g) = 0 \not\Longrightarrow f = g$. Consider two functions that are different in 1 point.

Exercise 6.1 (Rudin 6.1)

Suppose α increases on $[a,b]$, $a \leq x_0 \leq b$, α is continuous at x_0, $f(x_0) = 1$, and $f(x) = 0$ if $x \neq x_0$. Prove that $f \in \mathscr{R}(\alpha)$ and that $\int f\, d\alpha = 0$.

Solution.

Exercise 6.2 (Rudin 6.2)

Suppose $f \geq 0$, f is continuous on $[a,b]$, and $\int_a^b f(x)\, dx = 0$. Prove that $f(x) = 0$ for all $x \in [a,b]$. (Compare this with Exercise 1.)

Solution.

Exercise 6.3 (Rudin 6.3)

Define three functions $\beta_1, \beta_2, \beta_3$ as follows: $\beta_j(x) = 0$ if $x < 0$, $\beta_j(x) = 1$ if $x > 0$ for $j = 1,2,3$; and $\beta_1(0) = 0, \beta_2(0) = 1, \beta_3(0) = \frac{1}{2}$. Let f be a bounded function on $[-1,1]$.

(a) Prove that $f \in \mathscr{R}(\beta_1)$ if and only if $f(0+) = f(0)$ and that then

$$\int f\, d\beta_1 = f(0). \tag{899}$$

(b) State and prove a similar result for β_2.

(c) Prove that $f \in \mathscr{R}(\beta_3)$ if and only if f is continuous at 0.

(d) If f is continuous at 0 prove that

$$\int f\, d\beta_1 = \int f\, d\beta_2 = \int f\, d\beta_3 = f(0). \tag{900}$$

Solution.

Exercise 6.4 (Rudin 6.4)

If $f(x) = 0$ for all irrational x, $f(x) = 1$ for all rational x, prove that $f \notin \mathscr{R}$ on $[a,b]$ for any $a < b$.

Solution.

Exercise 6.5 (Rudin 6.5)

Suppose f is a bounded real function on $[a, b]$, and $f^2 \in \mathscr{R}$ on $[a, b]$. Does it follow that $f \in \mathscr{R}$? Does the answer change if we assume that $f^3 \in \mathscr{R}$?

Solution.

Exercise 6.6 (Rudin 6.6)

Let P be the Cantor set constructed in Sec. 2.44. Let f be a bounded real function on $[0, 1]$ which is continuous at every point outside P. Prove that $f \in \mathscr{R}$ on $[0, 1]$. *Hint:* P can be covered by finitely many segments whose total length can be made as small as desired. Proceed as in Theorem 6.10.

Solution.

Exercise 6.7 (Rudin 6.7)

Suppose f is a real function on $(0, 1]$ and $f \in \mathscr{R}$ on $[c, 1]$ for every $c > 0$. Define

$$\int_0^1 f(x)\, dx = \lim_{c \to 0} \int_c^1 f(x)\, dx \tag{901}$$

if this limit exists (and is finite).

(a) If $f \in \mathscr{R}$ on $[0, 1]$, show that this definition of the integral agrees with the old one.

(b) Construct a function f such that the above limit exists, although it fails to exist with $|f|$ in place of f.

Solution.

Exercise 6.8 (Rudin 6.8)

Suppose $f \in \mathscr{R}$ on $[a, b]$ for every $b > a$ where a is fixed. Define

$$\int_a^\infty f(x)\, dx = \lim_{b \to \infty} \int_a^b f(x)\, dx \tag{902}$$

if this limit exists (and is finite). In that case, we say that the integral on the left converges. If it also converges after f has been replaced by $|f|$, it is said to converge absolutely.
Assume that $f(x) \geq 0$ and that f decreases monotonically on $[1, \infty)$. Prove that

$$\int_1^\infty f(x)\, dx \tag{903}$$

converges if and only if

$$\sum_{n=1}^\infty f(n) \tag{904}$$

converges. (This is the so-called "integral test" for convergence of series.)

Solution.

Exercise 6.9 (Rudin 6.9)

Show that integration by parts can sometimes be applied to the "improper" integrals defined in Exercises 7 and 8. (State appropriate hypotheses, formulate a theorem, and prove it.) For instance show that

$$\int_0^\infty \frac{\cos x}{1+x}\,dx = \int_0^\infty \frac{\sin x}{(1+x)^2}\,dx. \tag{905}$$

Show that one of these integrals converges *absolutely*, but that the other does not.

Solution.

Exercise 6.10 (Rudin 6.10)

Let p and q be positive real numbers such that

$$\frac{1}{p} + \frac{1}{q} = 1. \tag{906}$$

Prove the following statements.

(a) If $u \geq 0$ and $v \geq 0$, then

$$uv \leq \frac{u^p}{p} + \frac{v^q}{q}. \tag{907}$$

Equality holds if and only if $u^p = v^q$.

(b) If $f \in \mathscr{R}(\alpha)$, $g \in \mathscr{R}(\alpha)$, $f \geq 0$, $g \geq 0$, and

$$\int_a^b f^p\,d\alpha = 1 = \int_a^b g^q\,d\alpha, \tag{908}$$

then

$$\int_a^b fg\,d\alpha \leq 1. \tag{909}$$

(c) If f and g are complex functions in $\mathscr{R}(\alpha)$, then

$$\left|\int_a^b fg\,d\alpha\right| \leq \left\{\int_a^b |f|^p\,d\alpha\right\}^{1/p} \left\{\int_a^b |g|^q\,d\alpha\right\}^{1/q}. \tag{910}$$

This is Hölder's inequality. When $p = q = 2$ it is usually called the Schwarz inequality. (Note that Theorem 1.35 is a very special case of this.)

(d) Show that Hölder's inequality is also true for the "improper" integrals described in Exercises 7 and 8.

Solution.

Exercise 6.11 (Rudin 6.11)

Let α be a fixed increasing function on $[a, b]$. For $u \in \mathscr{R}(\alpha)$, define

$$\|u\|_2 = \left\{\int_a^b |u|^2\,d\alpha\right\}^{1/2}. \tag{911}$$

Suppose $f, g, h \in \mathscr{R}(\alpha)$, and prove the triangle inequality

$$\|f - h\|_2 \leq \|f - g\|_2 + \|g - h\|_2 \tag{912}$$

as a consequence of the Schwarz inequality, as in the proof of Theorem 1.37.

Solution.

Exercise 6.12 (Rudin 6.12)

With the notations of Exercise 11, suppose $f \in \mathscr{R}(\alpha)$ and $\epsilon > 0$. Prove that there exists a continuous function g on $[a, b]$ such that $\|f - g\|_2 < \epsilon$.
Hint: Let $P = \{x_0, \ldots, x_n\}$ be a suitable partition of $[a, b]$, define

$$g(t) = \frac{x_i - t}{\Delta x_i} f(x_{i-1}) + \frac{t - x_{i-1}}{\Delta x_i} f(x_i) \tag{913}$$

if $x_{i-1} \leq t \leq x_i$.

Solution.

Exercise 6.13 (Rudin 6.13)

Define

$$f(x) = \int_x^{x+1} \sin(t^2)\, dt. \tag{914}$$

(a) Prove that $|f(x)| < 1/x$ if $x > 0$. *Hint:* Put $t^2 = u$ and integrate by parts, to show that $f(x)$ is equal to

$$\frac{\cos(x^2)}{2x} - \frac{\cos[(x+1)^2]}{2(x+1)} - \int_{x^2}^{(x+1)^2} \frac{\cos u}{4u^{3/2}}\, du. \tag{915}$$

Replace $\cos u$ by -1.

(b) Prove that

$$2xf(x) = \cos(x^2) - \cos[(x+1)^2] + r(x) \tag{916}$$

where $|r(x)| < c/x$ and c is a constant.

(c) Find the upper and lower limits of $xf(x)$, as $x \to \infty$.

(d) Does $\int_0^\infty \sin(t^2)\, dt$ converge?

Solution.

Exercise 6.14 (Rudin 6.14)

Deal similarly with

$$f(x) = \int_x^{x+1} \sin(e^t)\, dt. \tag{917}$$

Show that

$$e^x |f(x)| < 2 \tag{918}$$

and that

$$e^x f(x) = \cos(e^x) - e^{-1}\cos(e^{x+1}) + r(x), \tag{919}$$

where $|r(x)| < Ce^{-x}$, for some constant C.

Solution.

Exercise 6.15 (Rudin 6.15)

Suppose f is a real, continuously differentiable function on $[a,b]$, $f(a) = f(b) = 0$, and

$$\int_a^b f^2(x)\,dx = 1. \tag{920}$$

Prove that

$$\int_a^b xf(x)f'(x)\,dx = -\frac{1}{2} \tag{921}$$

and that

$$\int_a^b [f'(x)]^2\,dx \cdot \int_a^b x^2 f^2(x)\,dx > \frac{1}{4}. \tag{922}$$

Solution.

Exercise 6.16 (Rudin 6.16)

For $1 < s < \infty$, define

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}. \tag{923}$$

(This is Riemann's zeta function, of great importance in the study of the distribution of prime numbers.)
Prove that

(a) $\zeta(s) = s\int_1^\infty \frac{[x]}{x^{s+1}}\,dx$

(b) $\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \frac{x-[x]}{x^{s+1}}\,dx$,

where $[x]$ denotes the greatest integer $\leq x$.
Prove that the integral in (b) converges for all $s > 0$.
Hint: To prove (a), compute the difference between the integral over $[1, N]$ and the Nth partial sum of the series that defines $\zeta(s)$.

Solution.

Exercise 6.17 (Rudin 6.17)

Suppose α increases monotonically on $[a,b]$, g is continuous, and $g(x) = G'(x)$ for $a \leq x \leq b$. Prove that

$$\int_a^b \alpha(x)g(x)\,dx = G(b)\alpha(b) - G(a)\alpha(a) - \int_a^b G\,d\alpha. \tag{924}$$

Hint: Take g real, without loss of generality. Given $P = \{x_0, x_1, \ldots, x_n\}$, choose $t_i \in (x_{i-1}, x_i)$ so that $g(t_i)\Delta x_i = G(x_i) - G(x_{i-1})$. Show that

$$\sum_{i=1}^{n} \alpha(x_i)g(t_i)\Delta x_i = G(b)\alpha(b) - G(a)\alpha(a) - \sum_{i=1}^{n} G(x_{i-1})\Delta\alpha_i. \tag{925}$$

Solution.

Exercise 6.18 (Rudin 6.18)

Let $\gamma_1, \gamma_2, \gamma_3$ be curves in the complex plane, defined on $[0, 2\pi]$ by

$$\gamma_1(t) = e^{it}, \quad \gamma_2(t) = e^{2it}, \quad \gamma_3(t) = e^{2\pi i t \sin(1/t)}. \tag{926}$$

Show that these three curves have the same range, that γ_1 and γ_2 are rectifiable, that the length of γ_1 is 2π, that the length of γ_2 is 4π, and that γ_3 is not rectifiable.

Solution.

Exercise 6.19 (Rudin 6.19)

Let γ_1 be a curve in R^k, defined on $[a, b]$; let ϕ be a continuous 1-1 mapping of $[c, d]$ onto $[a, b]$, such that $\phi(c) = a$; and define $\gamma_2(s) = \gamma_1(\phi(s))$. Prove that γ_2 is an arc, a closed curve, or a rectifiable curve if and only if the same is true of γ_1. Prove that γ_2 and γ_1 have the same length.

Solution.

7 Sequences of Functions

As daunting as the behavior of sequences of functions can be for the first-timer, most of the theory of sequences of real numbers is pretty much solved. In real applications, we are primarily interested not in the convergence of *functions*. For example, in differential equations, we are solving for some function, and we may approximate the solution with a sequence of simpler functions. In probability theory, random variables—as we will see later—are simply functions, and studying any stochastic process (e.g. Markov chains), is the same as studying sequences of functions. Therefore, let's first establish what it means for a function to converge.

Definition 7.1 (Pointwise Convergence)

Let $E \subset \mathbb{R}$ and $f, f_n : E \to \mathbb{R}$. We say that f_n **converges pointwise** to f, denoted $f_n \to f$, if

$$\lim_{n\to\infty} f_n(x) = f(x) \text{ for all } x \in E \tag{927}$$

Example 7.1

Let $f_n(x) = x/n$. Then $f_n \to 0$ pointwise, where 0 is the 0 function. This is true since for every fixed x, we can set n so large that $x/n < \epsilon$ for any ϵ.

7.1 Modes of Escape

The next question to ask is whether properties of functions are preserved under the limit operations. For example, if the f_n's are continuous, differentiable, or Riemann integrable, is the same true for the limit function f? What are the relations between f_n' and f' or $\int f_n$ and $\int f$? To say that a function f is continuous at a point x means

$$\lim_{t\to x} f(t) = f(x) \tag{928}$$

For functions, the analogous result is whether

$$\lim_{t\to x} \lim_{n\to\infty} f_n(t) = \lim_{n\to\infty} \lim_{t\to x} f_n(t) \tag{929}$$

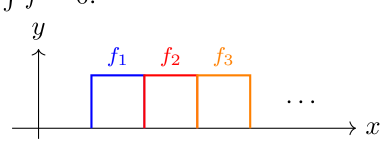
The answer is no, and there are a few general ways it may fail to do so.[11]

Example 7.2 (Escape to Horizontal Infinity)

The following function converges to 0 pointwise.

$$f_n : (0, +\infty) \to \mathbb{R}, \quad f_n := \mathbb{1}_{[n,n+1]} \tag{930}$$

However, note that $\int f_n = 1$, but $\int f = 0$.



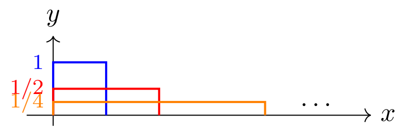
Example 7.3 (Escape to Width Infinity)

The following function converges to 0 pointwise.

$$f_n : (0, +\infty) \to \mathbb{R}, \quad f_n := \frac{1}{n}\mathbb{1}_{[0,n]} \tag{931}$$

[11] Thanks to terrytao.wordpress.com/2010/10/02/245a-notes-4-modes-of-convergence/.

However, note that $\int f_n = 1$, but $\int f = 0$.

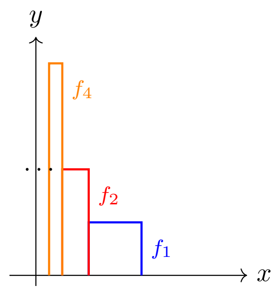


Example 7.4 (Escape to Vertical Infinity)

The following function converges to 0 pointwise.

$$f_n : (0, 1] \to \mathbb{R}, \quad f_n := n\mathbb{1}_{[\frac{1}{n}, \frac{2}{n}]} \tag{932}$$

However, note that $\int f_n = 1$, but $\int f = 0$.



Therefore, in general, the limits are not equal. Because derivatives, integrals, and continuity are simply limits or consequences of them, this implies that the properties are also violated!

Example 7.5 (Pointwise Limit of Continuous Functions is not Continuous)

Let $f_n(x) : [0, 1] \to \mathbb{R}$ defined $f_n(x) = x^n$. Then,

$$0 \leq x < 1 \implies x^n \to 0 \text{ as } n \to \infty \tag{933}$$
$$x = 1 \implies x^n \to 1 \text{ as } n \to \infty \tag{934}$$

So,

$$f_n \to f^*(x) := \begin{cases} 0 & \text{if } x < 1 \\ 1 & \text{if } x = 1 \end{cases} \tag{935}$$

Note that all f_n are continuous but f^* is discontinuous since

$$\lim_{x\to 1} \lim_{n\to\infty} f_n(x) = 0 \neq 1 = \lim_{n\to\infty} \lim_{x\to 1} f_n(x) \tag{936}$$

Example 7.6 (Pointwise Series Sum of Continuous Functions is not Continuous)

Consider

$$f_n(x) = \frac{x^2}{(1+x^2)^n} \tag{937}$$

and let

$$f(x) = \sum_{n=0}^{\infty} f_n(x) = \sum_{n=0}^{\infty} \frac{x^2}{(1+x^2)^n} \tag{938}$$

Since $f_n(0) = 0$, we have $f(0) = 0$. For $x \neq 0$, the series is a convergent geometric series with sum

$1 + x^2$. Therefore,

$$f(x) = \begin{cases} 0 & \text{if } x = 0 \\ 1 + x^2 & \text{if } x \neq 0 \end{cases} \tag{939}$$

Therefore the sum of a convergent series of continuous functions may not be continuous.

Example 7.7 (Cannot Exchange Integrals and Limits Under Pointwise Convergence)

Consider the function $f_n : [0, 1] \to \mathbb{R}$ defined by $f_n(0) = f_n(1/n) = 0$ and $f_n(1/2n) = 2n$, with everything else linearly interpolated. Then $f_n \to 0$ since it is constantly 0 at 0 and for every $x > 0$, there exists $1/N < x$ and so $f_n(x) = 0$ for all $n \geq N$. However, $\int_0^1 f_n(x)\,dx = 1$, so

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,dx \neq \int_0^1 \lim_{n\to\infty} f_n(x)\,dx \tag{940}$$

So integration is not continuous with respect to the topology induced by pointwise convergence.

The problem is not in the construction of the limits or the integral, but with the pointwise convergence. To get equality, we need to make a stronger assumption than simple existence of limits.

7.2 Conditions for Uniform Convergence

The problem seems to be that certain points may converge at a much slower rate than others. Therefore, we want to impose some sort of uniform condition, which says that the values of the function at every point must converge.

Definition 7.2 (Uniform Convergence)

Given $f_n : E \to \mathbb{R}$ of bounded functions, (f_n) is said to **converge uniformly** to a bounded function $f : E \to \mathbb{R}$ if $\forall \epsilon > 0$, there exists $N \in \mathbb{N}$ s.t.

$$n \geq N \implies |f_n(x) - f(x)| < \epsilon \text{ for all } x \in E \tag{941}$$

the "for all $x \in E$" is the uniform part, which is similar to uniform continuity.

Example 7.8

$f_n(x) = x^n$ does not converge uniformly to *any* function in $[0, 1]$. It suffices to find a sequence $(y_n) \subset [0, 1]$ s.t. $|f_n(y_n) - f(y_n)| \not\to 0$ as $n \to \infty$. Take $y_n = 1 - \frac{1}{n}$. Then $f_n(y_n) \to \frac{1}{e} \neq 0$, but $f(y_n) \to 1$.

Example 7.9

The triangle functions do not converge uniformly since f_n is unbounded, i.e. $f_n(\frac{1}{2n}) = 2n$, while $f_n(\frac{1}{n}) = 0$. So we can construct two sequences that

$$(1/2n) \to \infty, (1/n) \to 0 \text{ as } n \to \infty \tag{942}$$

Lemma 7.1 (Uniform Convergence Implies Pointwise Convergence)

Uniform convergence implies pointwise convergence.

Proof. Say $f_n \to f$ uniformly. Then for every $\epsilon > 0$, there exists a $N \in \mathbb{N}$ s.t.

$$n \geq N \implies |f_n(x) - f(x)| < \epsilon \text{ for all } x \in E \tag{943}$$

So just fix a point x, take any $\epsilon > 0$, and we have our δ due to uniform convergence.

Theorem 7.2 (Uniform Convergence iff Supremum of Difference Converges to 0)

This is an immediate consequence of the definition.

$$f_n \to f \text{ uniformly on } E \iff \lim_{n\to\infty} \sup_{x\in E} |f_n(x) - f(x)| = 0 \tag{944}$$

Proof. We can prove this with a bunch of iff statements. Let $f_n \to f$ uniformly, then this is equivalent to $\forall \epsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $n \geq N$ implies

$$|f_n(x) - f(n)| < \epsilon \quad \forall x \in E \tag{945}$$

$$\iff M_n \sup_{x\in E} |f_n(x) - f(x)| \leq \epsilon \tag{946}$$

$$\iff M_n \to 0 \text{ as } n \to 0 \tag{947}$$

which implies that $M_n \to 0$ as $n \to 0$.

Generally, to prove uniform convergence, you will need to find that $|f_n(x) - f(x)|$ is bounded by something that is independent of x, and it goes to 0 as $n \to \infty$. Here is an equivalent condition.

Definition 7.3 (Uniformly Cauchy)

A sequence $f_n : E \to \mathbb{R}$ is called **uniformly Cauchy** if $\forall \epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $\forall n, m \geq N$,

$$|f_n(x) - f_m(x)| < \epsilon \text{ for all } x \in E \tag{948}$$

Lemma 7.3 (Cauchy Criterion of Uniform Convergence)

(f_n) converges uniformly iff (f_n) is uniformly Cauchy.

Proof. We prove bidirectionally.

1. $(\to)$. Suppose (f_n) converges uniformly on E, and let f be the limit function. Then there exists $N \in \mathbb{N}$ s.t.

$$n \geq N \implies |f_n(x) - f(x)| < \frac{\epsilon}{2} \text{ for all } x \in E \tag{949}$$

Therefore,

$$n, m \geq N \implies |f_n(x) - f_m(x)| \leq |f_n(x) - f(x)| + |f(x) - f_m(x)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \tag{950}$$

for all $x \in E$.

2. $(\leftarrow)$. Suppose the uniform Cauchy criterion holds. For every $x \in E$, the sequence $(f_n(x))_n$ converges as a Cauchy sequence in $\mathbb{R}$. Call this limit $f(x)$. Thus the sequence (f_n) converges

pointwise to f.
Now let $\epsilon > 0$. Then, by uniform Cauchy, $\exists N \in \mathbb{N}$ s.t.

$$n, m \geq N \implies |f_n(x) - f_m(x)| < \epsilon \quad \forall x \in E \tag{951}$$

Now, if we fix n[a], we know that

$$m \geq N \implies |f_n(x) - f_m(x)| < \epsilon \tag{952}$$

Now we can take the limit as $m \to \infty$, and therefore the limit of the LHS must be bounded by that of the RHS.

$$\lim_{m \to \infty} |f_n(x) - f_m(x)| = |f_n(x) - f(x)| \leq \epsilon \tag{953}$$

Now this is true for all $n \geq N$, which basically is the definition of convergence.[b]

■

The wrong proof (that uniformly Cauchy implies uniform convergence) that I had in my first attempt was that I tried to directly bound the distances using some variant of the triangle inequality, like

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f(x)| + |f(x) - f_m(x)| \tag{954}$$

However, this doesn't really help since this is an *upper bound*. Perhaps we can use the reverse triangle inequality.

$$||f_n(x)| - |f_m(x)|| \leq |f_n(x) - f_m(x)| \tag{955}$$

but again, this inequality lacks $f(x)$ entirely.

[a]This is a classic trick used to convert Cauchy convergence to regular convergence. What we want to do is that since given some x_n, the rest of the points x_m must also be close to x_n, so the limit of the x_m must also be close to x_n, which is basically the definition of convergence.

[b]If you are dissatisfied with the $\leq$, just set $\epsilon/2$ to get strictly less than.

Example 7.10

We have uniform convergence

$$\frac{\sin(e^n x)}{n} \to 0 \text{ since } \left|\frac{\sin(e^n x)}{n}\right| \leq \frac{1}{n} \to 0 \tag{956}$$

Theorem 7.4 (Weierstrass M-Test)

Suppose (f_n) is a sequence of functions on E, and suppose

$$|f_n(x)| \leq M_n \text{ for all } x \in E \tag{957}$$

for each $n \in \mathbb{N}$. Then if $\sum_n M_n$ converges, $\sum_n f_n$ converges uniformly.

Proof. If $\sum_n M_n$ converges, then for arbitrary $\epsilon > 0$, we have

$$\left|\sum_{i=n}^{m} f_i(x)\right| \leq \sum_{i=n}^{m} M_n \leq \epsilon \tag{958}$$

provided $n, m \in \mathbb{N}$ are large enough. Therefore this is uniformly Cauchy, and so uniformly convergent.

Here is my wrong first attempt for a proof. Assume that $\sum_{k=1}^{\infty} M_k$ converges, and we wish to show that

$\forall \epsilon > 0$, $\exists N \in \mathbb{N}$ s.t.

$$n \geq N \implies \bigg| \underbrace{\sum_{k=1}^\infty f_k(x) - \sum_{k=1}^n f_k(x)}_{\text{not allowed}} \bigg| = \bigg| \sum_{k=n+1}^\infty f_k(x) \bigg| < \epsilon \tag{959}$$

The problem is that in our statement, we are assuming that we can actually subtract a finite term $\sum_{k=1}^n f_k(x)$ from a potentially divergent one: $\sum_{k=1}^\infty f_k(x)$. Therefore, we are assuming that this is convergent in the first place! This is exactly why we want to work with Cauchy sequences, which doesn't carry this kind of assumption, and so the sum from $i = n$ to m is guaranteed to be finite.

After this, the rest of the steps are fine, since I use the fact that absolutely convergent series are also convergent, and so

$$\bigg| \sum_{k=n+1}^\infty f_k(x) \bigg| \leq \sum_{k=n+1}^\infty |f_k(x)| \leq \sum_{k=n+1}^\infty M_k \tag{960}$$

Example 7.11 (Converse of Weierstrass M-test Not True)

The converse is clearly not true, since we can just take any uniformly convergent bounded sequence and set $M_n \to +\infty$ as $n \to \infty$.

The final theorem is a significant one. So far, we uniform convergence to prove that the limit of some sequence satisfies some property. In here, it's sort of the opposite.

Theorem 7.5 (Dini's Theorem)

Suppose K is compact, and

1. (f_n) is a sequence of continuous functions on K.
2. $f_n \to f$ pointwise on K.
3. f is continuous.[a]
4. $f_n \leq f_{n+1}$ for all $n \in \mathbb{N}$.

Then $f_n \to f$ uniformly on K.

[a]Note that the pointwise limit of continuous functions may not be continuous!

Proof. For convenience, let's define $g_n \coloneqq f - f_n \geq 0$, which is continuous on K. It suffices to prove that $g_n \to 0$ uniformly; that is, $\forall \epsilon > 0$, $\exists N \in \mathbb{N}$ s.t.

$$n \geq N \implies |g_n(x)| < \epsilon \tag{961}$$

Fix $\epsilon > 0$, and from compactness, we should already be thinking of trying to construct an open cover of K. Let

$$E_n \coloneqq \{x \in K \mid g_n(x) < \epsilon\} \tag{962}$$

Note three things.

1. First, E_n is open as the preimage of continuous g_n.
2. Second g_n is decreasing, so $g_n \geq g_{n+1}$, and so
$$E_n \subset E_{n+1} \quad \forall n \in \mathbb{N} \tag{963}$$
3. Third, $g_n \to 0$ pointwise, so at some point the E_n's must cover all of K.[a]
$$\bigcup_{n=1}^\infty E_n = K \tag{964}$$

We have constructed an open cover, and now we take the fact that K is compact to get a finite subcover $\{E_{n_k}\}_k$. But since the E_n's are nested, we can set $N = \max_k \{n_k\}$, which implies $E_N = K$. Therefore, as long as $n \geq N$, g_n is small enough so that it will map all of $E_N = K$ to a value less than ϵ. We have

$$n \geq N \implies g_n(x) = |g_n(x)| < \epsilon \tag{965}$$

[a] Remember that this is for *fixed* ϵ! This final point is not true if ϵ is not fixed.

Let's reflect on this proof and how we used the theorem's assumptions. First, monotonicity of f_n was needed to make sure that E_n was increasing. Second, compactness of K was needed to extract a finite subcover that we can exploit.

7.3 Consequences of Uniform Convergence

Now we will formalize and prove the manipulations that are unlocked by uniform convergence.

Theorem 7.6 (Limits are Swappable)

Suppose $f_n \to f$ uniformly over a set E of a metric space. Let $x \in E'$, and suppose that $\lim_{t \to x} f_n(t) = A_n$ for all n. Then,

1. (A_n) converges, and
2. we have

$$\lim_{t \to x} f(t) = \lim_{n \to \infty} A_n \iff \lim_{t \to x} \lim_{n \to \infty} f_n(t) = \lim_{n \to \infty} \lim_{t \to x} f_n(t) \tag{966}$$

Proof. Let $\epsilon > 0$, then by uniform continuity, there exists a $N \in \mathbb{N}$ s.t.

$$n, m \geq N \implies |f_n(t) - f_m(t)| < \epsilon \tag{967}$$

Now let $t \to x$, and since the limit exists, we have

$$n, m \geq N \implies |A_n - A_m| < \epsilon \tag{968}$$

and so (A_n) is by definition a Cauchy sequence in $\mathbb{R}$. Say that $A_n \to A$.
Now we wish to prove that $\lim_{t \to x} f(t) = A$. Take $\epsilon > 0$, and we wish to show that there exists some $\delta > 0$ s.t.

$$|t - x| < \delta \implies |f(t) - A| < \epsilon \tag{969}$$

Note that by the triangle inequality

$$|f(t) - A| \leq |f(t) - f_n(t)| + |f_n(t) - A_n| + |A_n - A| \tag{970}$$

Therefore, we can take $\frac{\epsilon}{3} > 0$.

1. By uniform convergence of $f_n \to f$, there exists a $N_1 \in \mathbb{N}$ s.t.

$$n \geq N_1 \implies |f_n(x) - f(x)| < \frac{\epsilon}{3} \text{ for all } x \in E \tag{971}$$

2. By convergence of $A_n \to A$, there exists a $N_2 \in \mathbb{N}$ s.t.

$$n \geq N_2 \implies |A_n - A| < \frac{\epsilon}{3} \tag{972}$$

3. Therefore, choose $N = \max\{N_1, N_2\}$, and for this N, by convergence of $f_n(t) \to A_n$ we can choose a δ s.t.

$$|x - t| < \delta \implies |f_n(t) - A_n| < \frac{\epsilon}{3} \tag{973}$$

This essentially bounds the three values, and so by choosing the δ in (3), we get

$$|t - x| < \delta \implies |f(t) - A| \leq \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon \tag{974}$$

Corollary 7.7 (Uniform Limits of Continuous Functions are Continuous)

If (f_n) is a sequence of continuous functions on E, and if $f_n \to f$ uniformly on E, then f is continuous on E.

Proof. Using the sequential definition of continuity, we claim that $\lim_{t \to x} f(t) = f(x)$. We know from the previous theorem that

$$\lim_{t \to x} \underbrace{\lim_{n \to \infty} f_n(t)}_{=f(t)} = \lim_{n \to \infty} \underbrace{\lim_{t \to x} f_n(t)}_{=f_n(x)} \tag{975}$$

where the LHS follows from (f_n) being uniformly convergent, which implies pointwise convergence, and the RHS follows from f_n being continuous.

However, the converse is not true. A sequence of continuous functions may converge to a continuous function though not uniformly. Now let's move onto integration.

Theorem 7.8 (Limits of Integrals are Integrals of Uniform Limits)

Let (f_n) be a sequence of Riemann integrable functions on $[a, b]$. If $f_n \to f$ uniformly on $[a, b]$, then $f \in \mathcal{R}([a, b])$ and

$$\lim_{n \to \infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx = \int_a^b \lim_{n \to \infty} f_n(x)\,dx \tag{976}$$

Proof. Let $\epsilon_n = \sup |f_n(x) - f(x)|$ with the supremum taken over $a \leq x \leq b$. Then,

$$f_n - \epsilon_n \leq f \leq f_n + \epsilon_n \tag{977}$$

so the upper and lower integrals of f satisfy

$$\int_a^b (f_n - \epsilon_n)\,dx \leq \int_{\bar{a}}^b f(x)\,dx \leq \int_a^{\bar{b}} f(x)\,dx \leq \int_a^b (f_n + \epsilon_n)\,dx \tag{978}$$

Hence, we have

$$0 \leq \int_a^{\bar{b}} f(x)\,dx - \int_{\bar{a}}^b f(x)\,dx \leq 2\epsilon_n(b-a) \tag{979}$$

and taking $n \to \infty$ sets $\epsilon_n \to 0$.

Corollary 7.9 (Series Function May be Integrated Term by Term)

If $f_n \in \mathcal{R}([a, b])$, and f is defined

$$f(x) \coloneqq \sum_{n=1}^\infty f_n(x) \tag{980}$$

with the series converging uniformly on $[a, b]$, then

$$\int_a^b f(x)\,dx = \sum_{n=1}^\infty \int_a^b f_n(x)\,dx \tag{981}$$

Now we move onto differentiation.

Theorem 7.10 (Limits of Derivatives)

Suppose (f_n) is a sequence of differentiable functions on $[a, b]$. If
1. there exists some $x_0 \in [a, b]$ such that $(f_n(x_0))_n$ converges, and
2. $(f_n') \to f'$[a] on $[a, b]$,

Then (f_n) converges uniformly to a function f, with

$$f'(x) = \lim_{n \to \infty} f_n'(x) \tag{982}$$

[a] At this point, f' is just notation since f isn't even defined.

Proof. Choose $N \in \mathbb{N}$ s.t. for $n, m \geq N$, the following hold

$$|f_n(x_0) - f_m(x_0)| < \frac{\epsilon}{2}, \qquad |f_n'(t) - f_m'(t)| < \frac{\epsilon}{2(b-a)} \tag{983}$$

which is possible due to convergence of $f_n(x_0)$ and uniform convergence of the derivative. Then applying the mean value theorem to the function $(f_n - f_m)$ gives

$$|f_n(x) - f_m(x) - f_n(t) + f_m(t)| = (f_n - f_m)'(c)|x - t| \tag{984}$$

for any $x, t \in [a, b]$ and $c \in (x, t)$. However, $(f_n - f_m)'$ is bounded, so

$$|f_n(x) - f_m(x) - f_n(t) + f_m(t)| \leq \frac{|x - t|\epsilon}{2(b-a)} \leq \frac{\epsilon}{2} \tag{985}$$

and therefore we can use the triangle inequality to get

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f_m(x) - f_n(x_0) + f_m(x_0)| + |f_n(x_0) - f_m(x_0)| \tag{986}$$

$$\leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \tag{987}$$

and so f_n is uniformly Cauchy $\iff$ (f_n) converges uniformly.
To show the equality of the limit, we fix a point $x \in [a, b]$ and define

$$\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x}, \qquad \phi(t) = \frac{f(t) - f(x)}{t - x} \tag{988}$$

for $t \in (a, b), t \neq x$. Then,

$$\lim_{t \to x} \phi_n(t) = f_n'(x) \tag{989}$$

and we can see from the above inequality that

$$|\phi_n(t) - \phi(t)| \leq \frac{\epsilon}{2(b-a)} \tag{990}$$

for $n, m \geq N$. So we can conclude

$$\phi(t) = \lim_{n \to \infty} \phi_n(t) \tag{991}$$

Proof. Here is an alternative shorter proof of the equality (but not uniform convergence of f_n!) where we assume $f_n \in C^1([a, b])$. Let's call the limit of f_n' to be g to avoid confusion. We know that since f_n' is continuous, it's integrable and by the fundamental theorem of calculus we have

$$f_n(x) - f_n(0) = \int_0^x f_n'(t)\, dx \tag{992}$$

Since $f_n' \to g$ uniformly and $f_n' \in C^0$, g is continuous and so we can define the function

$$f(x) := f(0) + \int_0^x g(t)\,dt \tag{993}$$

But we can see that by uniform convergence, we can swap integrals

$$\begin{aligned} f(x) - f(0) &= \int_0^x g(t)\,dt && (994)\\ &= \int_0^x \lim_{n\to\infty} f_n'(t)\,dt && (995)\\ &= \lim_{n\to\infty} \int_0^x f_n'(t)\,dt && (996)\\ &= \lim_{n\to\infty} f_n(x) - f_n(0) && (997) \end{aligned}$$

which implies that $f(x) - f(0) = \lim_{n\to\infty} f_n(x) - f_n(0)$ and so $f(x) = \lim_{n\to\infty} f_n(x)$. But since f' is continuous, the function f defined above is differentiable, with derivative $f'(x)$ to be whatever function is in the integral, i.e. $g(x)$. So

$$f'(x) = g(x) = \lim_{n\to\infty} f_n'(x) \tag{998}$$

7.4 Equicontinuous Families

Note that uniform convergence may not be met due to some counterexamples. In general, there are 3 ways that uniform convergence can fail to happen.

1. *Concentration.* Note that x^n as $n \to \infty$ almost converges except at one point.
2. *Translation.* Consider $f_n(x) = \sin(x - n)$. Then by increasing n we are shifting it to $+\infty$.
3. *Oscillation.* Consider $f_n(x) = \sin(nx)$. As n increases the function oscillates widely. This is sort of like the worst.[12]

We would like uniform convergence, so we want conditions to avoid lack of uniform convergence. Keep in mind to counterexamples. To avoid translation, work with compact space, or if not compact, have the functions decay uniformly. To avoid oscillation, we can bound the derivative, which is a restriction on each function $|f_n'(x)| \leq M$. The Cauchy criterion is too much. To avoid going to infinity, just bound f: $|f_n(x)| \leq M$ for all $n \in \mathbb{N}, x \in X$.

The bounding of derivatives can be a bit strong. We aren't always working with differentiable functions, so we introduce a similar concept.

Definition 7.4 (Equicontinuous Family)

A family of functions $\mathcal{F}$ on E is said to be **equicontinuous** if $\forall \epsilon > 0$ there exists a $\delta > 0$ s.t.

$$|x - y| < \delta \implies |f(x) - f(y)| < \epsilon \tag{999}$$

for all $x, y \in E, f \in \mathcal{F}$.

So this doesn't even depend on f. You can think of this as uniformly continuous for a class of functions that doesn't depend on f. The first class of equicontinuous functions you should know are those with bounded derivatives.

[12]It turns out that this is the same as (2) under the Fourier transform.

Lemma 7.11 (Functions with Bounded Derivatives Are Equicontinuous)

Fix $M \geq 0$. Then

$$\mathcal{F}_n \coloneqq \{f : [0,1] \to \mathbb{R} \mid |f'(x)| \leq M\} \tag{1000}$$

is an equicontinuous family.

Proof. For any $f \in \mathcal{F}$, the MVT $|f(x) - f(y)| = |f'(c)(x-y)|$ for some $c \in (x,y)$. But since $f'(c)$ is bounded by M, take $\delta = \epsilon/M$.

Example 7.12

$\mathcal{F} = \{\sin(nx)\}_{n\in\mathbb{N}}$ is not equicontinuous on $[0,1]$ since

$$\left|\sin\left(n\frac{\pi}{2n}\right) - \sin\left(n\frac{\pi}{n}\right)\right| = 1 \tag{1001}$$

for all n. So setting $x_n = \frac{\pi}{2n}, y_n = \frac{\pi}{n}$, we have $d(x_n, y_n) \to 0$ while $d(f(x_n), f(y_n)) \geq 1$. So this is not equicontinuous.

Definition 7.5 (Pointwise Bounded)

Given a sequence of functions (f_n) over E, we say the sequence is **pointwise bounded** if it satisfies each of the equivalent conditions.

1. There exists some function $\phi(x)$ s.t. $|f_n(x)| < \phi(x)$ for all $x \in E, n \in \mathbb{N}$.
2. For every $x \in E$, the sequence $(|f_n(x)|)_n$ is bounded.

Definition 7.6 (Uniformly Bounded)

Given a sequence of functions (f_n) over E, we say the sequence is **uniformly bounded** if there exists some M s.t. $|f_n(x)| \leq M$ for all $x \in E, n \in \mathbb{N}$.

Therefore, uniform boundedness is stronger, since this bound doesn't even depend on x.

Lemma 7.12 (Uniform Boundedness Implies Pointwise Boundedness)

Uniform boundedness implies pointwise boundedness.

Proof. Take $\phi(x) = M$.

We may wonder what the conditions are for the converse. The following theorem gives us those conditions.

Theorem 7.13 (Conditions for Uniform Boundedness)

If K is compact, with

1. f_n is continuous on K for each n
2. f_n pointwise bounded.
3. f_n equicontinuous on K

Then $\{f_n\}$ is uniformly bounded.

Proof.

If (f_n) is pointwise bounded on E and E_1 is a countable subset of E, it is always possible to find a subsequence (f_{n_k}) that converges for every $x \in E_1$. As an intuitive example, suppose

1. $(f_{2n}(q_1))$ converges
2. $(f_{3n}(q_2))$ converges
3. $(f_{5n}(q_3))$ converges
4. $(f_{7n}(q_4))$ converges

So combining the first two, we have that $(f_{6n}(q_i))$ converges for $i = 1, 2$. Continuing on, $(f_{30n}(q_i))$ converges for $i = 1, 2, 3$. But you can't do this infinitely. So if you want a single subsequence s.t. all the sequences converges, we can do

$$f_2, f_6, f_{30}, f_{210}, f_{2310}, \ldots \tag{1002}$$

Since

1. if you take out f_2, it is a subsequence of (f_{3n}) which converges for q_2, and
2. if you also take out f_6, it is a subsequence of (f_{6n}) which converges for q_1, q_2, and
3. if you also take out f_{30}, it is a subsequence of (f_{30n}) which converges for q_1, q_2, q_3

so $f_{n_k}(q_i)$ converges for all i. Now let's formalize this argument.

Lemma 7.14

Let (f_n) be a sequence of functions on $[0, 1]$ that's uniformly bounded. Let $\{q_m\}_{m=1}^\infty$ be a countable set of numbers in $[0, 1]$. Then $\exists$ a subsequence (f_{n_k}) for which $f_{n_k}(q_m)$ is convergent for all $m \in \mathbb{N}$.

Proof. Intuitively, if we find a sequence of functions, we want to look at each point—say 1—and look at $(f_n(1))_n$. $(f_n(1))$ is bounded and so contains a convergent subsequence $(f_{n_k}(1))_k$. Now with this subsequence, we look at $(f_{n_k}(0))_k$ which is bounded and therefore $(f_{n_{k_j}}(0))_j$ converges, and $(f_{n_{k_j}}(1))_j$ must converge as a subsequence of convergent $(f_{n_k}(1))_k$. Now do this for all q's, and we get a single subsequence that converges for all of them.
For ease of notation, let f_{ij} denote the jth term of the ith subsequence. Then there exists $(f_{n,1})_n$ s.t. $(f_{n,1}(q_1))_n$ converges. Take a subsequence $f_{n,2}$ of $f_{n,1}$ s.t. $(f_{n,2}(q_2))_n$ converges. Given $(f_{n,k})_n$, find a subsequence of it, called $(f_{n,k+1})_n$ for which $(f_{n,k+1}(q_{k+1}))_n$ converges. Now $(f_{n,n})_n$ is a subsequence of the original one (nth term of nth subsequence) for which $(f_{q,n})_n$ is eventually a subsequence of $(f_{n,j})_n$ for any fixed j.

Now the Ascolli's theorem gives us conditions to get rid of translation, oscillations, and infinity. To prove the second statement, we will need a lemma, so we state it now, along with providing a neat trick for constructing sequences.

Theorem 7.15 (Arzela-Ascolli's Theorem)

We claim the following.

1. If a sequence of continuous functions $f_n : [0, 1] \to \mathbb{R}$ (or more generally, over a compact set) converges uniformly, then they form an equicontinuous family.
2. If a sequence of functions $f_n : [0, 1] \to \mathbb{R}$ is equicontinuous and so uniformly bounded, then it has a uniformly convergent subsequence.

Proof. Assume (f_n) is uniformly convergent. Then it is uniformly Cauchy. To prove equicontinuity, given a $\epsilon > 0$ we need to find a $\delta > 0$ for all the functions. Since f_n is uniformly Cauchy, $\exists N$ s.t. if $n \geq N$, then

$$\sup_{x \in [0,1]} |f_n(x) - f_N(x)| < \epsilon/3 \tag{1003}$$

Consider the first N functions $f_1, \ldots, f_N$. They are all continuous on a compact set and so uniformly continuous. So for each f_i, there exists a δ_i s.t. $|x - y| < \delta_i \implies |f_i(x) - f_i(y)| < \epsilon$. So take $\delta = \frac{1}{3} \min_i \delta_i > 0$. So for all $1 \leq i \leq N$,

$$|x - y| < \delta \implies |f_i(x) - f_i(y)| < \epsilon/3 \tag{1004}$$

and for $n \geq N$,

$$|f_n(x) - f_n(y)| \leq \underbrace{|f_n(x) - f_N(x)}_{<\epsilon/3} + \underbrace{f_N(x) - f_N(y)}_{<\epsilon/3} + \underbrace{f_N(y) - f_n(y)}_{<\epsilon/3} < \epsilon \tag{1005}$$

For the second part, let $E = \mathbb{Q} \cap [0,1]$. It is a good thing that E is dense in $[0,1]$. Let (f_n) be an equicontinuous on $[0,1]$ and uniformly bounded. Due to the lemma, there exists a $(f_{n_k})_k$ so that the f_{n_k} converges pointwise on E (since E is countable). We will now use equicontinuity of $(f_{n_k})_k$ to prove it's uniformly Cauchy on $[0,1]$, which will imply that it's convergent. To make notation easier we will call $f_{n_k} = g_k$. Let $\epsilon > 0$. Since g_k is equicontinuous, $\exists \delta > 0$ s.t.

$$|x - y| < \delta \implies |g_k(x) - g_k(y)| < \epsilon \tag{1006}$$

Since $E = \{q_1, q_2, \ldots\}$ is dense in $[0,1]$, $\{B_\delta(q_i)\}_{i=1}^\infty$ is an open cover of $[0,1]$. Since $[0,1]$ is compact, there exists a finite subcover

$$[0,1] \subset \bigcup_{j=1}^N B_\delta(q_{i_j}) \tag{1007}$$

Since $(g_k(q_{i_j}))$ converges for each $1 \leq j \leq N$, there exists M_j s.t.

$$n, m \geq M_j \implies |g_n(q_{i_j} - g_m(q_{i_j}))| < \epsilon \tag{1008}$$

Take $M = \max_j M_j$. Now if $m, n \geq M$, given $x \in [0,1]$ $\exists q_i$ with $1 \leq i \leq N$ so that $x \in B_\delta(q_i)$, and so

$$|g_n(x) - g_m(x)| \leq \underbrace{|g_n(x) - g_n(q_i)|}_{<\epsilon} + \underbrace{|g_n(q_i) - g_m(q_i)|}_{<\epsilon} + \underbrace{|g_m(q_i) - g_m(x)|}_{<\epsilon} < 3\epsilon \tag{1009}$$

where the first and third inequalities come from equicontinuity, and the middle come from convergence on E. So by setting $\delta/3$ we are done.

Example 7.13

An important application is in the existence of minimizers/maximizers for optimization problems involving functions. To minimize

$$J(f) = \int_0^1 \sqrt{1 + f'(t)}\, dt \tag{1010}$$

is the length of the curve of $f : [0,1] \to \mathbb{R}$. To minimize the length of the curve, we must search over a set of functions. So to use EVT, you must know what the compact subsets of functions.

Ascolli's theorem exactly characterizes these compact subsets. These compact subsets of function spaces is the closure of equicontinuous functions.

Corollary 7.16

A set of functions $K \subset C([0,1])$ is compact iff it is, under the supremum metric $\sup_{x \in [0,1]}$,
1. closed
2. bounded
3. equicontinuous

The first two are needed for finite dimensions. The third condition is for function spaces.

Theorem 7.17 (Contraction Mapping Theorem)

Let (X, d) be a metric space with $J : X \to X$ and let there exist $c < 1$ s.t.

$$d(J(x), J(y)) \leq c d(x, y) \text{ for all } x, y \in X \tag{1011}$$

Then there exists a unique $x^* \in X$ s.t. $J(x^*) = x^*$.

7.5 The Stone-Weierstrass Theorem

The Stone-Weierstrass theorem is a bit more general, while the Weierstrass approximation theorem is for polynomials.

Theorem 7.18 (Weierstrass Approximation Theorem)

If $f \in C([a,b])$, there exists a sequence of polynomials (p_n) that converges uniformly to f.

Lemma 7.19

If $f : [0,1] \to \mathbb{R}$ is continuously differentiable on $[0,1]$, then

$$\lim_{n \to \infty} \int_0^1 f(x) \sin(nx)\, dx = 0 \tag{1012}$$

This means that as $n \to \infty$, the integral becomes small.

Proof.

Now we prove a stronger version.

Theorem 7.20

For every $f \in C([0,1])$,

$$\lim_{n \to \infty} \int_0^1 f(x)\, \sin(nx)\, dx = 0 \tag{1013}$$

Proof. Let $\epsilon > 0$. Then by the Weierstrass approximation theorem[a], there exists a polynomial $p : [0,1] \to \mathbb{R}$ for which

$$\sup_{x \in [0,1]} |p(x) - f(x)| < \epsilon \tag{1014}$$

Now consider

$$\left|\int_0^1 f(x)\,\sin(nx)\,dx\right| \leq \left|\int_0^1 p(x)\sin(nx)\,dx\right| + \left|\int_0^1 (f(x)-p(x))\sin(nx)\,dx\right| \tag{1015}$$

$$\leq \left|\int_0^1 p(x)\sin(nx)\,dx\right| + \epsilon \tag{1016}$$

$$\leq 2\epsilon \tag{1017}$$

where the first inequality is the triangle inequality, the second is due to the Weierstrass approximation theorem, and the third is due to $p(x)$ being infinitely differentiable, and so by the lemma above it is $\leq \epsilon$.

[a] aka, the set of polynomials is dense in the set of continuous functions with the supremum metric. Remember polynomial interpolation, which is for a finite number of points. This is a little different.

Theorem 7.21

Proof. Suppose for the sake of contradiction that $\sin(n_k x) \to g(x)$ uniformly for subsequence $(n_k)_k$. Then $g(x)$ must be continuous on $[0,1]$. Then

$$\int_a^b g(x)^2\,dx = \lim_{n\to\infty}\int_0^1 g(x)\sin(n_k x)\,dx = 0 \tag{1018}$$

due to the theorem, which implies that $g = 0$. But since $g(nx) = \pm 1$ for some x for all n, we have a contradiction.

7.6 Approximation of the Identity

Definition 7.7 (Approximation of the Identity)

A family of functions $\{\varphi_\epsilon\}$ parameterized by ϵ is called an **approximation of the identity** if

1. $\int_{-\infty}^{\infty} \varphi_\epsilon(y)\,dy = 1$
2. $\lim_{\epsilon\to 0}\int_{|y|>\delta}\varphi_\epsilon(y)\,dy = 0$ for all $\delta > 0$
3. $\varphi_\epsilon \geq 0$.[a]

[a] This condition is flexible, but it makes things a bit easier for now.

Example 7.14

Consider the functions f_ϵ satisfying $f(-\epsilon) = f(\epsilon) = 0$, $f(0) = 1/\epsilon$, and everything in between linearly interpolated. Then this is an approximation of the identity.

Example 7.15

Take any $\varphi \geq 0$ s.t. $\int_{-\infty}^{\infty}\varphi = 1$ and define

$$\varphi_\epsilon = \frac{1}{\epsilon}\varphi\left(\frac{x}{\epsilon}\right) \tag{1019}$$

which we can think of as squeezing the function horizontally to 0 and making the amplitude very large.

Then we see that
$$\int_{-\infty}^{\infty} \varphi_\epsilon(y)\,dy = \int_{-\infty}^{\infty} \frac{1}{\epsilon}\varphi\left(\frac{y}{\epsilon}\right) = \int_{-\infty}^{\infty} \varphi(x)\,dx = 1 \tag{1020}$$
Fix $\delta > 0$. Then
$$\int_{\delta}^{\infty} \varphi_\epsilon(x)\,dx = \int_{\delta}^{\infty} \frac{1}{\epsilon}\varphi\left(\frac{x}{\epsilon}\right)dx = \int_{\delta/\epsilon}^{\infty} \varphi(y)\,dy \to 0 \text{ as } \epsilon \to 0 \tag{1021}$$
Since δ is fixed, we have $\delta/\epsilon \to +\infty$ as $\epsilon \to 0$, and so this integral of the tail above converges to 0.

The approximation of the identity (AoI) has an amazing property.

Theorem 7.22

Let $\{\varphi_\epsilon\}$ be an AoI. Assume $f : \mathbb{R} \to \mathbb{R}$ is bounded and continuous. Then
$$\lim_{\epsilon \to 0} \int_{-\infty}^{\infty} \varphi_\epsilon(y)\, f(y)\,dy = f(0) \tag{1022}$$

Figure 40: The triangle has area 1. Now if you integrate the product of $\varphi_\epsilon(y)$ and $f(y)$, it's like taking the product and multiplying by f. But as $\epsilon \to 0$, the triangle's area is 1, and at the end you just multiply by $f(0)$.

Proof. We have
$$\begin{aligned}\left| \int_{-\infty}^{\infty} \varphi_\epsilon(y)\, f(y)\,dy - f(0)\right| &= \left| \int_{-\infty}^{\infty} \varphi_\epsilon(y)\,\big(f(y) - f(0)\big)\,dy\right| && (1023)\\ &\leq \left| \int_{-\delta}^{\delta} \varphi_\epsilon(y)\big(f(y) - f(0)\big)\,dy\right| + \left| \int_{|y|>\delta} \varphi_\epsilon(y)\big(f(y) - f(0)\big)\,dy\right| && (1024)\\ &\leq \sup_{y\in[-\delta,\delta]} |f(y) - f(0)| + 2M \int_{|y|>\delta} \varphi_\epsilon(y)\,dy && (1025)\end{aligned}$$
where the final step follows from the left integral is less than $\sup_{[-\delta,\delta]} |f(y) - f(0)|$, and for the right integral, we have $f(y) - f(0) \leq 2\sup|f| \leq 2M$. Since you don't know the limit exists, you take the limsup,
$$\limsup_{\epsilon\to 0} \left| \int_{-\infty}^{\infty} \varphi_\epsilon(y)\, f(y)\,dy - f(0)\right| \leq \sup_{y\in[-\delta,\delta]} |f(y) - f(0)| \to 0 \text{ as } \delta \to 0 \tag{1026}$$

Corollary 7.23 (Convolution)

If $\{f_\epsilon\}$ is an AoI with f bounded and continuous and $x \in \mathbb{R}$, we have
$$\lim_{\epsilon\to 0} \int_{-\infty}^{\infty} \varphi_\epsilon(y)\, f(x - y)\,dy = f(x) \tag{1027}$$
This is called the **convolution** of f with φ_ϵ.

Definition 7.8 (Dirac Delta Function)

$\varphi_\epsilon \to \delta_0$ as $\epsilon \to 0$, the Dirac delta function. This is a limit.

Theorem 7.24

Consider the functions

$$\varphi_n(x) = \begin{cases} c_n(1-x^2)^n & \text{if } x \in [-1,1] \\ 0 & \text{else} \end{cases}, \qquad c_n = \bigg(\int_{-1}^1 (1-x^2)^n \, dx \bigg)^{-1} \tag{1028}$$

Then,

$$\int_{-1}^1 \varphi_n(x) = \int_{-\infty}^\infty \varphi_n(x)\,dx = 1 \tag{1029}$$

so $\{\varphi_n\}$ is an approximation of the identity.

Proof. Note that c_n is chosen such that $\int_{-\infty}^\infty \varphi_n(x)\,dx = 1$ and $\varphi_n \geq 0$. We want to show

$$\int_{|x|>\delta} \varphi_n(x)\,dx \to 0 \text{ as } n \to \infty \text{ for all } \delta > 0 \tag{1030}$$

We claim that $c_n \leq 10\sqrt{n}$. since we wish to upper bound the multiplicative inverse of an integral, it suffices to lower bound the inverse—i.e. the integral itself.

$$\int_{-1}^1 (1-x^2)^n\,dx = 2\int_0^1 (1-x^2)^n\,dx \geq 2\int_0^{1/\sqrt{n}} (1-x^2)^n\,dx \tag{1031}$$

$$\geq 2\int_0^{1/\sqrt{n}} (1-nx^2)\,dx \tag{1032}$$

where the last inequality follows from the binomial inequality $(1-s)^n \geq 1 - sn$. Therefore, the final integral is now computable, so it equals

$$'' = 2\bigg(x - \frac{nx^3}{3}\bigg)\bigg|_0^{1/\sqrt{n}} = \frac{4}{3}\frac{1}{\sqrt{n}} \implies c_n \leq \bigg(\frac{4}{3}\frac{1}{\sqrt{n}}\big)^{-1} < \sqrt{n} \tag{1033}$$

Now we have

$$\int_{|x|>\delta} \varphi_n(x)\,dx = 2\int_\delta^{+\infty} \varphi_n(x)\,dx \tag{1034}$$

$$= 2\int_\delta^1 c_n(1-x^2)^n\,dx \tag{1035}$$

$$\leq 20\sqrt{n}\int_\delta^1 (1-x^2)^n\,dx \tag{1036}$$

$$\leq 20\sqrt{n}(1-\delta^2)^n \to 0 \text{ as } n \to \infty \tag{1037}$$

Note that we could have claim that the bound be $c_n \leq (10\sqrt{n})^{1000}$ and this would still be true.

Theorem 7.25 (Stone-Weierstrass Theorem)

Let $f \in C([a,b])$. Then $\forall \epsilon > 0$, $\exists$ a polynomial p s.t.

$$d(f,p) \coloneqq \sup_{x \in [a,b]} d\big(f(x), p(x)\big) < \epsilon \tag{1038}$$

This is equivalent to saying that $\mathbb{R}[x]$ is dense in $C([a,b])$, or that for any $f \in C([a,b])$, $\exists$ sequence (p_n) of polynomials s.t. $p_n \to p$ uniformly on $[a,b]$.

Proof. By translation and dilation, it suffices to take $[a,b] = [-1,1]$. This is because translation/dilations are automorphisms (?). It also suffices to consider only f for which $f(-1) > 0$ and $f(1) = k$ for some number k. This is because we can always replace f with $\tilde{f}$ defined by

$$\tilde{f}(x) \coloneqq f(x) - \left(\frac{(1+x)f(1) + (1-x)f(-1)}{2} \right) \tag{1039}$$

Let's extend f by 0 outside of $[-1,1]$. f is now a bounded continuous function $\mathbb{R}$ implying that f is uniformly continuous. Then we can take the integral.

$$\varphi_n(x) = \begin{cases} \frac{(1-x^2)^n}{\int (1-x^2)^n \, dx} & \text{if } x \in [-1,1] \\ 0 & \text{else} \end{cases} \tag{1040}$$

Since f is bounded, uniformly continuous on $\mathbb{R}$, and since $\{\varphi\}$ is an approximation of the identity,

$$\int_{-\infty}^{\infty} \varphi_n(t) f(x-t) \, dt \to f(x) \tag{1041}$$

and so p_n is defined on $[-\frac{1}{2}, \frac{1}{2}]$ with $p_n \to f$ uniformly.

Now we can use the exact same strategy to prove convergence of Fourier series.

Definition 7.9 (L^2 Inner Product)

The L^2 inner product is defined on $C([a,b])$ as

$$\langle f, g \rangle \coloneqq \frac{1}{b-a} \int_a^b f(x) \overline{g(x)} \, dx \tag{1042}$$

where f, g are complex valued.

We know that orthonormal bases behave nicely. We present one particularly important one.

Lemma 7.26

The functions $\{e^{inx}\}_{n \in \mathbb{Z}}$ are orthonormal in $C([0, 2\pi])$.

Proof. We have for $n, m \in \mathbb{Z}$

$$\langle e^{inx}, e^{imx} \rangle = \frac{1}{2\pi} \int_0^{2\pi} e^{inx} \overline{e^{imx}} \, dx = \frac{1}{2\pi} \int_0^{2\pi} e^{inx} e^{-imx} \, dx = \frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)x} \, dx \tag{1043}$$

So if $n = m$, then $\langle f, g \rangle = 1$. If not, then

$$\langle f, g \rangle = \frac{1}{2\pi i(n-m)} e^{i(n-m)x} \bigg|_0^{2\pi} = 0 \tag{1044}$$

and we are done.

Lemma 7.27

Define

$$\varphi_N(x) = \sum_{k=-N}^{N} e^{ikx} \tag{1045}$$

Then, the family $\{\varphi_N\}_{N=0}^{\infty}$ forms an AoI. This is called a *generalized AoI*.

With this, we can prove that any sufficiently smooth (i.e. C^1) functions can be approximated with Fourier series.

7.7 Exercises

Exercise 7.1 (Rudin 7.1)

Prove that every uniformly convergent sequence of bounded functions is uniformly bounded.

Solution.

Exercise 7.2 (Rudin 7.2)

If $\{f_n\}$ and $\{g_n\}$ converge uniformly on a set E, prove that $\{f_n + g_n\}$ converges uniformly on E. If, in addition, $\{f_n\}$ and $\{g_n\}$ are sequences of bounded functions, prove that $\{f_n g_n\}$ converges uniformly on E.

Solution.

Exercise 7.3 (Rudin 7.3)

Construct sequences $\{f_n\}$, $\{g_n\}$ which converge uniformly on some set E, but such that $\{f_n g_n\}$ does not converge uniformly on E (of course, $\{f_n g_n\}$ must converge on E).

Solution.

Exercise 7.4 (Rudin 7.4)

Consider

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{1 + n^2 x}. \tag{1046}$$

For what values of x does the series converge absolutely? On what intervals does it converge uniformly? On what intervals does it fail to converge uniformly? Is f continuous wherever the series converges? Is f bounded?

Solution.

Exercise 7.5 (Rudin 7.5)

Let

$$f_n(x) = \begin{cases} 0 & (x < \frac{1}{n+1}) \\ \sin^2 \frac{\pi}{x} & (\frac{1}{n+1} \leq x \leq \frac{1}{n}) \\ 0 & (\frac{1}{n} < x) \end{cases} \tag{1047}$$

Show that $\{f_n\}$ converges to a continuous function, but not uniformly. Use the series $\sum f_n$ to show that absolute convergence, even for all x, does not imply uniform convergence.

Solution.

Exercise 7.6 (Rudin 7.6)

Prove that the series

$$\sum_{n=1}^{\infty} (-1)^n \frac{x^2 + n}{n^2} \tag{1048}$$

converges uniformly in every bounded interval, but does not converge absolutely for any value of x.

Solution.

Exercise 7.7 (Rudin 7.7)

For $n = 1, 2, 3, \ldots, x$ real, put

$$f_n(x) = \frac{x}{1 + nx^2}. \tag{1049}$$

Show that $\{f_n\}$ converges uniformly to a function f, and that the equation

$$f'(x) = \lim_{n \to \infty} f_n'(x) \tag{1050}$$

is correct if $x \neq 0$, but false if $x = 0$.

Solution.

Exercise 7.8 (Rudin 7.8)

If

$$I(x) = \begin{cases} 0 & (x \leq 0) \\ 1 & (x > 0) \end{cases} \tag{1051}$$

if $\{x_n\}$ is a sequence of distinct points of (a, b), and if $\sum |c_n|$ converges, prove that the series

$$f(x) = \sum_{n=1}^{\infty} c_n I(x - x_n) \quad (a \leq x \leq b) \tag{1052}$$

converges uniformly, and that f is continuous for every $x \neq x_n$.

Solution.

Exercise 7.9 (Rudin 7.9)

Let $\{f_n\}$ be a sequence of continuous functions which converges uniformly to a function f on a set E. Prove that

$$\lim_{n\to\infty} f_n(x_n) = f(x) \tag{1053}$$

for every sequence of points $x_n \in E$ such that $x_n \to x$, and $x \in E$. Is the converse of this true?

Solution.

Exercise 7.10 (Rudin 7.10)

Letting (x) denote the fractional part of the real number x (see Exercise 16, Chap. 4, for the definition), consider the function

$$f(x) = \sum_{n=1}^{\infty} \frac{(nx)}{n^2} \quad (x \text{ real}). \tag{1054}$$

Find all discontinuities of f, and show that they form a countable dense set. Show that f is nevertheless Riemann-integrable on every bounded interval.

Solution.

Exercise 7.11 (Rudin 7.11)

Suppose $\{f_n\}$, $\{g_n\}$ are defined on E, and
(a) $\sum f_n$ has uniformly bounded partial sums;
(b) $g_n \to 0$ uniformly on E;
(c) $g_1(x) \geq g_2(x) \geq g_3(x) \geq \ldots$ for every $x \in E$.
Prove that $\sum f_n g_n$ converges uniformly on E. *Hint:* Compare with Theorem 3.42.

Solution.

Exercise 7.12 (Rudin 7.12)

Suppose g and $f_n (n = 1, 2, 3, \ldots)$ are defined on $(0, \infty)$, are Riemann-integrable on $[t, T]$ whenever $0 < t < T < \infty$, $|f_n| \leq g$, $f_n \to f$ uniformly on every compact subset of $(0, \infty)$, and

$$\int_0^\infty g(x)\, dx < \infty. \tag{1055}$$

Prove that

$$\lim_{n\to\infty} \int_0^\infty f_n(x)\, dx = \int_0^\infty f(x)\, dx. \tag{1056}$$

Solution.

Exercise 7.13 (Rudin 7.13)

Assume that $\{f_n\}$ is a sequence of monotonically increasing functions on R^1 with $0 \leq f_n(x) \leq 1$ for all x and all n.

(a) Prove that there is a function f and a sequence $\{n_k\}$ such that

$$f(x) = \lim_{k \to \infty} f_{n_k}(x) \tag{1057}$$

for every $x \in R^1$. (The existence of such a pointwise convergent subsequence is usually called Helly's selection theorem.)

(b) If, moreover, f is continuous, prove that $f_{n_k} \to f$ uniformly on compact sets.

Solution.

Exercise 7.14 (Rudin 7.14)

Let f be a continuous real function on R^1 with the following properties: $0 \leq f(t) \leq 1, f(t+2) = f(t)$ for every t, and

$$f(t) = \begin{cases} 0 & (0 \leq t \leq \frac{1}{3}) \\ 1 & (\frac{2}{3} \leq t \leq 1). \end{cases} \tag{1058}$$

Put $\Phi(t) = (x(t), y(t))$, where

$$x(t) = \sum_{n=1}^{\infty} 2^{-n} f(3^{2n-1} t), \quad y(t) = \sum_{n=1}^{\infty} 2^{-n} f(3^{2n} t). \tag{1059}$$

Prove that Φ is continuous and that Φ maps $I = [0, 1]$ onto the unit square $I^2 \subset R^2$. In fact, show that Φ maps the Cantor set onto I^2.

Solution.

Exercise 7.15 (Rudin 7.15)

Suppose f is a real continuous function on $R^1, f_n(t) = f(nt)$ for $n = 1, 2, 3, \ldots$, and $\{f_n\}$ is equicontinuous on $[0, 1]$. What conclusion can you draw about f?

Solution.

Exercise 7.16 (Rudin 7.16)

Suppose $\{f_n\}$ is an equicontinuous sequence of functions on a compact set K, and $\{f_n\}$ converges pointwise on K. Prove that $\{f_n\}$ converges uniformly on K.

Solution.

Exercise 7.17 (Rudin 7.17)

Define the notions of uniform convergence and equicontinuity for mappings into any metric space. Show that Theorems 7.9 and 7.12 are valid for mappings into any metric space, that Theorems 7.8 and 7.11 are valid for mappings into any complete metric space, and that Theorems 7.10, 7.16, 7.17, 7.24, and 7.25 hold for vector-valued functions, that is, for mappings into any R^k.

Solution.

Exercise 7.18 (Rudin 7.18)

Let $\{f_n\}$ be a uniformly bounded sequence of functions which are Riemann-integrable on $[a, b]$, and put

$$F_n(x) = \int_a^x f_n(t)\,dt \quad (a \leq x \leq b). \tag{1060}$$

Prove that there exists a subsequence $\{F_{n_k}\}$ which converges uniformly on $[a, b]$.

Solution.

Exercise 7.19 (Rudin 7.19)

Let K be a compact metric space, let S be a subset of $\mathscr{C}(K)$. Prove that S is compact (with respect to the metric defined in Section 7.14) if and only if S is uniformly closed, pointwise bounded, and equicontinuous.

Solution.

Exercise 7.20 (Rudin 7.20)

If f is continuous on $[0, 1]$ and if

$$\int_0^1 f(x)x^n\,dx = 0 \quad (n = 0, 1, 2, \dots), \tag{1061}$$

prove that $f(x) = 0$ on $[0, 1]$.

Solution.

Exercise 7.21 (Rudin 7.21)

Let K be the unit circle in the complex plane (i.e., the set of all z with $|z| = 1$), and let $\mathscr{A}$ be the algebra of all functions of the form

$$f(e^{i\theta}) = \sum_{n=0}^{N} c_n e^{in\theta} \quad (\theta \text{ real}). \tag{1062}$$

Then $\mathscr{A}$ separates points on K and $\mathscr{A}$ vanishes at no point of K, but nevertheless there are continuous functions on K which are not in the uniform closure of $\mathscr{A}$.

Solution.

Exercise 7.22 (Rudin 7.22)

Assume $f \in \mathscr{R}(\alpha)$ on $[a, b]$, and prove that there are polynomials P_n such that

$$\lim_{n \to \infty} \int_a^b |f - P_n|^2 \, d\alpha = 0. \tag{1063}$$

Solution.

Exercise 7.23 (Rudin 7.23)

Put $P_0 = 0$, and define, for $n = 0, 1, 2, \ldots,$

$$P_{n+1}(x) = P_n(x) + \frac{x^2 - P_n^2(x)}{2}. \tag{1064}$$

Prove that

$$\lim_{n \to \infty} P_n(x) = |x|, \tag{1065}$$

uniformly on $[-1, 1]$.

Solution.

Exercise 7.24 (Rudin 7.24)

Let X be a metric space, with metric d. Fix a point $a \in X$. Assign to each $p \in X$ the function f_p defined by

$$f_p(x) = d(x, p) - d(x, a) \quad (x \in X). \tag{1066}$$

Prove that $|f_p(x)| \leq d(a, p)$ for all $x \in X$, and that therefore $f_p \in \mathscr{C}(X)$. Prove that $\|f_p - f_q\| = d(p, q)$.

Solution.

Exercise 7.25 (Rudin 7.25)

Suppose ϕ is a continuous bounded real function in the strip defined by $0 \leq x \leq 1, -\infty < y < \infty$. Prove that the initial-value problem

$$y' = \phi(x, y), \quad y(0) = c \tag{1067}$$

has a solution.

Solution.

Exercise 7.26 (Rudin 7.26)

Prove an analogous existence theorem for the initial-value problem

$$\mathbf{y}' = \mathbf{\Phi}(x, \mathbf{y}), \quad \mathbf{y}(0) = \mathbf{c}, \tag{1068}$$

where now $\mathbf{c} \in R^k, \mathbf{y} \in R^k$, and $\mathbf{\Phi}$ is a continuous bounded mapping of the part of R^{k+1} defined by $0 \leq x \leq 1, \mathbf{y} \in R^k$ into R^k.

Solution.

8 Analytic Functions

Definition 8.1 (Analytic Functions)

We cannot assume that the Taylor series of an infinitely differentiable function converges to the function f within a neighborhood $U(x_0)$, nor can we assume that it converges at all! These types of "nice" functions that have a Taylor approximation within the neighborhood of x_0 are called **analytic functions** and can be written in the form

$$f(x) = f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \ldots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + r_n(x_0; x) \tag{1069}$$

where r is called the **remainder term**.

Example 8.1 (Infinitely Differentiable, Non-Analytic Function)

A example of a non-analytic function is

$$f(x) = \begin{cases} e^{-1/x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases} \tag{1070}$$

which looks like the following.

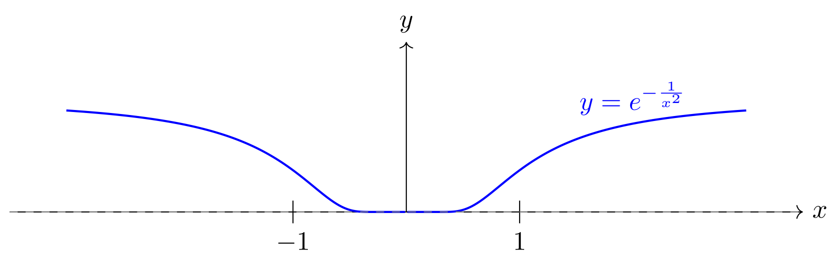


Figure 41: Graph of the function $y = e^{-\frac{1}{x^2}}$. This function equals 0 at $x = 0$ and approaches 1 as $|x|$ approaches infinity. One can verify that the derivative $f^{(k)}(0) = 0$ for all k and hence the Taylor series is identically equal to 0, while $f(x) \neq 0$ if $x \neq 0$.

The relationship between these different conditions is nicely summarized in the figure.

f infinitely differentiable at x_0 $\iff$ Taylor series of f exists at x_0

Taylor series converges at x_0

Taylor series converges to f at x_0 $\iff$ f is analytic

8.1 Exponential and Logarithmic Functions

8.2 Trigonometric Functions

8.3 Fourier Series

8.4 Gamma Function

8.5 Exercises

Exercise 8.1 (Rudin 8.1)

Define

$$f(x) = \begin{cases} e^{-1/x^2} & (x \neq 0), \\ 0 & (x = 0). \end{cases} \tag{1071}$$

Prove that f has derivatives of all orders at $x = 0$, and that $f^{(n)}(0) = 0$ for $n = 1, 2, 3, \ldots$.

Solution.

Exercise 8.2 (Rudin 8.2)

Let a_{ij} be the number in the ith row and jth column of the array

$$\begin{array}{cccccc} -1 & 0 & 0 & 0 & \cdots \\ \frac{1}{2} & -1 & 0 & 0 & \cdots \\ \frac{1}{4} & \frac{1}{2} & -1 & 0 & \cdots \\ \frac{1}{8} & \frac{1}{4} & \frac{1}{2} & -1 & \cdots \\ \cdots\cdots & & \cdots\cdots & & \cdots \end{array}$$

so that

$$a_{ij} = \begin{cases} 0 & (i < j), \\ -1 & (i = j), \\ 2^{j-i} & (i > j). \end{cases} \tag{1072}$$

Prove that

$$\sum_i \sum_j a_{ij} = -2, \quad \sum_j \sum_i a_{ij} = 0. \tag{1073}$$

Solution.

Exercise 8.3 (Rudin 8.3)

Prove that

$$\sum_i \sum_j a_{ij} = \sum_j \sum_i a_{ij} \tag{1074}$$

if $a_{ij} \geq 0$ for all i and j (the case $+\infty = +\infty$ may occur).

Solution.

Exercise 8.4 (Rudin 8.4)

Prove the following limit relations:
1. $\lim_{x \to 0} \frac{b^x - 1}{x} = \log b \quad (b > 0)$.
2. $\lim_{x \to 0} \frac{\log(1+x)}{x} = 1$.
3. $\lim_{x \to 0} (1+x)^{1/x} = e$.
4. $\lim_{n \to \infty} (1 + \frac{x}{n})^n = e^x$.

Solution.

Exercise 8.5 (Rudin 8.5)

Find the following limits:
1. $\lim_{x \to 0} \frac{e - (1+x)^{1/x}}{x}$.
2. $\lim_{n \to \infty} \frac{n}{\log n} [n^{1/n} - 1]$.
3. $\lim_{x \to 0} \frac{\tan x - x}{x(1 - \cos x)}$.
4. $\lim_{x \to 0} \frac{x - \sin x}{\tan x - x}$.

Solution.

Exercise 8.6 (Rudin 8.6)

Suppose $f(x)f(y) = f(x+y)$ for all real x and y.
1. Assuming that f is differentiable and not zero, prove that
$$f(x) = e^{cx} \tag{1075}$$
where c is a constant.
2. Prove the same thing, assuming only that f is continuous.

Solution.

Exercise 8.7 (Rudin 8.7)

If $0 < x < \frac{\pi}{2}$, prove that
$$\frac{2}{\pi} < \frac{\sin x}{x} < 1. \tag{1076}$$

Solution.

Exercise 8.8 (Rudin 8.8)

For $n = 0, 1, 2, \ldots$, and x real, prove that
$$|\sin nx| \leq n |\sin x|. \tag{1077}$$

Note that this inequality may be false for other values of n. For instance, $|\sin \frac{1}{2}\pi| > \frac{1}{2} |\sin \pi|$.

Solution.

Exercise 8.9 (Rudin 8.9)

1. Put $s_N = 1 + (\frac{1}{2}) + \cdots + (1/N)$. Prove that
$$\lim_{N\to\infty} (s_N - \log N) \tag{1078}$$
exists. (The limit, often denoted by γ, is called Euler's constant.)
2. Roughly how large must m be so that $N = 10^m$ satisfies $s_N > 100$?

Solution.

Exercise 8.10 (Rudin 8.10)

Prove that $\sum 1/p$ diverges; the sum extends over all primes.

Solution.

Exercise 8.11 (Rudin 8.11)

Suppose $f \in \mathscr{R}$ on $[0, A]$ for all $A < \infty$, and $f(x) \to 1$ as $x \to +\infty$. Prove that
$$\lim_{t\to 0} t \int_0^\infty e^{-tx} f(x)\, dx = 1 \quad (t > 0). \tag{1079}$$

Solution.

Exercise 8.12 (Rudin 8.12)

Suppose $0 < \delta < \pi$, $f(x) = 1$ if $|x| \leq \delta$, $f(x) = 0$ if $\delta < |x| \leq \pi$, and $f(x + 2\pi) = f(x)$ for all x.

1. Compute the Fourier coefficients of f.
2. Conclude that
$$\sum_{n=1}^\infty \frac{\sin(n\delta)}{n} = \frac{\pi - \delta}{2} \quad (0 < \delta < \pi). \tag{1080}$$
3. Deduce from Parseval's theorem that
$$\sum_{n=1}^\infty \frac{\sin^2(n\delta)}{n^2 \delta} = \frac{\pi - \delta}{2}. \tag{1081}$$
4. Let $\delta \to 0$ and prove that
$$\int_0^\infty \left(\frac{\sin x}{x}\right)^2 dx = \frac{\pi}{2}. \tag{1082}$$
5. Put $\delta = \pi/2$ in (c). What do you get?

Solution.

Exercise 8.13 (Rudin 8.13)

Put $f(x) = x$ if $0 \leq x < 2\pi$, and apply Parseval's theorem to conclude that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}. \tag{1083}$$

Solution.

Exercise 8.14 (Rudin 8.14)

If $f(x) = (\pi - |x|)^2$ on $[-\pi, \pi]$, prove that

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos nx \tag{1084}$$

and deduce that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \quad \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}. \tag{1085}$$

Solution.

Exercise 8.15 (Rudin 8.15)

With D_n as defined in (77), put

$$K_N(x) = \frac{1}{N+1} \sum_{n=0}^{N} D_n(x). \tag{1086}$$

Prove that

$$K_N(x) = \frac{1}{N+1} \cdot \frac{1 - \cos(N+1)x}{1 - \cos x} \tag{1087}$$

and that

1. $K_N \geq 0$,
2. $\frac{1}{2\pi} \int_{-\pi}^{\pi} K_N(x)\, dx = 1$,
3. $K_N(x) \leq \frac{1}{N+1} \cdot \frac{2}{1-\cos\delta}$ if $0 < \delta \leq |x| \leq \pi$.

If $s_N = s_N(f; x)$ is the Nth partial sum of the Fourier series of f, consider the arithmetic means

$$\sigma_N = \frac{s_0 + s_1 + \cdots + s_N}{N+1}. \tag{1088}$$

Prove that

$$\sigma_N(f; x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t) K_N(t)\, dt, \tag{1089}$$

and hence prove Fejér's theorem.

Solution.

Exercise 8.16 (Rudin 8.16)

Prove a pointwise version of Fejér's theorem: If $f \in \mathscr{R}$ and $f(x+), f(x-)$ exist for some x, then

$$\lim_{N \to \infty} \sigma_N(f; x) = \frac{1}{2}[f(x+) + f(x-)]. \tag{1090}$$

Solution.

Exercise 8.17 (Rudin 8.17)

Assume f is bounded and monotonic on $[-\pi, \pi]$, with Fourier coefficients c_n.

1. Use Exercise 17 of Chap. 6 to prove that $\{nc_n\}$ is a bounded sequence.
2. Combine (a) with Exercise 16 and with Exercise 14(e) of Chap. 3, to conclude that

$$\lim_{N \to \infty} s_N(f; x) = \frac{1}{2}[f(x+) + f(x-)] \tag{1091}$$

   for every x.
3. Assume only that $f \in \mathscr{R}$ on $[-\pi, \pi]$ and that f is monotonic in some segment $(\alpha, \beta) \subset [-\pi, \pi]$. Prove that the conclusion of (b) holds for every $x \in (\alpha, \beta)$.

Solution.

Exercise 8.18 (Rudin 8.18)

Define

$$f(x) = x^3 - \sin^2 x \tan x \tag{1092}$$
$$g(x) = 2x^2 - \sin^2 x - x \tan x. \tag{1093}$$

Find out, for each of these two functions, whether it is positive or negative for all $x \in (0, \pi/2)$, or whether it changes sign. Prove your answer.

Solution.

Exercise 8.19 (Rudin 8.19)

Suppose f is a continuous function on R^1, $f(x + 2\pi) = f(x)$, and α/π is irrational. Prove that

$$\lim_{N \to \infty} \frac{1}{N} \sum_{n=1}^{N} f(x + n\alpha) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t)\, dt \tag{1094}$$

for every x. *Hint: Do it first for* $f(x) = e^{ikx}$.

Solution.

Exercise 8.20 (Rudin 8.20)

The following simple computation yields a good approximation to Stirling's formula. For $m = 1, 2, 3, \ldots$, define

$$f(x) = (m + 1 - x) \log m + (x - m) \log(m + 1) \tag{1095}$$

if $m \leq x \leq m + 1$, and define

$$g(x) = \frac{x}{m} - 1 + \log m \tag{1096}$$

if $m - \frac{1}{2} \leq x < m + \frac{1}{2}$. Conclude that

$$\frac{7}{8} < \log(n!) - (n + \frac{1}{2}) \log n + n < 1 \tag{1097}$$

for $n = 2, 3, 4, \ldots$. Thus

$$e^{7/8} < \frac{n!}{(n/e)^n \sqrt{n}} < e. \tag{1098}$$

Solution.

Exercise 8.21 (Rudin 8.21)

Let $L_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_n(t)|\, dt \quad (n = 1, 2, 3, \ldots)$. Prove that there exists a constant $C > 0$ such that $L_n > C \log n$, or, more precisely, that the sequence

$$\left\{ L_n - \frac{4}{\pi^2} \log n \right\} \tag{1099}$$

is bounded.

Solution.

Exercise 8.22 (Rudin 8.22)

If α is real and $-1 < x < 1$, prove Newton's binomial theorem

$$(1 + x)^\alpha = 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha - 1) \cdots (\alpha - n + 1)}{n!} x^n. \tag{1100}$$

Show also that

$$(1 - x)^{-\alpha} = \sum_{n=0}^{\infty} \frac{\Gamma(n + \alpha)}{n! \Gamma(\alpha)} x^n \tag{1101}$$

if $-1 < x < 1$ and $\alpha > 0$.

Solution.

Exercise 8.23 (Rudin 8.23)

Let γ be a continuously differentiable closed curve in the complex plane, with parameter interval $[a, b]$, and assume that $\gamma(t) \neq 0$ for every $t \in [a, b]$. Define the index of γ to be

$$\text{Ind}(\gamma) = \frac{1}{2\pi i} \int_a^b \frac{\gamma'(t)}{\gamma(t)} \, dt. \tag{1102}$$

Prove that $\text{Ind}(\gamma)$ is always an integer.

Solution.

Exercise 8.24 (Rudin 8.24)

Let γ be as in Exercise 23, and assume in addition that the range of γ does not intersect the negative real axis. Prove that $\text{Ind}(\gamma) = 0$.

Solution.

Exercise 8.25 (Rudin 8.25)

Suppose γ_1 and γ_2 are curves as in Exercise 23, and $|\gamma_1(t) - \gamma_2(t)| < |\gamma_1(t)|$ for $a \leq t \leq b$. Prove that $\text{Ind}(\gamma_1) = \text{Ind}(\gamma_2)$.

Solution.

Exercise 8.26 (Rudin 8.26)

Let γ be a closed curve in the complex plane (not necessarily differentiable) with parameter interval $[0, 2\pi]$, such that $\gamma(t) \neq 0$ for every $t \in [0, 2\pi]$. Prove that there exists a common value $\text{Ind}(\gamma)$ and that the statements of Exercises 24 and 25 hold without any differentiability assumption.

Solution.

Exercise 8.27 (Rudin 8.27)

Let f be a continuous complex function defined in the complex plane. Suppose there is a positive integer n and a complex number $c \neq 0$ such that

$$\lim_{|z| \to \infty} z^{-n} f(z) = c. \tag{1103}$$

Prove that $f(z) = 0$ for at least one complex number z.

Solution.

Exercise 8.28 (Rudin 8.28)

Let $\bar{D}$ be the closed unit disc in the complex plane. Let g be a continuous mapping of $\bar{D}$ into the unit circle T. Prove that $g(z) = -z$ for at least one $z \in T$.

Solution.

Exercise 8.29 (Rudin 8.29)

Prove that every continuous mapping f of $\bar{D}$ into $\bar{D}$ has a fixed point in $\bar{D}$. (This is the 2-dimensional case of Brouwer's fixed-point theorem.)

Solution.

Exercise 8.30 (Rudin 8.30)

Use Stirling's formula to prove that

$$\lim_{x \to \infty} \frac{\Gamma(x+c)}{x^c \Gamma(x)} = 1 \tag{1104}$$

for every real constant c.

Solution.

Exercise 8.31 (Rudin 8.31)

In the proof of Theorem 7.26 it was shown that

$$\int_{-1}^{1} (1-x^2)^n \, dx \geq \frac{4}{3\sqrt{n}} \tag{1105}$$

for $n = 1, 2, 3, \ldots$. Use Theorem 8.20 and Exercise 30 to show the more precise result

$$\lim_{n \to \infty} \sqrt{n} \int_{-1}^{1} (1-x^2)^n \, dx = \sqrt{\pi}. \tag{1106}$$

Solution.